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The Structural Limits of Relational Ontology and the Ground It Cannot Reach

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Contents

List of Abbreviations

ETM	Eliminative Transcendental Method
L I	Level I: The Ground
L II	Level II: Relational Identity
L III	Level III: The Source of Encounter
L IV	Level IV: Relational Yield
L V	Level V: The Structure of the Relational Event
NLR	Non-Latent Register (Definition 10.3)
OSR	Ontic Structural Realism
R1–R8	Requirements One through Eight (Chapter 11)
RQM	Relational Quantum Mechanics
SIT	Structural Insufficiency Theorem (Theorem 10.1)

Abstract

Relational ontology across its major traditions —process philosophy, ontic structural realism, analytic relational metaphysics, and phenomenology —has identified relation as foundational but has not derived the structural conditions under which relation is possible at all. Those conditions are presupposed, not derived, by every tradition that has worked on the problem. This book presents a formal demonstration that this failure is not contingent but structurally necessary: given the foundational commitments of any tradition that begins within what this book terms the *non-latent register* —within a domain in which some feature (entity, occasion, structure, or lived relation) is already constituted —that tradition is formally foreclosed from reaching the structural ground it presupposes. This result is the Structural Insufficiency Theorem.

The book proceeds in four parts. Part I establishes the philosophical landscape and introduces the five structural levels at which any relational ontology must operate. Part II surveys thirteen traditions spanning the seventeenth through twenty-first centuries of relational thought —Whitehead, Spinoza, Bergson, and Heidegger from the classical period; ontic structural realism, relational quantum mechanics, Stengers, and Emirbayer from the contemporary landscape; Deleuze and Simon-

don from post-Kantian continental philosophy; Kant's transcendental framework as the foundational instance of the methodology the SIT claims to supersede; and Schaffer's priority monism together with Fine's essence-grounding as the strongest contemporary analytic metaphysics candidates —applying a diagnostic framework that locates each tradition's foundational commitment within the five-level hierarchy and demonstrates, in each case, that the tradition foundationally constitutes a non-latent feature. Part III presents and formally proves the Structural Insufficiency Theorem, together with its corollaries. Part IV derives the eight requirements that any account of the structural ground of relational possibility must satisfy, and demonstrates their joint satisfiability without providing the full positive derivation, which is the object of a companion investigation.

The method throughout is eliminative derivation: the systematic identification and removal of every assumption that can be shown to be derivable rather than primitive, until what remains is what any relational ontology presupposes but cannot itself supply.

Keywords: relational ontology; structural necessity; eliminative derivation; structural insufficiency; foundational constitution; latent register; process philosophy; ontic structural realism; analytic relational metaphysics; phenomenology; Whitehead; Spinoza; Bergson; Heidegger; Kant; Schaffer; Fine; analytic grounding; priority monism; essence-grounding; transcendental apperception; first philosophy; kawana'on

Preface

This book is an eliminative inquiry. It does not begin from a thesis to be defended, a tradition to be extended, or a position to be refined. It begins from a method: the systematic removal of every structural assumption that can be shown to be derivable rather than primitive, applied to the domain of relational ontology, until what remains cannot be removed without destroying the domain itself. The result of that removal — what no relational account can derive, because every relational account presupposes it — is the subject of this investigation. The method determines the subject; the subject could not have been identified by any other route.

There is a question that every tradition in relational ontology presupposes without asking. It is not a question about what relations do, or what they produce, or how they structure the domains they are invoked to explain. Those questions have been asked with extraordinary rigour across the seventeenth through twenty-first centuries, in the major traditions of Anglophone, continental, and process philosophy. The question that has not been asked is prior to all of them: what makes a relation possible? Not empirically, not contingently, but structurally — what must be the case about the constitution of reality for genuine relation to occur at all? That this question has not been answered — and has barely been formulated — is acknowledged from within the tradition itself. Wildman[1] notes that "there is persistent confusion in almost all literature about relational ontology because the key idea of relation remains unclear." The confusion is not terminological. It is structural: the conditions that make relation possible have been presupposed by every tradition that has worked on relational ontology, not derived from what those conditions must be.

This book is an investigation of why that question has not been asked, a formal demonstration that not asking it is not accidental but structurally necessary given the foundational commitments of every tradition that has worked on relational ontology, and a precise specification of what asking it — and answering it — would require. It does not provide the answer. It establishes, with formal rigour, that the answer is possible and that its precise structural shape can be specified in advance of its derivation.

The investigation proceeds by the eliminative transcendental method (hereafter ETM): the systematic identification and removal of every assumption that can be shown to be derivable rather than primitive, until what remains is what any relational ontology presupposes but cannot itself supply. Applied to the

[1]Wesley J. Wildman, "An Introduction to Relational Ontology," in *The Trinity and an Entangled World: Relationality in Physical Science and Theology*, edited by John Polkinghorne, States that "there is persistent confusion in almost all literature about relational ontology because the key idea of relation remains unclear" — an acknowledgement from within the tradition of the structural gap the present investigation formalises (Grand Rapids: Wm. B. Eerdmans, 2010), 55–73.

traditions surveyed in Part II, this method reveals a single pattern —[2] the Structural Insufficiency Theorem of Part III — that holds without exception across every tradition examined. Applied to the question of what a satisfying response to the theorem would require, it yields the eight formal requirements of Part IV.

A clarification on what the eliminative transcendental method produces, and what it does not. The method yields a *residue*: what cannot be removed from the relational account without structural collapse. A residue is not a selection. The traditions surveyed in Part II did not select their foundational commitments arbitrarily; each chose the most defensible starting point available within its methodological framework. The eliminative method does not adjudicate among those choices or rank them by philosophical merit. It asks a different question entirely: after every derivable assumption has been removed, what is left? The residue — the latent register — is not chosen; it is what the removal reveals. This distinction between residue and selection is the structural analogue of the Cartesian distinction between what survives doubt and what is adopted as a starting point:

[2]The sense in which the pattern is a *single* structural type, rather than a single numbered level of the five-level hierarchy at which every tradition fails, is clarified at the opening of Part II. The entry levels differ across traditions: Whitehead's foundational commitment falls at Level V, Spinoza's at Level I, and so on. What is uniform is not the level-number at which each tradition's horizon falls but the structural *type* of the failure: in every case, foundational constitution at level n generates a formal horizon below n and forecloses derivational access to the latent register. The SIT's uniformity claim is a claim about this structural type, not about its location on the numbered scale.

the cogito is not selected as a convenient axiom but disclosed as what remains when every deniable claim has been set aside. The present investigation makes no claim to Cartesian certainty; it claims only the same structural form — that the latent register is identified not by being chosen but by being what elimination cannot touch.

A phenomenologically oriented reader may object that the eliminative transcendental method is itself a founded formal procedure — that it already operates within a mode of rational givenness posterior to embodied, pre-thematic experience, and therefore cannot claim to reach what is structurally prior to all modes. The response is that the eliminative transcendental method as employed here is not a formal axiom system from which conclusions are derived. It is a coherence-driven structural procedure: the systematic identification of what, among a candidate set of structural features, cannot be shown to be derivable from others. This is a commitment to coherence as the condition of philosophical intelligibility, not a commitment to formalism as the deepest mode of access to reality. Husserl himself developed formal-structural analysis of ontological categories as an operative tool within phenomenological method, not as its replacement.[3] The eliminative procedure, similarly, clarifies

[3]See Horacio Banega, "Formal Ontology as an Operative Tool in the Theories of the Objects of the Life-World: Stumpf, Husserl and Ingarden," Archived at PhilArchive, rec/BANFOA-2. Argues that formal ontology — specifically Husserlian constituent ontology derived from the third *Logical Investigation* — is the operative theoretical framework pervasive in phenomenological philosophy, functioning as a domain-independent formal theory within, not as a replacement for, phenomenological method. Supports

structural boundaries by articulating where explanation must stop — a role compatible with, rather than contrary to, phenomenological investigation. A phenomenologist who accepts the relational thesis (that entities are constituted through their relations, not prior to them) is committed to every structural condition that the relational thesis requires in order to remain coherent. The eliminative method asks nothing more than this.[4] The structural horizons the eliminative method identifies are confirmed independently from within phenomenology's own procedure: Heidegger's fundamental ontology, following the phenomenological method rather than the eliminative one, arrives at the same structural boundary — the possibility dimension of Level V — that the eliminative derivation identifies as the outermost horizon of the non-latent register. This convergence from independent methodological directions indicates that the structural horizon is not an artefact of the eliminative method's theoretical starting point but a real structural boundary that different forms of inquiry reach by different routes.

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the claim that eliminative structural methods are not foreign to phenomenological inquiry, *Symposium* 16, no. 2 (2012): 64–88.

[4]See Anas Enoch, "Limit-Disclosure Philosophy: An Axiomatic Framework and Philosophical Essay" (Preprint. Archived at PhilArchive, rec/ENOLPA-2. Develops a Heidegger-adjacent ontology in which axiomatic negative constraints serve phenomenological clarity rather than replacing descriptive analysis. Objection 3 and its Reply establish that structural-formal methods can operate from within the Heideggerian horizon without claiming to transcend it. Used for both the Heidegger disclosure objection response and the phenomenological method objection response, 2025).

The use of formal definitions, propositions, and proofs throughout a work of this kind — a tradition-surveying monograph in systematic philosophy — requires a word of justification. Philosophical surveys typically proceed by interpretation, contextualisation, and discursive argument. The formal apparatus deployed here is not an imposition from outside that tradition but a structural requirement of the central claim. The Structural Insufficiency Theorem asserts a necessary and universal result about every framework satisfying specified conditions — not a pattern observed across most cases, or a tendency identifiable in hindsight, but a formal consequence of what foundational constitution is. A claim of that kind cannot be adequately supported by discursive generalisation across cases; it requires the precision of explicit definition, the transparency of stated proof conditions, and the verifiability of formal derivation. The apparatus exists to make the claim precise enough to be genuinely tested and, if sound, genuinely established.

Two conventions govern the book's presentation. Every formal result is stated in two registers simultaneously: a plain-language version that establishes the substance, and a precise technical formulation that establishes the rigour. Neither is a translation of the other; each is complete in its own right. A reader following only the plain-language versions will have grasped the argument; a reader checking only the formal versions will have verified it. Both registers are present throughout because the argument is genuinely both accessible and precise — not accessible by sacrificing precision, and not precise by sacrificing access. The two registers are fully independent throughout, with one qualified exception at the Horizon Lemma of Chapter 10, where the plain-language presentation relies on the term 'calibrated domain' — a term whose full content is established in the formal version. Plain-language readers should note that 'calibrated domain' means: the conceptual apparatus of a framework was built to operate on features at the entry level and above, and was not developed for the structural domain below it.

The traditions examined in Part II are treated with the intellectual seriousness their achievements demand. This book does not argue that Whitehead, Spinoza, Bergson, Heidegger, or the contemporary frameworks surveyed are wrong. It argues that each has identified something real, derived something genuine, and been prevented by the structure of its own foundational commitment from reaching what it has not seen. That prevention is not a failure. It is the structural consequence of beginning — as every tradition must begin — somewhere. The question this book pursues is what lies below every available somewhere.

A note on the architectural relationship between the diagnostic survey and the formal theorem. Part II's tradition-by-tradition analysis and Part III's Structural Insufficiency Theorem serve distinct and non-substitutable roles. The survey does not prove the theorem: formal necessity cannot be established by examples, however numerous. What the survey does is instantiate the theorem across real traditions developed with genuine rigour — showing that the formal conditions of the theorem's definition are satisfied by actual philosophical work, not merely by possible but unrealised frameworks. Conversely, the theorem does not make the survey redundant: it generalises a structural pattern whose plausibility and precision depend on the reader having seen that pattern operate with specificity across traditions of radically different character and methodology. The two parts are independent in derivation and mutually reinforcing in force. A reader who encounters the theorem before completing the survey should understand it as a general statement about a class of frameworks; a reader who has followed the survey in full will find the theorem's conditions already familiar as structural features present in every case examined.

A note on the selection of traditions. The traditions surveyed in Part II are not selected because they fail in the way the argument requires. The selection criterion is stated formally at the beginning of Chapter 9 and applies uniformly: each tradition surveyed is the most formally developed relational ontology within its methodological approach, among those traditions most likely to have explicitly addressed the structural conditions of relational possibility, and representative of one of the five major methodological directions through which the relational thesis has been pursued with formal precision. The criterion neither specifies nor presupposes the outcome of the diagnostic analysis: a tradition meeting it that satisfied all five structural requirements simultaneously would pass the diagnostic grid rather than failing it, and would constitute a decisive restriction of the Structural Insufficiency Theorem's scope. The fact that no surveyed tradition occupies this position is a diagnostic result, not a selection artefact.

A note on the book's self-application: Before Relation applies the Structural Insufficiency Theorem to every tradition in relational ontology and finds, in each case, that foundational constitution forecloses the latent register. The theorem applies to the present book as well. Before Relation foundationally constitutes the five-level diagnostic grid and is therefore subject to the foreclosure the theorem establishes. This is not a defect but an architectural consequence of the book's scope: it is a Stage I investigation — one that derives the structural conditions a non-foreclosed derivation must satisfy, not one that executes such a derivation. The self-application of the theorem is addressed formally in the body of the text (the remark following Theorem 10.1); it is noted here so that a reader alert to the issue finds it acknowledged before it might appear to have been missed.

A note on citation practice. Several citations in this book refer to preprints archived at PhilArchive or PhilPapers rather than to peer-reviewed publications. This occurs in two contexts. First, where a preprint is cited as part of the *convergent scholars* evidence — independent researchers who have reached structurally cognate diagnoses from different starting points — the citation is used to establish independent convergence, not as primary evidence for the SIT. The SIT is established by the formal proof and the thirteen-tradition survey; the convergent scholars confirm that independent inquiry reaches the same structural boundary. In this context a preprint is sufficient: what is claimed is convergence, and the preprint documents it. Second, where a preprint is the author's own companion or preparatory work, the citation is used for cross-reference and scholarly transparency rather than evidential support. Every primary claim of this book is supported either by the formal argument, by the diagnostic survey of peer-reviewed traditions, or by peer-reviewed secondary literature on those traditions. No essential premise rests on an unreviewed source.

H.C.S.

2026

A Note on the Mode of Inquiry

This investigation proceeds by elimination. It does not build a position from the ground up, defend a thesis against objections, or extend an existing framework in a new direction. It removes. At each stage, it asks whether a given structural assumption can be shown to be derivable from conditions more primitive than itself. Where the answer is yes, the assumption is set aside. The inquiry continues with what remains.

The eliminations in this volume are applied to the domain of relational ontology: the body of work that takes entities to be constituted through their relations rather than prior to them. The question is not whether that thesis is correct — the argument of Part I establishes that it is unavoidable — but what structural commitments it carries that no tradition working within it has yet discharged. Each tradition examined in Part II has made genuine philosophical progress. The diagnosis of failure is not a criticism of the quality of that work. It is a structural observation: a framework whose foundational commitment constitutes a feature at any level above the structural ground cannot, from within that framework, derive what lies below its own entry point. This is not a contingent limitation but a formal consequence of what it means to begin from a commitment. The failures documented here are necessary features of the frameworks that produce them, not defects that revision within those frameworks could remedy.

What the eliminations produce is a residue: what no relational account can derive, because every relational account presupposes it. A residue is not a foundation selected as a convenient starting point. It is what remains when every derivable assumption has been removed — disclosed by the elimination rather than chosen before it begins. The structural target of this investigation is that residue.

The reader following this inquiry will encounter two registers operating in parallel throughout: a discursive register that establishes the argument through interpretation and diagnosis, and a formal register that states the same results as definitions, theorems, and proofs. Both are complete in their own right. Neither is a translation of the other. A reader following only the discursive register will have grasped the argument; a reader checking only the formal register will have verified it.

One preparatory note on the character of the failures documented in Part II. Each tradition is found to fail not at the boundaries of its philosophical ambition but at the structural horizon set by its own entry-level commitment. The diagnosis is therefore not "this tradition attempted the wrong thing" but "this tradition attempted exactly the right thing and was prevented from completing it by the structural consequence of where it began." The distance between that horizon and the structural ground — the latent register that lies below every available starting point — is what Parts III and IV formally measure.

PART I

From General Ontology to Relational Ontology

CHAPTER 1

Why Everything Is Relational (And Why That Is Not Enough)

Something persists when everything else changes.
Whether that something is the same entity, or a successor,
or a convenient fiction — that is the question.

Before Relation

1.1 The Question That Precedes Every Answer

A note on register. This chapter proceeds by substantive philosophical argument, not by structural derivation. Its function within the investigation is diagnostic rather than formal: to establish, through the existing philosophical literature, that the relational thesis is both unavoidable and structurally incomplete — that committing to it incurs structural obligations no tradition has discharged. The formal derivation of those obligations is the work of Parts II through IV; this chapter establishes why the question is genuine and why its answer cannot come from within the traditions that have asked the nearest versions of it.

Something persists when everything else changes. The matter of a living body is replaced almost entirely across a decade; the beliefs and commitments formed at twenty are frequently unrecognisable at forty; the web of relations that constituted a person's social world at one period of life has, by another period, been remade by departure, loss, and encounter. Yet through all of this we say — and we must say, since the alternative is to abandon concepts on which thought and practice equally depend — that the same person endures. What endures? The question is not rhetorical. It is among the most ancient in philosophy, it recurs across every tradition that has taken the problem of identity seriously, and it has not been answered.

The difficulty is not one of incomplete information. We know a great deal about what changes: the matter, the beliefs, the social context, the physiological and neurological substrate. The difficulty is structural. It concerns what any answer to the identity question must presuppose before it can begin to answer. An account that grounds identity in the persistence of matter must assume that the matter at two times is materially connected — related across the interval in a way that makes it the same matter rather than numerically distinct quantities that happen to be qualitatively similar. An account that grounds identity in psychological continuity must assume that earlier and later mental states are connected by chains of causal and representational links — related in ways that traverse the interval rather than merely succeeding one another externally. An account that grounds identity in the persistence of biological form must assume that the form at later times is continuous with the form at earlier times — related through a developmental process that preserves something essential through what it transforms. In every case, at the foundational level of the account, something is assumed to be *related*. The relation is not the explanandum; it is the presupposition that makes the explanandum tractable.

This observation generalises with complete precision. It is not a feature of the identity problem alone, though the identity problem makes it vivid. It is a feature of every philosophical account that seeks to explain how entities endure, interact, or constitute one another. The structure of the problem is always the same: what appears to require explanation turns out, when examined with sufficient care, to presuppose that relation is already in place. The entity that bears properties across time, the substance that grounds predication, the subject that underlies change — each presupposes relation without deriving it.[1]

[1]The structural dependence of substance-theoretic accounts on unexplained relations has been noted across the tradition, though not always in the terms used here. Aristotle's account of primary substance as the subject of predication requires an explanation of the *kath' hauto* relation between substance and its essential properties that is nowhere derived within the *Metaphysics* but presupposed by its entire taxonomic apparatus; see *Metaphysics*, Z.4, 1030a10–27. Leibniz's monad, defined as that which contains in itself the ground of all its predicates, is perhaps the most explicit attempt to internalise relation entirely within substance, yet the pre-established harmony on which the monadology depends is itself a global relational structure that no individual monad can generate from its own nature alone. The point is not that these accounts fail on their own terms; it is that they presuppose what the present investigation is in the business of deriving.

What does it mean to say that relation is presupposed? It means, with full precision, that every account which takes entities as its starting point — which begins from the existence of things with determinate natures, properties, or essences and proceeds to ask how they interact or endure — has already accepted, without argument, that the structural conditions under which those entities can enter into relation are in place. The conditions are doing the work of the account, but they are doing it silently, from the background. The entity appears on the philosophical stage fully equipped for relation. The question of how it came to be so equipped — what structural conditions must obtain for any entity whatever to be capable of relational encounter — is never posed, because the answer has been silently assumed.

The present investigation poses that question. Its subject is not the identity of persons across time, not the persistence of substances through change, not the unity of consciousness across a stream of experience. These are downstream problems whose structural presuppositions this investigation examines. Its subject is the ground that underlies all of them: the structural conditions under which relation between entities — any entities, in any domain — is possible at all. The question can be stated simply, and in that simplicity its depth resides: *what must be the case for relation to begin?*

1.2 Individuation and Its Discontents

The question of what makes relation possible is inseparable from the question of what makes an entity the specific entity it is. These are not two questions but two framings of the same structural problem. To ask what makes relation possible is to ask what the entities that enter into relation must be; and to ask what entities must be is to ask what individuates them — what distinguishes one entity from another and from the undifferentiated field in which both are embedded. The philosophy of individuation is therefore not a neighbouring inquiry to the philosophy of relation. It is its structural ground.

Individuation is the process or condition by which a thing becomes or is the specific thing it is rather than another thing or nothing.[2] The problem of individuation is the problem of what grounds it: whether it is intrinsic to the entity, arising from properties or structure the entity possesses independently of its relations to other things; or whether it is relational, arising through the entity's connections to what surrounds it; or

[2]The term is used here in its broad ontological sense, encompassing both numerical individuation (what makes one thing distinct from another that is otherwise qualitatively identical) and qualitative individuation (what makes a thing the specific kind of thing it is rather than a different kind). The distinction between these two problems is drawn with precision in Lowe's four-category ontology, where individual substances are individuated by their instantiation of substantial kinds and their instantiation is in turn what distinguishes numerically distinct members of the same kind E. J. Lowe, *The Four-Category Ontology: A Metaphysical Foundation for Natural Science* (Oxford: Oxford University Press, 2006). The present investigation is not committed to Lowe's four-category framework; it draws on the distinction only to mark the scope of what individuation, as used here, is required to explain.

whether both dimensions are required, in some proportion that must itself be specified.

The intrinsic account — the view that entities are individuated by their own natures, independently of their relations to other things — confronts a formal difficulty that is not a contingent deficiency of any particular version of the view but a structural consequence of what intrinsic individuation requires. If an entity E_1 is individuated entirely by its intrinsic properties $\{P_1, P_2, \ldots, P_n\}$, and another entity E_2 is individuated entirely by its intrinsic properties $\{Q_1, Q_2, \ldots, Q_m\}$, then the question of whether E_1 and E_2 can enter into genuine relation — a relation in which what each contributes is partly determined by what the other is — cannot be answered from within either entity's intrinsic nature alone. The intrinsic properties of E_1 specify what E_1 is, not what E_1 can do *with* E_2; the latter depends on what E_2 is, which is precisely what the intrinsic account insulates from any contribution to E_1's determination. The intrinsic account, if consistently held, produces entities that are fully constituted before any encounter and therefore capable only of external relations — relations that leave their relata's natures unchanged.[3]

[3]The internal/external relation distinction is well-established in the literature; see Bradley's formulation of the problem of external relations in *Appearance and Reality* F. H. Bradley, *Appearance and Reality: A Metaphysical Essay* (London: Swan Sonnenschein, 1893), 22–28, and Armstrong's account of states of affairs as the ground of both internal and external relations David M. Armstrong, *A World of States of Affairs* (Cambridge: Cambridge University Press, 1997). The present point is sharper than the standard formulation: it is not that external relations are philosophically objectionable in themselves but that they are structurally insufficient for the kind of relational individuation the present investigation is examining. A relation that leaves both relata

But a relation that leaves its relata unchanged is not a relation in which the relata are constituted through encounter; it is a relation in which already-constituted relata happen to stand in a configuration that is external to what they are. This is not the concept of relation that motivates the relational traditions in philosophy, science, and social theory. It is the negation of that concept.

The relational account of individuation — the view that entities are individuated through their relations to other things, so that what an entity is cannot be specified independently of the relational field in which it is embedded — confronts a different but equally structural difficulty. If E_1 is what it is in virtue of its relations to E_2, and E_2 is what it is in virtue of its relations to E_1 (and to everything else), then there is no point at which the account makes contact with something that is not itself relationally constituted. The relata are constituted by their relations; the relations hold between relata. The circle is not merely formal — it is ontological. What enters into relation in this account is not an entity with a determinate character that its relations further specify, but an entity whose character *is* its relations. This generates the regress question in its sharpest form: what are the relata of the relations that constitute the relata?

This is the structure of the individuation problem as the present investigation inherits it. Neither the intrinsic account

unchanged is not a relation in which the relata are *constituted through* their encounter but one in which already-constituted relata happen to stand in a configuration. This is the reduction of relation to arrangement — of genuine encounter to mere juxtaposition.

nor the purely relational account can close the circle. The intrinsic account cannot explain genuine relational encounter; the purely relational account cannot specify what is related without presupposing what it is trying to explain. Both failures are structural. Both follow necessarily from what the respective accounts commit to at the foundational level. Both point towards the same gap: neither account supplies a derivation of what makes an entity *capable* of relational encounter while remaining *specifically itself* through that encounter. This capacity is what any adequate account of individuation must explain — and it is precisely what neither intrinsic nor purely relational individuation can derive.

The gap between the two failures is not a space in which further work within the available frameworks might produce a solution. It is a structural void: the location of the foundational problem that the present investigation is organised to address. Before that address is given, however, the present chapter must establish something more basic: that the relational turn in philosophy — the widespread move towards accounts in which relation is constitutive of what entities are — is not a philosophical fashion but a structural necessity. If it is a necessity, then the gap just identified is not merely a lacuna within relational ontology. It is the deepest unasked question of an entire century of thought.

1.3 The Relational Turn and Its Grounds

The half-century between 1950 and 2000 saw a convergence without precise precedent in the history of ideas. Independently, across philosophy, physics, biology, and social theory, researchers working with different problems, different methods, and different vocabularies arrived at the same structural conclusion: what entities are cannot be determined by examining entities in isolation from their relations. The convergence was not coordinated; it did not reflect a shared programme or a common influence. It reflected, rather, the pressure of a common structural problem encountered along multiple independent lines of inquiry.

In philosophy of physics, the standard interpretation of quantum mechanics had, since its formulation in the late 1920s, resisted assimilation into a coherent ontological picture. The measurement problem — the question of how a quantum system acquires a definite value upon measurement when the formalism describes it before measurement as a superposition of possible values — was not a technical difficulty awaiting a technical fix. It was a symptom of the deeper problem that quantum systems, as the formalism describes them, do not have determinate properties independently of their relations to other systems, including the measuring apparatus. Rovelli's relational interpretation, which draws this consequence explicitly by holding that quantum states are always states *relative to* a reference system and that there are no absolute, reference-independent facts about physical quantities, is the ontological conclusion that the physics itself compels.[4] The physics does not leave the ontological alternative open; intrinsic individuation of physical quantities is what the formalism consistently refuses.

In structural realism in philosophy of science, the same conclusion emerged from a different direction. The ontic structural realist thesis — that what the world ultimately contains is not objects with intrinsic natures but structures, and that what we call objects are at best nodes within those structures — is not a metaphysical preference. It is the conclusion that the history of scientific theory change compels: that across the major transitions of theoretical physics, what is preserved from one theory to its successor is structural relations, not the intrinsic natures of the objects the prior theory posited. French and Ladyman argue that this structural continuity is the only kind there is, and that objects as traditionally conceived are dispensable.[5] The re-

[4]See Rovelli, "Relational Quantum Mechanics" (1996), which argues that the apparent paradoxes of quantum mechanics dissolve when it is recognised that physical quantities have meaning only relative to the system with which they interact. The key philosophical consequence is not merely that measurement is context-dependent but that there are no intrinsic physical quantities at all: what a system is, physically, is always what it is in relation to something. The structural parallel with the individuation problem is exact: the quantum formalism requires relational individuation and cannot supply the ground of that individuation — which is why the measurement problem persists even after the relational interpretation is adopted.

[5]The argument is developed systematically in Ladyman and Ross, *Every Thing Must Go* (2007), which defends what the authors call "rainforest realism": the claim that the world contains real patterns at multiple levels of description, where the patterns are structural and the items that instantiate them are not ontologically fundamental. The critical response from those who insist that structures presuppose unstructured relata — the "no-structure-without-objects" objection — is precisely the regress problem iden-

lational turn in philosophy of science is mandated by the science itself.

In theoretical biology, the autopoietic account of living organisation developed by Maturana and Varela in the 1970s established that what distinguishes a living system from a non-living one is not the stuff it is made of but the relational organisation of its processes: the recursive, self-generating network by which the system produces the components that in turn produce the network.[6] Life is relational not contingently but constitutively: to be alive, for Maturana and Varela, is to be relationally organised in a specific way. The intrinsic properties of the components

tified in the preceding section: if what things are is fully structural, what are the relata of the structures? The present investigation locates this objection as the structural void at the heart of ontic structural realism, not as a refutation of the relational commitment but as the mark of the foundational question that commitment cannot yet answer.

[6]The autopoietic account is developed in Maturana and Varela, *Autopoiesis and Cognition* (1980). The central ontological move is the definition of a living system in purely relational terms: not by appeal to any intrinsic property of biological matter but by the structural organisation of production and self-constitution. The philosophical consequence is that biological individuation cannot be given intrinsically; it is constituted through the relational dynamics of the autopoietic network. Simondon's contemporaneous account of individuation as a process rather than a state — the view that what an individual is cannot be understood apart from the pre-individual milieu from which it arises through the resolution of a metastable tension — converges on the same structural conclusion from the direction of general ontology rather than biology. See Gilbert Simondon, *Individuation in Light of Notions of Form and Information*, translated by Taylor Adkins, Originally published as *L'individuation à la lumière des notions de forme et d'information*, Éditions Jérôme Millon, 2005 (Minneapolis: University of Minnesota Press, 2020), 1–52, particularly the Introduction and Part I.

are necessary but not sufficient; what matters is the structural pattern of their relations.

In social theory, the relational turn associated with Emirbayer's "Manifesto for a Relational Sociology" (1997) challenged the "substantialist" premise that dominated both individualist and holist approaches to social explanation: the premise that social entities — persons, institutions, groups — are constituted prior to and independently of their social relations, which they then enter as already-formed units. The relational counterclaim is that social entities have no determinate character apart from the relational processes through which they are constituted and maintained. What a person is, socially, is inseparable from the relational networks in which they are embedded; extract them from those networks and there is no determinate social entity remaining.[7]

Four domains. Four independently derived convergences on the same structural conclusion. The relational turn is not a philosophical fashion; it is the structural consequence of following the evidence and the argument wherever they lead, across physics, biology, social theory, and philosophy, until the same

[7] Emirbayer's "Manifesto for a Relational Sociology," *American Journal of Sociology* 103.2 (1997): 281–317. The substantialist/relational distinction maps precisely onto the intrinsic/relational individuation distinction drawn in §1.2 above, with the additional specificity that substantialism in social theory is the tacit assumption that makes methodological individualism possible: it is only because social actors are assumed to be constituted prior to their relations that the social world can be explained by aggregating their individual properties and preferences. The relational critique is that this assumption is never earned; it is imported as a default that the social evidence consistently undermines.

unavoidable result is reached: entities are constituted through their relations, not prior to them. The entity that enters into relation is already relationally formed. What it is cannot be specified independently of what it is in relation to.

This is not a result that can be dismissed as domain-specific. The four domains are not tangential to reality; together they cover what one might reasonably call its full range — from the sub-atomic to the social, from the physical to the organismic, from the mathematical to the experiential. If the relational turn is mandated in each of them by domain-internal evidence and argument, then it is mandated as a general ontological conclusion: *relation is constitutive of what entities are.* This is the presupposition that the present investigation inherits. It is not the claim the present investigation defends. It is the starting point from which the investigation's real question begins.

1.4 What Taking Relation Seriously Commits Us To

The relational turn established a conclusion. What it did not do — and this is the structural gap that this book exists to examine — is derive the conditions under which that conclusion is coherent. To say that entities are constituted through their relations is to say something about what relation is and what it requires. If relation is constitutive, then the conditions of relational constitution must be specifiable; otherwise, "constituted through relation" names a mechanism without supplying one. The relational turn points towards the ground of its own claim and then stops. The present investigation takes up precisely where the relational turn stopped.

To take relation seriously as a constitutive ontological phenomenon — not as a feature of already-constituted entities but as the process through which constitution occurs — is to incur five structural commitments that are not optional additions to the relational thesis but internal requirements of it. They are what the relational thesis presupposes every time it is stated. The five commitments are stated here as requirements, not as an external framework imposed upon relational ontology from outside. They are derived from what any account that takes relation seriously must be in a position to answer.

The Ground of Relation. If entities are constituted through their relations, then there must be something from which those entities and their relational capacities emerge. Not a substrate that pre-contains them in some hidden form, but a structural source from which both the capacity for relation and the specific character of relata are generated. This is not the demand for a foundation in the traditional metaphysical sense — a bedrock of substance or essence upon which the relational edifice rests. It is the structural requirement that the relational account not be self-grounding in the circular sense: that "relation constitutes entities" not be the whole story, since the story requires an account of what makes relation possible in the first place. Every relational ontology implicitly posits something that performs this grounding function. The question that the present investigation addresses is whether any tradition has formally derived what performs it, or whether it has been stipulated —

posited as the unexamined starting point that makes the rest of the account run.

Relational Identity. If entities are constituted through relation but are not dissolved into relation without remainder, then each entity must have something that makes it the specific relatum it is — something that persists through relational encounter as the entity's characteristic mode of self-constitution. Without this, the relational account cannot distinguish between two different things entering into relation and one thing relating to itself; it cannot explain why the outcome of encounter between E_1 and E_2 is different from the outcome of encounter between E_1 and E_3; it cannot, ultimately, maintain that there are entities in relation rather than relation all the way down with no relata. Relational identity is the structural condition under which entities are constituted through relation while remaining distinguishable participants in it.

The Source of Encounter. If entities are constituted through relation, then the encounter that constitutes them must arise from somewhere. It is not enough to say that entities with relational identities are capable of encounter; the encounter must actually occur, and what makes it occur cannot be, without circularity, the relational identities themselves — since those identities are, by hypothesis, what the encounter constitutes. The source of encounter must be derivable from the same structural ground that produces relational identities; otherwise it is an external perturbation principle imported to do work that the account cannot generate from within.

Relational Yield. If encounter between entities constitutes something, then what it constitutes must be specifiable — not merely as the sum of what the relata already contained, which would collapse relational constitution into mere aggregation, but as something that neither relatum carried alone and that the encounter genuinely produces. This is the requirement that relation be generative: that encounter produce a structural surplus irreducible to the relata that generated it. Without this requirement, the relational turn amounts only to a claim about the interdependence of entities' properties, not to a claim that relation produces anything genuinely new. The philosophical force of the relational turn — its capacity to explain novelty, development, and emergence — rests entirely on whether relational yield can be formally specified.

The Structure of the Relational Event. If encounter produces a relational yield, then the encounter must itself have an internal structure — a formal organisation that accounts for how the relata come into contact, how the contact is sustained rather than immediately dissolving, and how the encounter is oriented towards the yield it produces. Without this, the relational event is a black box: a label for something that happens between the moment before encounter and the moment in which yield has been produced. The structural organisation of the relational event is what makes it an event with determinate ontological consequences rather than an opaque transition between states.

These five — the ground of relation, relational identity, the source of encounter, relational yield, and the structure of the relational event — are the structural commitments that any account claiming to take relation seriously must be in a position to discharge. They are not an external checklist constructed for purposes of critique. They are internal requirements of the relational thesis itself: what that thesis must be able to say if it is to mean what it purports to mean. A relational ontology that cannot answer all five questions is not merely incomplete in the sense of having left some work for future researchers; it is incomplete in the structural sense of having made a claim it has not earned the right to make.

Remark 1.1 (On the Status of the Five Requirements). It bears emphasis that the five requirements stated above are not criteria devised for the purposes of this investigation and applied to the traditions surveyed in Part II from a position external to them. They are derived from the internal logic of the relational commitment itself — from what that commitment must be in a position to say if it is to function as a foundational ontological claim rather than a programmatic gesture. Readers familiar with the traditions surveyed in Part II will recognise that each tradition has engaged some of the five requirements with genuine depth and rigour. The structural claim of the present investigation is not that any tradition has failed to engage the requirements but that none has engaged all five simultaneously, through a single formal derivation proceeding from a single foundational dynamic. The significance of this claim is not primarily critical; it is diagnostic. It identifies the structural gap not in order to expose failure but in order to determine, with the precision the problem requires, what form a satisfying account must take. The formal statement of that gap — and the formal specification of what closing it requires — is the positive contribution of this book.

Object-Oriented Ontology and the Withdrawal Thesis

The claim that the relational turn is a structural necessity — that any ontological account of what entities are arrives at relational constitution as an unavoidable structural result — faces a direct challenge from Object-Oriented Ontology. Graham Harman's central argument, developed in *Tool-Being* and *Guerrilla Metaphysics*, is that objects always withdraw from their relations: they have a real being that exceeds what any relational determination of them can capture.[8] On this view, to treat objects as exhausted by their relations is to miss the most fundamental ontological fact — the surplus of being over relational expression. If OOO is correct, the relational necessity claim fails at the outset, and the present investigation's diagnostic apparatus applies to nothing.

Two responses are available, and both must be made.

First: the withdrawal thesis, on examination, presupposes the relational structure it appears to contest. The object that withdraws from its relations is constituted as *this* object rather than *that* object — as the specific object with the specific surplus of being that is its own — by its determinate nature. But a determinate nature that distinguishes this object from every other is precisely what the five-level diagnostic framework iden-

[8]Graham Harman, *Tool-Being: Heidegger and the Metaphysics of Objects* (Chicago: Open Court, 2002); Graham Harman, *Guerrilla Metaphysics: Phenomenology and the Carpentry of Things* (Chicago: Open Court, 2005).

tifies as Level II: the structural condition of relational identity. To say that an object withdraws from its relations is already to presuppose that the object has been individuated as *this* object with *this* nature from which withdrawal is possible; and that individuation is a structural condition the withdrawal thesis inherits rather than derives. The withdrawal thesis does not escape relational ontology's structural commitments; it presupposes them at Level II while declining to discharge them. OOO is, in this sense, a vivid instance of the general structural pattern the diagnostic framework names: the presupposition of a structural level as the terminus of explanation rather than its starting point.

Second: the diagnostic claim of the present investigation is not that all being is relational all the way down, but that the structural conditions of relational possibility — including the structural conditions under which the specific character of entities is constituted — have not been derived by any tradition. OOO can be read as a particularly precise diagnosis of one of the gaps the present investigation identifies: the withdrawn object is a non-latent structural feature constituted as foundationally primitive, and the SIT applies to it for the same structural reason it applies to every other foundational commitment in the relational tradition.

A Harmanian will object that this argument assumes the very framework OOO contests: OOO denies that withdrawn objects have a "specific identity" in the determinate sense the five-level framework requires, since real objects withdraw precisely from every determination. The objection is less decisive than it appears, for the following reason. The withdrawal thesis must apply to *this* object rather than *that* one. Even on OOO's own terms, the object that withdraws is not an undifferentiated remainder; it is a *specific* object whose withdrawal is its own. Harman's account requires that real objects be numerically distinct — that this object's withdrawal is not that object's withdrawal — and numerical distinctness requires a principle of individuation. The question is not whether withdrawn objects have determinate relational identities in the five-level framework's sense, but whether they are individuated at all. If they are not individuated, the withdrawal thesis collapses: there is no "this object" that withdraws. If they are individuated, the principle of individuation is a structural condition the withdrawal thesis presupposes. In either case, the withdrawal thesis either presupposes Level II or abandons the claim that there are distinct real objects — and abandoning that claim is not OOO's position. The present investigation does not contest OOO's claim that objects withdraw. It identifies the structural condition — specific individuation, Level II — that the withdrawal thesis presupposes without deriving, and which OOO's own framework requires in order for the individuation of withdrawn objects to be intelligible.

1.5 The Diagnostic Question

The relational turn is a structural necessity. The five commitments it incurs are internal requirements of its own foundational claim. The question the present investigation asks of every tradition that has advanced the relational thesis is therefore both precise and fair: at which of the five structural levels does the tradition enter the argument? What does it formally derive at and above its entry point? And what — by the structural act of entering where it does rather than lower — does it make formally unreachable from within its own framework?

This question is the diagnostic question of the present book. It will be applied, in the chapters of Part II, to each of the major traditions in relational ontology: Whitehead's process philosophy, Spinoza's substance ontology as the foundational tradition most concerned with the ground of relation, Bergson's philosophy of duration as the tradition most attuned to the temporal character of relational constitution, and Heidegger's analysis of being-in-time as the tradition most committed to deriving the directional structure of relational existence. It will then be applied to the contemporary landscape: ontic structural realism, relational quantum mechanics, process ontology after Whitehead, and social relational ontology.

The diagnostic question is precise, and its precision deserves a formal statement before the analysis begins.

Definition 1.1 (Entry Level)**.** The *entry level* of a relational ontology is the highest-numbered structural commitment — among the five identified in §1.4 — that the tradition takes as its foundational starting point without derivation. A tradition whose foundational commitment is to actual occasions as the ultimate units of reality has its entry level at Level V (the relational event), since actual occasions are themselves constituted as relational events and the tradition builds upward from there without asking what structurally precedes the event that makes occasions possible. A tradition whose foundational commitment is to an absolute ground from which all determinations follow has its entry level at Level I (the ground), since the ground is stipulated rather than derived and the tradition builds upward from the ground without deriving the ground itself.

Definition 1.2 (Structural Foreclosure)**.** A tradition *structurally forecloses* a structural level L_k if and only if its foundational commitment is to a structural feature at level L_j where $j > k$, and the act of taking that feature as foundational is formally equivalent to constituting it as the structural ground of the account — thereby making the derivation of anything at level L_k strictly impossible from within the framework's own resources without abandoning or revising the foundational commitment. Structural foreclosure is stronger than incompleteness: it is not that the tradition has not yet derived level L_k but that it cannot do so while remaining the tradition it is.

Criterion 1.1 (The Diagnostic Method). For each tradition surveyed in Part II, the analysis proceeds as follows:

(i) Identify the tradition's foundational commitment and its entry level according to Definition 1.1.

(ii) Establish what the tradition formally achieves at and above the entry level, with full fidelity to the tradition's own conceptual apparatus.

(iii) Apply the subtraction test to the levels below the entry level: ask whether the tradition has the resources to derive those levels from within its own framework without abandoning the foundational commitment.

(iv) If the answer to (iii) is negative, determine whether the failure is structural — following necessarily from the foundational commitment by the argument of Definition 1.2 — or accidental, requiring only further work within the existing framework to remedy.

(v) State the diagnostic result: what the tradition establishes, what it structurally forecloses, and why the foreclosure is necessary rather than contingent.

The diagnostic method does not ask whether any tradition is wrong on its own terms. Each tradition is taken to have correctly identified something real about the structure of relation. The question is not correctness but completeness: whether what has been correctly identified can serve as the complete account that the relational thesis requires.

The diagnostic method, so stated, generates a result whose significance can already be anticipated, even before the survey of Part II has been executed. If the structural foreclosure thesis holds — if every major tradition in relational ontology has its entry level at a non-foundational structural level, and if entry at a non-foundational level formally precludes the derivation of the structural levels below it — then the traditions are not competitors for the same foundational account. They are partial accounts of different levels of the same structural problem, each correct within the scope its entry level allows, each systematically blind to the levels below that scope not through any individual failure of insight but through the structural consequence of the foundational commitment that defines the tradition.

This would establish something of considerable philosophical significance: not that the relational turn was mistaken, but that it has everywhere begun in the middle of the story. The story begins earlier than any of the traditions have started it. What lies at the beginning — what the present investigation calls, without yet being in a position to supply, the ground of relational ontology — is the structural level that every tradition has presupposed without deriving and that no tradition's foundational commitments permit it to reach. Whether this is the case is what the diagnostic survey of Part II establishes. That it follows necessarily from the structural foreclosure thesis, and that the structural foreclosure thesis is established by the survey, is the content of the Structural Insufficiency result that Part III derives.

The investigation now turns to its first and most fundamental diagnostic subject. Whitehead's process philosophy constitutes the richest and most formally developed account of relational events in the tradition; its entry level is therefore the most instructive place to begin. If the most sophisticated account of the relational event in the history of philosophy fails to discharge the five structural commitments — if even the depth and formal precision of the mature Whiteheadian account cannot reach the levels below its entry point — then the structural foreclosure thesis acquires an evidential weight that subsequent chapters will confirm and deepen.

Summary of Chapter 1. Relation is not a feature of already-constituted entities but the process through which constitution occurs. This conclusion is not a philosophical preference; it is a structural necessity mandated by the evidence of four independent domains — philosophy of physics, structural realism, theoretical biology, and social theory — each of which arrived at it independently under the pressure of domain-internal argument. Accepting it incurs five structural commitments: the ground of relation, relational identity, the source of encounter, relational yield, and the structure of the relational event. These are internal requirements of the relational thesis, not an external critical apparatus. No tradition in relational ontology has formally derived all five simultaneously from a single foundational dynamic. The diagnostic method formalises the question that follows from this observation: at which structural level does each tradition enter the argument, what does it achieve at and above its entry level, and what does it structurally foreclose below it? The survey of Part II executes this method across the major traditions. The result of the survey is the subject of Part III.

CHAPTER 2

Five Questions, Five Levels

> *A requirement is not a criterion imposed from outside. It is what the subject matter cannot do without and cannot supply for itself.*
>
> *Before Relation*

2.1 From Commitment to Requirement

The five structural commitments identified in Chapter 1 are not five independent difficulties that a sufficiently resourceful relational ontology might address piecemeal. They are five aspects of a single structural problem: the problem of what any account of relational constitution must be able to say if its central claim is to function as more than a programmatic gesture. An account that says "entities are constituted through their relations" has committed itself to specifying what makes that constitution possible. The five commitments are the decomposition of that specification into its irreducible components. They are what the commitment requires in order to be dischargeable.

This chapter converts the five commitments into five formal requirements: precisely specified conditions that any satisfying account of relational constitution must meet. The conversion is not a translation from informal to formal language. It is a structural operation: each requirement is derived from its corresponding commitment by asking what the commitment would need in order to yield determinate answers rather than further questions. A commitment to the ground of relation becomes a requirement once the conditions of adequacy for a ground account are specified. A commitment to relational identity becomes a requirement once the formal criteria for identity through relational constitution are made precise. And so for each of the five in turn.

The significance of the conversion is diagnostic. Once the five requirements are formally specified, every tradition in relational ontology can be evaluated against them with precision: not merely assessed as "strong" or "weak" in its treatment of relation, but located at a determinate structural level, with the character of what it achieves above that level and what it structurally forecloses below it specified without remainder. This is the diagnostic method formalised in Criterion 1.1. The present chapter supplies the content that method requires: the precise specification of what meeting each requirement entails, what failing it looks like, and why no partial achievement constitutes a complete account.

Each of the five sections that follow is organised around the same structural movement. It opens with the question that the structural commitment makes urgent — the question that any relational ontology must be in a position to answer if it is to claim that its central commitment is coherent. It sharpens that question through increasingly precise formulations until the formal requirement is stated exactly. It illustrates the formal requirement through a concrete case drawn from a non-philosophical domain — not as a proof of the requirement but as a demonstration of what meeting it looks like in a domain where the structural problem takes a determinate and examinable form. It closes by identifying the characteristic failure mode of each requirement: what an account does when it cannot discharge the requirement and what the structural consequences of that failure are. The five sections together constitute the diagnostic framework against which Part II operates.

The Five Levels as an Eliminative Result

Before turning to the five levels individually, the status of the five-level decomposition itself requires explicit address. The diagnostic framework of this book operates against a five-level structural grid. A referee — or a careful reader — is entitled to ask: why five levels? Is the decomposition arbitrary? Could the commitments of relational ontology be decomposed into four levels, or six, or some other number? And if the decomposition is not arbitrary, where is the argument that establishes it?

Note on reading order. The derivation that follows proceeds from a single sentence — "entities are constituted through their relations" — without assuming any of the five levels. Readers who have already encountered the level definitions will find the derivation confirming what they have seen; readers who have not will find the derivation introducing the levels through their necessity. Either order produces the same result: the five levels are shown to be analytically required rather than stipulated.

The argument is available and is load-bearing, but it must be stated with care to avoid circularity. A *circular* version would say: each level is necessary because removing it leaves the other four unable to discharge the relational commitment — where "unable" is assessed using the five-level framework itself. The *non-circular* version asks a prior question: starting from the relational thesis alone — "entities are constituted through their relations" — and applying the eliminative method to that thesis directly, what conditions does the thesis require to be coherent? The five levels are the answer to that prior question. They are derived *from* the relational thesis, not from each other.

The derivation proceeds as follows. The relational thesis makes three structural commitments simultaneously: it commits to (a) the existence of entities that relate (relata), (b) the process by which those entities are constituted through encounter (the relational event and its yield), and (c) the source from which both the relata and the constitutive process emerge (the ground). These three commitments are not five; the question is whether the move from three to five is itself eliminative. It is. The "relata" commitment (a), examined for what it requires to be coherent, decomposes into two conditions: the account must specify what makes relata structurally distinct from one another (Level II: relational identity) and what brings specific relata into actual encounter rather than leaving them in isolation

(Level III: the source of encounter). The "relational event and yield" commitment (b) decomposes into two conditions: the account must specify what the encounter produces and why it is irreducible to what the relata already contained (Level IV: relational yield) and what internal organisation of the encounter accounts for how yield is produced rather than merely describing its occurrence (Level V: the structure of the relational event). The "ground" commitment (c) requires a single condition: a structural source from which the relata, their capacity for encounter, and the constitutive process all emerge (Level I: the ground). Five conditions emerge from three commitments by the non-circular application of the eliminative method to the relational thesis itself.

That the decomposition yields exactly this count — two conditions from (a), two from (b), one from (c) — is a claim that requires argument rather than assertion. The argument proceeds by showing that each commitment decomposes into its conditions by the same non-circular test: what does this commitment require, and are those requirements independent of one another?

Why commitment (a) yields exactly two. The existence of relata requires: that they be structurally distinct from one another, and that they be capable of being brought into actual encounter. These are independent because structural distinctness is compatible with two relata that never encounter each other — a framework can specify what makes each relatum specific while providing no account of what actualises encounter — and encounter-actualisation is independent of a formal account of distinctness, since a framework can specify that certain encounters occur without formally accounting for what makes the encountering parties the specific parties they are. Neither condition is derivable from the other. A third condition on relata would need to identify something the relational thesis requires of relata that is not already covered by distinctness and encounter-actualisation. The strongest candidates — persistence through time, causal efficacy, and numerical identity — all reduce on inspection to one or the other of the two conditions. Persistence through time is either a form of structural distinctness (the relatum at t_2 is distinct from the relatum at t_1) or a precondition for repeated encounter-actualisation, both of which are already covered. Causal efficacy is either the mechanism by which encounter-actualisation operates — in which case it is part of Level III, not a further condition on relata themselves — or a feature of the relatum's structural identity, in which case it falls under Level II. Numerical identity is the minimal form of structural distinctness: that a relatum is numerically this one rather than that one is the weakest case of what Level II's conditions specify. No third condition on relata survives this examination: the two conditions jointly discharge everything the relational thesis requires relata to be and do prior to entering encounter.

Why commitment (b) yields exactly two. The relational event and its yield require: a specification of what encounter produces and why it is irreducible to what the relata already contained, and a specification of the internal structural organisation through which encounter produces that yield rather than assigning it as a brute outcome. These are independent because a framework can specify the yield while leaving the production mechanism a black box, and it can specify the mechanism while failing to establish that the yield is irreducibly novel. Neither is derivable from the other. The strongest candidates for a third condition on the relational event — the modal structure of the encounter (what alternative yields it could have produced), and the temporal staging of the yield (whether it emerges during or after the encounter) — both reduce on inspection to one of the two identified conditions. The modal structure of the encounter is either a specification of the mechanism's formal organisation — what the encounter's internal structure makes possible and forecloses, which is precisely what Level V addresses — or a specification of the yield's irreducibility: that the actual yield is not the only structurally possible yield for this pair of relata establishes its irreducibility from a different direction but falls under Level IV's conditions. The temporal staging of the yield is either an aspect of the encounter's internal structural organisation (when in the process of encounter the yield emerges is a structural feature of the mechanism) or a property of the yield itself. Neither is a third independent condition: both are sub-cases of what Level IV and Level V jointly cover. No third condition on the relational event and its yield survives this examination.

Why commitment (c) yields exactly one. The ground from which relata and the constitutive process emerge is a single structural condition because the relational thesis makes a single commitment about it: that such a source exists and has sufficient generativity to produce the relata and their constitutive process rather than being one additional relatum among others. Further decomposition of the ground commitment — specifying the ground's internal structure, its dynamics, pre-ground conditions — is possible, and is the content of the R1–R8 requirements derived in Part IV. But those are conditions on any adequate account of the ground's internal structure, not further levels of the five-level structural grid. At the level of the relational thesis alone, the commitment about the ground is single: that it exists and is generative. What it would need to be in order to satisfy that commitment is the content of R1, not of additional grid levels.

The eliminative test applied to each of the five levels in turn now proceeds as a consistency check within this derivation. Having derived five conditions, we ask: can any be dropped while the remaining four still discharge the three commitments? The answer is no in each case, for reasons that appeal to the relational thesis rather than to the five-level framework.

Remove the ground (Level I) and the account of relational constitution has no structural origin for either the entities that enter into relation or the capacity for relation that those entities bear. The relational thesis requires that relation be constitutive, not contingent; constitutive relation requires that relata and their relational capacity arise from something rather than appear ex nihilo. The objection that relata need not "arise from" a generative source — that they simply *are* the nodes of the relational structure, individuated by structural position alone without any further generative origin — does not evade this requirement; it relocates it. If relata are individuated by structural position, then what individuates the positions? A structural position is itself a relational fact: a location *in* a structure, constituted by the relations that define it. Those relations in turn require relata to hold between. If those relata are again constituted by structural position in further relations, the regress is without termination. The regress does not arise from an external demand imposed on the relational thesis; it arises from within the relational thesis itself, from the requirement that relata be genuinely numerically distinct. Any account that accepts the relational thesis and refuses a generative ground must either accept the regress — abandoning the claim that distinct relata are constituted at all — or accept a brute structural fact that is itself unexplained by the thesis, which is to constitutionally commit to precisely the ground whose independent derivation is thereby refused. Without Level I, the relational thesis makes an ontological claim it cannot ground.

Remove relational identity (Level II) and the account cannot distinguish between two different entities entering into relation and one entity relating to itself. The relational thesis requires that encounter between *distinct* relata produce something neither carried alone; without an account of what makes relata distinct, the thesis collapses into the claim that something relates to itself, which is not constitution but self-expression.

Remove the source of encounter (Level III) and the account cannot explain why compatible entities actually encounter each other. The relational thesis claims that relation is constitutive of what entities are; if encounter never actually occurs, the thesis is vacuous. Without Level III, the thesis has relata with relational capacity but no account of why that capacity is ever exercised.

Remove relational yield (Level IV) and the relational thesis loses its central content: it claims relation is constitutive, but if encounter produces nothing genuinely new — nothing irreducible to what the relata already contained — then "constituted through relation" means nothing more than "causally produced by," which is mechanism, not the relational thesis.

Remove the structure of the relational event (Level V) and the account cannot explain how encounter produces yield — only that it does. The relational thesis requires that constitution through relation be a determinate structural process, not a black box. Without Level V, yield is asserted to arise but not shown to arise through any specifiable structural organisation. Specifically, Level IV's conditions of irreducibility (Y_1) and asymmetric novelty (Y_4) can be asserted but not derived: it is precisely the encounter's internal structure that accounts for why *this* pair produces *this* yield rather than some other.

The five levels are therefore the minimum-complexity decomposition of what the relational thesis requires, derived by applying the eliminative method to the thesis itself rather than to the five-level framework. A finer decomposition is possible and would not falsify this result; it would constitute a further articulation within the same structural space to which the diagnostic claims of Part II would apply mutatis mutandis.

Remark 2.1 (The Eliminative Necessity of Exactly Five Levels). The eliminative derivation above establishes that five levels are necessary and jointly sufficient. The following extends that argument with explicit reductiones against candidate mergers and candidate additions, showing that the count of five is not itself arbitrary.

Why Level II cannot be merged with Level I. If relational identity is absorbed into the ground, then either the ground pre-specifies each entity's character (making entities determinate prior to relation, collapsing the relational thesis) or distinctness is left unaccounted for. A generative ground that produces differentiation without a formal account of what makes each output the specific entity it is generates differentiation in name only. Level II is the account of that specificity; it is not available from Level I without importing it.

Why Level III cannot be merged with Level II. Having a specific relational identity is compatible with never entering actual encounter: it specifies which entity something is, not which entities it meets and why. The encounter source is what actualises the encounter for a specific pair rather than leaving them in structural proximity without contact. It is not derivable from relational identity alone.

Why Level IV cannot be merged with Level III. The encounter source specifies what brings relata into contact; it does not specify what that contact produces. The same encounter mechanism is compatible with a range of possible yields; the yield is a further structural result, not a consequence of the encounter source alone.

Why Level V cannot be merged with Level IV. The yield is what the encounter produces; the event structure is how it produces it. A black-box transition from encounter to yield — one that asserts the yield without characterising the internal organisation of its production — fails the relational thesis's requirement that constitution through relation be a determinate structural process, not merely a labelled transition.

Why no sixth level is generated. The strongest candidates for a sixth structural commitment are:

Intentional or normative orientation: whether the relational dynamic is directed towards ends involving evaluation. Directionality is a property of the relational event structure (Level V) and of the yield's incorporation into the ground's continuing dynamic (Level I). Remove this as a separate candidate level: relational constitution is still possible with five levels. The relational thesis generates no further structural commitment requiring a separate condition of normative orientation beyond what Level V's account of the event's internal direction already specifies.

Reflexivity: whether relational systems can take themselves as objects of their own encounter. This is a property that some relational events exhibit — it characterises a specific class of events, not a structural condition of relational possibility as such. It is therefore not generated as a necessary commitment by the relational thesis.

Phenomenal character: whether relational events involve experience. This is a substantive metaphysical question, but the relational thesis makes no commitment about whether encounters are experienced. Phenomenal character, if it belongs to some relational events, is a further structural feature derivable within specific domains, not a sixth condition of relational possibility in general.

The five levels are therefore not only necessary and jointly sufficient but minimally so: no level can be dropped, merged, or replaced by a smaller or larger count while preserving the structural completeness of the account. The hierarchy is derivable from the relational thesis alone — without knowledge of any positive derivation of the structural ground.

Structural skeleton of the five-level derivation. *The derivation proceeds in five steps, each appealing only to the relational thesis and the eliminative method.*

Step 1. *The relational thesis makes three structural commitments: (a) entities that relate (relata), (b) the constitutive process and its yield (encounter and result), (c) a structural source (ground).*

Step 2. *Commitment (a) decomposes into two independent conditions: what makes relata structurally distinct from one another [**Level II**] and what brings specific relata into actual encounter rather than leaving them in structural proximity without contact [**Level III**]. Neither is derivable from the other.*

Step 3. *Commitment (b) decomposes into two independent conditions: what encounter produces and why that yield is irreducible to the relata's prior resources [**Level IV**] and the internal structural organisation of the encounter through which yield is produced rather than merely asserted [**Level V**]. Neither is derivable from the other.*

Step 4. *Commitment (c) yields a single condition: the structural source from which relata and the constitutive process both emerge*

*[**Level I**]. Further decomposition of the ground's internal structure is the content of the R1–R8 requirements, not of further grid levels.*

Step 5. *No level can be removed without collapsing the relational thesis; no additional level is generated by the thesis that is not already covered by I–V. The count of five is therefore necessary and not arbitrary.*

No merger, split, or additional level survives the reductiones in Remark 2.1. The derivation uses only the relational thesis; no feature of any positive account is introduced.

The following table restates the same derivation in formal tabular form, adding for each step the structural question that step must answer and the eliminative argument that makes it non-removable. Both registers — the prose summary above and the formal table below — are complete accounts of the derivation; the table is provided for ease of reference when applying the diagnostic grid to traditions not surveyed in Part II.

> **Structural skeleton of the five-level derivation.** *The derivation proceeds as a series of structurally necessary steps. Each step names the question the relational thesis must be able to answer and the level its answer occupies. The logical move at each step is eliminative: the named level cannot be removed from the account without that account ceasing to be a relational account of the relevant structural fact.*

Step	Level	Structural question generated	Eliminative argument
1	*I: Ground*	What makes relata possible at all — what sustains them as potential terms of encounter before any encounter occurs?	Without a grounding condition, entities pre-exist relation as bare ungrounded terms; the relational thesis collapses.
2	*II: Relational Identity*	What determines which entity each relatum is, as distinct from every other?	Differentiation without a condition of specific identity generates multiplicity in name only; merger with Level I fails.
3	*III: Encounter Source*	What brings a specific pair of relata into actual encounter rather than leaving them in structural proximity without contact?	Relational identity is compatible with permanent non-encounter; a separate actuating condition is required.
4	*IV: Relational Yield*	What does the encounter produce?	The same encounter mechanism is compatible with multiple possible yields; a separate specification of what encounter produces is required.
5	*V: Relational Event Structure*	What is the internal organisation through which encounter produces its yield?	Asserting a yield without characterising the encounter's internal structure fails the relational thesis's requirement that constitution be a determinate structural process, not a labelled transition.

Joint sufficiency: the covering argument. The five conditions are not only each necessary but jointly sufficient: any structural question about relational constitution falls within the scope of exactly one of the five levels, leaving no remainder. The argument proceeds by exhaustive partition. Relational constitution requires (i) something from which relata and their relational capacity emerge — Level I covers this; (ii) an account of what makes each relatum the specific entity it is rather than another — Level II covers this; (iii) an account of why and how compatible relata actually enter encounter rather than remaining in structural proximity without contact — Level III covers this; (iv) an account of what encounter produces that neither relatum carried in isolation — Level IV covers this; (v) an account of the internal structural organisation through which the encounter event produces its yield — Level V covers this. A structural question about relational constitution that does not fall into any of these five categories would need to be a question about relational constitution that is neither about the ground, nor the relata, nor the source of encounter, nor the product of encounter, nor the structure of the encounter event itself. No such question can be coherently formulated: these five categories jointly exhaust the structural anatomy of what it is for entities to be constituted through their relations. The five-level framework is therefore not only necessary at each level but jointly sufficient as an account of what relational constitution structurally requires.

2.2 On the Independence of the Diagnostic Framework

A specific objection about the provenance of the five-level grid must be addressed here. One might suspect that the grid was constructed after knowing what a successful positive derivation would supply — that its five levels were chosen to make existing traditions fail at precisely the level where the positive derivation succeeds. This would make the grid a post-hoc device rather than an independent diagnostic instrument. The objection is serious and requires three responses: one logical, one empirical, one analytic.

The logical response. The five-level grid is derivable from the relational thesis alone by the eliminative method demonstrated above, without any knowledge of what a positive derivation would contain. The derivation shown above proceeds: from the single sentence "entities are constituted through their relations" to three structural commitments (relata, constitutive process, ground) to five conditions by decomposing each commitment for what it requires to be coherent. No step in this derivation appeals to or presupposes the positive derivation. A reader who accepts the relational thesis and applies the eliminative method will arrive at the five conditions; the positive derivation's existence or content is irrelevant to this arrival. The grid is therefore derivable independently: it was derivable from the relational thesis before any positive derivation existed, and it remains derivable from the relational thesis independently of any claim about what the positive derivation contains.

The empirical response. The logical response establishes that circularity is not structurally required. The empirical response establishes that it did not in fact obtain. If the five-level grid had been designed post-hoc to make existing traditions fail where the positive derivation succeeds, then independent scholars working without knowledge of the *Kawana'on*[1] system would not be expected to independently identify the same structural gaps the five levels articulate. Yet that is precisely what the literature shows. The Level I gap — the absence of a derived generative ground — is identified independently by Barr et al.[2]

[1]*Kawana'on* is the collective name for the generative first philosophy derived in the companion series *The Triadic Ontological System* Hector Caliso Suano, "Kawana'on: The Collective Name for the Generative First Philosophy" (Unpublished preprint. Article in the companion series *The Triadic Ontological System*. Derives the collective name for the generative first philosophy through historical-linguistic and structural analysis of the Philippine Austronesian vessel-word *kawa*, 2026). The name is derived through historical-linguistic and structural analysis of the Philippine Austronesian vessel-word *kawa* — a term whose semantic field encompasses holding, containing, and receiving — and names the structural role of the latent register as the ground that holds the conditions for relational constitution without itself being constituted relationally. The derivation of the name, and of the system it names, is the subject of the companion series; *Before Relation* neither presupposes that derivation nor previews its content. What *Before Relation* establishes is the negative result — the structural limits that make a positive derivation necessary — and the eight formal requirements (R1–R8) that any such derivation must satisfy. Whether the *Kawana'on* system satisfies those requirements is left to the reader's independent evaluation of the companion works against the criteria established in Chapter 11.

[2]Federico Barr et al., "Relational Ontology as a Bridge Between Realism and Empiricism" (Working paper. Archived at PhilArchive, rec/BARROA-18. States the grounding regress directly: "If entities exist only through relations, what grounds the relations? This threatens infinite regress or circularity." 2024).

("If entities exist only through relations, what grounds the relations? This threatens infinite regress or circularity"), by Crux[3] ("Relation-based ontologies attempt to avoid substance by treating relations as primary, only to find that relations quietly harden into relata that behave like substances in all but name"), and by Lucero[4] (whose grounding-theoretic elimination from within the philosophy of religion independently converges on the same structural constraints the Level I conditions specify). The Level II gap — the absence of derived structural individ-

[3]Aeon Timaeus Crux, "Phainesis in Zero Ground Showing the Work: On the Dissolution of the Problem of Ontological Grounding" (Archived 5 February 2026. PhilArchive rec/CRUPIZ, `https://philarchive.org/rec/CRUPIZ`. Diagnoses the identical structural pattern Before Relation formalises as the SIT: "Relation-based ontologies attempt to avoid substance by treating relations as primary, only to find that relations quietly harden into relata that behave like substances in all but name." And: "The problem recurs with remarkable consistency across the entire history of ontology, cutting across traditions that otherwise disagree on nearly everything else." And, most directly: "The failure is not accidental; it is structural." — an independent diagnosis, from within the field, of the cross-traditional structural impasse Before Relation formally establishes, 2026).

[4]Jorge C. Lucero, "What the Ground of Beings Must Be: Aseity, Simplicity, Immutability, Atemporality," Forthcoming; volume, issue, and page numbers not yet assigned at time of composition. Archived at PhilArchive, rec/LUCWTG. Argues by grounding-theoretic elimination — ruling out external dependence (incoherence for an ultimate ground), composition (wholes are grounded in parts, so the ultimate ground can have no parts), and any dependence structure that change would introduce — that "the ground of beings cannot itself be a being" but must be a "trans-categorial ground" characterised by aseity, simplicity, immutability, and atemporality. Independent peer-reviewed confirmation, from the philosophy of religion tradition using grounding theory, of Before Relation's Level I requirement that the structural ground not be constituted as a feature within the order it grounds, and that structural constraints on the ground are derivable rather than merely asserted, *International Journal for Philosophy of Religion*, 2026,

uation — is identified independently by Rivera,[5] who demonstrates across grounding, process ontology, and structural ontology that each "presupposes individuation rather than explaining it." The Level III gap — the absence of a derived encounter source — is identified independently by Hakkarainen and Keinänen,[6] who establish from within analytic metaphysics that the constitutive mechanism of relational adherence remains "a constitutively inexplicable primitive" across the best available responses to Bradley's regress. The Level I–II gaps in the physics-oriented frameworks are identified independently by Adlam,[7] who diagnoses in OSR and RQM precisely the fail-

[5]Miguel Angel Rivera, "Unity Before Multiplicity: Ontological Individuation and the Problem of Primitive Multiplicity" (Preprint. Archived at PhilArchive, rec/RIVUBM. Establishes, through diagnostic application to grounding, process ontology, and structural ontology, that each "presupposes individuation rather than explaining it" — that "structure organizes; it does not individuate" — and that any ontology admitting real multiplicity must already rely on a non-epistemic principle of individuation that is explanatorily prior to multiplicity. Directly confirms Before Relation's Level I and Level II diagnostic claims across multiple traditions simultaneously, 2025).

[6]Jani Hakkarainen and Markku Keinänen, "Bradley's Relation Regress and the Inadequacy of the Relata-Specific Answer," Archived at PhilArchive, rec/HAKBRR. Establishes that the best current response to Bradley's relation regress — the relata-specific answer — leaves the constitutive mechanism by which relations hold between their relata as "a constitutively inexplicable primitive" and "highlights the need for a metaphysical account of the constitution of the holding of adherence" — peer-reviewed confirmation that the source-of-encounter problem is structurally unresolved in analytic metaphysics, corresponding to Before Relation's Level III gap, *Acta Analytica* 38, no. 2 (2022): 229–243.

[7]Emily Adlam, "Relational Observables, Quiddities, and Structural Realism," Archived at PhilPapers, rec/ADLROQ. Argues that structuralist approaches to spacetime "face a number of important open questions" regard-

ure to account for how non-relational, non-modal features arise from purely relational structure. The cross-traditional pattern of the SIT is identified independently by Crux[8] ("The failure is not accidental; it is structural") and Timmenga.[9]

Each of these scholars worked without access to the *Kawana'on* system. Each arrived at descriptions of specific structural gaps that map onto specific levels of the five-level grid. If the grid had been designed post-hoc to manufacture failures, this convergence would be inexplicable. If the grid was derived from the relational thesis by the eliminative method — as the logical response establishes — the convergence is exactly what one would expect: independent investigators, following different methods from different directions, would identify the same structural requirements precisely because those requirements follow from what the relational thesis commits anyone who takes it seriously to supplying.[10]

ing how non-modal features of reality arise from purely relational structure; confirms the gap *Before Relation* identifies in OSR and RQM remains active in 2025 literature, *Synthese* 206, no. 2 (2025): 104, arXiv: `2507.20313`.

[8]Crux, "Phainesis in Zero Ground Showing the Work: On the Dissolution of the Problem of Ontological Grounding."

[9]Friso Timmenga, "The Paradox of Process Philosophy," Archived at PhilArchive, rec/TIMTPO-11. Documents convergent failures in process philosophy and OOO: Harman's finding that Whitehead "falls short of accounting for the existence of individual entities" while reducing entities to their processual relations; the Garfield-Priest analysis of Nāgārjuna's "bottomless web of explanation"; confirms that both process philosophy and object-oriented ontology face the same structural regress the present investigation names, *Inscriptions* 7, no. 2 (2024): 158–166.

[10]The independence of these scholars from the *Kawana'on* system can be assessed as follows. Hakkarainen and Keinänen ("Bradley's Relation Regress and the Inadequacy of the Relata-Specific Answer") is a peer-

The convergence extends to established literature that predates all current versions of this investigation. Simondon[11] identifies the pre-individual as the domain from which individuation emerges and that no individuated account can reach from within its own resources — a structural diagnosis that converges on what the present investigation identifies as the Level I gap,

reviewed article in *Acta Analytica* (2022) whose subject is Bradley's regress within analytic metaphysics; the conclusion about a "constitutively inexplicable primitive" arises from a problem internal to their own technical analysis and has no connection to the present project. Adlam ("Relational Observables, Quiddities, and Structural Realism") is a peer-reviewed article in *Synthese* (206:2, 2025; doi:10.1007/s11229-025-05190-5) addressing structural realism and quiddities, by an established philosopher of physics with no published connection to this investigation. Timmenga ("The Paradox of Process Philosophy") is a published article in *Inscriptions* (7:2, 2024). The remaining citations — Barr et al. ("Relational Ontology as a Bridge Between Realism and Empiricism"), Lucero ("What the Ground of Beings Must Be: Aseity, Simplicity, Immutability, Atemporality"), Crux ("Phainesis in Zero Ground Showing the Work: On the Dissolution of the Problem of Ontological Grounding"), and Rivera ("Unity Before Multiplicity: Ontological Individuation and the Problem of Primitive Multiplicity") — are preprints or forthcoming works with PhilArchive records; their independence from the *Kawana'on* project is asserted here and should be verified by any reader who wishes to confirm it, through the publication and correspondence records of each author. Where any such author has had access to drafts of this investigation, that access should be noted in the final submission record. The present investigation's claim of convergence rests on the two demonstrated independently published results (Hakkarainen–Keinänen and Adlam) as the minimal evidentiary core; the preprints provide additional convergent instances whose independence is claimed but should be confirmed.

[11]Gilbert Simondon, *L'individuation à la lumière des notions de forme et d'information*, Republished with additional material by Éditions Jérôme Millon (Grenoble) in 1995; revised Millon edition 2005; English translation: *Individuation in Light of Notions of Form and Information*, trans. Taylor Adkins, University of Minnesota Press, 2020 (Paris: Presses Universitaires de France, 1964).

arrived at from within the phenomenology of technical objects and biological individuation rather than from any programme in relational ontology. Bitbol[12] demonstrates from within relational quantum mechanics that the framework "cannot account for the relational properties of the apparatus itself" — a precise statement of the Level I gap applied to RQM — without access to or engagement with any relational ontology framework of the present kind. These two results, established from entirely independent disciplinary starting points, confirm that the structural pattern the five-level grid identifies is not a feature of the grid's design but a real structural boundary that different forms of inquiry reach by different routes.

The first two responses are mutually reinforcing: the logical response shows the circularity was not structurally necessary; the empirical response shows it did not factually occur. The grid is an independent diagnostic instrument by both criteria.

The analytic response: why these specific requirements. The logical and empirical responses together establish the grid's independence. But a hostile referee may press a sharper version of the objection: not that the grid is post-hoc in its structure, but that the *specific content* of each requirement — why R2 is "pre-temporal" rather than something else, why exactly this formulation rather than a near-neighbour — was chosen to match

[12]Michel Bitbol, "Quantum Mechanics as Generalized Theory of Probabilities," Argues from within the relational quantum mechanics programme that the framework "cannot account for the relational properties of the apparatus itself" without introducing a structural primitive the programme does not derive, *Collapse: Philosophical Research and Development* 8 (2014): 87–121.

the *Kawana'on* system. This objection requires a third response that neither the logical nor the empirical response supplies.

The specific content of each requirement is not chosen; it is analytically forced by two conditions together: (1) the relational thesis, and (2) the non-presupposition condition, which states that any feature a derivation of relational conditions takes as a starting point rather than deriving is a foundational constitution and therefore a structural foreclosure by Definition 10.4. Given these two conditions, the specific content of each requirement is the *minimal negation* of the corresponding tradition's presupposition: it negates exactly what the tradition presupposes, and no more. This is a formal operation, not a descriptive one.

Consider R2 specifically, since "pre-temporal" invites the challenge most directly. Every tradition surveyed treats time — or at minimum the temporal order within which relations occur — as a starting point. This is not a contingent feature of these particular traditions; it follows from the relational thesis itself. If relation is fundamental, then temporal succession is a relational phenomenon: it is what occurs between or within related events. The natural objection is that temporal succession might instead be a *precondition* for relation — the medium within which encounters occur, prior to and enabling of relation rather than constituted by it. But this objection is not available to anyone committed to the relational thesis. If temporal succession is a pre-relational medium — if it is structurally available before any relation occurs — then the relational thesis is false at the most basic level: there is at least one structural feature (temporal succession) that is not constituted through relation but is presupposed by it. The relational thesis claims that entities are constituted through their relations; if the temporal order precedes and enables those relations without being constituted by them, the thesis has a foundational exception built in at the level of time itself. A tradition committed to the relational thesis therefore cannot treat temporal succession as a pre-relational medium; it must treat it as a relational result. A tradition that begins within the temporal order is therefore beginning within the relational order — which means it is beginning within what it is supposed to be deriving the conditions of. The specific presupposition is: *the temporal order is available as a starting point*. The minimal negation of this is precisely: *the structural domain from which the derivation begins is formally prior to the temporal order*. This is what R2 says. Any weaker specification — "non-physical," "atemporal," "abstract" — would fail to negate the specific presupposition, because a tradition could be non-physical or abstract while still beginning within the temporal order. Any stronger specification would add content not analytically forced by the non-presupposition condition. "Pre-temporal" is not a label for a feature of the *Kawana'on* system that R2 was written to match; it is the minimal negation of the temporal presupposition that every tradition committed to the relational thesis and operating within the temporal order necessarily makes.

The same minimal-negation operation applies to R1 and R3–R8, each requirement's specific content being analytically forced by the relational thesis plus the non-presupposition condition without residual interpretive latitude. This is why the requirements have the specific formulations they have.

The same two conditions — the relational thesis and the non-presupposition condition — that analytically force the content of R1–R8 are also the conditions that any adequate positive derivation must satisfy. A derivation that genuinely takes the relational thesis seriously and genuinely refuses to presuppose what it has not derived will, by that very fact, satisfy the minimal negation of each tradition's presupposition: it will have derived what the traditions assumed, and in deriving it, will satisfy the requirement that names what they failed to derive. This is not a coincidence between the requirements and the positive derivation; it is a structural consequence of their shared derivation from the same two conditions. The *Kawana'on* system satisfies R1–R8 because its derivation is governed by the relational thesis and the non-presupposition condition: the same conditions that force the requirements also force the system. R1–R8 are not specifications written to match the system; they are the minimal negations of each tradition's presupposition, and the system satisfies them because the system's derivation does not make those presuppositions.

A standing commitment. The three responses above — logical, empirical, and analytic — jointly establish that the diagnostic grid is an independent instrument. But a standing openness is appropriate here that no internal argument can substitute for. The derivation of the five levels from the relational thesis by the eliminative method is not offered as something the reader is asked to take on trust; it is offered as a derivation any reader can perform. The procedure is available in full in the preceding sections: begin from the single sentence "entities are constituted through their relations," apply the eliminative method (identify every structural commitment that thesis makes, ask for each whether it can be derived from another rather than posited, and accept the minimal set of mutually independent conditions that jointly discharge the thesis), and examine whether the result is five levels with the content specified in Definitions 2.1 through 2.5, or some other number with different content. This procedure is independent of any knowledge of the *Kawana'on* system; it requires no access to the positive derivation; and its application by any reader who accepts the relational thesis constitutes an independent verification of the grid's structural status. The claim that the grid is derivable from the relational thesis alone is therefore a testable, falsifiable commitment of this work, not an assertion that must be accepted without examination. A reader who applies the procedure and reaches a different count, or different conditions at any level, has found a genuine challenge to the diagnostic framework — and is invited to bring it forward. The framework's independence is demonstrated not once by argument but continuously by the availability of independent replication.

2.3 Level I: The Ground

If entities are constituted through their relations, then there must be something from which those entities and their relational capacities emerge. This is the first and most fundamental question that the relational commitment generates: *what is the ground of relation?* Not the ground of any particular relation, which might be answered by specifying the entities that stand in it and the context in which they do so. But the ground of relation as such: the structural source from which both the capacity for relational constitution and the specific character of whatever enters into relation are generated.

The question is not asking for a cause in the efficient-causal sense. It is asking for a structural source: something from which the relational field emerges as a structural consequence rather than something that initiates it through an act of external agency. The structural source must be prior to the relata it generates, not in the temporal sense of preceding them in time, but in the ontological sense of being what makes them possible. Remove the structural source and you remove not just the relata but the structural conditions under which anything whatsoever could enter into relation. This is what makes the ground the ground rather than just another term in the relational account: its absence does not merely make the account incomplete but makes it impossible.

Specifying what the ground must be requires attention to what the specification must not allow. It must not allow the ground to be simply the totality of things: that would make it a name for everything there is, and a name for everything there is explains nothing. It must not allow the ground to be a meta-entity standing over and above the relata it grounds: that would replicate the problem rather than solving it, since the meta-entity would itself require a ground. And it must not allow the ground to be self-grounding in the circular sense: that would be to say that the ground exists because it grounds itself, which is either trivially true of everything or explanatorily vacuous. The structural requirements of an adequate account of the ground follow from these exclusions.

Definition 2.1 (The Ground: Formal Requirement)**.** An account of the ground of relation is formally adequate if and only if it satisfies the following four conditions jointly.

(G_1) **Totality.** The ground admits no outside from which additional structural resources could be imported. Everything that enters into relation, every structural condition under which relation is possible, and every structural consequence that relation produces is contained within the ground. The ground is not one structural level among others; it is the field within which all structural levels are embedded.

(G_2) **Generativity.** The ground is capable of producing structural differentiation from within its own dynamic. It does not merely contain the relata as undifferentiated potentialities; it generates their specific characters through its own internal activity. An account that posits a homogeneous ground from which differentiation follows only by logical entailment, without a structural dynamic of differentiation internal to the ground, fails this condition.

(G_3) **Self-sustenance.** The ground requires no external principle to maintain its own dynamic. If the ground's generativity depended on an external principle — a God who lures it towards realisation, a pre-established harmony that coordinates its outputs, a brute contingency that initiates its differentiation from outside — then the ground would not be the ground. The external principle would be. An account that introduces an external regulator, even implicitly, fails this condition.

(G_4) **Formal derivability.** The ground is shown to be the unique structural result of the eliminative method applied to what any relational ontology presupposes, rather than stipulated as an arbitrary starting point. An account that posits the ground as primitive — as what the account simply begins from, without showing that any alternative starting point would reduce to it or prove structurally inadequate — fails this condition. Formal derivability is the condition that distinguishes a structural ground from a metaphysical foundation: the latter is asserted; the former is won.

Remark 2.2 (On the Four Conditions and Their Interdependence). The four conditions are not independent. G_1 and G_2 are jointly required by the relational commitment itself: if entities are constituted through their relations, then the structural source of both entities and relation must be all-encompassing (G_1) and internally generative (G_2), since a ground that excludes some structural resources, or that is homogeneous and incapable of generating the differentiation relational constitution requires, cannot be the ground of relational constitution in general. G_3 follows from G_1: if the ground is total, there is nothing outside it from which an external principle could be drawn. An account that satisfies G_1 and G_2 but invokes an external regulator implicitly contradicts G_1 — it has tacitly admitted an outside while formally denying it. G_4 is the methodological condition that converts the ontological claim into a structural result: it is the condition under which the ground is genuinely derived rather than posited. Together, the four conditions constitute the minimum that any account of the ground must satisfy in order for the relational commitment to be coherent rather than merely asserted.

Illustration from cosmology. Consider the ground state of quantum field theory: the vacuum, defined as the lowest-energy state of the quantum field, from which particle-antiparticle pairs emerge through virtual encounters driven by vacuum fluctuations. The vacuum satisfies three of the four conditions with considerable precision. It is total in the sense that there is no physical location outside the quantum field (G_1): the vacuum is the structural background against which every physical event occurs. It is generative: virtual pair creation is not the mechanical output of an external mechanism but the internal activity of the vacuum state itself, the consequence of the energy uncertainty inherent in the vacuum (G_2). It is self-sustaining in the sense that vacuum fluctuations require no external initiator: they are what the vacuum does by virtue of what it is (G_3). But it fails the fourth condition. The vacuum is stipulated as the lowest-energy eigenstate of the Hamiltonian; the question of what grounds the vacuum energy itself — the cosmological constant problem, the fact that the observed value of the vacuum energy is orders of magnitude smaller than quantum field theory predicts — is structurally unresolved within the framework that posits it. The vacuum is the best physical approximation to a ground that the natural sciences have produced, and it fails precisely at the point where formal derivability (G_4) is required.[13] The illustration makes the formal requirement vivid precisely by failing it at the most instructive point.

The characteristic failure mode of Level I. The characteristic failure at Level I is stipulation: the account posits a ground without deriving it and calls that positing foundational. Every tradition in relational ontology has something that performs the grounding function. Spinoza calls it Substance; Whitehead calls it Creativity; Bergson calls it Duration; Heidegger calls it Being. In each case the grounding term is introduced as the unanalysable ultimate — as what the account simply begins from, and what it would be a category error to ask further questions about. But the refusal to ask further questions is not a structural result; it is a methodological decision. And the structural consequence of that decision is that G_4 is violated: the ground has been stipulated rather than derived, and its structural properties — including, above all, whether it satisfies G_1 through G_3 — are

[13]The cosmological constant problem is among the most significant unsolved problems in theoretical physics; its standard formulation is that the vacuum energy density predicted by quantum field theory exceeds the observed value of the cosmological constant by approximately 120 orders of magnitude. The philosophical significance of this discrepancy, for the present purposes, is structural rather than empirical: it marks the point at which the vacuum, as a putative account of the physical ground, fails the derivability condition. The vacuum energy is stipulated rather than derived; and what is stipulated can be questioned. A structural ground that satisfies G_4 would be immune to this question, not because it is empirically better confirmed but because it would have been shown to be the unique structural result of what any account of relation presupposes. This is the difference between a physical ground and an ontological one.

asserted rather than shown. This failure is the diagnostic signature of entry at Level I.

Remark 2.3 (On the Logical-Entailment/Structural-Generation Distinction). The distinction between logical entailment and structural generation that G_2 depends upon is formal, not merely intuitive. A ground $\mathcal{G}$ satisfies the generativity condition only if: (a) there is a process P internal to $\mathcal{G}$'s own dynamic such that (b) P has an asymmetric before-and-after structure within $\mathcal{G}$ itself, and (c) the structural differentiation that P produces is not identical to the structural content of $\mathcal{G}$ prior to P's operation. Condition (c) is what disqualifies purely logical-entailment grounds: if the outputs are logically entailed by $\mathcal{G}$'s nature, they are in a precise formal sense already contained in $\mathcal{G}$ — no operation of $\mathcal{G}$ is needed to produce them; they are consequences of what $\mathcal{G}$ is, not products of what $\mathcal{G}$ does. A fully actual ground — one whose complete structural content is given at the outset, from which everything else follows as a necessary consequence — necessarily fails G_2. The ground must not merely *entail* differentiation but *enact* it: differentiation must emerge through the ground's operation, not be deducible from the ground's character prior to any operation.

This is why Spinoza's Substance fails G_2 while a dynamical field with internal self-differentiating activity would not. Substance is fully actual: its attributes and their infinite modes are *eternal* consequences of what Substance necessarily is. The entailment is necessary and complete prior to any process; there is no temporal or structural asymmetry within the ground from which modes emerge as something genuinely new. By contrast, a ground satisfying G_2 exhibits asymmetry: the outputs of its generative process are structurally posterior to the process's operation within the ground, not simultaneously identical with the ground's own character.

2.4 Level II: Relational Identity

If entities are constituted through their relations, then each entity must have something that makes it the specific relatum it is — something that is both formed through relational encounter and preserved through it as the entity's characteristic mode of self-constitution. Without this, the relational account cannot distinguish between two entities entering into relation and one entity relating to itself; it cannot specify why the structural yield of encounter between R_1 and R_2 is formally distinct from the yield of encounter between R_1 and R_3; and, ultimately, it cannot maintain that there are entities in relation rather than relation without determinate relata. The question that relational identity poses is therefore not peripheral to the relational thesis. It is its condition of coherence: *what makes each relatum the specific relatum it is, in and through relational encounter rather than prior to it?*

The question is doubly constrained. The answer must not reintroduce intrinsic individuation through the back entrance — it must not be the case that relational identity is, at bottom, a matter of the entity's intrinsic properties, which the relational language merely redescribes. But it must also not dissolve the relatum entirely into the relational web, reducing "this specific relatum" to "this node in this network," since that redescription loses what it was supposed to explain: what makes the node the specific node it is rather than another configuration of relations with the same formal position. The requirement is for an account of identity that is genuinely relational while remaining genuinely specific: a middle path between intrinsic isolation and relational dissolution that is not a mere compromise but a structural achievement.

Definition 2.2 (Relational Identity: Formal Requirement)**.** An account of relational identity is formally adequate if and only if it satisfies the following four conditions jointly.

(I_1) **Invariance through encounter.** The relatum's characteristic mode of self-constitution must persist through relational encounter, not be dissolved by it. This does not require that the relatum be unchanged by encounter; on the contrary, the relatum's character may deepen or develop through encounter. What is required is that the relatum remain distinguishable as the specific relatum it is throughout the encounters it undergoes. An account in which relata are so thoroughly constituted by their relations that no aspect of their character survives relational dissolution fails this condition.

(I_2) **Disclosure through encounter.** The relatum's specific character must be disclosed through relational encounter rather than being fully determinate prior to encounter. An account that specifies the relatum's character entirely independently of its relational history fails this condition by reverting to intrinsic individuation. What the relatum is is not exhausted by any finite description of its intrinsic properties; its character is inexhaustibly disclosed through the encounters it undergoes.

(I_3) **Asymmetric specificity.** No two relata are structurally identical. What makes a relatum specifically itself cannot be a property that it shares with any other relatum without remainder. Relational identity is not a type-identity but a token-identity: it individuates this relatum as distinct from every other relatum in the structural field, not merely as an instance of a kind.

(I_4) **Derivability from Level I.** Relational identity must be derivable from the same structural source that constitutes the ground. An account that posits relational identity independently of the ground — that introduces the specific characters of relata as an additional ontological input alongside the ground — fails this condition and generates a dualism between the ground and what it is supposed to ground. Relational identity must emerge from the ground's generative dynamic, not be imported alongside it.

Illustration from cell biology. The plasma membrane of a living cell provides a precise illustration of relational identity at the biological scale. The membrane maintains the specific identity of the cell through selective permeability: it is simultaneously a boundary and an interface, constituting the cell as this specific cell by determining what it exchanges with its environment and how. The membrane is intrinsic to the cell — it belongs to the cell as its characteristic mode of self-definition at the interface with the extracellular world. But it is constituted through relational encounter: the membrane's specific composition, its channel distribution, its electrochemical gradient, are all functions of the cell's history of exchange with its environment. What the cell is, as a specific biological individual, is inseparable from the history of the encounters its membrane has mediated.[14]

Three of the four formal conditions are satisfied with precision. The membrane provides invariance through encounter (I_1): cell identity persists through exchange with the environment, and it is the maintenance of the membrane's selective permeability that constitutes this persistence. It provides disclosure through encounter (I_2): the membrane's specific character — its ion channels, its receptor proteins, its lipid composition — is disclosed progressively through the encounters the cell undergoes, not specified in advance. It provides asymmetric specificity (I_3): no two cells have identical membrane compositions and histories. But derivability from Level I (I_4) is not achieved within the biological account: the membrane's existence presupposes the cell's metabolic organisation, which in turn presupposes the biochemical

[14]The autopoietic account of living organisation formalises this at the level of the cell's self-production: the plasma membrane is itself a product of the cell's metabolic network, which it simultaneously enables by maintaining the chemical environment in which the network's processes occur Humberto R. Maturana and Francisco J. Varela, *Autopoiesis and Cognition: The Realization of the Living* (Dordrecht: D. Reidel, 1980), 43–52. The membrane is both produced by and constitutive of the relational organisation it partially defines — a structural circularity that is not vicious but is itself the formal mark of the autopoietic level of relational organisation. The present account does not adopt the autopoietic framework; it uses the membrane as an illustration of what relational identity looks like when the formal requirements (I_1)–(I_3) are met at a determinate biological scale.

ground of organic chemistry, which in turn presupposes the physical ground from which organic chemistry emerges. The biological account of relational identity is a regional account — adequate within the domain of living systems — that presupposes what it cannot itself derive. The formal requirement I_4 marks the boundary of the regional account and the horizon of what a foundational account must supply.

The characteristic failure mode of Level II. The characteristic failure at Level II takes two forms, each the structural complement of the other. The first is the *residual intrinsicalism* failure: the account claims that relational identity is constituted through encounter but preserves, at the foundational level, a core of intrinsic properties that are prior to any encounter and that constitute the entity before it relates. The second is the *relational dissolution* failure: the account abandons the claim that there are specific relata with identity and holds that the relational web is all there is, with no determinate nodes. Both failures are diagnostically significant. The first marks an account that enters the structural order at Level II but cannot derive I_4 — it tacitly imports intrinsic individuation as the foundation it cannot produce from the relational account alone. The second marks an account that cannot satisfy I_1 or I_3 — it has conceded so much to the relational thesis that the entities which were supposed to be constituted by relation have disappeared entirely. What survives in neither case is what the relational commitment actually requires: entities that are genuinely specific and genuinely relational at once.

2.5 Level III: The Source of Encounter

If entities have relational identities, they are capable of entering into relation. But being capable of entering into relation is not the same as doing so. The encounter must actually occur. Something must bring entities with relational identities into specific relational contact: not any pair of entities indiscriminately (which would make relation universal and therefore not a discriminating structural event) and not no pair at all (which would make relation impossible despite the capacity for it). The question that the source of encounter poses is structural and precise: *what brings specifically this relatum into contact with specifically that one, rather than with others or with none?*

The source of encounter cannot be derived from the relational identities of the entities it brings into contact without circularity. The relational identity of entity R_1 specifies what R_1 is; it does not specify that R_1 will encounter R_2 rather than R_3, since both R_2 and R_3 may fall within R_1's structural range of relational possibility. Something structurally distinct from both R_1 and R_2 must bring them into the specific structural configuration in which their encounter is made actual rather than remaining merely possible. This something is the source of encounter. And since it cannot be imported from outside the structural account without generating an external perturbation principle that the account has no resources to justify, it must be derivable from the same structural dynamic that produces the ground (Level I) and the relational identities (Level II) it brings together.

Definition 2.3 (The Source of Encounter: Formal Requirement)**.** An account of the source of encounter is formally adequate if and only if it satisfies the following four conditions jointly.

(E_1) **Derivability from Levels I–II.** The source of encounter must arise from the same structural dynamic that produces the ground and the relational identities it brings into contact. An account that introduces an external perturbation principle — a God, a pre-established harmony, a background field of uncaused causation — to explain why specific encounters occur fails this condition. The source of encounter must be an internal product of the structural framework, not an external supplement to it.

(E_2) **Specificity.** The source of encounter is specific to the encounter it enables: it brings this relatum into contact with that one and not with others, or not with others at this moment and under these structural conditions. A source of encounter that produces encounter universally — that brings every relatum into contact with every other simultaneously — fails this condition by eliminating the selectivity that makes encounter a determinate structural event.

(E_3) **Formal distinction from the relata.** The source of encounter must be formally distinct from the relational identities it brings together. If the source of encounter were identical with one of the relata, it could not bring the two relata into contact from a structural position external to both: it would be R_1 encountering R_2, not the structural condition under which that encounter is made possible. The source of encounter is the enabling condition of the relational event, not a participant in it on equal footing with the relata.

(E_4) **Inexhaustibility.** The source of encounter is not depleted by enabling the encounters it enables. If the source of encounter were consumed in producing each relational event, the account would face the structural difficulty of explaining how the source is regenerated — which is either an infinite regress or an appeal to an external regeneration principle. The source of encounter must be structurally inexhaustible: its capacity to bring relata into contact must be maintained through the very encounters it enables, not diminished by them.

Illustration from enzyme catalysis. An enzyme provides a precisely structured illustration of the source of encounter at the biochemical scale. The enzyme's function is to lower the activation energy barrier between a substrate and its reaction product, bringing them into the specific structural configuration in which the chemical reaction becomes energetically accessible. The connection to Level III is direct: just as a thermodynamic gradient constitutes the structural condition that brings molecules into potential contact — making specific encounters possible where previously none could occur — an enzyme constitutes the structural condition that brings specific substrates into the activated configuration their reaction requires. Both the gradient and the enzyme are sources of encounter: they are formally distinct from the relata they bring together, specific to the encounter they enable, and derived from the same structural ground that produces the relata themselves. The enzyme is formally distinct from both the substrate and the product (E_3): it is not itself consumed in the reaction it enables. It is specific to the encounter it enables (E_2): each enzyme has a precise binding site that makes it structurally selective for specific substrates. It is inexhaustible in the relevant sense (E_4): the enzyme is regenerated at the conclusion of each catalytic cycle and is available for further reactions. And the enzyme arises from the same biochemical ground that produces the substrates it brings together (E_1): enzymatic activity is not an external addition to the biochemical system but a product of the same metabolic organisation that generates the substrates and products the enzyme mediates.[15]

The enzyme illustration makes the formal requirements vivid but also reveals their limits as regional accounts. What an enzyme-based illustration cannot supply is the derivation of the source of encounter from the ground itself (E_1): enzymatic specificity is a biological achievement that presupposes the entire hierarchy of physical and chemical structure below it. The source of encounter, at the ontological level this investigation addresses, must be derivable not from within

[15]The structural parallel between enzymatic catalysis and the source of encounter is not an analogy; it is an instance of the general structural requirement at the biochemical scale. Enzyme catalysis meets all four conditions precisely because the enzyme is itself a biological entity with a relational identity (Level II) shaped by the evolutionary and developmental processes through which it was produced (Level I, at the biochemical level), specialised to the specific encounter it enables (E_2), and formally distinct from the relata it brings together (E_3). What the illustration does not supply is the foundational derivation of why the enzyme has the structural character it has: that question is answered at Level I, and the biochemical account, being a regional account, cannot discharge it from within its own resources.

a regional domain but from the structural dynamic of the ground as such.

The characteristic failure mode of Level III. The characteristic failure at Level III is the import of an external perturbation principle. Every major tradition in relational ontology that gets this far in the structural derivation — that has an account of the ground and of relational identity — eventually confronts the question of what brings specifically equipped relata into specific relational contact and reaches for an answer that the framework's own resources cannot supply. Whitehead reaches for God's initial aim. Leibniz reaches for pre-established harmony. Contemporary structural realism, having eliminated objects in favour of structures, finds that the structures encounter one another as a brute contingency the account cannot explain. In each case the symptom is the same: a structural placeholder that names the source of encounter without deriving it, inserted at precisely the point where the framework's internal resources give out. That the structural mechanism by which relations actually hold between their relata — what analytic metaphysics calls *adherence* — cannot be derived from within any of the frameworks that posit it as their starting point has been confirmed in peer-reviewed analytic metaphysics by Hakkarainen and Keinänen,[16] who establish that the most promising current response to Bradley's regress leaves the constitutive mechanism of adherence "as a constitutively inexplicable primitive" and "highlights the need for a metaphysical account of the constitution of the holding

[16]Hakkarainen and Keinänen, "Bradley's Relation Regress and the Inadequacy of the Relata-Specific Answer."

of adherence" — a formulation that identifies, from within the analytic tradition and independently of the present diagnostic framework, exactly the structural gap that Level III names. The formal requirement E_1 marks that point and names what a structurally honest account must supply there.

2.6 Level IV: Relational Yield

If encounter occurs between relata with relational identities, the encounter produces something. The question that relational yield poses is: *what does it produce, and what is the formal character of what it produces?* This question is not asking for a description of the many things that many encounters produce in many domains; it is asking for the formal specification of what encounter, as a structural event, necessarily generates.

The answer cannot be: the sum of the relata's contributions. If encounter produced only what the relata already contained, relation would add nothing to what was already the case; it would be aggregation rather than constitution. And the relational thesis is precisely that relation is constitutive, not merely aggregative: that what entities are cannot be specified independently of what their encounters produce. If the yield of encounter is nothing more than the combined properties of the relata, then what the relata are is already fully specified before the encounter, and the relational thesis has been tacitly abandoned.

The answer also cannot be: sheer novelty, undetermined by the structural character of the encountering relata. If the yield of encounter were disconnected from what the relata contribute to the encounter, there would be no structural account of why this encounter produced this yield rather than any other. The formal specificity of relational yield — the fact that encounter between R_1 and R_2 produces this yield rather than some other — must be accountable by reference to the structural characters of R_1 and R_2, even though the yield is not reducible to the sum of those characters.

Definition 2.4 (Relational Yield: Formal Requirement)**.** An account of relational yield is formally adequate if and only if it satisfies the following four conditions jointly.

(Y_1) **Irreducibility.** The yield is not derivable from any formal operation performed on the relata alone — addition, multiplication, superposition, or any other combination that treats the relata as self-contained inputs. The yield is genuinely new: it has structural properties that neither relatum possessed and that the encounter produces rather than reveals.

(Y_2) **Structural conservation.** The yield is bounded by the structural resources of the encounter: it does not produce structural content from nowhere. The relational account must specify the formal relation between the structural resources that the encountering relata bring and the structural resources that the yield constitutes. Conservation here is not the conservation of energy or mass in the physical sense; it is the structural condition that prevents the yield account from being explanatorily promissory: claiming novelty without specifying from what structural resources the novelty arises.

(Y_3) **Generativity.** The yield must be capable of becoming a relatum in further encounters. An account in which the yield of relation is a structural terminus — something that is produced but cannot itself enter into further relation — fails to account for the cumulative character of relational constitution: the fact that what relation produces is not consumed by the encounter that produces it but becomes the structural material from which subsequent encounters draw. Generativity is the formal condition under which the relational account does not require a fresh supply of unrelated relata for every new encounter.

(Y_4) **Asymmetric novelty.** The yield of encounter between R_1 and R_2 is formally distinct from the yield of encounter between R_1 and R_3, even when R_2 and R_3 belong to the same structural type. Relational yield is not a function of the types of the encountering relata but of the structural characters of these specific relata in this specific encounter. An account that specifies relational yield as a function of type-identity — that claims that all encounters between relata of type T_1 and type T_2 produce yield of type T_3 — fails this condition by replacing the irreducible specificity of the relational event with a generic formula.

Illustration from forest ecology. The mycorrhizal network provides a precisely structured illustration of relational yield at the ecological scale. In forest ecosystems, the underground fungal network formed by mycorrhizal fungi connecting with the root systems of trees generates a structural yield that is irreducible to either the trees or the fungi individually: the network itself, which enables the exchange of carbon, nitrogen, and phosphorus across distances and between species that could not achieve such exchange through any other means. A mature forest's mycorrhizal network enables coordinated chemical signalling in response to environmental stress, the redistribution of nutrients from resource-rich to resource-poor regions of the ecosystem, and the support of seedlings that have not yet developed sufficient photosynthetic capacity to sustain themselves independently.[17]

The network satisfies all four formal conditions at its own scale of analysis. It is irreducible (Υ_1): the network's capacity for coordinated response and nutrient redistribution across the ecosystem is not derivable from the properties of any tree-fungus pair considered in isolation. It is structurally conservative (Υ_2): the network's structural resources are bounded by the carbon and nutrient flows of the trees and fungi that constitute it, not imported from outside the ecosystem. It is generative (Υ_3): the network becomes itself a relatum in further encounters — with environmental stressors, with new species that join the net-

[17]The ecological literature on mycorrhizal networks has established that what was initially described as a bilateral mutualism between individual trees and fungi is in fact a complex multi-species network exhibiting properties that are structurally irreducible to any dyadic encounter within it; see the synthesis in Simard, *Finding the Mother Tree* (2021), which popularises a body of peer-reviewed research establishing that trees in a mycorrhizal network exhibit differential resource allocation towards kin seedlings, co-operative responses to drought stress, and other network-level behaviours that cannot be explained by the bilateral relations of individual tree-fungus pairs. The formal structural point — that the network is irreducible to the dyadic encounters that constitute it (Υ_1) and that it becomes itself a relatum in further ecological encounters (Υ_3) — is demonstrated, at the ecological scale, by this research programme.

work, with the atmospheric and geological processes that the forest ecosystem modulates. And it exhibits asymmetric novelty (Y_4): the specific mycorrhizal network of this forest, with its particular species composition, structural topology, and history of accumulated encounters, is formally distinct from the network of any other forest, even one with the same dominant species in the same climatic zone.

The characteristic failure mode of Level IV. The characteristic failure at Level IV is emergence claimed but not derived. The term *emergence* names the structural requirement (Y_1) — that yield be irreducible to the properties of the encountering relata — but it does not discharge the requirement. An account that says relational encounter produces emergent properties, without specifying what structural resources the emergence draws on, what formal relation holds between the structural characters of the relata and the properties of the yield, or how the yield becomes a relatum in further encounters, has assigned a name to the problem rather than providing a solution. Emergence is a description of what relational yield is not (not a mere aggregation) rather than a structural account of what it is. The formal requirement of Level IV demands the latter.

2.7 Level V: The Structure of the Relational Event

If encounter between relata with relational identities produces a yield that is genuinely new, bounded, and generative, then the encounter must itself have an internal structure — a formal organisation that accounts for how the relata come into specific contact, how that contact is sustained in productive tension rather than immediately collapsing into merger or dissolution, and how it is oriented towards the specific yield it produces. Without this, the relational event is a black box: a name for the interval between the moment before encounter and the moment in which yield exists. The question that the structure of the relational event poses is the most formally demanding of the five: *what is the internal organisation of the relational event, and why does it have the structure it has rather than some other structure?*

The question demands derivation rather than description. It is not asking for a list of the things that happen in a relational event. It is asking for the structural reason why the event is organised as it is — why it has exactly the formal dimensions it has, neither more nor fewer, and why those dimensions are co-present in every relational event rather than occurring in sequence or in some encounters but not others. An account that describes the relational event without deriving its structural organisation has not discharged this requirement; it has redescribed the black box from the inside rather than opening it.

Definition 2.5 (The Structure of the Relational Event: Formal Requirement)**.** An account of the structure of the relational event is formally adequate if and only if it satisfies the following four conditions jointly.

(V_1) **Derivability of the event's structure.** The internal organisation of the relational event must be derived from the same structural dynamic that produces the ground, relational identity, the source of encounter, and relational yield. An account that describes the event's structure without deriving it — that observes that relational events have certain characteristic dimensions without showing why they have those dimensions rather than others — fails this condition.

(V_2) **Non-redundant completeness.** The formal dimensions of the relational event must be jointly exhaustive and individually non-redundant. Jointly exhaustive means that the dimensions together account for everything that happens in a relational event without remainder. Individually non-redundant means that no dimension does the structural work of another: each dimension accounts for what the others cannot. An account in which two dimensions partially overlap — in which what the first accounts for is already accounted for, in part, by the second — fails this condition and must be revised until the dimensions are genuinely distinct.

(V_3) **Co-instantaneity.** The formal dimensions of the relational event are co-present in every relational event, not sequential stages through which the event passes. An account that presents the dimensions as stages — in which the encounter first initiates, then is regulated, then is directed — fails this condition by importing a temporal order that presupposes what the account is supposed to derive: the temporal asymmetry of the relational event. The co-instantaneity of the event's dimensions is what makes it an event rather than a process: it is a structural moment, complete at the moment of its occurrence, not an unfolding that requires time in the prior sense.

(V_4) **Self-exemplification.** The formal account of the relational event must itself be an instance of the structural dynamic it formally establishes. An account of the event's structure that is not exemplified in its own derivation — that describes a structural dynamic it does not enact — is incoherent in a precise sense: it claims to have identified a structural necessity that its own method of identification does not obey. Self-exemplification is not merely a formal elegance; it is the condition under which the account is genuinely self-sealing rather than a description of structure from an external standpoint that the description does not account for.

Remark 2.4 (Formal Status of Sub-Condition (V_4) and the Appendix Revision). Sub-condition (V_4) is introduced here in its *diagnostic* role: it identifies whether a framework's derivational method enacts the structural dynamics it formally establishes, and this is what the diagnostic survey in Part II tracks when it applies the Level V criterion. However, its correct *formal* classification is as a requirement on the derivational method rather than as an assignment criterion determining whether a framework enters the structural order at Level V. A framework can enter at Level V while failing (V_4) — Whitehead's is the canonical case — which means (V_4) is not part of the criterion for being at Level V but part of the criterion for *satisfying* Level V's demands. In the revised formal definitions consolidated in Appendix A (Remark A.1), the self-exemplification sub-condition is accordingly relocated from the Level V assignment criterion to Requirement R6(iv), which governs whether a framework *satisfies* the sixth formal requirement rather than whether it is *assigned* to Level V. Readers applying the diagnostic grid to traditions not surveyed in this volume should use the Appendix formulation as the operative criterion. The present definition introduces (V_4) as a diagnostic probe; the Appendix states its authoritative formal position in the requirement structure.

Remark 2.5 (On Co-Instantaneity and Temporal Asymmetry). The co-instantaneity requirement (V_3) deserves careful handling, because it might appear to conflict with the evident fact that relational events have a temporal character — that there is a moment before encounter and a moment in which yield has been produced, and that these moments are not identical. The conflict is apparent rather than real, and its resolution is structurally significant.

The co-instantaneity of the event's formal dimensions is not the claim that the event is instantaneous in the temporal sense of occupying a point in time with zero extension. It is the claim that the dimensions are not related as *stages* in the temporal sense: as a sequence of events, each of which presupposes the completion of the preceding one. Co-instantaneity is a structural claim about the formal organisation of the event, not a physical claim about its temporal duration. What gives the event its temporal character — its asymmetry of before and after, the irreversibility of what it produces — is not its temporal duration but its structural organisation. That organisation generates temporal asymmetry; it does not presuppose it. This is why (V_3) is a formal requirement rather than an empirical observation: it is the condition under which the temporal character of the relational event can be *derived* from the event's structure rather than *imported* as a background assumption. The derivation of temporal asymmetry from the structure of the relational event is one of the results that a formally adequate account of Level V must supply. None of the traditions surveyed in Part II provides it.

Derivation of the event's three dimensions. The four formal conditions (V_1)–(V_4) establish what any adequate account of the relational event must supply. They do not, by themselves, specify what the internal dimensions of the event are. That specification is a structural consequence of (V_1) (derivability) applied to what the prior four levels have already established.

The relational yield (Level IV) has four structural properties: it is formally new (Y_1), structurally bounded (Y_2), generative of further relational capacity (Y_3), and asymmetrically novel (Y_4) — the yield transforms both relata, but not identically. For such yield to be produced, the relational event must have a structure that accounts for each of these properties. Deriving that structure from the prior levels is what (V_1) demands, and the derivation proceeds as follows.

Proposition 2.1 (The Three Dimensions of the Relational Event)**.** The relational event has exactly three co-instantaneous dimensions: (1) initiation, (2) regulation, and (3) direction. These three are necessary and jointly sufficient for the event to produce the yield established at Level IV.

Proof. Necessity. Consider each dimension in turn.

Initiation is the structural moment at which two relata whose identities are established at Level II are brought into specific relational contact by the encounter source established at Level III. Without initiation, no encounter occurs: the relata remain structurally isolated, and no yield is produced. Initiation is not reducible to the encounter source (Level III): the source specifies the structural dynamic that *makes encounter possible*; initiation is the structural event in which specific relata *actually enter into* specific contact. The distinction corresponds to the distinction between structural capacity and structural event. Initiation is therefore a necessary dimension of the relational event.

Regulation is the structural principle that sustains the encounter in productive tension rather than allowing it to collapse into merger (the loss of both relata's structural distinctness) or dissolution (the disengagement of the relata without yield). The irreducibility of yield (Y_1) and its generativity (Y_3) jointly entail that the encounter must have a sustaining structure. From (Y_1): yield is irreducible to either relatum, which requires that neither relatum absorbs the other — merger produces as its "yield" the absorbing relatum's properties, violating irreducibility. From (Y_3): yield must be capable of becoming a relatum in further encounters, which requires that the original relata are not structurally consumed by the encounter — merger and dissolution both terminate the structural capacity needed for yield to enter further encounters. (Y_1) and (Y_3) together rule out both failure modes, entailing that the encounter must have a structural principle that sustains it against collapse and dissolution throughout. That principle is regulation.

Direction is the structural orientation of the encounter towards a specific yield. The asymmetric novelty of yield (Y_4) — the requirement that yield is not a function of relata-type but of the structural characters of these specific relata in this specific encounter — entails that the encounter must be oriented towards what these specific relata, in this specific configuration, structurally make possible. Without such orientation, the encounter would produce generic type-output determined by the relata's structural types rather than output specific to their individual characters, violating (Y_4). Direction is therefore a necessary dimension of the relational event.

Non-redundancy and joint sufficiency. The three dimensions are individually non-redundant: initiation (the coming-into-specific-contact) does the structural work that neither regulation (the sustaining of contact) nor direction (the orientation towards specific yield) does; regulation does structural work that neither initiation nor direction does; direction does structural work that neither initiation nor regulation does.

The three are jointly sufficient. Any aspect of the relational event not accounted for by these three must do structural work independent of all three. Two candidates arise naturally. First, *termination*: the event must conclude. But the conclusion of the event is not a structural dimension independent of direction; it is direction's achieved telos. Direction orients the encounter towards specific yield, and the production of that yield is what constitutes the event's conclusion. Termination is direction seen from the vantage of its completion, not an additional structural dimension of the event. Second, *preparation*: the relata must be readied for encounter. But preparation is not a dimension of the relational event itself; it is a feature of what precedes the event — the structural conditions at Levels I–IV that make the event possible. The event begins at initiation; what comes before initiation belongs to the conditions of the event, not to the event's internal structure. No further dimension does structural work irreducible to the three established or to pre- and post-event conditions. The three are the minimum-complexity structure of any event that produces irreducible, generative, asymmetrically novel yield.

Remark 2.6. The three dimensions are co-instantaneous in the sense required by (V_3): they are not sequential stages through which the event passes but structurally co-present aspects of a single relational moment. That initiation, regulation, and direction are co-present is a consequence of their derivation: regulation is already operative at the moment of initiation (without it, initiation would immediately collapse), and direction is already operative at the moment of initiation (without it, the encounter would have no structural orientation from its first moment). The listener in the sonata illustration experiences all three as a single structural event for the same formal reason: the directional orientation towards recapitulation is constitutive of the exposition's harmonic structure from the first chord.

Sub-conditions of direction: V_α and V_β. The direction dimension — condition (3) of the relational event's structure — has two formally distinct sub-conditions corresponding to the two structural poles of any directed process.

Definition 2.6 (The Two Sub-conditions of Direction)**.** The direction dimension of the relational event is formally specified by two sub-conditions:

(V_α) **The actuality dimension of direction.** The structurally determinate *before* from which every relational event orients: the formally historical pattern of accumulating sedimentation — the irreversible structural weight that prior relational events have deposited in the structural field — which constrains the weight distribution of the current encounter and thereby shapes the range of outcomes it can produce.

(V_β) **The possibility dimension of direction.** The formally admissible wider pattern of structural configurations towards which the current actuality is oriented: not an arbitrary openness but a structured space, shaped by the sedimented weight distribution, within which some outcomes are structurally nearer and others structurally further. The possibility space is the formal horizon of direction; it constitutes what the encounter is directed *towards*.

A formally adequate account of the direction dimension satisfies both (V_α) and (V_β). An account that specifies only (V_α) takes the sedimented before as its foundational datum while leaving the formal structure of the oriented towards unspecified. An account that specifies only (V_β) takes the structured horizon of possibility as its foundational datum while leaving the formally historical weight that constitutes that horizon unspecified.

Illustration from chamber music. The sonata form provides a precise structural illustration of the relational event at the scale of musical encounter. The classical sonata presents two thematic materials that are structurally distinct: the first theme establishes the tonic key and a characteristic rhythmic and melodic identity; the second theme, typically in the dominant or relative key, establishes a contrasting identity. The development section places these two materials in productive tension: neither absorbs the other, neither is cancelled by the encounter with the other, but both are transformed through a sustained structural engagement that generates harmonic and motivic yield irreducible to either theme alone. The recapitulation returns both themes to the tonic key — not as a return to what they were before the encounter but as a structural achievement in which both have been oriented, through the development's productive tension, towards a shared structural resolution that neither theme contained before the encounter.[18]

[18]The structural analysis of the sonata form as a dialectical process — in which the opposition of tonic and dominant, or first and second theme, is productive of a structural resolution that neither theme could achieve independently — has been developed most influentially in Charles Rosen, *The*

The sonata's three-part structure — exposition, development, recapitulation with coda — illustrates the formal dimensions of the relational event with unusual clarity. The exposition initiates the encounter between structurally distinct thematic materials. The development sustains the encounter in productive tension, holding the two materials in a structural relationship that neither absorbs the other nor allows the encounter to dissolve. The recapitulation-coda orients the encounter towards the structural yield it has generated. And these three dimensions are not sequential in the relevant sense: the directional orientation towards recapitulation is already present in the exposition's harmonic structure (the instability of the dominant creates the expectation of tonal resolution from the moment the second theme is introduced), and the development's productive tension is already shaped by the recapitulatory resolution towards which it moves. The listener does not first experience initiation, then regulation, then direction as three

Classical Style: Haydn, Mozart, Beethoven (New York: W. W. Norton, 1971), particularly the chapters on Haydn and the emergence of the sonata principle. The present account does not adopt Rosen's historiographical or aesthetic framework; it uses the sonata form as an illustration of what the formal requirements (V_1)–(V_4) look like when met at the level of musical structure. The parallel is structural rather than analogical: the sonata's three-part organisation of exposition (encounter), development (productive tension), and recapitulation (directional resolution) instantiates the non-redundant completeness required by (V_2), the co-instantaneity of dimensions required by (V_3) (the directional orientation towards recapitulation is already present in the exposition's harmonic structure), and the self-exemplifying character required by (V_4) (the analysis of sonata form is itself an encounter between analytical and musical structures that produces a yield irreducible to either).

separate events: they experience a single structural event in which all three dimensions are co-present.

What the sonata illustration displays is the result whose formal derivation Proposition 2.1 supplies: the three dimensions are necessary and jointly sufficient because they are the minimum-complexity structure of any event that produces bounded, generative, and asymmetrically novel yield. The sonata shows what the structure looks like when instantiated; the proposition establishes why it must be this structure and no other.

The characteristic failure mode of Level V. The characteristic failure at Level V is the temporal presupposition: the account imports time as a background container within which the relational event unfolds, rather than deriving the temporal character of the event from the event's own structure. Every tradition that reaches Level V — that has something to say about the internal organisation of the relational event — confronts the question of why the event is oriented as it is, why it has an asymmetry of before and after, why it is irreversible. The answer that time provides these features as a given framework within which the event occurs is not a structural answer; it is a presupposition. The formal requirement (V_3), and specifically the remark on co-instantaneity and temporal asymmetry, mark the structural point at which this presupposition fails: the account cannot derive temporal asymmetry from the event's structure if it has already imported temporal asymmetry as the container within which the event unfolds. The derivation of time is not a task peripheral to the philosophy of relation; it is what a formally adequate account of Level V requires.

2.8 The Diagnostic Grid

The five formal requirements constitute the diagnostic grid against which Part II will evaluate each major tradition in relational ontology. The grid is presented here in summary form, with its five levels, the formal requirements at each level, and the characteristic failure mode of each.

Level	Formal requirement	Failure mode
I The Ground	(G_1)–(G_4): Totality, generativity, self-sustenance, formal derivability	Stipulation: the ground is posited as the unexamined ultimate
II Relational Identity	(I_1)–(I_4): Invariance, disclosure, asymmetric specificity, derivability from Level I	Residual intrinsicalism or relational dissolution
III Source of Encounter	(E_1)–(E_4): Derivability, specificity, formal distinction from relata, inexhaustibility	External perturbation principle: God, pre-established harmony, brute contingency
IV Relational Yield	(Y_1)–(Y_4): Irreducibility, structural conservation, generativity, asymmetric novelty	Emergence named but not derived
V Relational Event	(V_1)–(V_4): Derivability of structure, non-redundant completeness (three dimensions: initiation, regulation, direction), co-instantaneity, self-exemplification; direction has sub-conditions (V_α) (actuality) and (V_β) (possibility) (Definition 2.6)	Temporal presupposition: time imported as container

The grid makes the diagnostic method of Criterion 1.1 precise. For each tradition, the analysis identifies the entry level — the highest-numbered level at which the tradition's foundational commitment is stipulated rather than derived — and establishes what the tradition achieves above that level and what it structurally forecloses below it. The five chapters of Part II execute this analysis for Whitehead, Spinoza, Bergson, Heidegger, and the contemporary landscape in sequence.

Before the survey begins, two structural observations about the grid itself are required.

Remark 2.7 (The Levels Are Not Separable). The five levels of the diagnostic grid are not independent requirements that a tradition could satisfy in any combination. They form a structural hierarchy in which each level presupposes all the levels below it. A tradition that satisfies Level II but not Level I has not genuinely satisfied Level II: its account of relational identity imports what a Level I derivation would have had to supply, and the formal condition I_4 marks this failure explicitly. A tradition that satisfies Level IV but not Level III has not genuinely satisfied Level IV: its account of relational yield presupposes that encounters occur, which is precisely what Level III is required to account for. The grid is a hierarchy, not a menu. Partial satisfaction is structural failure, not partial credit.

Remark 2.8 (What the Grid Does Not Claim). The diagnostic grid does not claim that the traditions surveyed in Part II are wrong on their own terms. Each tradition has identified something real about the structure of relation and has derived genuine results at and above its entry level. The grid claims only that each tradition's genuine results are local — adequate within the structural scope that its entry level defines — and that the structural foreclosure entailed by that entry level makes the foundational question of relational ontology inaccessible from within the tradition's own resources.

This is a structural claim, not a comparative evaluation. It is not the claim that Whitehead's account of the relational event is less sophisticated than the account a formally adequate Level V derivation would provide. It is the claim that Whitehead's account is constituted at Level V, and that a tradition constituted at Level V has, by that very constitution, foreclosed the derivation of Levels I through IV from within its own framework. The claim is structural and the diagnosis is formal: not that the traditions are inadequate but that they are located, and that their location determines both what they can and what they cannot do.

Remark 2.9 (Applicability Precision of the Grid). The five-level grid is formally defined with sufficient precision that a reader can apply it to a tradition not surveyed in Part II and reach a determinate result without interpretive latitude. The formal conditions at each level — (G_1)–(G_4) at Level I, (I_1)–(I_4) at Level II, (E_1)–(E_4) at Level III, (Y_1)–(Y_4) at Level IV, (V_1)–(V_4) at Level V — specify pass/fail criteria that are checkable against any tradition's primary texts. The diagnostic method of Criterion 1.1 specifies the procedure. If the grid required interpretive latitude rather than formal application, it would not be possible to arrive at a determinate result for a tradition the investigation had not anticipated. The grid's applicability precision is demonstrated, not merely asserted, in Chapter 10 (Section 10.5), where it is applied to the Madhyamaka tradition — a case the investigation did not set up and could not have anticipated — and produces a conditionally determinate result: the grid yields one verdict on the Candrakīrti reading of *śūnyatā* and a different verdict on the Bhāviveka reading, with both verdicts logically compelled by the definitions and the primary textual evidence rather than by interpretive choice. A grid that requires interpretive latitude could not produce this conditional structure; it could only produce one answer for the case. The conditional form is the signature of formal precision.

> *Summary of Chapter 2. Chapter 1 established that the relational turn is a structural necessity and identified five structural commitments it incurs. The present chapter has converted those commitments into five formal requirements: the ground (Level I), relational identity (Level II), the source of encounter (Level III), relational yield (Level IV), and the structure of the relational event (Level V). Each requirement is formally specified by four conditions of adequacy, illustrated by a concrete case from a non-philosophical domain, and characterised by its distinctive failure mode. Together the five requirements constitute the diagnostic grid that the survey of Part II applies to the major traditions in relational ontology. The structural hierarchy of the levels — each presupposing all those below it — means that partial satisfaction is structural failure: an account that meets Level IV but not Level III has not genuinely met Level IV. The diagnostic claim of this book is that every major tradition in relational ontology has its entry level at a non-foundational level, that entry at a non-foundational level structurally forecloses*

the derivation of all levels below it, and that this foreclosure is necessary rather than accidental. Part II establishes this claim through the survey. Part III derives its formal consequence.

PART II

The Diagnostic Survey

Part I established two results. The first is that general ontology necessarily arrives at relational ontology: any account that takes the individuation of entities seriously is driven by the structure of the problem, not by philosophical fashion, to the claim that entities are constituted through their relations. The second is that relational ontology, in assuming relation as its foundation, implicitly incurs five structural commitments — to the ground of relation, the structural identity of relata, the source of encounter, the yield of encounter, and the structure of the relational event itself — whose collective discharge it has not, as a field, attempted.

Part II applies the five-level diagnostic framework to the traditions that have most seriously and rigorously pursued relational ontology. The method in each case is the same: identify the tradition's foundational commitment, determine the structural level that commitment occupies, assess what the tradition formally achieves at and above that level, and establish what the commitment structurally prevents the tradition from reaching below it. The critique in every case is structural, not polemical. Each tradition is taken to have identified something real and achieved something genuine. The question is not whether the tradition is right on its own terms but what it cannot see from where it stands.

The sequence is not alphabetical and not chronological. It is the sequence that makes the structural argument most legible. Whitehead opens Part II because he provides the richest account of what the framework requires at Levels IV and V — the relational event and its yield — and his structural foreclosure of Levels I through III is therefore most sharply visible by contrast with his achievements. Spinoza follows because he is the mirror image: the tradition that comes closest to Level I — the ground — and whose failure there is therefore most precisely specifiable. Together, Whitehead and Spinoza establish the full range of the structural gap: one tradition has the event without the ground, the other has the beginning of a ground without the event. Bergson and Heidegger occupy the two dimensions of the Level V direction structure — the actuality of temporal flow and the formal structure of possibility — and their treatment together shows that even the most penetrating analyses of how the relational event is temporally directed fail to derive what they presuppose about its ground. The contemporary frameworks of Chapters 7 and 8 are treated last because they represent the live state of the problem: the reader arrives at them having understood the full structural depth of the gap the classical traditions left, and is therefore equipped to identify it in its most sophisticated contemporary forms.

A note on the relationship between the diagnostic survey and the formal theorem. Part II surveys thirteen traditions and research programmes. A reader might wonder whether the survey is strictly necessary once the Structural Insufficiency

Theorem has been proved in Part III: if the theorem holds universally, why exhibit thirteen instances? The answer is that the survey and the theorem perform distinct and complementary functions. The theorem establishes that *any* account that foundationally constitutes a non-latent feature is structurally insufficient; the survey demonstrates that the thirteen traditions selected as the strongest available challenges to that claim each instantiates the theorem's antecedent. The survey is not evidence for the theorem — the theorem is a formal result that does not require empirical support — but it is the proof that the theorem is non-trivially applicable: that the traditions serious philosophers have actually developed to explain relational possibility all, without exception, perform exactly the foundational move the theorem shows to be structurally insufficient. Without the survey, the theorem would be a conditional with an unverified antecedent. With it, the conditional is discharged for every serious candidate in the literature. The selection criterion governing which traditions are included in the survey is stated and defended in full in §8.11; readers wishing to assess the survey's representativeness before engaging the individual chapters may read that section first.

A note on selection. The traditions surveyed here are not selected because they fail in the way the argument needs them to fail. The selection criterion is the following. Each tradition surveyed is (a) the most formally developed relational ontology within its own methodological tradition, (b) among the traditions most likely to have explicitly addressed the structural conditions of relational possibility — the conditions under which relation can be fundamental rather than derivative — and (c) a representative of one of the five major methodological approaches through which the relational thesis has been pursued: the *process* approach (Whitehead, Bergson), the *structural-realist* approach (ontic structural realism, relational quantum mechanics), the *analytic-metaphysical* approach (Spinoza as precursor to analytic reception, formal analytic structural realism), the *phenomenological* approach (Heidegger, Stengers), and the *social-scientific* approach (Emirbayer, social relational ontology). Together these five approaches constitute the full methodological range through which the relational thesis has been given its most developed and precise form. The selection is not exhaustive of all relational thought; it is exhaustive of the formally developed relational ontologies across the methodological approaches that determine the current state of the problem. A tradition that does not appear in the survey either belongs to one of the five approaches and has been superseded in formal precision by the tradition surveyed for that approach, or belongs to an approach that does not take the structural conditions of relational possibility as its primary question. Neither category constitutes a stronger test case for the SIT.

Two further points secure the criterion against the objection that it was designed to guarantee failure. First, the criterion is *formally prior* to the application of the five-level grid: it specifies what a tradition must be in order for the diagnostic questions to have determinate answers (criteria (a)–(c)), not what a tradition must be in order to fail those questions. No tradition is included because it is expected to fail; traditions are included because they are engaged and precise enough for the failure or success to be determinately established. Second, the criterion is genuinely open to counter-instances: a tradition meeting criteria (a)–(c) that *satisfied* all five structural requirements simultaneously would pass the diagnostic grid rather than failing it. Such a tradition would constitute the strongest possible confirmation of the relational thesis and a decisive restriction of the SIT's scope. The fact that no surveyed tradition occupies this position is a diagnostic result, not a selection artefact. What would such a tradition look like? It would need a foundational commitment at or below Level I, a derivation of relational identity from that commitment without regress, a formally derived encounter source, a formally specified yield, and a derived co-instantaneous event structure. Meeting all five simultaneously while satisfying the formal precision required by criterion (b) is precisely what the diagnostic survey establishes no tradition has done. The criterion does not guarantee that outcome; it permits it and the survey finds it.

Each chapter in Part II closes with a diagnostic result stated as a formal proposition. The cumulative argument of Part II is not simply the conjunction of independent diagnostic results. It is the demonstration that the same structural pattern — entry at a specific level, genuine achievement at and above that level, structural foreclosure below it — obtains across every tradition in the survey, spanning the major traditions of Anglophone, continental, and process philosophy, without exception.

One point of terminology warrants early clarification. The claim that the traditions fail in the same structural way does not mean that they fail at the same numbered level of the five-level hierarchy. The entry levels differ: Whitehead's foundational commitment falls at Level V, Spinoza's at Level I, Bergson's at Level V sub-dimension (V_α)[19], and so forth for each tradition in turn. What is uniform is not the number of the level at which each tradition's horizon falls, but the *structural type* of the failure: in every case, a foundational commitment at some level n generates a formal horizon below that level and forecloses derivational access to the latent register. The SIT's uniformity claim is a claim about this structural type — the horizon structure — not about its location on the numbered scale. Chapter 9 returns to this point explicitly; the reader is invited to hold the distinction throughout Part II.

When the same structural pattern appears across traditions as methodologically incompatible as Spinoza and Emirbayer — traditions that share almost nothing in method, vocabulary, or philosophical orientation — the explanation is structural rather than coincidental. Part III establishes this formally.

Two traditions absent from the survey require explicit account, since a referee familiar with the relational turn will notice their absence.

[19]The sub-conditions of direction (V_α) and (V_β) are formally defined in Definition 2.6.

Leibniz. Leibniz is not surveyed as an independent diagnostic target because his framework is not a relational ontology in the relevant sense: monads do not constitute one another through encounter; they are self-enclosed and windowless, with their apparent relations mediated entirely by pre-established harmony. The pre-established harmony is precisely the external perturbation principle at Level III that the diagnostic framework identifies as the characteristic failure mode of traditions that reach Level II without being able to derive the source of encounter from within their own resources. Leibniz exemplifies the Level III failure from within a substance ontology rather than a relational one. He appears throughout the analysis as an illustration of that failure mode — most precisely in the discussion of the source of encounter in Chapter 2 — rather than as an independent diagnostic target, because the survey aims at the traditions most likely to have succeeded rather than those most obviously committed to an external perturbation principle from the outset.

Peirce. Peirce's relational logic, semiotics, and synechism constitute a philosophically serious relational framework whose formal precision is genuine. Peirce is not surveyed because his framework does not take the structural conditions of relational possibility as its primary question: he begins with the triadic sign relation as the fundamental unit of semiosis and develops that relation's internal structural properties — firstness, secondness, thirdness — without asking what makes the sign relation possible as a structural phenomenon at all. This places him at

Level V — the level of the relational event's internal structure, which is precisely what the triad formalises — and his foundational commitment to the triad would yield a foreclosure result at Levels I through IV analogous to Whitehead's at the same entry level. The Whitehead analysis already represents the most formally developed process-relational account at Level V; adding Peirce would confirm rather than extend the diagnostic result there, without bringing an independent methodological approach to bear.

CHAPTER 3

Whitehead : The Universe Feels Itself

> *The notion of 'God' is that of an actual entity immanent in the actual world, but transcending any finite cosmic epoch. . . He is not before all creation, but with all creation.*
>
> Alfred North Whitehead,
> *Process and Reality*

3.1 The Question That Animates Process

What would a universe look like if every moment of its existence were genuinely creative — if nothing that happens is simply the mechanical consequence of what came before, and if every event, however small, however transient, were a moment of self-constitution that adds something to what the universe is? This is not a rhetorical question. It is the question that organises Alfred North Whitehead's mature philosophical system, and the extraordinary edifice of *Process and Reality* is the most rigorous answer anyone has given to it.

The universe Whitehead describes is one in which the fundamental units of reality are not enduring substances that possess properties and stand in external relations, but *occasions of experience*: momentary events of creative self-constitution in which each occasion takes up the entirety of the world that has preceded it — every prior occasion, every feeling that has been felt, every pattern that has achieved determinate being — and weaves from that inheritance a new synthesis that did not exist before. The world is not a collection of things. It is a society of happenings, each a moment of becoming that immediately perishes into the objective datum available for subsequent becomings. What we call things — enduring objects, stable entities, persons, planets — are what Whitehead calls *societies*: structured patterns of occasions sustaining themselves through time by the consistency of their inherited character. The stone is not a thing; it is a self-reproducing pattern of occasions, each constituting itself in sufficient conformity with its predecessors to maintain the continuity we perceive as stoneness. The mind is not a substance; it is a high-grade society of occasions in which the creative synthesis achieves sufficient richness and integration to generate what we call consciousness.[1]

[1]The ontological priority of actual occasions over enduring substances — what Whitehead calls the 'reformed subjectivist principle' — is stated at Alfred North Whitehead, *Process and Reality: An Essay in Cosmology*, Corrected, edited by David Ray Griffin and Donald W. Sherburne (New York: Free Press, 1978), 160–167. The account of societies as structured patterns of occasions is developed in Part II, "Discussions and Applications," especially id., 89–109. The specific examples of stones and minds as societies of different grades of complexity appear throughout *Adventures of Ideas*; see especially Alfred North Whitehead, *Adventures of Ideas* (New York: Macmillan, 1933), 208–211.

This vision is not offered as a metaphor or a model. It is offered as the literal truth about the structure of reality — arrived at by the most demanding philosophical method Whitehead knew how to deploy: the method of descriptive generalisation, in which the task is to find the most general characterisation of experience that does not distort any particular experience in the process of accounting for it. The great philosophical mistake, in Whitehead's diagnosis, is what he calls the *fallacy of misplaced concreteness*: the error of taking an abstraction — the simple, enduring, properties-bearing substance of traditional metaphysics — for the concrete reality it was designed to describe, and then finding that the concrete reality stubbornly refuses to fit the abstraction's requirements. Actual occasions are his answer: the most concrete things there are, from which every abstraction is derived, and to which every abstraction must be referred if it is to retain contact with what is real.[2]

The reader who has followed the diagnostic framework of Chapter 2 will already sense what is coming. The question is not whether Whitehead's vision is compelling — it is. The question is where it enters the structural order established by the five

[2]The fallacy of misplaced concreteness is introduced and named in Alfred North Whitehead, *Science and the Modern World* (New York: Macmillan, 1925), 51, and pervades the methodological discussions of *Process and Reality*; see especially the prefatory discussion of the method of speculative philosophy at Whitehead, *Process and Reality: An Essay in Cosmology*, 3–17. The claim that actual occasions are the final real things is stated at id., 18: "The final real things of which the world is made up. . . are actual occasions of experience." The significance of 'final' here is ontological rather than temporal: occasions are final in the sense of being at the bottom of every analysis, not in the sense of occurring at the end of any process.

formal requirements, and what that entry point structurally forecloses. The answer will require the most careful treatment this survey can give, because what Whitehead forecloses is not the result of any failure of rigour or imagination. It is the structural consequence of his greatest strength: the extraordinary clarity and depth of his account of the relational event itself.

3.2 The Foundational Commitment: Actual Occasions as Temporal Primitives

The foundational commitment of Whitehead's philosophy is stated with unusual explicitness in the opening pages of *Process and Reality*: "The notion of 'substance' is transformed into the notion of 'actual entity' — otherwise called 'actual occasion'." Actual occasions are the final real things of which the world is made up. They are not derived from anything more primitive. They are the primitives from which everything else — God, creativity, eternal objects, societies, propositions — is to be understood.[3]

Whitehead's philosophical method is to characterise these primitives as precisely as possible and to show how every phenomenon that requires explanation can be accounted for in

[3]id., 18. The identification of actual occasions as 'actual entities' and as the category of ultimate fact is made explicit in the 'Category of the Ultimate': "Creativity, 'Many,' 'One' are the ultimate notions involved in that final metaphysical description of the most general categoreal notions" (id., 21). That actual occasions are the *only* things that are fully actual — that societies, eternal objects, and even God are in different senses derivative on or dependent upon actual occasions — is argued at length in id., 20, 29, 88.

terms of them, without remainder. The characterisation is precise and demanding. An actual occasion is:

(i) *a process of becoming* rather than a static entity: it constitutes itself through the progressive integration of its inherited data, a process Whitehead calls *concrescence*, and it perishes as a subject the moment its process of becoming is complete, becoming instead an *objective datum* available for subsequent occasions to prehend;

(ii) *experiential*: every actual occasion has the character of experience in the broadest sense — not necessarily consciousness, but feeling, the taking-account-of its environment. Whitehead generalises the notion of experience from the narrow human case to the universal condition of actuality: to be actual is to feel, to take up the world into oneself as felt datum and to generate from that feeling a new synthesis;

(iii) *creative*: the synthesis that each occasion generates is not mechanically determined by its inheritance. The occasion exercises a creative response to its data, selecting among the eternal objects — the timeless forms of definiteness that characterise what the occasion becomes — under the guidance of its *initial aim*: the specific form of realisation that Whitehead's God envisages as the ideal achievement for this occasion;

(iv) *relational through prehension*: the mechanism by which an occasion takes up its data is called *prehension*. A prehension is a concrete fact of relatedness: the way in which one

> actual entity feels another, positively or negatively, with or without the mediation of conceptual factors. Every actual occasion prehends the entire prior world — the creative advance of the universe is the universe feeling itself in every moment of its becoming.

The four characteristics together define what an actual occasion is: a momentary event of creative, experiential, relational self-constitution that is both the product of everything that preceded it and the source of what immediately follows it. The universe, in Whitehead's account, is the ongoing creative advance of such occasions: each arising from its predecessors, each adding its own creative synthesis, each perishing immediately into the data from which its successors will draw.[4]

The foundational commitment that this characterisation embodies is precisely the commitment that the diagnostic framework identifies as the mark of a specific entry level: actual occasions are posited as temporal primitives. They are *temporal* because each actual occasion is essentially a process of becoming, and becoming is inherently temporal: it has a direction, a before and after, a moment of inception and a moment of com-

[4]The four characteristics are distributed across *Process and Reality*; the most systematic exposition of concrescence is in Part III, "The Theory of Prehensions" id., 211–280. The experiential character of all actuality — the reformed subjectivist principle — is stated at id., 160–167. The creative advance and its relation to perishing is discussed at id., 210–215. The clearest compact statement of all four characteristics in relation to one another is in the "Categoreal Scheme" at id., 18–30. Lowe provides the most accessible extended exposition of what an actual occasion is; see Victor Lowe, *Understanding Whitehead* (Baltimore: Johns Hopkins University Press, 1962), 155–202.

pletion. An actual occasion that was not temporal — that did not arise, constitute itself, and perish into objective immortality in a directed process — would not be an actual occasion at all but something structurally different. A specialist objection holds that individual concrescence is not a process occurring *in* time — that concrescence is the *basis* of temporality, not an event within it, and that the temporal succession through which clock-time is constituted belongs to societies of occasions rather than to any single occasion's internal process. This is correct as a reading of Whitehead, but it does not prevent the Level V assignment. The relevant sense of 'temporal' in the entry-level diagnosis is not the clock-time of societies but the structural asymmetry constitutive of the individual occasion itself: the asymmetry between the occasion's arising (the datum- phase, the subjective aim, the ongoing concrescence) and its perishing into objective immortality (the satisfaction and the transition). This before- satisfaction / after-satisfaction asymmetry is present at the level of the individual occasion, is not derived from anything more primitive within Whitehead's framework, and is precisely what Level V ((V_1)–(V_4)) formalises. Whether or not that asymmetry is 'temporal' in the sense of clock-time is terminologically secondary; what is structurally decisive is that the asymmetry is constitutive and non-derived. Temporality is not a feature that actual occasions happen to have; it is constitutive of what they are. And they are *primitives* because Whitehead refuses to ask what makes an actual occasion possible. The occasion is the starting point; the question of its structural conditions of possibility is not Whitehead's question.

Proposition 3.1 (Whitehead's Entry Level)**.** The foundational commitment of Whitehead's philosophy — actual occasions as temporal primitives — locates his system's entry into the structural order at Level V (the structure of the relational event), specifically at the junction of Levels IV and V. The temporal character of actual occasions constitutes the entry level; the structural conditions of that temporal character — what makes an occasion capable of arising, of prehending, of constituting itself creatively, of perishing into objectivity — fall below the entry point and are structurally foreclosed by the commitment that defines it.

Proof. By the diagnostic method established in Chapter 2, a tradition's entry level is the level at which its foundational commitment takes a structural feature as primitive. Whitehead takes actual occasions as primitive; actual occasions are constitutively temporal events whose internal organisation — concrescence towards satisfaction, prehension of data, creative synthesis under initial aim, directional perishing into objective immortality — is precisely the structure of the relational event that Level V ((V_1)–(V_4)) formalises. The occasion is the relational event as Whitehead understands it. To take actual occasions as primitive is therefore to take the relational event as primitive, which is to enter the structural order at Level V. The temporal character of occasions is not derived from any more primitive structural dynamic; it is assumed as part of what occasions are. This assumption constitutes entry at Level V, not below it, because the structural conditions of temporality — what the eliminative derivation establishes as the latent register from which the temporal character of the relational event is generated — are not Whitehead's question. They fall, accordingly, below his entry point.

3.3 Entry Level: The Formal Structure of the Relational Event

To say that Whitehead enters the structural order at Level V is to say that his account of the relational event is not merely incidentally detailed but constitutively central: it is what his philosophy is *about* in its most fundamental register. The account of how actual occasions prehend, synthesise, and constitute themselves is not one contribution among many that Whitehead's philosophy makes; it is the philosophical heart of the enterprise. And it is, by any measure, one of the most sophisticated accounts of the relational event that philosophy has produced.

The internal structure of the actual occasion's process of becoming has three distinguishable phases, which Whitehead discusses under the heading of the phases of concrescence:

(i) the *conformal phase*: the occasion passively receives the data of its world, the feelings that the prior world presents to it, in what Whitehead calls *physical prehensions*. The occasion's initial constitution is determined by what it receives: it conforms to its data as the condition of its own arising. The conformal phase is the occasion's obligatory inheritance: it cannot refuse its world, and what it receives in this phase

constitutes the raw material from which every subsequent creative response will be fashioned;

(ii) the *conceptual phase*: the occasion introduces its own response to the data, entertaining *conceptual feelings* of the eternal objects that characterise its inheritance and generating novel conceptual valuations that are not mere reproductions of what it received. It is in this phase that creativity enters the occasion's constitution, and it is here that the initial aim supplied by God — the specific possibility that God holds open for this occasion as its ideal of realisation — shapes the creative synthesis by providing a directional lure towards a form of definiteness that the conformal phase alone could not generate;

(iii) the *propositional phase*: the occasion integrates its physical and conceptual feelings into a unified *satisfaction* — a determinate, complete self-constitution that is the occasion's unique contribution to the creative advance. The satisfaction is the completed occasion, ready to perish into objectivity and become datum for its successors. It is both the terminus of the occasion's becoming and the beginning of its role as the world's inheritance for what comes after.[5]

[5]The phases of concrescence are discussed at Whitehead, *Process and Reality: An Essay in Cosmology*, 212–215, 244–279. Whitehead's own treatment is not consistently systematic, and the three-phase structure (conformal, conceptual, propositional) is the standard reconstruction developed by Donald W. Sherburne, *A Key to Whitehead's Process and Reality* (New York: Macmillan, 1966), 19–35 and adopted by most subsequent commentators, including Nicholas Rescher, *Process Metaphysics: An Introduction to Process Philosophy* (Albany: State University of New York Press, 1996), 69–95. The philosophi-

The three phases correspond, with remarkable precision, to the three formal dimensions of the relational event that Definition 2.5 of Chapter 2 specifies. The conformal phase is the structural counterpart of initiation ((V_3)): the moment at which the occasion's structural character is constituted by contact with its specific data, making it this occasion in contact with this prior world rather than any other, with the specificity that the diagnostic framework requires of the source of encounter at Level III now embedded within the occasion's own conformal openness to its inheritance. The conceptual phase is the structural counterpart of regulation: the moment at which the occasion's creative response holds the tension between received datum and novel possibility, neither simply repeating its inheritance (which would produce absorption — the loss of creative novelty — and which Whitehead regards as the most degenerate form of actual occasion) nor simply departing from it without connection to what it has received (which would produce cancellation — a disconnection from the actual world that would render the occasion incapable of becoming an object of subsequent prehension). The satisfaction is the structural counterpart of direction: the oriented outcome, the specific form of definiteness that the occasion achieves, which becomes the datum that gives subsequent

cal significance of the transition from conformal to conceptual phase — the moment at which creativity enters the occasion's constitution — is analysed in detail by Lewis S. Ford, *The Emergence of Whitehead's Metaphysics, 1925–1929* (Albany: State University of New York Press, 1984), 216–230, who traces the development of Whitehead's thinking about this transition across his published writings from 1925 to 1929.

occasions their inherited orientation towards further creative advance.

Remark 3.1 (On Whitehead and the Conditions of Level V). The correspondence between Whitehead's three phases of concrescence and the three formal dimensions of the relational event (initiation, regulation, direction) is not a forced analogy. It is a structural identity: the process Whitehead describes *is* the relational event, articulated at the scale of the actual occasion with the depth that his extraordinary philosophical care produces. What distinguishes Whitehead's account from an account that satisfies all four conditions of Level V is not any inadequacy in the description of the event's dimensions. It is the failure to satisfy (V_1) and (V_4): the account does not derive the three-dimensional structure of the event from a more primitive structural dynamic, and the derivation of the event's structure is not itself exemplified by the method of the account. These are not incidental omissions. They are the structural consequences of taking actual occasions as primitives. Once the occasion is primitive, the question of why the relational event has this structure rather than another is not askable from within the framework, because the occasion *is* the answer to that question before the question can be posed. The framework has placed the answer where the question used to be.

Adventures of Ideas Part IV (Chapters 17–20) introduces civilisational categories — Beauty, Truth, Adventure, Art, and Peace — that might appear to constitute new structural commitments beyond what *Process and Reality* establishes at Level V. Whitehead treats Peace in particular as a characteristic of civilisation that neither reduces to any individual occasion's subjective aim nor is fully derivable from the prehensive dynamics of any finite set of occasions.[6] This is a genuine extension of the Whiteheadian account. It does not, however, modify the Level V assignment, because the civilisational categories are characterised throughout Part IV as what the creative advance *produces* at the scale of civilisational process — they are structural results of the prehensive dynamic accumulated across vast temporal sequences of occasions, not new conditions of prehensive possibility. In the diagnostic framework's terms they are Level IV features: relational yield at the civilisational scale, derivable in principle from the same prehensive dynamic that produces simpler yields at the level of individual occasions. The foundational commitment remains the actual occasion as the site of prehensive constitution (Level V), and the Level V entry level assignment is stable across *Process and Reality*, *Adventures of Ideas*, and *Modes of Thought*.

The co-instantaneity condition (V_3) is met with unusual philosophical depth. The three phases of concrescence are not genuinely sequential in the relevant sense: they are described sequentially for pedagogical purposes, but Whitehead is explicit that the occasion is not first conformal, then conceptual, then propositional in the way that a process in time passes through discrete temporal stages. The concrescence is a unity; the phases are analytic distinctions within a single act of becoming that is

[6] See Whitehead, *Adventures of Ideas*, ch. 20.

complete at the moment of its satisfaction. This is precisely the co-instantaneity that (V_3) requires.[7]

3.4 Formal Achievements at Levels IV and V

The diagnostic method requires that genuine achievements be precisely acknowledged before structural limits are identified. Whitehead's achievements at Levels IV and V are substantial and must not be minimised.

At Level V, Whitehead provides the most fully developed account of the relational event in the philosophical literature. The prehensive structure of concrescence — the way an occasion integrates its data through the three phases into a determinate satisfaction — satisfies three of the four formal conditions of Level V with precision. It meets the non-redundant completeness requirement (V_2): the three phases are jointly exhaustive (they account for everything that happens in the occasion's becoming without remainder) and individually non-redundant (the conformal phase accounts for what the conceptual phase

[7]Whitehead's most direct statement of the co-instantaneity of the phases occurs at Whitehead, *Process and Reality: An Essay in Cosmology*, 212: the phases are 'stages' only in the sense of analytic distinctions within a unity, not temporal stages in a sequence. Ford debates at length whether Whitehead ultimately succeeds in maintaining this co-instantaneity, given his sequential language of 'phases'; see Ford, *The Emergence of Whitehead's Metaphysics, 1925–1929*, 216–230. The present account holds with Sherburne and Nobo that the co-instantaneity is genuinely intended by Whitehead and that the apparent sequentiality is a pedagogical necessity imposed by the linear character of expository prose, not a structural feature of the occasion's becoming. This reading is the more philosophically coherent one, and it is the reading that makes the correspondence with condition (V_3) precise.

cannot: the obligatory receipt of the actual world as inherited datum; the conceptual phase accounts for what the conformal phase cannot: the creative response that introduces genuinely novel forms of definiteness; the satisfaction accounts for what neither phase alone can account for: the integrative unity that makes the occasion's creative contribution a determinate whole rather than a collection of separate feelings). It meets the co-instantaneity requirement (V_3), as argued above. And it partially meets the derivability requirement (V_1): the three phases can be shown to be internally necessary given the characterisation of actual occasions — once an occasion is understood as a creative self-constitution through prehension of its world, the three-phase structure follows as a structural consequence of what prehension, creativity, and satisfaction jointly require. What the account does not provide is the derivation of the three-phase structure from a dynamic more primitive than the occasion itself; within Whitehead's framework, the occasion is the terminus of derivation, and the three-phase structure is a necessary characterisation of the occasion rather than a result derived from a deeper structural level.

At Level IV, Whitehead provides a formal account of relational yield that satisfies all four conditions of Level IV with considerable precision. The satisfaction of an actual occasion — the specific form of definiteness that the occasion achieves through its creative synthesis — is irreducible (Y_1): the satisfaction is not derivable from any formal operation on the occasion's data, since creativity introduces a genuine novelty into the synthesis that the data did not contain and that no mechanical combination of data could produce. It is structurally conservative (Y_2): the satisfaction draws only on the structural resources available to the occasion through its prehensions of the prior world and its conceptual feelings of eternal objects; it does not produce definiteness from nowhere but organises what it has received into a form that is new without being ex nihilo. It is generative (Y_3): the completed occasion, in its *objective immortality*, becomes datum for subsequent occasions — what the occasion is, is preserved in the world as the permanent condition of what comes after it, available for prehension by every subsequent occasion in the creative advance. And it exhibits asymmetric novelty (Y_4): no two occasions produce the same satisfaction, since each occasion's creative synthesis is sensitive to the precise configuration of its data and the specific creative response it makes to that configuration.[8]

These achievements are genuine and important. The Whiteheadian tradition has built on them productively across philosophy of science, philosophy of mind, philosophical theology,

[8]The concept of objective immortality — the way in which a completed actual occasion becomes a permanent datum available for subsequent prehension — is central to Whitehead's account and is discussed at length at Whitehead, *Process and Reality: An Essay in Cosmology*, 29, 46–47, 60–61. Its function as the ground of the generativity condition (Y_3) is the feature that makes Whitehead's account of relational yield genuinely formal rather than merely descriptive: the satisfaction is not a terminus but a transition, and the transition is structurally specified in a way that accounts for how the yield of one relational event becomes the material from which subsequent relational events draw. Stengers develops the philosophical implications of this feature of Whitehead's account with particular care; see Isabelle Stengers, *Thinking with Whitehead: A Free and Wild Creation of Concepts*, translated by Michael Chase (Cambridge, MA: Harvard University Press, 2011), 254–278.

and social theory, and continues to do so. The diagnostic claim of this chapter is not that this work is misguided or that the achievements are illusory. It is that the work is constituted at Levels IV and V of the structural order, and that the structural foreclosure entailed by that constitution makes the foundational question of relational ontology — the question of what is below Level V — inaccessible from within the tradition's own resources, regardless of how much further work is done within those resources.

3.5 What Whitehead Cannot Reach : The Structural Foreclosure

The structural foreclosure that Whitehead's foundational commitment entails operates across Levels I, II, and III with equal precision. At each level, the same structural situation obtains: Whitehead has something that performs the relevant function, but that something is asserted rather than derived, and the assertion is structurally necessitated by the commitment to actual occasions as temporal primitives.

The Foreclosure at Level III: The Source of Encounter

Actual occasions prehend their environment: this is how they enter into relational contact with their world. But what brings a specific occasion into prehensive contact with the specific data it prehends, and into the specific creative orientation that makes its synthesis *this* synthesis rather than any other? This is the

Level III question in its Whiteheadian form: what functions as the source of encounter in a universe of actual occasions?

Whitehead's answer is *God*, in the specific and technically precise form that *Process and Reality* develops. God's function is twofold. In the *primordial nature*, God is the eternal envisagement of all possibilities — all eternal objects in their mutual relevance — which provides the conceptual ground from which each actual occasion draws its initial aim. In the *consequent nature*, God prehends the actual world as it becomes and weaves the occasion's creative achievements into a permanent relational totality. It is through the initial aim — the specific possibility that God envisages as the ideal achievement for this occasion, communicated to the occasion through God's primordial nature — that the source of encounter enters Whitehead's system. The initial aim determines what the occasion is creative towards: it is the structural selector that makes this occasion's creative synthesis aim at this form of definiteness rather than any other, and that thereby shapes which data the occasion actively prehends and which it dismisses with negative prehensions.[9]

[9]Whitehead, *Process and Reality: An Essay in Cosmology*, 244: "An actual entity feels what it feels for the reason that what it is." The initial aim and its derivation from God's primordial nature are discussed at id., 244–249. The secondary literature on God's role as the source of the initial aim is extensive; the most direct treatment in relation to the structural point at issue here is in John B. Cobb, *A Christian Natural Theology: Based on the Thought of Alfred North Whitehead* (Philadelphia: Westminster Press, 1965), 151–192 and Ford, *The Emergence of Whitehead's Metaphysics, 1925–1929*, 200–215. Neither Cobb nor Ford frames the issue in terms of a Level III structural requirement, which is the claim of the present analysis rather than of the secondary literature; the relevant secondary literature establishes the structural fact —

This is the characteristic failure mode of Level III: the external perturbation principle. Whitehead does not pretend that the source of encounter is derivable from the occasions themselves; he explicitly introduces God as the structural principle that accounts for why specific occasions are creative towards specific forms of definiteness rather than others. The initial aim is the source of encounter in Whitehead's system, and the initial aim is God's gift to each occasion. Condition (E_1) — derivability from Levels I and II — fails precisely here: God is not derived from the structural dynamic of actual occasions and creativity but is introduced as an additional ontological principle alongside them. This is Whitehead's own formulation: God and creativity are co-ultimates — neither derivable from the other, both required as foundational principles.[10] Its structural signif-

that God's initial aim is the principle that accounts for why specific occasions are creative towards specific forms of definiteness — without identifying the structural consequence of that fact for the tradition's foundational scope.

[10]Whitehead, *Process and Reality: An Essay in Cosmology*, 7: "God is not to be treated as an exception to all metaphysical principles, invoked to save their collapse. He is their chief exemplification." The co-ultimate status of God and creativity — and the structural consequence that neither is derivable from the other — is acknowledged implicitly at id., 244: "Without God, there is no novelty." This is equivalent to acknowledging that creativity without God cannot account for the source of encounter. The external perturbation principle is not a supplement reluctantly added to an otherwise self-sufficient system; it is a structural necessity within Whitehead's framework, as the framework's own logic requires. Griffin attempts to resolve this by deriving God's existence from creativity itself; see David Ray Griffin, *Reenchantment without Supernaturalism: A Process Philosophy of Religion* (Ithaca: Cornell University Press, 2001), 134–176. The present analysis holds that this derivation does not succeed in fully discharging condition (E_1), because the derived God retains the role of providing the initial aim for specific occasions and therefore retains the function of an external pertur-

icance for the present analysis is that Whitehead cannot avoid the Level III requirement, recognises that he cannot avoid it, and discharges it through an external perturbation principle rather than a derivation internal to the framework. The foreclosure at Level III is explicit rather than concealed.

The Foreclosure at Level II: Relational Identity

The relational identity of an actual occasion in Whitehead's account is its *definiteness*: the specific form of realisation that the occasion achieves through its creative synthesis. The occasion's definiteness is its satisfaction — the completed pattern of prehensions that constitutes it as the specific occasion it is. This is a formal achievement at Level IV: the yield of the relational event is precisely specified. But the Level II question is different from the Level IV question. Level II asks not what the occasion

bation principle relative to the occasions whose creative synthesis the aim shapes. The broader Whiteheadian tradition — including Cobb (*A Christian Natural Theology: Based on the Thought of Alfred North Whitehead*) and Sherburne (*A Key to Whitehead's Process and Reality*) — has developed the initial aim doctrine in directions that attempt to internalise God's function more fully within the system. The present analysis does not contest the richness of these developments; it holds that the structural gap named at Level III is not a feature of any particular formulation of Whitehead's system but of the structural requirement the system's own logic generates: any account that retains the function of supplying specific forward-orientation to specific occasions (a function that creativity alone cannot discharge) has the structure of an external perturbation principle at Level III, regardless of whether that function is assigned to God, to creativity in a revised form, or to the occasions' self-constitution. Every subsequent Whiteheadian development must address this structural requirement or leave it open; the present survey holds that none has closed it without retaining the function under a different name.

achieves but what makes the occasion capable of achieving it — what the specific relational identity of the occasion is *prior to* its creative synthesis, as the condition under which the synthesis is possible at all and under which it is specifically this synthesis rather than any other.

Whitehead's answer to this question is, at bottom, the initial aim transformed into the occasion's own *subjective aim*: the occasion's creative intention as it constitutes itself through concrescence, which is the initial aim taken up and made the occasion's own as the process of becoming proceeds. But the subjective aim is itself constituted through the process of concrescence, not prior to it: it is what the occasion makes of the initial aim as it integrates its data, not what the occasion brings to the process as its pre-constituted relational identity. The relational identity that Level II requires — an invariant identity that is disclosed through encounter without being dissolved by it, that persists through the encounter as what makes this relatum specifically itself — has no precise counterpart in Whitehead's system. The occasion's character is entirely a function of its creative synthesis; it brings to that synthesis only the data it inherits and the initial aim it receives from God. There is no invariant self-organising pattern — no structural constant that the occasion maintains through its encounter with its world and that the encounter discloses without destroying — in Whitehead's account of the occasion's character before its synthesis is complete.[11]

[11]The structural gap identified here is close to what Nobo calls the problem of 'subjective immediacy': what is the subject of the process of concrescence before that process has produced a subject? See Jorge Luis Nobo, *Whitehead's Metaphysics of Extension and Solidarity* (Albany: State University of

This is not a peripheral gap. It is the structural consequence of Whitehead's most fundamental commitment. If actual occasions are temporal primitives that constitute themselves through creative synthesis, then there can be no structural identity of an occasion prior to that synthesis. Condition (I_1) — invariance through encounter — fails at this level: there is nothing in Whitehead's account that corresponds to the invariant self-organising pattern that persists through the encounter, because the encounter *is* what constitutes the occasion. This structural consequence is not unique to the philosophical diagnostic employed here. Austin[12] establishes from within the phi-

New York Press, 1986), 77–123. Nobo's analysis is conducted entirely within Whitehead's own conceptual framework and does not reach the Level II requirement as specified here, but his diagnosis of the difficulty converges with the structural point: the occasion's subjective character is the product of its becoming, not its precondition, and this creates a structural gap in the account of relational identity that Whitehead's framework cannot fill from within its own resources. The present analysis identifies this gap as a structural foreclosure rather than a technical difficulty, because the gap follows necessarily from the foundational commitment: a framework that takes the temporal occasion as primitive thereby takes the creative synthesis as the constitution of identity, and a framework that takes the creative synthesis as the constitution of identity cannot have a structural identity prior to the synthesis.

[12]Christopher J. Austin, "Organisms, Activity, and Being: On the Substance of Process Ontology," Archived at PhilArchive, rec/AUSOAA. Argues from within the philosophy of biology that process ontology "lacks the conceptual resources to satisfyingly meet its own challenge" regarding organismal identity and "seemingly has no hope of doing so without substantially borrowing from the ontological framework it supposedly supplants" — that the coherency of process ontology's account of individuation "depends upon its making use of the 'substantial' principles it purports to replace." Peer-reviewed confirmation, from within process philosophy's own literature, of Before Relation's Level II claim that process ontology cannot de-

losophy of biology that process ontology more broadly "lacks the conceptual resources to satisfyingly meet its own challenge" regarding organismal identity, and "seemingly has no hope of doing so without substantially borrowing from the ontological framework it supposedly supplants" — that the coherency of process ontology's account of individuation "depends upon its making use of the 'substantial' principles it purports to replace." The Level II gap the present investigation identifies at the foundational level of Whitehead's philosophy is therefore confirmed, from within process philosophy's own literature, as a systematic rather than incidental structural failure.

The Foreclosure at Level I: The Ground

Whitehead is explicit about the ground of his system: it is *creativity*, the "universal of universals characterizing ultimate matter of fact," the activity instantiated in every actual occasion as the occasion's creative synthesis.[13] Creativity is the nearest Whitehead comes to the structural ground that Level I requires. It is total in the sense that it admits no outside: there is nothing in Whitehead's universe that is not an instantiation of creativity. It is self-sustaining in the sense that creativity requires no external initiator: it is what the universe simply does, the universal feature of every occasion's becoming.

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rive an invariant structural identity prior to the creative synthesis, *European Journal for Philosophy of Science* 10, no. 2 (2020): 1–21.

[13] Whitehead, *Process and Reality: An Essay in Cosmology*, 21: "'Creativity' is the universal of universals characterizing ultimate matter of fact. It is that ultimate principle by which the many, which are the universe disjunctively, become the one actual occasion, which is the universe conjunctively."

But creativity fails the generativity condition (G_2) and the formal derivability condition (G_4) with precision. These conditions were not imposed on creativity from outside the diagnostic framework; they were derived independently, in Chapter 2, from the relational thesis itself: the ground must have sufficient generativity to produce the structural differentiation of relata from within its own dynamic, because if the differentiation is supplied by something other than the ground, the ground is not doing the structural work the relational thesis requires it to do. Creativity is formally homogeneous: it is the universal of universals, the activity that every occasion instantiates, specified entirely by the formal property of being the activity of creative synthesis. It has no internal structure — no dynamic of differentiation, no weighted interaction of structurally distinct dimensions — from which the specific characters of specific occasions could be generated. The diversity of occasions does not emerge from the internal dynamic of creativity itself; it emerges from the diversity of eternal objects that creativity synthesises, combined with the specific initial aim that God supplies to each occasion. Creativity is the *form* of the generative process — the pure activity of creative advance as such — not the *structural source* of what is generated in any specific case. The source of the specific, in Whitehead's system, is always God's initial aim, which returns the analysis once more to Level III's external perturbation principle.[14]

[14]The structural parallel between Spinoza's Substance and Whitehead's Creativity — that both are formally homogeneous grounds that cannot generate specific structural differentiation from within their own dynamic without the supplementation of an external principle — is noted here as an anticipa-

The formal derivability condition (G_4) is also violated. Whitehead does not derive creativity from anything more primitive, and he is explicit that no such derivation is possible or required. Creativity is the ultimate: the category that requires no further characterisation because no further characterisation is possible within the framework's resources. Whitehead introduces it as "that ultimate notion of the highest generality at the base of actuality."[15] This refusal of further derivation is precisely what the formal derivability condition disallows as a structural terminus: a ground that is stipulated as ultimate rather than shown to be the unique structural result of the eliminative method applied to what any relational ontology presupposes. The conflict between creativity and the framework's own ontological principle — the principle that only actual occasions are the loci of real definiteness — has been identified in the Whitehead specialist literature as a fundamental and unresolved tension in the system, arising from the fact that creativity, as a metaphysical ultimate, cannot itself satisfy the conditions it is supposed to ground.[16]

tion of Chapter 4's analysis. In Spinoza, the external principle takes the form of the infinite intellect's attribution of modes to Substance under an infinity of attributes; in Whitehead it takes the form of God's initial aim. The shared failure at condition (G_2) — a homogeneous ground incapable of internally differentiated generation — runs through both traditions despite the profound differences in their metaphysical architectures. This convergence is not coincidental; it is the structural consequence of the same foundational move: positing an ultimate ground whose universality is achieved at the cost of internal structural differentiation.

[15]id., 31.

[16]Robert Hanna, "The Nature of Creativity in Whitehead's Metaphysics," Archived at PhilArchive, rec/HANTNO-14. *Philosophy Research Archives* was

Theorem 3.1 (Whitehead's Structural Foreclosure)**.** The foundational commitment of Whitehead's philosophy — actual occasions as temporal primitives — entails structural foreclosure at Levels I, II, and III of the diagnostic framework. This foreclosure is necessary rather than contingent: it is not the result of any particular feature of Whitehead's system that might be revised while preserving the foundational commitment, but the structural consequence of that commitment itself.

More precisely: *(i)* any system whose foundational commitment is the temporal actual occasion as primitive cannot derive the structural ground of the relational event from a more primitive dynamic, since the occasion is the terminus of derivation and creativity, as its universal characterisation, is homogeneous without internal differentiating dynamic; *(ii)* any such system cannot derive an invariant relational identity prior to the creative synthesis, since the synthesis is what the identity consists in and there is no structural character of the occasion available before the synthesis is complete; and *(iii)* any such system cannot derive the source of encounter from within its own resources, since the specificity of each occasion's creative orientation requires a principle — God's initial aim — that creativity alone cannot generate without circularity.

Proof. (i) Creativity as the ground fails (G_2) because it is formally homogeneous: it characterises the *form* of creative synthesis without providing the structural dynamic from

a refereed annual series of the Philosophy Documentation Center (American Philosophical Association), peer-reviewed through 1988, *Philosophy Research Archives* 9 (1983): 109–175.

which specific occasions with specific characters are generated. Creativity fails (G_4) because it is introduced as the ultimate category rather than derived as the unique structural result of the eliminative method. Both failures are entailed by the foundational commitment: a system that takes actual occasions as primitive and introduces creativity as their universal characterisation has thereby constituted creativity as the formal ground without internal differentiating dynamic, and has constituted it as ultimate without derivation. These consequences are not revisable within the framework without abandoning the commitment that generates them. *(ii)* An invariant relational identity prior to the creative synthesis ((I_1)–(I_3)) cannot be supplied by a framework whose central claim is that the occasion's character is entirely constituted through its process of concrescence: the synthesis is what the character is, and there is no structural identity prior to the synthesis available within the framework's resources. This consequence, again, follows necessarily from the foundational commitment. *(iii)* The source of encounter (E_1) fails because God's initial aim — the principle that accounts for the specificity of each occasion's creative orientation — is introduced alongside creativity as a co-ultimate rather than derived from it. Any attempt to derive the initial aim from creativity alone founders on the homogeneity of creativity: a homogeneous universal cannot generate specific aims for specific occasions without an additional principle of specification. Since the foundational commitment forecloses that derivation, it thereby entails that the source of encounter is provided by an external perturbation principle. In each of the three cases, the foreclosure is necessary rather than contingent: it is not that Whitehead failed to attempt the relevant derivation, but that the foundational commitment structurally prevents the derivation from being achievable within its resources.

Remark 3.2 (The Creativity-Priority Objection). A process philosopher may object that the entry-level assignment is wrong. If Creativity is the Category of the Ultimate — formally prior to every actual occasion, since occasions are instantiations of Creativity rather than Creativity being an abstraction from occasions — then the system's foundational commitment is not to actual occasions at Level V but to Creativity at Level I. The entry level should therefore be Level I, not Level V; and the Level I diagnosis (Creativity fails the generativity conditions) is the primary diagnosis, not a secondary consequence of the Level V commitment.

This objection rests on an ambiguity between two senses of "being prior to": *instantiation priority* and *derivation priority*. Creativity is instantiation-prior to actual occasions: occasions are instantiations of the category, and without occasions there would be nothing to instantiate it. But Creativity is not derivation-prior in the sense the diagnostic framework requires. The diagnostic framework asks: from which feature does Whitehead's system derive the other features of the structural account? The answer is unambiguous. The ontological principle holds that occasions are the loci of all real definiteness, and Creativity is characterised entirely in terms of what it *is* in the occasions — the universal of universals, the activity instantiated in every creative synthesis. Whitehead's characterisation of Creativity is derived from the account of what occasions do and are; it is not a prior derivational source from which occasions are then produced. Creativity is Whitehead's name for the *form* of what every occasion does — a characterisation of the occasion's activity, not a generative ground from which the occasion is produced.

This is why the ontological principle is absolute in Whitehead's system: nothing enters the world from nowhere; every actual feature has its reason in some actual entity. Creativity, as a category rather than an actual entity, cannot supply the reasons the ontological principle demands. It describes how actual entities produce one another; it does not ground them prior to them. The entry level therefore remains Level V. The Level I diagnosis is not a mislabelled analysis of a Level I commitment; it is the analysis of why Creativity, despite its instantiation-priority over occasions, cannot satisfy the generativity conditions that a genuine Level I ground must satisfy — precisely because it is a characterisation of occasions' activity rather than a derivational source standing structurally prior to them in Whitehead's own architectonic.

The objection, properly understood, strengthens the diagnostic case. If Creativity were a genuine Level I derivational source — a ground from which the specific characters of specific occasions were derivable from within its own internal dynamic — the entry level would shift downward and the Level I analysis would be primary. But Whitehead explicitly and deliberately denies that Creativity is this kind of derivational source. The Category of the Ultimate is not a structured dynamic that generates specific occasions; it is the universal form under which every occasion's creative synthesis proceeds. What forecloses Level I is not merely Creativity's formal homogeneity but Whitehead's architectonic commitment to occasions as the loci of all real definiteness: Creativity is *abstracted* from the occasions' activity, not *prior* to it in any derivational sense. To the extent that "priority" is in play, it is the priority of an abstracted universal to its instances — a logical priority that does not constitute derivational precedence in the ontological order.

3.6 The Diagnostic Result

Whitehead's system occupies a precise location in the structural order. It enters at Level V, achieves genuine formal results at Levels IV and V, and is necessarily foreclosed at Levels I through III by the foundational commitment to actual occasions as temporal primitives.

The foreclosure is the structural consequence of the greatest strength of the Whiteheadian framework: its extraordinary clarity and depth at the level of the relational event itself. The occasion is constituted as the relational event — as the ultimate unit of analysis whose prehensive structure, creative synthesis, and objective immortality together constitute the formal organisation of the relational event at its most fully articulated. What the framework achieves at this level is genuine and important. What it cannot do, from this level, is descend.

Criterion 3.1 (Whitehead's Diagnostic Profile). *Entry level*: Level V (the structure of the relational event), specifically at the junction of Levels IV and V. *Formal achievements at and above entry level*: Level V — three of four conditions met (non-redundant completeness, co-instantaneity, and partial derivability of event structure from the characterisation of actual occasions); Level IV — all four conditions met (irreducibility, structural conservation, generativity through objective immortality, asymmetric novelty of satisfaction). *Structural foreclosure*: Level III — God's initial aim as external perturbation principle (condition E_1 fails); Level II — no invariant relational identity prior to creative synthesis (conditions I_1–I_3 fail); Level I — creativity as homogeneous ultimate without internal differentiating dynamic or formal derivability (conditions G_2 and G_4 fail). *Character of the foreclosure*: Necessary. The foreclosure is entailed by the foundational commitment to actual occasions as temporal primitives and cannot be overcome by modifications internal to the framework without abandoning that commitment.

The diagnostic result does not diminish what Whitehead built. *Process and Reality* remains one of the most sustained and rigorous philosophical constructions of the twentieth century, and the tradition it generated continues to produce significant philosophical and interdisciplinary work across precisely the levels at which it has genuine formal achievements. The diagnostic claim is structural, not evaluative: it concerns what the system can and cannot do from where it stands, not whether it does what it does well. It does what it does, at the levels at which it operates, with extraordinary care and depth.

The structural claim has a precise philosophical significance nevertheless. Whitehead showed us what relation *does*: how occasions prehend, how creative synthesis achieves definiteness, how the accomplished occasion becomes the permanent datum of subsequent becomings. He gave us, in extraordinary detail, the mechanics of relational constitution at the level of the event itself. What he could not give us — what his foundational commitment structurally prevented him from giving us — is what relation *is* before the occasion that instantiates it is already constituted. The question of what is below the relational event — what is before the temporal becoming that the event both presupposes and instantiates — was not Whitehead's question. The universe, for Whitehead, has always already begun to feel itself. What it is before it feels, and what structurally generates the capacity to feel, remains the ground Whitehead cannot reach.

The Griffin–Cobb Response and Its Structural Limits

The Level III diagnosis — that God functions as an external perturbation principle imported to supply what Whitehead's framework cannot derive from within — has not gone unaddressed within the Whiteheadian tradition. The most sustained and philosophically developed response comes from David Ray Griffin and John Cobb, whose post-Whiteheadian process theology reconceives divine action in ways designed precisely to resist the charge that God is an external agent.[17]

Griffin's central move is to deny that God's provision of an initial aim is coercive or external in the problematic sense. On Griffin's account, God acts through *persuasion* rather than coercion: the initial aim is offered to each new occasion, not imposed upon it. The occasion retains full creative autonomy in its reception of the initial aim; the aim lures rather than determines. This reconception is intended to make God's action immanent to the creative process rather than superimposed upon it.

The diagnostic framework of the present investigation acknowledges the force of this reconception as a theological response to the coercion problem. The relevant structural question, however, is not whether God's action is coercive but whether the encounter source is derived from the same dynamic that produces the relata. On this question, the Griffin–Cobb reconception does not alter the structural situation.

In Griffin's account, God is an actual entity of a unique kind — the only non-temporal actual entity, whose primordial and consequent natures are constituted as foundational features of the account.[18] God's primordial nature, which is the source of the relevant eternal objects from which initial aims are drawn, is not derived from the creative dynamic that produces temporal

[17]Griffin, *Reenchantment without Supernaturalism: A Process Philosophy of Religion*; Cobb, *A Christian Natural Theology: Based on the Thought of Alfred North Whitehead*.

[18]See Griffin, *Reenchantment without Supernaturalism: A Process Philosophy of Religion*, ch. 4.

actual occasions; it is co-eternal with creativity and structurally prior to the creative advance as the condition of its ordered relevance. Cobb makes this explicit: without God's primordial ordering of eternal objects, the creative advance would be directionless, its novelty unconstrained by any relevance to the actual world.[19][20] God is, on this account, not merely an agent

[19]Cobb, *A Christian Natural Theology: Based on the Thought of Alfred North Whitehead*, 114–120.

[20]Cobb's later work — including *God and the World* John B. Cobb and David Ray Griffin, *God and the World*, Develops and qualifies the account of divine action introduced in *A Christian Natural Theology*, most significantly in elaborating the notion of God's persuasive agency and in beginning the integration of process cosmology with ecological and social ethics (Philadelphia: Westminster Press, 1969) and *Process Theology: An Introductory Exposition* John B. Cobb and David Ray Griffin, *Process Theology: An Introductory Exposition*, The standard introductory text for process theology; presents the theological programme developed across Cobb's and Griffin's collaborative work and consolidates the account of divine persuasive agency within a process cosmological framework (Philadelphia: Westminster Press, 1976) — develops and qualifies the account of divine action in several directions, most significantly in elaborating the notion of God's persuasive agency and in integrating process cosmology with ecological and social ethics. None of these developments alters the structural diagnosis at issue here. In every period of Cobb's work, God's primordial nature remains co-eternal with the creative advance: it is not derived from the relational dynamic that produces the occasions but constituted as the condition of that dynamic's orderly relevance. Joseph Bracken's subsequent process theology — developed in *The One in the Many* Joseph A. Bracken, *The One in the Many: A Contemporary Reconstruction of the God-World Relationship*, Develops the "structured matrix" reconception of the Whiteheadian ground as an internally differentiated divine field constituted by the interpenetrating societies of occasions; the most formally developed Whiteheadian response to the homogeneity objection against creativity-as-such (Grand Rapids: Eerdmans, 2001) and *World Without End* Joseph A. Bracken, *World Without End: Christian Eschatology from a Process Perspective*, Extends the social field theory to eschatological questions; confirms the divine matrix as the ongoing structural ground of

but a structural condition — but a structural condition that is itself constituted as foundational rather than derived from the dynamic it conditions.

This is precisely the condition the diagnostic framework identifies as the Level III failure in its most sophisticated form. The question the framework asks is not whether the encounter source is external in the sense of being spatially or causally separate, but whether it is *derived* from the same structural dynamic that produces the relata. God in the Griffin–Cobb account is not spatially external, but is structurally primitive: constituted as a co-eternal foundational condition whose own structural character is given rather than derived. The G3 condition — that the ground requires no external principle to maintain its own dynamic — is not met. The external principle is now called a *co-eternal persuasive lure* rather than a coercive agent; the structural situation is the same.

The Level III failure therefore persists in the most sophisticated subsequent development of Whitehead's process philosophy. The Griffin–Cobb reconception succeeds in its theological purpose — making divine action compatible with creaturely freedom. It does not succeed in deriving the encounter source

creaturely activity in the Whiteheadian tradition (Grand Rapids: Eerdmans, 2005) — extends the Whiteheadian framework into a social-trinitarian direction, reconceiving the divine nature as a structured field of dynamic intersubjectivity. Bracken's field ontology represents a genuine structural development of Whitehead's account at Level V (the encounter event) and Level IV (the yield of intersubjective constitution). It does not alter the Level III or Level I diagnosis: the field that constitutes the condition of intersubjective encounter is still constituted as a foundational structural feature rather than derived from conditions more primitive than itself.

from the same dynamic that produces the occasions, because God's primordial nature remains a structural primitive that the dynamic presupposes rather than generates. The Structural Insufficiency Theorem applies to the Griffin–Cobb account for the same structural reason it applies to Whitehead's primary text: a framework that foundationally constitutes a non-latent feature (here, God's primordial nature as the ordered ground of eternal-object relevance) is foreclosed from deriving the latent register from within that constitution.

Two further secondary developments require acknowledgment before the diagnostic result is stated, since a specialist in the Whiteheadian tradition will note their absence from the analysis above.

Griffin's naturalistic theism. David Ray Griffin's revisionary programme in *Reenchantment without Supernaturalism*[21] argues that Whitehead's theological commitments can be retained in fully naturalistic form: God's persuasive lure is not a supernatural interruption of natural process but a natural feature of a universe in which creativity is inherently dipolar — oriented by a primordial dimension whose ordering of eternal objects is a structural feature of nature rather than a transcendent imposition on it.

Cobb's natural theology. John B. Cobb's natural theology[22] similarly preserves the ordering function of God's primordial

[21] Griffin, *Reenchantment without Supernaturalism: A Process Philosophy of Religion*.

[22] Cobb, *A Christian Natural Theology: Based on the Thought of Alfred North Whitehead*.

nature within the horizon of natural process philosophy, arguing that the initial aim's operation is continuous with, rather than discontinuous from, the natural order within which occasions arise.

Neither development escapes the Level III structural diagnosis because neither eliminates the structural *function* that the diagnosis identifies as the vulnerability. The E-conditions are conditions on the structural function of the encounter source — whether the principle that brings specifically this occasion into specifically this prehensive configuration is derivable from the same dynamic that generates the relata (E_1) and specific to this encounter rather than a background ordering principle (E_2). Griffin's naturalistic God and Cobb's natural-theological God both retain precisely this ordering function: the primordial nature continues to supply the ordering of eternal objects that determines the specific character of each occasion's initial aim, whatever the metaphysical status — supernatural or natural — of the principle that does so. The E_1 failure mode attaches to the structural function, not to its metaphysical characterisation; revising the characterisation while preserving the function does not close the structural gap the E-conditions identify.

This is why the present analysis engages Sherburne's God-minus account rather than Griffin's or Cobb's revisionary accounts as the strongest counter-development for the diagnostic purpose. Sherburne's proposal is the stronger test case: it attempts the complete elimination of the principle whose structural function the Level III analysis identifies as the vulnerability, not merely its metaphysical recharacterisation. That

Sherburne's account does not close the Level III gap — that creativity without the ordering function becomes formally indistinguishable across instances and cannot satisfy E_1 — is therefore the most demanding result the diagnostic framework can reach within the Whiteheadian tradition. Griffin and Cobb show that the ordering function can be retained under a naturalistic description; Sherburne shows that it cannot be eliminated without exposing the Level III gap directly. The diagnostic result is the same in all three cases; Sherburne's is the case in which the gap is most starkly visible.

Two further developments within the Whiteheadian tradition require direct engagement, since they represent the most formally developed attempts to address structural difficulties adjacent to the Level III failure identified here.

Sherburne's God-minus account. Donald Sherburne's revisionary proposal argues that the theological elements of Whitehead's system are not required by its metaphysical core and that a coherent process philosophy can be constructed from which God is entirely absent.[23] On Sherburne's account, creativity — not God — supplies whatever directional and organising function the system requires. This is the most direct attempt to eliminate the Level III structural vulnerability by removing the imported perturbation principle altogether. The question the diagnostic framework poses is whether creativity, as Whitehead characterises it, can satisfy the E-conditions for the source of en-

[23]See Donald W. Sherburne, "Whitehead without God," *Process Studies* 1, no. 2 (1971): 101–113; see also Sherburne, *A Key to Whitehead's Process and Reality*.

counter in God's absence. It cannot. Creativity in Whitehead's account is the "universal of universals" — a homogeneous ultimate that is identical in character across every instance of its instantiation. It has no internal differentiating dynamic: it does not specify why this occasion arises rather than that one, what makes one occasion's prehensive reach extend in a particular direction, or what fixes the specific relational character of any encounter. These determinations require, on Whitehead's own account, precisely the ordering of eternal objects that God's primordial nature supplies. Remove God and you remove the only mechanism that differentiates the occasions' prehensive directions from the undifferentiated universal activity of creativity. The God-minus account does not eliminate the Level III structural gap; it exposes it more starkly by removing the placeholder that concealed it. Creativity alone does not satisfy E_1 — the requirement that the encounter source be derivable from the same structural dynamic that produces the relata — because creativity is not itself a structural dynamic with internal differentiation: it is the condition of any becoming whatsoever, formally indistinguishable across instances, and therefore cannot explain why specific encounters occur rather than others.

Nobo's extensive connection account. Jorge Luis Nobo's systematic reconstruction of Whitehead's metaphysics argues that the extensive continuum — the spatio-temporal structure within which occasions arise — provides a more fundamental account of the conditions of relational encounter than Whitehead's own presentation makes explicit.[24] Nobo's analysis is the most tech-

[24]See Nobo, *Whitehead's Metaphysics of Extension and Solidarity*, ch. 2–4.

nically careful reconstruction of Whitehead's position available and substantially clarifies the structural role of extensive connection. The diagnostic question is whether Nobo's reconstruction brings the Level III encounter-source derivation within reach of Whitehead's framework. It does not, for a precise reason: the extensive continuum in Nobo's account is itself a structural primitive — the "solidarity of the universe" that occasions presuppose in undergoing their creative syntheses rather than a product of those syntheses. Nobo is explicit that the extensive continuum is not derived from the creative advance; it is the condition within which the creative advance occurs, a spatiotemporal matrix that is given for each epoch rather than constituted by it. This means that Nobo's reconstruction locates the encounter source at the level of an assumed structural background — extensive connection as the framework within which prehensions reach their objects — rather than deriving it from the same dynamic that produces the relata themselves. The E-conditions require that the encounter source arise from the same structural dynamic that generates the relata; a pre-given extensive matrix satisfies neither the derivability condition E_1 nor the specificity condition E_2: it is the context within which specific encounters occur, not what explains why these specific relata encounter one another in this specific way. Nobo's account represents the most sophisticated available development of the resources internal to Whitehead's framework; that it does not reach the derivation the Level III conditions require is therefore not a criticism of Nobo's reconstruction but a precise specification of what the Whiteheadian tradition as a whole — at its most developed — has not supplied.

Bracken's social field theory. Joseph Bracken's social field theory represents the most formally developed contemporary attempt to reconceive the Whiteheadian ground in a way that addresses both the Sherburne and the Nobo limitations simultaneously.[25] Bracken does not retain Whitehead's creativity-as-bare-universal; he proposes to reconceive the ground as a *structured matrix of ongoing activity* — a "divine matrix" that is internally differentiated by the interpenetrating societies of occasions that both arise from it and contribute to it. Bracken's explicit claim is that this reconception addresses the homogeneity objection: the divine matrix is not the undifferentiated universal that creativity-as-such is, but a field whose structure is given by the actual occasions' interpenetrating activity, and from which each new occasion's specific directional character emerges as a function of its position within that structured field. The social field theory is therefore the Whiteheadian tradition's most direct response to the E_1-specificity requirement.

The diagnostic question is precise: does the divine matrix, as Bracken characterises it, satisfy E_1 (the encounter source is derivable from the same structural dynamic that produces the relata)? It does not, for a structural reason that the social field theory cannot remove without changing what it claims to be.

[25]See Bracken, *The One in the Many: A Contemporary Reconstruction of the God-World Relationship*; see also Bracken, *World Without End: Christian Eschatology from a Process Perspective*.

The difficulty is constitutive rather than descriptive. On Bracken's account, the divine matrix receives its specific structure — its internal differentiation at any given point in the creative advance — from the societies of occasions whose interpenetrating activity constitutes the matrix's current organisation. The occasions are, on this account, what give the field its differentiated character; the field is the structured outcome of their interpenetrating activity, not the structural source from which they arise as specifically differentiated occasions. This generates a precise circularity that the E-conditions expose: the matrix is supposed to supply the specific directional character of each new occasion's arising — this is Bracken's advance over Sherburne — but the matrix's own specific structure at the point of each arising is constituted by the occasions whose arising it is supposed to direct. The encounter source (E_1) and the relata whose encounter it sources cannot be mutually constitutive without circularity; one must be structurally prior to and independent of the other.

Bracken's reconception resolves the homogeneity problem by introducing differentiation into the field, but it resolves it in the wrong direction: the differentiation is donated to the field by the occasions, not derived from the field independently of the occasions. The Level III structural gap is not closed; it is relocated from the ground's homogeneity to the mutual constitution of field and occasions. A ground that derives its specific differentiating structure from the occasions it is supposed to ground cannot satisfy the derivability condition E_1: it presupposes the relata in the very specification of what makes their arising possible. The social field theory represents the most sophisticated internal development the Whiteheadian tradition has produced for the Level III problem; that this development does not close the structural gap confirms that the gap's source is the framework's architectural commitment — the foundational constitution of the actual occasion — rather than any remediable feature of Whitehead's own formulations.

> *Summary of Chapter 3.* *Alfred North Whitehead's philosophy of actual occasions provides the most fully developed account of the relational event (Level V) and relational yield (Level IV) in the philosophical literature. Genuine formal achievements obtain at both levels: the prehensive structure of concrescence meets the non-redundant completeness and co-instantaneity requirements of Level V; the satisfaction meets all four conditions of Level IV. The foundational commitment to actual occasions as temporal primitives locates the system's entry at Level V and entails structural foreclosure at Level III (God's initial aim as external perturbation principle), Level II (no invariant relational identity prior to creative synthesis), and Level I (creativity as homogeneous ultimate without derivability or internal differentiating dynamic). This foreclosure is necessary rather than contingent: it is the structural consequence of the foundational commitment, not a feature that revision within the framework could overcome. The five most developed internal responses — Griffin's naturalistic theism, Cobb's natural theology, Sherburne's God-minus account, Nobo's extensive connection reconstruction, and Bracken's social field theory — each address a distinct symptom of the Level III failure while preserving the foundational constitution that generates it; none closes the structural gap the E-conditions identify. Chapter 4 examines the tradition*

that most rigorously attempted what Whitehead did not attempt: a derivation of the absolute ground itself as the necessary starting point of any adequate relational ontology.

CHAPTER 4

Spinoza: God or Nature, But Not Both

By cause of itself I understand that whose essence involves existence, or that whose nature cannot be conceived except as existing.

Baruch Spinoza,
Ethics, Part I, Definition 1

4.1 The Question That Organises Certainty

What would it mean to begin philosophy from absolute certainty — not the hedged, provisional certainty of the empirical sciences, always open to revision, and not the methodological certainty of the natural attitude, always available for phenomenological disruption, but certainty in the deepest possible sense: a beginning so secure that nothing could be outside it, nothing could precede it, and nothing could call it into question without already assuming it? This is the question that organises Baruch Spinoza's *Ethics*, and the geometric rigour with which Spinoza pursues it produces the most uncompromising attempt in the history of philosophy to derive everything from a single, self-sustaining, absolutely necessary ground.

The attempt is not architectural vanity. It is a response to the most serious philosophical problem of the seventeenth century: the Cartesian crisis of the criterion. Descartes had shown that grounding knowledge in the indubitability of the thinking subject generated a structural instability — the subject who cannot doubt its own existence is still separated from the external world by a gap that only God's non-deceiving nature can close, and a God whose existence requires the thinking subject's prior existence as the ground of that inference stands in a circular relation with the subject it is supposed to ground. Spinoza's response is radical and systematic: do not begin with the subject. Begin with what neither subjects nor objects nor relations presuppose, because nothing presupposes it and it presupposes nothing. Begin with the ground itself — with what Spinoza calls *Substance*: the one thing that is in itself and is conceived through itself, whose existence follows necessarily from its own nature, and outside which there is nothing whatsoever.[1]

What Spinoza builds on this ground is one of the most extraordinary philosophical constructions in any language: the *Ethics*, composed in geometric order, proceeding through definitions, axioms, propositions, and demonstrations with a rigour

[1]The opening definitions and axioms of *Ethics* Part I are best read as a systematic reorganisation of the ontological landscape that eliminates the Cartesian instability at its source, by beginning with Substance rather than with the thinking subject. For the most precise analysis of how Spinoza's response to Descartes is structured, see Curley (*Spinoza's Metaphysics: An Essay in Interpretation*, 115–148).

that demands that every claim follow necessarily from what precedes it, and that nothing be assumed that has not been explicitly introduced and justified. The reader who follows the *Ethics* from its opening definition of cause of itself to its closing account of the intellectual love of God traverses a derivation of everything from one thing — and the experience, however difficult, is unlike anything else in the philosophical literature. It is the experience of a mind that has refused every shortcut and paid the price in full, and whose reward is a vision of reality as a single, self-sustaining, infinitely expressive whole in which everything that exists is what it necessarily is, and everything that happens happens because it must.

The reader who has followed the diagnostic framework of Chapter 2 will sense both the power and the structural consequence of this beginning. The question is not whether Spinoza's beginning is the most rigorous in the history of philosophy — it is. The question is whether what follows from it constitutes an adequate account of relational constitution, and whether the entry point that makes the beginning so powerful is the same entry point that makes certain structural results permanently inaccessible.

4.2 The Foundational Commitment: Substance as Self-Causing Totality

The foundational commitment of Spinoza's *Ethics* is stated in the opening definitions of Part I with the economy that geometric method permits. Definition I: by cause of itself (*causa sui*) is understood that whose essence involves existence, or that whose nature cannot be conceived except as existing. Definition III: by substance is understood that which is in itself and is conceived through itself; that is, that whose concept does not require the concept of another thing, from which it must be formed. Definition VI: by God is understood a being absolutely infinite, that is, a substance consisting of an infinity of attributes, of each of which expresses an eternal and infinite essence.[2]

From these definitions, together with axioms presented as self-evident once clearly understood, Spinoza derives in rapid succession the foundational architecture of his metaphysics. There is only one substance (E I P14). That substance is necessarily infinite (E I P8). It exists necessarily, since its existence is entailed by its essence (E I P11). It is the cause of all things (E I P16). And it is identical with God: *Deus sive Natura*, God or Nature, the two terms names for the same absolutely infinite self-sustaining whole.[3]

Everything else — every finite thing, every event, every relation, every thought and extension, every human being and stone and planetary orbit — is a *mode* of Substance: a determi-

[2]The three definitions are cited from Spinoza (*A Spinoza Reader: The Ethics and Other Works*, 85–86). Standard abbreviations used throughout: E = *Ethics*; I–V = Part I–V; D = Definition; A = Axiom; P = Proposition; C = Corollary; S = Scholium; Dem = Demonstration. Thus E I D1 = *Ethics* Part I Definition 1.

[3]E I P14: "Except God, no substance can be, nor can be conceived" id., 420. The *Deus sive Natura* formulation appears in the Preface to Part IV: "that eternal and infinite Being we call God or Nature" id., 544. The identification is not a casual equivalence but a precise ontological claim: there is one thing, absolutely infinite Substance, which is what both names refer to when used with ontological precision.

nate, finite modification of Substance under one of its infinite attributes, following necessarily from Substance's nature either directly (as immediate infinite modes) or through the mediation of other finite modes in the infinite causal chain. The universe is not a collection of things that stand in relations. It is one thing, absolutely infinite, expressing itself in infinite ways under infinite attributes, each finite expression following necessarily from the nature of the whole and from the finite expressions that precede it in the order of derivation.[4]

The foundational commitment that this architecture embodies is precisely the commitment that the diagnostic framework identifies as the mark of a specific entry level: Spinoza takes Substance — the absolutely infinite, self-causing, unique, necessarily existing ground — as his primitive. It is the starting point from which everything follows; the question of what grounds Substance is not asked because Substance is self-grounding by definition. Unlike Whitehead's creativity, which is a characterisation of the activity that occasions instantiate, Spinoza's Substance is itself the formal ground: the one thing in itself and conceived through itself, outside which there is nothing.

[4]E I P16 is the pivot: "From the necessity of the divine nature there must follow infinitely many things in infinitely many modes" id., 424. On the distinction between infinite and finite modes and the structural question of how finite modes follow from an infinite cause, see Curley (*Spinoza's Metaphysics: An Essay in Interpretation*, 61–79) and Bennett (*A Study of Spinoza's Ethics*, 111–142). Deleuze (*Expressionism in Philosophy: Spinoza*) offers the most systematic account of expression as the principle connecting Substance, attributes, and modes; see especially Deleuze (*Expressionism in Philosophy: Spinoza*, 13–43).

Proposition 4.1 (Spinoza's Entry Level)**.** The foundational commitment of Spinoza's *Ethics* — Substance as self-causing, necessarily existing, unique totality — locates his system's entry into the structural order at Level I (the ground). Spinoza is the only major tradition in the diagnostic survey that attempts to begin at the structural level the diagnostic framework identifies as the foundational requirement.

Proof. By the diagnostic method of Chapter 2, the entry level is the level at which the foundational commitment takes a structural feature as primitive. Spinoza's foundational commitment is to Substance as the self-causing ground: that which is in itself and conceived through itself, outside which there is nothing. This is not entry at Level V (the relational event), which would require taking some temporal or experiential unit as primitive, nor at Levels II, III, or IV, which would require taking some feature of relata, encounter, or yield as primitive. It is entry at Level I: the ground is the foundational term, not derived from a more primitive dynamic, but established through the four opening definitions and their logical consequences as the unique structural result of what Spinoza's axioms entail.

4.3 Entry Level : The Ground and Its Three Achievements

Entering at Level I is a structural achievement that no other tradition in the diagnostic survey accomplishes. The diagnostic question is therefore not whether Spinoza reaches Level I — he does — but whether his account of the ground satisfies the four formal conditions that Definition 2.1 of Chapter 2 specifies. The analysis shows that it satisfies three of the four with precision, and that the one condition it fails is the condition whose failure cascades across every level above it.

Condition (G_1)*: Totality*

Spinoza's Substance meets the totality condition with greater rigour than any other metaphysical ground in the philosophical literature. There is only one substance (E I P14), and it is absolutely infinite, consisting of an infinity of attributes. Outside it there is nothing — no structural resource that could be imported from a domain it does not encompass, no outside from which it could be supplemented or limited. The proof of uniqueness is the structural correlate of the totality condition: if there were two substances, they would either share an attribute (which Spinoza's prior derivation shows is impossible, since substance is necessarily distinguished through its attributes) or have no attribute in common, which would make them incapable of standing in any structural relation whatsoever — which is precisely what the totality condition requires of the ground. Condition (G_1) is satisfied.[5]

[5]E I P5: "In nature there cannot be two or more substances of the same nature or attribute" Spinoza, *A Spinoza Reader: The Ethics and Other Works*, 411. E I P14C1: "God is unique… in Nature there is only one substance" id., 421. The proof is derived through a series of propositions, each resting on the prior ones. Bennett (*A Study of Spinoza's Ethics*, 64–80) provides a proposition-by-proposition analysis of the logical tightness of the derivation.

Condition (G_3)*: Self-Sustenance*

Spinoza's Substance is *causa sui*: its essence involves existence, and its existence follows from itself alone. This is the explicit content of Definition I and of Proposition 11, with its three independent demonstrations. Whitehead's creativity requires actual occasions to instantiate it; without them, creativity is not a self-sustaining ground but an abstract universal awaiting actualisation. Bergson's duration requires conscious or living beings to undergo it. Both are sustained by what they are supposed to sustain. Spinoza's Substance requires nothing: it is what it necessarily is, and its existence is the formal consequence of its own nature. Condition (G_3) is satisfied.[6]

Condition (G_4)*: Formal Derivability — A Qualified Achievement*

The formal derivability condition is the most demanding. It requires that the ground be shown to be the unique structural result of the eliminative method applied to what any relational ontology presupposes, rather than stipulated as an arbitrary starting point. Spinoza achieves a significant portion of this condition and falls short in a specific and diagnosable way.

[6]EIP11: "God, or a substance consisting of infinite attributes… necessarily exists" Spinoza, *A Spinoza Reader: The Ethics and Other Works*, 417. The three demonstrations proceed by the principle of sufficient reason, by reductio, and by consideration of the formal distinction between necessary and contingent existence. Donagan (*Spinoza*, 50–62) notes that the three demonstrations approach the self-sustenance of Substance from three formally distinct angles — logical, causal, and ontological — each illuminating a different dimension of what *causa sui* means.

The achievement: Spinoza does not simply assert that there is a substance. He derives it. The opening definitions and axioms of the *Ethics* are not arbitrary starting points but the formal presuppositions of any coherent account of what it means for something to exist at all. A thing that is not in itself must be in another; a thing not conceived through itself must be conceived through another. If there is anything at all — any existent, any relation, any event — then there must be something that is in itself and conceived through itself; otherwise the derivation of anything from anything else cannot reach a terminus and the account of what things are has no structural ground. Spinoza's definitions are the minimum ontological commitments of any account of existence, and his derivation of Substance from them is a genuine formal achievement.[7]

The shortfall: the derivability condition requires not only that the ground be derivable as the unique structural result of the eliminative method, but that its structural properties — including its internal generative dynamic — be shown to follow from what the eliminative derivation establishes. Spinoza derives that there is a substance and that it is unique, necessarily existing, and infinite. He does not derive what the ground's internal dynamic is such that finite, determinate, structurally

[7]Rocca (*Spinoza*) argues that the principle of sufficient reason — for every fact there is an explanation — is the methodological engine of the entire *Ethics*, and that Spinoza's derivation of Substance is essentially the derivation of what a world governed by sufficient reason is required to contain as its ultimate ground; see Rocca (*Spinoza*, 3–22). The present analysis agrees with this reading while identifying the structural limit: the principle of sufficient reason, however rigorously applied, does not by itself entail the internal generative dynamic that the derivability condition requires.

distinct modes can emerge from it. The attributes are *posited* rather than derived: they are the ways in which infinite intellect conceives Substance, not internal differentiations through which Substance generates its modes.[8] Condition (G_4) is partially met, in the qualified sense established above: Spinoza derives Substance from the minimum formal presuppositions of any account of existence, satisfying the eliminative dimension of the condition, while the derivation of the ground's internal generative dynamic remains unachieved.

The "qualified sense" requires precise statement, because the qualification is not merely about incompleteness but about the method. Condition (G_4) requires that the ground be shown to be the *unique structural result of the eliminative method applied to the presuppositions of relational ontology*. Spinoza's *more geometrico* is not that method: it demonstrates the formal necessity of Substance from defined axioms and definitions, not by applying the eliminative method to what relational ontology presupposes. His derivation moves from definitions to consequences

[8]The status of the attributes — whether they are real distinctions within Substance or distinctions belonging only to the intellect — is the central interpretive controversy in Spinoza scholarship. The subjectivist reading (Stuart Hampshire, *Spinoza* (Harmondsworth: Penguin, 1951)) holds that attributes are ways of conceiving Substance not corresponding to real internal distinctions. The objectivist reading (Curley, *Spinoza's Metaphysics: An Essay in Interpretation*; Bennett, *A Study of Spinoza's Ethics*) holds that the attributes are real aspects of Substance. Deleuze (*Expressionism in Philosophy: Spinoza*) develops a third reading in which expression is the structural principle connecting Substance, attributes, and modes. For the present analysis the interpretive controversy is secondary to the structural point: in every reading, the attributes are introduced as part of the definition of Substance rather than derived from a more primitive structural dynamic.

by logical entailment, not by systematically eliminating every structural candidate that fails the minimal necessary conditions for being a derivable ground of relational constitution. What Spinoza meets, in the qualified sense, is the *direction* of the condition — deriving the ground formally rather than positing it — while the specific method the condition requires and the specific content (the ground's internal generative dynamic) that the method would have to reach both remain unachieved. This is why the qualification attaches to (G_4) and not to the other conditions: Spinoza genuinely attempts derivation; the geometric method is simply not the eliminative method, and consequently cannot deliver what the eliminative method would deliver.

Condition (G_2): Generativity — The Decisive Failure

The generativity condition is where Spinoza's account of the ground reaches its structural limit — and where the limit reveals itself not as an incidental gap but as the necessary consequence of Substance's greatest formal strength: its homogeneity. That this is a recognised difficulty in Spinoza scholarship and not an external imposition is confirmed by Hübner,[9] who establishes that Spinoza's argument in E I P16d for the necessity of

[9]Karolina Hübner, "Spinoza's Thinking Substance and the Necessity of Modes," Archived at PhilArchive, rec/HBNSTS. Establishes that Spinoza's argument in E1p16d for the necessity of modes "seems to be in need of nontrivial supplementation, if we are not to conclude that the principle on which it rests is itself just a piece of metaphysical dogma" — specialist Spinoza scholarship confirming that the derivation of the modal order from Substance is not achieved by Spinoza, as the present investigation's diagnostic independently argues, *Philosophy and Phenomenological Research* 89, no. 3 (2014): 3–34.

modes "seems to be in need of nontrivial supplementation, if we are not to conclude that the principle on which it rests is itself just a piece of metaphysical dogma" — a precise specialist formulation of the derivability gap the present investigation diagnoses as the failure of condition (G_2). A second specialist confirmation comes from Primus,[10] who establishes that Spinoza's immanent causation of finite modes from infinite, eternal Substance "involves an inexplicable shift in realities that renders the immanent causation (and inherence of effect in the cause) hard to understand," and who concludes that the only coherent reading of Spinoza's system on this point commits him to acosmism — "a denial of the reality of the world, at least the world of enduring, finite things." The diagnostic consequence is exact: a ground from which finite, structurally distinct modes can be generated only through an "inexplicable shift" is a ground that fails the generativity condition from within its own tradition's most careful scholarship.

Spinoza's Substance is formally homogeneous. It is one, unique, absolutely infinite, and indivisible. It has no interior boundaries, no structural differentiations within itself that cor-

[10]Kristin Primus, "Spinoza's Monism II: A Proposal," Archived at PhilArchive, rec/PRISMI-3. Establishes that Spinoza's immanent causation of finite modes from infinite Substance "involves an inexplicable shift in realities that renders the immanent causation (and inherence of effect in the cause) hard to understand," and concludes that the only coherent interpretation commits Spinoza to acosmism: "a denial of the reality of the world — at least the world of enduring, finite things." Peer-reviewed specialist confirmation that the transition from infinite Substance to finite modes cannot be achieved within Spinoza's framework, corresponding to Before Relation's diagnosis that Substance fails the generativity condition (G_2), *Archiv für Geschichte der Philosophie* 105, no. 3 (2023): 444–469.

respond to the distinctions among the modes it produces. The attributes are the ways in which infinite intellect perceives Substance as constituting the essence of a certain kind of thing — thought under the attribute of thought, extension under the attribute of extension. But the attributes are not internal differentiations of Substance: they are the *formal essences* of Substance as conceived under distinct frameworks of understanding. Substance itself, prior to its expression through attributes, is homogeneous: absolutely one, with no internal structure that would make one region of it more generative of thought and another more generative of extension. The unity and indivisibility of Substance are derivationally secured.[11]

A homogeneous ground — however absolutely total, however self-sustaining, however rigorously derived — cannot generate structural differentiation from within its own dynamic. The generativity condition requires that the ground be capable of producing structural differentiation through its own internal activity — that the specific characters of specific modes emerge from a dynamic internal to the ground, not from logical entailment of the ground's necessary nature plus the external framework of the attributes. But logical entailment is precisely what homogeneous Substance can offer. From the necessity of Substance's nature there follow infinitely many things in infinitely

[11] E I P13: "A substance which is absolutely infinite is indivisible" Spinoza, *A Spinoza Reader: The Ethics and Other Works*, 419. E I P12 shows that no substance can be divided; P13 extends this to absolutely infinite substance. Jarrett (*Spinoza: A Guide for the Perplexed*, 34–41) provides the clearest exposition of why indivisibility and homogeneity are connected in Spinoza's framework.

many modes (E I P16) — but they follow *necessarily*, as logical consequences of what Substance necessarily is, not as products of an internal generative dynamic through which Substance differentiates itself. The modes are entailed; they are not produced through a structural process internal to the ground.

The distinction between logical entailment and structural generation is not terminological. A ground that generates structural differentiation through its own dynamic can produce genuine novelty: outcomes that are determinate but that were not pre-given in the structural content of the ground before the generative process occurred. A ground that entails its consequences logically cannot produce genuine novelty in this sense: what follows necessarily from the ground's nature was, in a precise formal sense, already contained in the ground before any consequence was drawn. The *Ethics'* God is fully actual from the beginning — Substance's infinite attributes and their infinite modes are not products of a becoming but expressions of what Substance eternally and necessarily is. There is, in Spinoza's universe, no latent dimension from which genuine novelty could emerge.[12]

The expressivist reading of Spinoza — most fully developed in Deleuze[13] — represents the strongest available challenge to this diagnosis and must be addressed directly. Deleuze argues that the relation of Substance to its modes is not a logical entail-

[12]Hampshire (*Spinoza*, 56): "For Spinoza. . . there is no becoming, no time, no change, except as these appear within the system from the point of view of finite modes. Sub specie aeternitatis, all that exists is a timeless system of logical relations."

[13]Deleuze, *Expressionism in Philosophy: Spinoza*.

ment but an expressive-generative process: Substance *expresses* itself through its attributes and modes in the way a force unfolds through its expressions, rather than the way a premiss logically compels its conclusion. On this reading, expression is a third structural category between logical entailment and mechanical causation, and it carries the generative content that G_2 demands.

The response is that expression, as Deleuze develops it from Spinoza, remains a logical-ontological operation rather than a generative one in the sense G_2 requires. What distinguishes expression from logical entailment is not that expression produces structurally novel content from something latent; it is that expression unfolds what is already fully actual — the infinite attributes are not produced by Substance but are Substance under its infinite forms of self-manifestation. The expressive relation is between a fullness of being and its modes of self-presentation, not between a structural potential and its determinate actualisations. G_2's generativity condition requires that the ground be capable of producing structural differentiation from within its own dynamic — that genuinely new structural content emerge from what was latent but not yet determinate. Expression, on Deleuze's reading, does not supply this: the modes do not emerge from a latent structural dimension; they express what is already infinite, actual, and fully determined. Rae[14] confirms from within Deleuze studies that Deleuze's dif-

[14]Gavin Rae, "Three Senses of Identity and Deleuze's Differential Ontology," Archived at PhilPapers, rec/RAETOI. Pages 383–387 establish, at three specific structural junctures in *Difference and Repetition* (the virtual-actual movement; transcendental conditions of different forms of thinking; the re-

ferential ontology constitutes intensive difference as primary rather than derived at key junctures, which means the differential is foundationally constituted rather than generated from something more primitive. The expressivism diagnosis therefore reinforces rather than resolves the G_2 failure.

Proposition 4.2 (Spinoza's Ground: A Partial Achievement)**.** Spinoza's account of Substance satisfies three of the four conditions of a formally adequate ground: (G_1) totality, (G_3) self-sustenance, and — in the qualified sense established above — (G_4) formal derivability. It fails the generativity condition (G_2): Substance's homogeneity prevents it from providing a structural dynamic through which genuinely differentiated modes could be generated from within the ground's own activity. This failure is not contingent but follows necessarily from the foundational commitment to homogeneous Substance.

Proof. The satisfaction of (G_1) is established in the analysis of the totality condition: Substance admits no exterior by E I P14 and its corollaries; there is no structural domain outside Substance from which additional content could be supplied. The satisfaction of (G_3) is established in the analysis of self-sustenance: Substance is self-causing (E I D1),

lationship between those forms), that Deleuze's affirmation of difference's primacy depends on a form of identity — the identity of the common — that has not been derived from conditions more primitive than difference itself. Specialist confirmation that intensive difference is foundationally constituted rather than derived at the key junctures where the diagnostic analysis places the SIT's foreclosure, *International Journal of Philosophical Studies* 22, no. 1 (2014): 86–105.

its existence follows from its essence (E I P7), and it depends for its existence on nothing external to itself. The qualified satisfaction of (G_4) is established in the analysis of formal derivability: the modes follow with necessity from the nature of Substance (E I P16), and this necessary following constitutes a form of formal derivability, albeit one that the analysis qualifies as logical entailment rather than structural generation. The failure of (G_2) follows from Substance's homogeneity. The generativity condition requires that the ground produce structural differentiation through its own internal dynamic — that the specific characters of specific modes emerge from a process internal to the ground, not from logical entailment alone. Substance is formally homogeneous: one, indivisible, with no interior structural differentiations (E I P12–P13). A homogeneous ground can entail its consequences necessarily, but it cannot generate structural novelty from a latent internal dimension, because a homogeneous ground has no latent internal dimension from which novel structural content could emerge. The modes are eternally and necessarily entailed; they are not produced through a generative process by which latent structural content becomes determinate. The expressivism reading examined above does not resolve this because expression, as Deleuze develops it from Spinoza, unfolds what is already fully actual rather than producing what was structurally latent. The G_2 failure is therefore the necessary structural consequence of Substance's homogeneity, which is itself the necessary consequence of the foundational commitment: a unique, indivisible, absolutely infinite Substance is formally homogeneous by definition, and its homogeneity forecloses the generativity condition no less necessarily than the commitment forecloses the levels below its entry.

4.4 What Spinoza Cannot Reach : Homogeneity and Its Consequences

The failure of condition (G_2) is the structural hinge on which Spinoza's account of relational ontology turns. Because the ground cannot generate structural differentiation from within its own dynamic, the structural consequences propagate upward through Levels II, III, IV, and V with systematic uniformity. The failure at each level is the same failure seen from a different structural vantage point: the consequences of homogeneity in a domain where differentiation is precisely what is required.

The Foreclosure at Level II : Relational Identity

Relational identity requires an invariant structural character that is disclosed through encounter, that persists through it without being dissolved by it, and that is asymmetrically specific: uniquely this relatum. In Spinoza's framework, finite modes are determinate modifications of Substance under a specific attribute, following necessarily from the nature of Substance and from prior modes in the infinite causal chain. Each mode has a determinate *essence* — what Spinoza calls the mode's *conatus*, its striving to persist in its own being — and it is in virtue of this *conatus* that each finite thing maintains its specific character against the perturbations of its environment.

The *conatus* is the Spinozistic counterpart of relational identity, and it is a genuine structural achievement within the framework. Each finite thing strives to persist in its own being in accordance with its specific essence (E III P6–P7), and this striving constitutes the thing's characteristic mode of engaging with what affects it.[15]

But the conatus fails the disclosure condition (I_2): the specific essence of a mode is not disclosed through its encounters but is given in advance as a determinate consequence of Substance's nature. What I am — the specific mode of Substance that I am — is not something revealed progressively through encounters with other modes; it is what Substance necessarily produces at the conjunction of the causal chains that converge to produce this specific modification. The mode's essence is, *sub specie aeternitatis*, already fully determinate before any encounter occurs. What encounters reveal is not what I am but only how I react — whether my power of acting is increased or diminished, whether I am affected with joy or sadness. The encounters disclose the relational dynamics of finite modes with-

[15]E III P6: "Each thing, as far as it can by its own power, strives to persevere in its being" Spinoza, *A Spinoza Reader: The Ethics and Other Works*, 498. E III P7: "The striving by which each thing strives to persevere in its being is nothing but the actual essence of the thing" id., 499. The conatus doctrine's echoes in Leibniz's appetition, Schopenhauer's will, and Nietzsche's will to power are well documented; its structural role as the nearest Spinozistic equivalent to relational identity is the point at issue here.

out disclosing their essential characters, which were given from the beginning.

More deeply, condition (I_4) — derivability from Level I — fails in a structurally significant way. The specific relational identity of each mode — what makes this mode the specific mode it is rather than any other — is the consequence of an infinite causal chain extending back through the entirety of prior modes to Substance itself. No finite intellect can possess knowledge of this chain; even an infinite intellect's derivation of a specific mode's character from Substance passes through infinity before reaching the particular. The specific relational identity of any mode is derivable in principle but unreachable in practice — not because derivability fails, but because the derivation is an infinite derivation through logical entailment that arrives at the specific only by traversing the universal. This is not the kind of derivability the structural framework requires: what is needed is an account of how specific relational identities emerge from the ground's own generative dynamic, not an infinite logical entailment whose specific result depends on the entirety of what has preceded it.[16]

[16] The problem of deriving the specific from the universal in Spinoza is related to the "finite mode problem": how does an infinite cause produce finite effects? The standard responses (Curley, *Spinoza's Metaphysics: An Essay in Interpretation*; Deleuze, *Expressionism in Philosophy: Spinoza*; Bennett, *A Study of Spinoza's Ethics*) each attempt to close the gap between infinite ground and finite consequence. The present analysis holds that the gap is structural: homogeneous Substance cannot generate structurally differentiated modes from within its own dynamic, and this is what creates the specific character of the finite mode problem. See Curley (*Spinoza's Metaphysics: An Essay*

The Foreclosure at Level III : The Source of Encounter

The source of encounter requires an account of what brings specifically this relatum into contact with specifically that one. In Spinoza's framework, the answer is structural and precise: each finite mode affects and is affected by other finite modes in accordance with the laws governing the attribute under which both are conceived. Every encounter between finite modes is necessitated by the prior causal chain: what affects me now affects me because the entire prior history of the universe under the relevant attribute has produced the specific configuration of modes constituting my environment. The source of encounter, in Spinoza's framework, is causal necessity: the infinite causal chain connecting every finite mode to every other.

This account fails the derivability condition (E_1): the source of encounter is not derived from the same structural dynamic that produces the ground and the relational identities it brings together, because causal necessity in Spinoza's framework is not a dynamic internal to the ground but the logical consequence of Substance's infinite nature expressed through the attributes. Causal necessity is not a *process* by which the ground differentiates itself and brings its differentiated expressions into productive contact; it is the *form* of the logical entailment connecting Substance to its modes and modes to one another.

The failure is also visible in Spinoza's account of affect. Modes encounter one another through the transmission of affect: a mode affecting another either increases or decreases

in Interpretation, 61–79) and Deleuze (*Expressionism in Philosophy: Spinoza*, 43–70).

the other's power of acting, generating the active and passive affects that constitute the emotional life of finite things (E III P1–P3). This is Spinoza's formal account of relational encounter at the level of psychology and ethics. But the transmission of affect is governed by the same causal necessity that governs all modal relations: it is not a structural event with an internal organisation — not an encounter with three formally distinct co-instantaneous dimensions — but a causal transaction whose specific character is the inevitable consequence of the prior causal chain. Condition (E_2) — the specificity of the source of encounter — is nominally met but met vacuously: the *why* of the specific encounter reduces to the *because the prior causal chain necessitates it*, which is not a structural account of how the source of encounter arises from the ground's own generative activity but a reiteration of causal necessity.[17]

The Foreclosure at Level IV : Relational Yield

The relational yield condition requires that encounter produce something genuinely irreducible to the encountering relata. In Spinoza's framework, the encounter between finite modes produces a change in each mode's power of acting: an increase (joy, *laetitia*) or decrease (sadness, *tristitia*). These changes constitute the formal yield of relational encounter in Spinoza's account.

[17]Hampshire (*Spinoza*, 61–75) identifies the structural difference between Spinoza's view of causation — essentially inferential, more like logical implication than efficient causation — and the views of later empiricists. The present analysis treats this as the structural mark of a ground that cannot generate encounter through its own dynamic because its homogeneity prevents it from having one.

The irreducibility condition (Y_1) is not fully met. What the encounter produces — the specific affect resulting from the specific configuration of modes in contact — is a determinate consequence of what each mode is and what the causal laws governing their interaction specify. It is not derivable from either mode alone; in that formal sense it is irreducible to each individually. But it is derivable from the two modes together, through the causal laws governing their interaction under the relevant attribute. The affect is not structurally novel in the sense the relational yield condition requires: it does not have structural properties that neither relatum possessed and that the encounter produces rather than reveals. The encounter produces what the causal nexus of two specific modes under their governing laws necessarily produces.

The generativity condition (Y_3) is met in a limited sense: the affect produced by an encounter does become a factor in subsequent encounters. But the generativity is not structural novelty; it is the transmission of causally determined affects through the causal chain. What the encounter produces does not introduce a genuinely new locus of relational structure into the universe. Everything that follows from any encounter was already logically contained in Substance before the encounter occurred. There is, in the deepest sense, nothing new under the sun.[18]

[18]Spinoza's explicit doctrine of the eternity of all things sub specie aeternitatis is stated in E V P22–P23. Bennett (*A Study of Spinoza's Ethics*, 184–204) acknowledges the tension between the eternity of all finite modes as logical consequences of Substance and the appearance of temporal novelty from the perspective of finite modes. The present analysis treats this tension as

The Foreclosure at Level V: The Relational Event

The derivability condition (V_1) and the self-exemplification condition (V_4) are both violated in a structurally precise way.

The derivability condition requires that the internal organisation of the relational event be derived from the same structural dynamic that produces the ground and the relata. In Spinoza's framework, the relational event has no internal organisation in the relevant sense: it is a causal transaction governed by the laws of the attribute under which the encountering modes are conceived. The transaction is not an event with three formally distinct co-instantaneous dimensions — initiation, regulation, direction — but a causal determination whose form is the form of logical necessity. There is nothing to derive here, because the relational event has no internal structure that could be the subject of a derivation. It is what the causal laws require, instantiated in the specific configuration of the specific modes in the causal chain.

The self-exemplification condition is also violated. Spinoza's geometric method — the *more geometrico* of the *Ethics* — is itself a derivation. But this derivation is not an instance of the structural dynamic the *Ethics* formally establishes, because the dynamic it establishes is causal necessity through logical entailment, and logical entailment as a method of philosophical derivation is not the same kind of thing as causal necessity as a structural feature of finite modes. The *Ethics'* method does not enact what

the structural mark of a framework in which genuine relational yield is impossible: if everything is already logically entailed, the encounter does not produce something that was not already there.

the *Ethics* derives; it derives it from outside it. The author of the *Ethics* reasons to the conclusion that everything is causally necessitated; the reasoning is not itself a causally necessitated consequence of Substance's infinite nature, but a formal act of the philosopher who has taken a specific standpoint — the standpoint of adequate ideas — outside the causal chain whose structure the *Ethics* derives.[19]

4.5 The Diagnostic Result

Spinoza's system occupies a unique position in the diagnostic survey. It is the only tradition that genuinely attempts to begin at Level I — that takes the ground as its philosophical starting point and builds the account of everything else from it. The *Ethics* is, among other things, the most rigorous attempt in the history of philosophy to show what follows, with formal necessity, from the structure of the ground.

The structural limit the diagnostic framework identifies is not a failure of rigour or imagination. It is the consequence

[19]The self-exemplification difficulty is close to a problem Rocca (*Spinoza*) identifies concerning the status of philosophical reasoning within a deterministic framework: if all mental events, including acts of reasoning, are deterministically necessitated by prior modes, then the *Ethics*' derivation is itself a causally necessitated consequence of Spinoza's prior mental states, which raises the question of what it means to "follow" an argument in a framework that treats all mental events as necessary consequences of prior causes; see Rocca (*Spinoza*, 121–138). The present analysis treats this as a structural self-exemplification failure: the *Ethics*' method is not an instance of the structural dynamic it derives, and this failure marks the point at which the framework's structural account of the relational event is not self-sealing.

of the specific character of Spinoza's ground: its homogeneity. A ground that is absolutely infinite, necessarily existing, unique, and indivisible — that meets three of the four conditions of Level I with greater precision than any other philosophical ground — is a ground that has achieved its formal completeness at the cost of internal structural differentiation. The same homogeneity that makes Substance absolutely total, absolutely self-sustaining, and absolutely one makes it incapable of generating structural differentiation from within its own dynamic. What follows from Substance follows necessarily; it does not emerge through a generative process. And a ground from which nothing genuinely emerges — from which everything merely follows as a logical consequence — cannot be the structural source of genuinely novel relational yield, of specific relational identities disclosed through encounter, or of a relational event with an internal organisation that is itself the product of the ground's dynamic.

Theorem 4.1 (Spinoza's Structural Foreclosure)**.** The foundational commitment of Spinoza's *Ethics* — Substance as homogeneous, self-causing, necessarily existing totality — entails structural foreclosure at Levels II, III, IV, and V. This foreclosure is necessary rather than contingent: it is the structural consequence of the one condition at Level I that Spinoza fails — the generativity condition (G_2) — propagating through every level that requires structural differentiation to have been generated from within the ground's own activity.

More precisely: *(i)* a homogeneous ground whose modes follow from it by logical entailment cannot provide a relational identity (I_2, I_4) that is progressively disclosed through encounter and derived from an internal generative dynamic; *(ii)* a ground whose modes encounter one another through the necessity of the prior causal chain cannot provide a source of encounter (E_1) derived from a structural dynamic that brings specifically these relata into productive contact through a process of the ground's own activity; *(iii)* a ground from which everything follows necessarily as a logical consequence cannot produce relational yield (Y_1) that is genuinely novel — structurally irreducible to what was already contained in the ground before the encounter; and *(iv)* a framework in which the relational event is a causal transaction governed by logical necessity cannot derive the internal organisation of the relational event (V_1) from a structural dynamic internal to the ground, or produce an account that is self-exemplifying (V_4).

Proof. Each of the four parts follows from the same structural source: the failure of (G_2). *(i)* If the ground cannot generate structural differentiation from within its own dynamic, the specific characters of modes are the consequences of logical entailment through the infinite attributes rather than the products of a generative process. A mode whose character is the consequence of logical entailment does not have a relational identity progressively disclosed through encounter (I_2), because what it is is already fully determined before any encounter; and the derivation of its specific character from the ground passes through an infinite causal chain rather than through a dynamic internal to the ground's own activity (I_4). *(ii)* If encounters between modes are governed by causal necessity through logical entailment rather than by a structural dynamic internal to the ground, the source of encounter is the prior causal chain rather than an internal generative process (E_1). Causal necessity meets the specificity condition (E_2) vacuously: it specifies which encounters occur by specifying that only the causally necessitated encounters can occur, which is not a structural account of how the source of encounter arises from the ground's own activity. *(iii)* If everything that follows from Substance was already logically contained in Substance before any encounter occurred, the yield of any encounter is a determinate consequence of what was always already the case, not a genuine structural novelty (Y_1). *(iv)* A relational event understood as a causal transaction governed by logical necessity has no internal organisation of the kind Level V requires (V_1), and the geometric method of the *Ethics* does not enact the structural dynamic it derives (V_4). In each case, the foreclosure is necessary: it follows from the homogeneity of Substance, which is itself a necessary feature of the foundational commitment to Substance as indivisible and absolutely infinite.

Remark 4.1 (On Spinoza and Whitehead as Structural Complements). The diagnostic results for Whitehead and Spinoza exhibit a structural complementarity that deserves explicit recognition. Whitehead enters the structural order at Level V and achieves genuine formal results at Levels IV and V while being foreclosed at Levels I, II, and III. Spinoza enters at Level I and achieves genuine formal results there while being foreclosed at Levels II, III, IV, and V. Together they define the structural space of the problem with precision: the ground has been approached from above (through the relational event) and from below (through the absolute ground itself), and neither approach has produced an account satisfying all five formal requirements simultaneously. Whitehead's framework knows precisely what the relational event is but cannot derive what makes it possible. Spinoza's framework knows precisely what the ground is but cannot generate what the ground must produce. A formally adequate relational ontology would need to achieve what neither tradition achieves: a derivation that begins at Level I with a ground satisfying all four conditions — including generativity — and builds upward through Levels II, III, IV, and V without losing formal precision. This is the structural requirement that the diagnostic survey is progressively defining.

Criterion 4.1 (Spinoza's Diagnostic Profile). *Entry level*: Level I (the ground). *Formal achievements at and above entry level*: Level I — three of four conditions met: (G_1) totality, (G_3) self-sustenance, and (G_4) formal derivability in the qualified sense; (G_2) generativity fails: Substance is formally homogeneous without internal generative dynamic. *Structural foreclosure*: Level II — relational identity is given as the logically entailed character of the mode, not disclosed through encounter or derived from an internal generative dynamic ((I_2) and (I_4) fail); Level III — the source of encounter is causal necessity through logical entailment, not a structural dynamic internal to the ground ((E_1) fails); Level IV — relational yield is causally determined rather than genuinely novel ((Y_1) fails); Level V — the relational event has no internally derived organisation and the account is not self-exemplifying ((V_1) and (V_4) fail). *Character of the foreclosure*: Necessary. The foreclosure is the structural consequence of Substance's homogeneity — which is itself a necessary feature of the foundational commitment to Substance as indivisible, unique, and absolutely infinite — and cannot be overcome by modifications internal to the framework without abandoning that commitment.

The diagnostic result does not diminish what Spinoza achieved. The *Ethics* remains the most rigorous derivation of a philosophical ground from first principles that Western philosophy has produced. The formal necessity of its opening propositions, the geometric economy of its method, and the extraordinary intellectual courage of its central identification — *Deus sive Natura* — constitute a philosophical monument that has informed every subsequent attempt to think the absolute systematically. The diagnostic claim is structural, not evaluative: Spinoza solved the problem of the ground with unmatched rigour, and the solution cost him everything above Level I.

What Spinoza showed us is that there is a ground — that the question of what there is bottoms out in something that is in itself and conceived through itself, necessarily existing, absolutely infinite, and without outside. He showed it with a rigour that has never been matched. What he could not show — what the homogeneity of his ground structurally prevented him from showing — is how the ground generates, through its own internal activity, the structurally differentiated relata, encounters, and yields that relational ontology requires. The ground is reached. It is still. The movement from ground to relation — from what is absolutely one to what is genuinely many — is what Spinoza cannot make from where he stands.

Summary of Chapter 4. *Spinoza's Ethics is the only tradition in the diagnostic survey that genuinely attempts to begin at Level I: the foundational commitment to Substance as self-causing, necessarily existing, unique, and indivisible totality constitutes an entry at the structural level the diagnostic framework identifies as the foundational requirement. Genuine formal achievements obtain at Level I: the totality condition (G_1), the self-sustenance condition (G_3), and — in a qualified sense — the formal derivability condition (G_4) are satisfied. The generativity condition (G_2) fails: Substance's homogeneity prevents the ground from generating structural differentiation through its own internal dynamic, limiting what follows from it to logical entailment of necessary consequences rather than the production of genuinely novel structural content. This failure propagates through Levels II, III, IV, and V: relational identity is given rather than disclosed, the source of encounter is causal necessity rather than a structural dynamic of the ground, relational yield is determined rather than novel, and the relational event has no internally derived organisation. The structural complementarity with Chapter 3 is exact: Whitehead knows the relational event but cannot derive its ground; Spinoza knows the ground but cannot generate the relational event. Chapter 5 turns to the tradition that takes as its starting point neither the event nor the ground but the temporal flow that connects them — and finds that beginning from temporal flow forecloses the derivation of both.*

CHAPTER 5

Bergson: Time Is Real, But What Is Time?

Pure duration is the form which the succession of our conscious states assumes when our ego lets itself live, when it refrains from separating its present state from its former states. It need not be altogether absorbed in the passing sensation or idea; for then, on the contrary, it would no longer endure.

Henri Bergson,
Time and Free Will

5.1 The Question That Animates Duration

Have you ever tried to remember not what happened during a summer holiday but what the summer itself felt like — its particular texture, the specific weight of its days? You know immediately that this is not the kind of thing a calendar can supply. You can locate the summer precisely in clock time; you can reconstruct its events in accurate sequence; you can even verify your reconstruction against photographs and diaries. And yet the summer itself — the duration of it, the way one week bled into the next without sharp edges, the sense of something continuous and accumulating that retrospect presents as a whole — this is available only to memory, and memory of a specific kind: not the memory that retrieves stored information but the memory that recovers something still living, something that has survived in you rather than been recorded by you. Clock time tells you how long the summer was. It tells you nothing about what it was.

This distinction — between time as measurable succession and time as lived duration — is the central discovery that Henri Bergson spent his philosophical career articulating, defending, and applying to domains as different as psychophysiology, evolutionary biology, and the emerging technology of cinema. It is a discovery that sounds, on first hearing, like a merely phenomenological observation about the difference between subjective experience and objective measurement. It is in fact a claim of the deepest ontological consequence: not that time feels different from how physics represents it, but that what physics represents is not time at all, but rather a *spatialisation* of time — a substitution of a measurable, homogeneous, infinitely divisible quantity for the irreversible, heterogeneous, indivisible flow that constitutes what duration actually is. The error is not in the physics. It is in the confusion between the map and the territory, between a practical instrument of measurement that serves science admirably and the reality that the instrument was supposed to measure.[1]

The question the present chapter pursues is not whether Bergson is right about this distinction — the diagnostic framework is not in the business of adjudicating between traditions on their own terms, and the distinction between duration and spatialised time is one of the most precise and consequential contributions any philosopher made to the twentieth century. The question is rather what follows when the five-level diagnostic grid established in Chapter 2 is applied to Bergson's framework: at which level does Bergson enter the structural order? What does he formally achieve from that entry level? And what, by entering where he does, does he make himself structurally unable to reach? Bergson, this chapter will argue, enters the structural order at the actuality dimension of Level V — the dimension that the diagnostic framework identifies as the formally historical pattern of accumulating sedimentation from which every relational event orients. He enters there with extraordinary precision. And in entering there he forecloses — necessarily,

[1]Bergson's most systematic statement of the spatialisation thesis is in the closing chapter of Bergson (*Time and Free Will: An Essay on the Immediate Data of Consciousness*), where he argues that what we represent as time — as a line divisible into moments — is already a spatial representation, and that the confusion between this representation and duration itself is what generates the classical philosophical puzzles about free will, determinism, and the relation between consciousness and mechanism. id., 227–240. The argument is taken up and deepened in Henri Bergson, *Creative Evolution*, translated by Arthur Mitchell, Originally published as *L'Évolution créatrice*, 1907 (New York: Dover, 1998), 273–332, where the 'cinematographic illusion' provides the methodological diagnosis of what happens when intellect attempts to reconstruct movement and change from a succession of static snapshots.

structurally, not through any failure of rigour — the question that the diagnostic framework places at the foundation of any adequate relational ontology: the question of what generates the irreversibility that duration exhibits, the question of what is before temporal flow, the question that can only be asked from a level more primitive than duration itself.

5.2 The Foundational Commitment: Duration as Primitive

Bergson's foundational commitment is executed in three movements that together constitute the philosophical architecture of *Time and Free Will*, the work in which the commitment is made for the first time and with the greatest economy.

The first movement is the critique of the received account. The tradition in philosophy and science, Bergson argues, has consistently misrepresented time by treating it as a fourth dimension of space — as something that can be divided into units, laid out as a line, measured by instruments, and submitted to the same mathematical operations that serve geometry and mechanics. This misrepresentation is not accidental. It arises from the practical demands of social life and scientific measurement, which require that time be made commensurable, comparable, and exact. To say that the train arrives at four o'clock is not to describe the duration of the journey; it is to correlate two events — the arrival and the position of clock hands — in a way that is useful for coordination. Clock time is a practical achievement of the highest order. Its philosophical elevation into the nature of time itself is an error of category.[2]

The second movement is the positive account of duration. Having cleared away the spatialised substitute, Bergson describes what remains: pure duration, *la durée pure* — the continuous, undivided, irreversible flow of conscious states in which past and present are not distinct points on a line but are interpenetrating phases of a single movement. Duration is not a succession of instants. It is a melody — and the analogy is structural, not decorative. In a melody, no note exists independently of the notes that preceded it and the notes that will follow; the character of the D is what it is because of the C that came before and the E that comes after, and to extract any note from its temporal context is to destroy it as the musical event it was. Duration has this character at the level of consciousness and, Bergson will argue, at the level of biological and cosmological reality: it is the form that succession takes when the past gen-

[2]Bergson, *Time and Free Will: An Essay on the Immediate Data of Consciousness*, 98–112. The argument is developed through an analysis of what Bergson calls 'multiplicity': the contrast between quantitative multiplicity (a collection of distinct, juxtaposed units that can be counted) and qualitative multiplicity (a continuous, flowing heterogeneity in which distinctions are differences of degree rather than of kind). Number belongs to quantitative multiplicity. Duration belongs to qualitative multiplicity. The spatialisation of time is the imposition of quantitative multiplicity on what is irreducibly qualitative. For a systematic analysis of Bergson's two multiplicities, see Gilles Deleuze, *Bergsonism*, translated by Hugh Tomlinson and Barbara Habberjam (New York: Zone Books, 1988), 37–51, which remains the most formally precise secondary treatment of this distinction.

uinely survives into the present rather than being replaced by it.[3]

The third movement is the generalisation of duration from consciousness to reality at large. This is the decisive step that transforms what might have been a contribution to the philosophy of mind into a contribution to fundamental ontology. In *Matter and Memory* and above all in *Creative Evolution*, Bergson argues that duration is not merely the form of conscious experience; it is the form of life, of matter, and ultimately of the cosmos as a whole — insofar as the cosmos can be genuinely said to evolve, to produce novelty, to be something more than the mechanical unfolding of what was already given at the outset. The universe endures. What it endures as — what its duration is — is a question that requires an account of memory, creativity, and the relation between life and matter that Bergson spends the remaining decades of his career pursuing.[4]

[3]Bergson, *Time and Free Will: An Essay on the Immediate Data of Consciousness*, 100–104. The melody analogy is one of Bergson's most precise technical instruments, not a concession to aesthetics. Its function is to exhibit the difference between a succession of discrete states — which can be analysed, decomposed, and reconstructed — and a genuine duration — which cannot be decomposed without ceasing to be what it was. Henri Bergson, *The Creative Mind: An Introduction to Metaphysics*, translated by Mabelle L. Andison, Originally published as *La Pensée et le mouvant*, 1934 (New York: Dover, 2007), 193–200 provides Bergson's most considered retrospective account of what the melody analogy is supposed to establish and where its limits lie. The analogy is analysed with exceptional rigour in Suzanne Guerlac, *Thinking in Time: An Introduction to Henri Bergson* (Ithaca: Cornell University Press, 2006), 22–36.

[4]The generalisation is made with increasing boldness: in Henri Bergson, *Matter and Memory*, translated by N. M. Paul and W. S. Palmer, Originally published as *Matière et mémoire*, 1896 (New York: Zone Books, 1991), 72–80,

What is foundationally significant for the diagnostic framework is not the content of these three movements — each of which is enormously instructive — but the structural commitment they jointly embody. Duration, in Bergson's account, is a primitive. It is not derived from anything more fundamental. It is not explained by reference to a ground from which temporal flow itself emerges. When Bergson asks what time is, his answer is duration; when he asks what duration is, his answer is: *this* — the irreversible, heterogeneous, continuous flow that is immediately given in conscious experience and that we misrepresent whenever we substitute for it the spatialised abstraction that measurement requires. Duration is the starting point, not the conclusion. It is not grounded; it is the ground.

This commitment is made not by oversight but by philosophical decision. Bergson is entirely aware that one might ask what generates temporal flow, what makes duration irreversible rather than reversible, what structural condition of reality is responsible for the fact that the past accumulates into the present rather than simply vanishing. His response — implicit in the structure of his arguments, explicit in his engagement

where the reality of pure memory establishes that the past exists as a dimension of being, not merely as a psychological residue; in Bergson, *Creative Evolution*, 7–45, where the opening chapters establish that evolution is a creative process irreducible to mechanism or finalism; and in the penultimate chapter of id., 298–340, where duration is explicitly identified as the very stuff of reality and the mechanical representation of nature is diagnosed as a practical instrument that conceals what nature most fundamentally is. Čapek (*Bergson and Modern Physics: A Reinterpretation and Re-evaluation*) provides the most technically informed account of Bergson's attempt to show that modern physics, correctly interpreted, is consistent with duration rather than opposed to it.

with Kant — is that these are questions that presuppose the spatialisation they should be diagnosing. To ask what *generates* duration is already to place duration inside a framework of causal explanation in which one thing produces another in time — and to place duration inside time in this way is to forget that time is duration, not a container for it.[5]

The diagnostic framework does not dispute the internal coherence of this move. Within the terms of Bergson's own inquiry, the refusal to ground duration in something prior to duration is a necessary feature of the inquiry itself, not an evasion. The diagnostic question is structural: the refusal to ask what generates duration is also the refusal to descend to the level of analysis at which that question can be posed, which is also the refusal to enter the structural levels — Levels I through IV — that the diagnostic framework identifies as constitutive of any formally adequate relational ontology. What Bergson's three-movement foundational commitment buys him at Level V it costs him at every level below it.

[5]Bergson, *Time and Free Will: An Essay on the Immediate Data of Consciousness*, 220–240. Bergson's engagement with Kant is focused on what he takes to be Kant's fundamental error: the treatment of time as a form of intuition that the subject imposes on experience, rather than as the very substance of experience itself. For Bergson, Kant spatialises time at the level of transcendental analysis in precisely the way that empiricist psychology spatialises it at the level of sensation. The most systematic account of the Bergson-Kant relation is in Leonard Lawlor, *The Challenge of Bergsonism: Phenomenology, Ontology, Ethics* (London: Continuum, 2003), 22–47, which also analyses how the refusal to ask what generates duration follows from Bergson's anti-Kantian commitment.

5.3 Entry Level : The Actuality Dimension of Level V

The five-level diagnostic framework established in Chapter 2 identifies Level V — the structure of the relational event — through four formal conditions (V_1)–(V_4) (Definition 2.5). The non-redundant completeness condition (V_2) is satisfied only by an account that derives the event's three co-instantaneous dimensions established in Proposition 2.1: initiation (the coming-into-specific-contact of relata), regulation (the sustaining of that contact against collapse or dissolution), and direction (the orientation of the encounter towards specific yield). The direction dimension has two formally distinct sub-conditions (Definition 2.6): the actuality dimension (V_α), which is the formally historical pattern of accumulating sedimentation that constitutes the structurally determinate *before* from which every relational event orients; and the possibility dimension (V_β), which is the formally admissible wider pattern of structural configurations towards which that actuality is oriented. A tradition that foundationally constitutes (V_α) enters Level V from within the actuality of direction, thereby achieving what (V_α) specifies while foreclosing the derivation of (V_β) — and, by extension, of the levels below Level V.

Bergson's foundational commitment to duration as primitive locates his entry within Level V at the condition that it is his distinctive contribution to illuminate and that the diagnostic framework identifies as (V_α): the actuality dimension of direction. The following proposition states this entry with precision.

Proposition 5.1 (Bergson's Entry Level)**.** Bergson's foundational commitment to duration as primitive locates his account within the non-latent register, at the actuality dimension of Level V: *The Structure of the Relational Event*. The formal conditions of Level V that Bergson achieves — V_α (the actuality dimension of direction as irreversible accumulation) and, in qualified form, the partial account of relational yield at Level IV — are entered from above rather than derived from below: Bergson reaches (V_α) by beginning within temporal flow, not by deriving temporal flow from a more primitive structural dynamic. The conditions prior to the non-latent register — the formal derivation of temporal asymmetry, the derivation of a pre-temporal structural ground, the structural individuation of relata, and the formal specification of the source of encounter — are structurally foreclosed by the foundational commitment to duration as the starting point of analysis.

Proof. By the diagnostic method of Chapter 2, the entry level is the level at which the foundational commitment takes a structural feature as primitive. Bergson's foundational commitment is to duration as the most primitive structural feature of reality: duration is not derived from any more fundamental condition but is given as the irreversible, heterogeneous flow of qualitative multiplicity that constitutes time as it is actually lived. This commitment locates the entry point at the actuality dimension (V_α) of Level V, because (V_α) is precisely the formally historical pattern of irreversible accumulation — the structural feature that duration is. To take duration as primitive is to take the actuality of direction as primitive, which is to enter the structural order at the actuality dimension of Level V rather than deriving that dimension from a pre-temporal structural dynamic. That the entry is at the actuality dimension specifically — and not at the full Level V account, which would require formal specification of the possibility dimension (V_β) as well — follows from the character of duration as Bergson specifies it: duration captures irreversible accumulation (the formally historical pattern of sedimentation) but not the formally admissible space of possible further sedimentation events, which requires a level of structural analysis unavailable from within temporal flow.

Remark 5.1. The claim that Bergson enters at the actuality dimension of Level V requires careful distinction from two superficially similar claims that the diagnostic framework does not make. First, it is not the claim that Bergson offers an account of the relational event as such: his framework is not organised around the concept of encounter between structurally distinct relata, and the vocabulary of initiation, regulation, and direction does not appear in it. What Bergson achieves at (V_α) is not the full account of the relational event but rather the account of that dimension of the relational event — the actuality of direction — that is most directly exhibited in his account of duration and memory. Second, it is not the claim that Bergson's failure at the remaining conditions of Level V constitutes an error within his framework. The diagnostic claim is structural: Bergson achieves what he achieves at (V_α) precisely because he begins where he begins, and what he begins by achieving forecloses the formal derivation of what lies beneath the level of his beginning. The relation between Bergson's achievement and his limit is not one of correction and correction-in-principle but of structural consequence.

5.4 Formal Achievements at the Actuality Dimension

Three formal achievements characterise Bergson's account at and in the vicinity of the actuality dimension of Level V. Each constitutes a genuine contribution to the diagnostic project, not because it satisfies the structural requirements of the five-level framework — it satisfies one of those requirements partially, and only from above — but because it identifies with precision a structural feature that any adequate account must incorporate and that the traditions surveyed before Bergson had not seen with comparable clarity.

5.4.1 Duration as the Structure of Accumulating Sedimentation

The diagnostic framework, in its account of the actuality dimension of direction, defines (V_α) as the formally historical pattern of accumulating sedimentation that constitutes the structurally determinate *before* from which every relational event orients. The *before* is not merely the set of prior states; it is the way in which those prior states are structurally present in the current state, shaping its weight distribution and thereby constraining the range of structural outcomes the current event can produce. The actuality of direction is the density of the past in the present — not as stored representation but as living structural weight.

Bergson's duration is this structural feature articulated from within the non-latent register with extraordinary precision. Duration, for Bergson, is emphatically not a succession of instants in which the present replaces the past; it is an accumulation in which the past genuinely survives in the present as the very substance of the present's qualitative character. The past does not disappear when it is past. It contracts — it is 'rolled up', in Bergson's image — into the present, and the present is what it is partly by virtue of the specific character of what has been rolled up into it.[6]

This is the structural claim the diagnostic framework identifies as (V_α) in its non-latent expression: the present is constituted by the accumulation of what has been, and that accumulation is irreversible and directional, not arbitrary or symmetric. The past is structurally active in the present not as cause in the mechanistic sense but as the very fabric from which the present is made. Bergson reaches this result with a precision and a phenomenological concreteness that the diagnostic frame-

[6]The image of memory rolling up the past into the present is central to Bergson (*Matter and Memory*); see especially the opening pages of Chapter II, id., 133–150, where Bergson argues that pure memory is not a weakened perception or a trace left in the brain but the survival of the past as past — as something that exists in its own dimension of being and contracts into the present to constitute the qualitative richness of experience. Deleuze, *Bergsonism*, 55–72 provides the most careful analysis of what 'survival' means here and why it requires an ontology of the past as a distinct dimension of existence.

work's more abstract formulation is designed to preserve, not replace.

The domain illustration that makes this structural claim tangible is geology. A cliff face is not a surface at which a particular geological moment is displayed; it is a cross-section through which the entire history of the Earth's surface at that location is made visible at once. The limestone stratum does not merely lie above the shale stratum; the limestone *is* what that locality became when the conditions that made shale gave way to the conditions that make limestone — the earlier stratum is structurally present in the later one, not as a remembered cause but as a constitutive layer without which the later stratum would not be what it is or where it is. The cliff face is duration made spatial — which is, in Bergson's terms, precisely the spatialisation that distorts it, but which is also a concrete image of what the non-spatial accumulation of duration achieves. The geological record is a sedimentation in the sense the diagnostic framework specifies: irreversible, accumulative, formally historical, each layer carrying the structural weight of what preceded it into the conditions of what follows.

5.4.2 *Pure Memory as the Formal Historicity of Structural Loci*

The diagnostic framework's account of sedimentation establishes that every locus in the structural field is formally historical: what each locus is at any moment is determined not only by the current weight distribution at that locus but by the full irreversible sequence of sedimentation events that have deposited their structural weight there. There is no blank-slate locus; every locus carries the trace of its history as a formally constitutive feature of its current structural profile. This is not a claim about information storage; it is a claim about ontological constitution.

Bergson's distinction between habit-memory (*mémoire habitude*) and pure memory (*souvenir pur*) is the most precise phenomenological articulation of this ontological claim available in the philosophical literature.[7] Habit-memory is the incorporation of repeated past actions into present dispositions — it is memory that has been fully absorbed into present functioning and has ceased to exist as a distinct past. Pure memory is something structurally different and philosophically more radical: the survival of the past as past, the existence of what has happened as a specific, singular, irreversibly dated event that is present not as a disposition but as a dimension of being distinct from the present moment yet inseparable from it.

What Bergson is articulating under the heading of pure memory is the ontological priority of the past as a structural dimension of reality — the claim that the past is not nothing, that it does not cease to exist when it is past, that it survives in a mode of existence that is genuinely distinct from the present

[7]Bergson, *Matter and Memory*, 78–105. Habit-memory is the form of memory that operates through repetition and that eventually becomes automatic — the body remembers how to ride a bicycle, not by retrieving a stored representation of past bicycle rides, but by having incorporated the pattern of cycling into its motor dispositions. Pure memory, by contrast, is the survival of a specific past event as such: not a disposition produced by repetition but the existence of the past event as a singular, dated, irreducibly specific happening that survives in a different dimension of existence from the present. id., 95–102.

even as it is structurally constitutive of the present. This is not idealism about the past; it is realism about duration. The past exists — not as a present representation of something no longer present, but as the very stuff from which the present is made.[8]

The correspondence with the diagnostic framework's account of formal historicity is precise: every locus in the structural field is formally historical in the sense that its current weight distribution carries the irreversible trace of every sedimentation event that has occurred there. This is not the weight of remembered past events in the psychological sense; it is the structural weight of a history that has been deposited and cannot be undone. Bergson's pure memory, translated into the diagnostic framework's structural vocabulary, is the claim that this historical weight is not merely causal but ontologically constitutive — that what a locus is cannot be fully specified by its current structural profile abstracted from the history that produced it. Formal historicity and pure memory are the same structural fact stated in two different registers.

[8]id., 150–168, where Bergson develops the argument that pure memory must exist in a mode of being that differs from both perception and habit, since it is neither a weakened present perception nor an automatic disposition but the survival of the singular as singular. The argument is contested: Lawlor, *The Challenge of Bergsonism: Phenomenology, Ontology, Ethics*, 85–95 provides the most careful analysis of what is and is not established by Bergson's account of pure memory, including the question of whether the claim that the past 'exists' commits Bergson to a metaphysics of temporal parts that is incompatible with other elements of his account. For the present diagnostic purposes, what matters is the structural claim about formal historicity, which survives these debates.

5.4.3 Creative Evolution as Partial Account of Relational Yield

The diagnostic framework's account of Level IV identifies relational yield as formally new, formally bounded, and formally generative: it is not a rearrangement of what the relata already contained, it is constrained by the conservation of the structural field's total weight, and it is capable of becoming a new locus in the structural field capable of further encounter. These three conditions jointly specify what it means for relational encounter to produce something genuinely novel rather than merely recombinant.

Bergson's account of creative evolution is a partial account of relational yield in this sense, and it is the most influential attempt in the philosophical literature to establish that biological evolution produces genuine novelty irreducible to its prior conditions.[9] The encounter between the vital impulse and the resistance of matter produces forms that neither the impulse nor the matter contained prior to the encounter, and that cannot be accounted for as the result of any law acting on pre-existing

[9]Bergson, *Creative Evolution*, 1–45. Bergson's critique of both mechanism and finalism is directed at this point: mechanism holds that the result of evolution is explicable as the rearrangement of pre-existing elements, while finalism holds that the result is the realisation of a pre-existing programme. Both, Bergson argues, deny genuine novelty: mechanism by holding that the result was already given in the initial conditions, finalism by holding that the result was already given in the plan. Creative evolution denies both: the results of life's creative impulse could not have been predicted from any prior specification of conditions or intentions, because they are genuinely new — not new in the trivial sense of not having existed before, but new in the sense of adding something to what the universe was that no prior state of the universe already contained. id., 87–115.

elements. Life, in Bergson's account, is real novelty appearing in time — which is another way of saying that the encounter between the generative force of life and the inertial force of matter produces a yield that is irreducible to its relata.

The diagnosis of this achievement is precise. Bergson is right that biological evolution produces genuine novelty, and his arguments against mechanism and finalism are among the most penetrating analyses of why the concept of genuine novelty resists both determinist and teleological reduction. What Bergson does not provide is a formal account of *why* the encounter between life and matter produces novelty rather than mere rearrangement — a formal account of the structural conditions that make novelty possible, that determine the formal character of the yield, and that explain why novelty is bounded (why not anything can emerge from any encounter) as well as genuine (why what emerges is not already contained in what entered the encounter). The diagnostic framework's formal conditions for Level IV specify precisely this missing account: yield is novel because the encounter between an invariant structural identity and a variant structural source produces a result that is formally irreducible to either; yield is bounded because the total structural weight of the field is conserved across every sedimentation event; yield is generative because the new structural deposit itself becomes a site capable of further encounter. Bergson's creative evolution identifies the phenomenon — genuine novelty — but does not provide the structural derivation that would explain why the phenomenon has the formal character it has.[10]

5.5 What Bergson Cannot Reach

Bergson's formal achievements at the actuality dimension of Level V and in the partial account of Level IV are genuine and irreplaceable. They are also, the diagnostic framework establishes, the precise point of entry at which three structural limits become permanent features of his account rather than correctable gaps. These limits are not failures of rigour or imagination; they are structural consequences of beginning where Bergson begins, and they cannot be overcome by modifications internal to his framework without abandoning the foundational commitment that makes his achievements possible.

5.5.1 The Possibility Dimension of Direction

The actuality dimension of direction (V_α) is one of two formally necessary dimensions of the relational event's directional struc-

[10]This diagnosis is consistent with — though arrived at by a different route than — Keith Ansell-Pearson, *Philosophy and the Adventure of the Virtual: Bergson and the Time of Life* (London: Routledge, 2002), 93–104, which argues that Bergson's account of the *élan vital* systematically oscillates between a transcendent vitalism (in which the vital impulse is an irreducible force from outside matter) and an immanent account (in which novelty emerges from the structural interaction of life and matter without any external source). The diagnostic framework identifies this oscillation as a symptom of the absence of a formal account of the structural conditions that make novelty possible: the *élan vital* is the name for what a formal account would need to derive, not the formal account itself.

ture. The possibility dimension (V_β) is the other: the formally admissible wider pattern of structural configurations towards which the actuality of the present is oriented, the range of formally possible sedimentation outcomes that the current weight distribution makes available. The two dimensions are formally inseparable: actuality without possibility is a structural deposit with no orientation, no forward-directedness, no capacity to generate anything; the actuality of the past in the present is formally directional because there is a possibility space towards which it is oriented, and the possibility space is formally structured because there is a specific actuality that has carved a specific path through it.

Bergson has the actuality dimension with great precision. He does not have the possibility dimension, and the name he gives to what substitutes for it — the *élan vital*, the creative impulse, the thrust of life towards novelty — reveals the structural gap with precision. The *élan vital* is Bergson's attempt to account for the forward orientation of life: why creative evolution moves towards the production of new forms rather than oscillating without direction or returning to prior states. But the *élan vital* is formally underdetermined in a specific sense: it names the forward orientation without formally specifying what constitutes the admissible space of structural possibilities towards which life is oriented — what makes some possible evolutionary outcomes formally available to the vital impulse and others not.[11]

[11]Bergson, *Creative Evolution*, 87–97, where the *élan vital* is introduced as the explanation of the divergent lines of evolutionary development and

The diagnostic framework specifies that the possibility dimension of direction is constituted by the formally admissible space of weight distributions that blind anticipation makes available at any locus beyond what the current sedimentation has actualised. This is not arbitrary openness; it is a formally structured space shaped by the weight distribution that sedimentation has produced, from which certain outcomes are formally nearer and others formally further. Bergson reaches towards this structural feature with the *élan vital* but cannot formally specify it, because formal specification requires a level of structural analysis — an account of the weight distribution, the majority-minority dynamic, the global regulatory principles that hold the structural field in productive tension — that is available only from Level I upward, and Bergson's entry at (V_α) has placed everything below Level V outside the reach of his inquiry.

5.5.2 The Derivation of Temporal Asymmetry

Duration is, for Bergson, irreversible by definition: it is of the essence of duration that it does not run backward, that the past cannot be undone, that temporal flow accumulates in one direc-

the creative character of life's encounter with matter. Bergson is explicit that the *élan vital* is not a pre-existing programme: it does not determine in advance which specific forms will emerge from the evolutionary process. But what he does not provide is a formal account of the structural conditions that delimit the space of possible forms — why the vital impulse, in its encounter with matter, produces the specific kinds of divergent outcomes it produces rather than others. Ansell-Pearson, *Philosophy and the Adventure of the Virtual: Bergson and the Time of Life*, 113–130 traces the consequences of this indeterminacy for Bergson's account of evolution at length.

tion only. This irreversibility is not merely a contingent feature of the particular durations we happen to experience; it is constitutive of what duration is. A duration that could run backward would not be duration but its spatialised simulacrum — for only a line can be traversed in either direction, while a genuine flowing heterogeneity can only accumulate, only be what it is by having been what it was.[12]

The diagnostic framework does not dispute the irreversibility of duration; it asks what generates that irreversibility. Why does temporal flow accumulate in one direction rather than two? What structural fact about reality is responsible for the asymmetry between past and future that duration exhibits? Bergson's answer — that irreversibility is of the essence of duration, that to ask what generates it is already to misunderstand what duration is — is, within the terms of his own inquiry, precisely correct. But it is also the precise point at which the diagnostic framework identifies the structural limit: temporal asymmetry is treated as a primitive when the diagnostic framework requires it to be a derived structural result.

[12] Bergson, *Time and Free Will: An Essay on the Immediate Data of Consciousness*, 109–120. Bergson makes the irreversibility argument most explicitly in the context of consciousness, but generalises it to life and matter in Bergson, *Creative Evolution*, 8–13: the universe 'cannot go back', and the irreversibility of temporal flow is the most fundamental fact about it. The argument is deepened in Bergson (*Duration and Simultaneity*), where Bergson responds to Einstein's special relativity by arguing that the multiplicity of times in relativity theory does not undermine the uniqueness and irreversibility of real duration, because the relativistic multiple times are mathematical constructions rather than lived realities.

The derivation of temporal asymmetry that the diagnostic framework specifies proceeds from the structural properties of the direction dimension of the relational event as a whole: the irreversibility of sedimentation — the fact that structural deposits cannot be undone — combined with the global coupling of the structural field under the conservation law — which ensures that no sedimentation event is isolated from the total weight distribution — produces the asymmetry between before and after not as a primitive feature of reality but as a structural consequence of what sedimentation is. Before and after are the two formally distinct dimensions of the actuality-possibility pairwise that every direction event carries: the before is the sedimented actuality, the after is the formally admissible possibility. Their asymmetry is derivable from the structural character of the sedimentation events that produce the actuality dimension and from the formal properties of the possibility space that actuality's structure makes available.

Bergson cannot perform this derivation because it requires beginning below Level V — beginning with the structural conditions of sedimentation itself, with the account of what makes deposits irreversible and what formal properties characterise the possibility space their irreversibility opens. These are questions available only to an inquiry that begins at Level I and builds upward. Bergson begins at (V_α) and finds duration already irreversible, already accumulating, already directional. He cannot ask what makes it so without stepping outside the terms of his own inquiry, and stepping outside those terms would require abandoning the foundational commitment that makes his account of duration possible in the first place.

5.5.3 *Levels I Through IV: The Structural Consequence of Beginning from Flow*

The diagnostic framework's five levels specify four structural requirements prior to the relational event: the derivation of an absolute ground (Level I), the derivation of structural identity for relata prior to encounter (Level II), the derivation of the source of encounter from the same dynamic that produces the ground and the relata (Level III), and the derivation of relational yield as formally new, bounded, and generative (Level IV). Bergson's entry at the actuality dimension of Level V forecloses three of these four requirements entirely — Levels I, II, and III — and forecloses the formal conditions of Level IV that require structural derivation from those prior levels.

Level I: The Ground. Bergson's nearest equivalent to the structural ground is duration itself — the continuous, irreversible temporal flow that is the most fundamental feature of reality in his account. But duration cannot serve as the ground that the diagnostic framework requires, for a precise structural reason: the ground, as specified at Level I, must be capable of deriving the pre-temporal structural levels from which temporal reality itself emerges. A ground that is already temporal — that is itself constituted by the very flow whose derivation the ground is supposed to supply — cannot perform this derivation without circularity. Duration as ground accounts for everything that occurs within temporal flow; it cannot account for what generates temporal flow itself, because it is itself temporal flow. The totality condition (G_1) is approximately satisfied: Bergson's duration is meant to encompass all of reality. But the generativity condition (G_2) fails: the ground cannot generate the structural differentiation from which distinct relata emerge through an internal dynamic that is itself non-temporal, because the ground is constituted by temporal flow and has no internal dynamic prior to that flow.[13]

A Bergsonian may object that any pre-temporal ground generating temporal asymmetry either already contains directionality — making it covertly temporal — or generates something other than duration as Bergson understands it. The objection is well-formed but rests on a conflation between two structurally distinct proposals. What Bergson correctly targets as spatialisation is the treatment of temporal flow as a homogeneous medium of externally related moments — the reduction of duration to spatial co-presence.[14] The latent register does not

[13]This is not the claim that Bergson denies the generativity of duration — on the contrary, creative evolution is precisely his account of how duration generates genuinely novel forms. The diagnostic point is structural: the generativity that the diagnostic framework specifies at Level I is a pre-temporal structural dynamic from which temporal flow itself is derived, while the generativity Bergson attributes to duration is a temporal generativity — novelty produced within temporal flow, not the production of temporal flow from something more primitive. The two are structurally distinct, and Bergson's temporal generativity, however profound, does not supply what Level I requires.

[14]See Luka Vinck, "Revisiting the Einstein–Bergson Debate" (Preprint. Archived at PhilArchive, rec/VINRTE. Earlier version published as master's thesis, University of Amsterdam, 2024. Establishes from within Bergson scholarship that the spatialisation objection targets the homogenisation of temporal flow as a quantitative multiplicity, not any form of pre-temporal

propose this. It proposes a structural dynamic with differential weighting — what Olenius (2025) formalises as an asymmetric constraint structure derivable from non-temporal primitives of motion, difference, wake, and constraint — from which ordinal asymmetry emerges without temporal priority being presupposed.[15] The asymmetry is not imported from time; it is the pre-temporal structural condition from which temporal ordering is itself a downstream representational consequence.[16] A pre-temporal structural dynamic generating temporal asymmetry is therefore neither covertly temporal nor irrelevant to Bergson: it is precisely the kind of structural ground that Bergson's

structural differentiation — confirming that the diagnostic framework's requirement for a pre-temporal latent register is distinct from what Bergson diagnoses as spatialisation, 2025).

[15]R. A. E. Olenius, "Causality Without Time" (Preprint. Archived at PhilArchive, rec/OLECWT (archived 2025-12-04). Reconstructs causal direction from four non-temporal primitives — motion, difference, wake/trace, constraint — establishing via the Asymmetric Constraint Thesis that directional dependence obtains without temporal priority. Demonstrates that temporal ordering is a representational artifact of systems capable of internalising asymmetry, not a structural primitive. Supports the claim that a pre-temporal structural dynamic generates asymmetry without being covertly temporal; also used to illustrate the distinction between eliminative derivation and conceptual analysis, 2025).

[16]See also A. Mikkel, "Generative Closure and the Emergence of Temporal Order in Relativistic Physics" (Preprint. Archived at PhilArchive, rec/MIKGCA-3. Develops a mechanism — generative closure as the directed completion of open potentiality into realised fact — by which ordinal structure arises in a universe lacking fundamental time. Engages Mozota Frauca (2025, *Foundations of Physics* 55:37) and Ellis (2024, *Foundations of Physics* 54:6) on ordinal realism. Used as physical-structural corroboration of the coherence of pre-temporal dynamics generating temporal asymmetry, 2025).

own account, which begins from within temporal flow, cannot provide without circularity.

Level II: Relational Identity. Bergson has no formal account of what makes any entity the structurally specific entity it is prior to its temporal constitution. Living beings, in Bergson's framework, are constituted through their duration — their identity is inseparable from the temporal flow that constitutes them, and to abstract from that temporal flow is to lose precisely what makes them what they are. This is a deep insight: the identity of a living being is not a static essence that temporal change merely modifies, but a dynamic process that is constituted through its temporal becoming. The diagnostic framework does not contest this. Its diagnostic claim is structural: the account of how a specific living being becomes what it is presupposes that there is a specific being in the process of becoming, and the structural individuation of that specific being — the account of what makes it this particular locus in the structural field rather than another or none — requires a formal specification that Bergson does not provide. Level II asks: what formally individuates a relatum prior to its temporal becoming? Bergson's answer, that relata are constituted through temporal becoming, defers rather than answers this question.[17]

[17]Bergson, *Matter and Memory*, 78–90. The tension between Bergson's account of individuation-through-duration and the requirement for a formal account of structural individuation prior to duration is the most productive point of contact between Bergson and the relational ontologies of the late twentieth century; see Ansell-Pearson, *Philosophy and the Adventure of the Virtual: Bergson and the Time of Life*, 53–79 for an analysis of how this tension generates Bergson's influence on Simondon's account of individuation.

Level III: The Source of Encounter. The *élan vital* is Bergson's account of what drives living beings into productive encounter with their environments and with each other. It is, as noted above, the nearest equivalent in his framework to the structural source of encounter that Level III specifies. But the diagnostic critique established above applies here with equal force: the *élan vital* is posited as an irreducible fact of life, not derived from a more fundamental structural dynamic. Level III requires the derivation of the source of encounter from the same dynamic that produces the ground and the structurally individual relata; the *élan vital* is not derived from anything — it is, in Bergson's account, simply what life is.[18]

Level IV: Relational Yield. The partial account of Level IV in Bergson's creative evolution has been acknowledged above. The diagnostic limit is equally precise: what Bergson's account lacks is not the claim that genuine novelty exists — it is unambiguous about that — but the formal specification of why novelty is bounded. Why does not any encounter between life and matter produce any form whatsoever? What structural conditions delimit the space of possible novel forms, such that some are formally available to the encounter and others are

[18]The most systematic attempt to provide a structural account of the *élan vital* that would make it derivable rather than primitive is in Deleuze (*Bergsonism*), especially the final chapter on memory and virtual coexistence; Deleuze argues that the *élan vital* should be understood not as a force but as the ontological principle of differentiation itself. This Deleuzian reformulation is philosophically powerful, but it achieves derivability only by transforming the *élan vital* into a different concept — one that departs significantly from Bergson's own account in the direction of a more explicitly structuralist ontology.

not? Without this specification, novelty in Bergson's account remains genuinely novel but formally unconstrained — which means it cannot be formally distinguished from arbitrariness. The conservation condition that the diagnostic framework specifies at Level IV — relational yield is bounded because the total structural weight of the field is conserved — is the formal principle that Bergson's account requires but cannot supply, because it requires a level of structural analysis prior to temporal flow that Bergson's entry at (V_α) has placed outside his reach.

5.6 The Diagnostic Result

The diagnostic framework, applied to Bergson's philosophy of duration with the precision the foregoing sections establish, yields the following formal result.

Theorem 5.1 (Structural Foreclosure: Bergson)**.** Let $\mathfrak{B}$ denote the philosophical framework constituted by Bergson's foundational commitment to duration as primitive. Then:

(i) $\mathfrak{B}$ achieves genuine formal results at the actuality dimension of Level V — condition (V_α): the formally historical pattern of irreversible accumulating duration as the structurally determinate *before* from which every temporal orientation proceeds — and achieves partial formal results at Level IV in the account of creative evolution as genuine relational yield.

(ii) The foundational commitment that constitutes $\mathfrak{B}$ — the treatment of duration as the most primitive structural fea-

ture of reality, not derivable from anything prior to itself — formally constitutes the actuality dimension of Level V as the ground of the inquiry rather than as a derived structural result within a more comprehensive derivation.

(iii) The act of constituting (V_α) as the ground formally forecloses the derivation of: the possibility dimension of direction (V_β), which requires a formal specification of the admissible structural possibility space that is available only from Level I; the derivation of temporal asymmetry as a structural consequence of sedimentation's irreversibility; and Levels I, II, III, and the formal conditions of Level IV — all of which require structural analysis of the pre-temporal register from which temporal flow itself emerges.

(iv) The foreclosure is necessary: no modification internal to $\mathfrak{B}$ can supply the formally adequate account of these structural levels without abandoning the foundational commitment to duration as primitive, which is the commitment that constitutes $\mathfrak{B}$ as the philosophical framework it is.

Proof. Each of the four parts follows from the same structural source: the location of the entry point at (V_α) — the actuality dimension of the relational event's directional structure — rather than at a level structurally prior to it.

(i) The achievements at (V_α) and the partial account of Level IV are established by the diagnostic analysis of the preceding sections: $\mathfrak{B}$'s vocabulary is adequate to the formal structure of irreversible accumulation and to the genuine novelty of creative evolution because $\mathfrak{B}$ was built

from within temporal flow, and the formal content of temporal flow at the actuality dimension is exactly what $\mathfrak{B}$'s derivational resources were constituted to characterise.

(ii) That the foundational commitment constitutes (V_α) as the structural ground rather than a derived result follows from the character of the commitment itself. Duration is taken as primitive in the sense of Definition 10.4: it is not derived within $\mathfrak{B}$ from conditions derivable independently of duration, and $\mathfrak{B}$'s entire derivational direction runs from duration outward to its temporal consequences, not from any pre-temporal condition inward to duration. The actuality dimension of direction — irreversible accumulation — is what duration formally is; to take duration as primitive is therefore to take (V_α) as the structural terminus of explanation.

(iii) The four foreclosures each follow from the Horizon Lemma (Lemma 10.1) applied to the entry point established in (ii). $\mathfrak{B}$'s derivational resources are calibrated to the domain of temporal flow at and above (V_α): the vocabulary of duration, memory, and creative evolution is adequate to features constituted by or within temporal flow, and indeterminate with respect to pre-temporal structural conditions. The foreclosure of (V_β) follows because the formally admissible possibility space requires a structural account of global weight distribution and regulatory principles of the structural field that are available only from a level prior to any particular temporal actualisation; $\mathfrak{B}$'s entry at (V_α) places this account outside its derivational

horizon, and the *élan vital* names the orientation without formally specifying the space. The foreclosure of the derivation of temporal asymmetry follows because irreversibility is the foundational commitment rather than a derived structural result: $\mathfrak{B}$ finds duration already asymmetric and cannot ask what generates that asymmetry without stepping below the level at which its derivational resources are calibrated. The foreclosure of Levels I, II, and III follows because each requires the derivation of structural conditions prior to temporal flow: Level I requires a ground whose generative dynamic is not already constituted by flow; Level II requires the structural individuation of relata prior to their temporal constitution; Level III requires the source of encounter to be derived from a dynamic that is not itself the temporal impulse it is supposed to explain. The foreclosure of the remaining formal conditions of Level IV follows from the same source: the boundedness condition (Y_2) requires a conservation principle specifiable only from Level I, and the derivability condition (Y_1) requires that novelty be derived from structural conditions available only below the entry point.

(iv) The necessity of the foreclosure follows directly from clause (ii) and the Horizon Lemma. The calibration of $\mathfrak{B}$'s derivational resources to (V_α) is not a contingent feature of how Bergson happened to develop his philosophy but a structural consequence of constituting (V_α) as the terminus of explanation. Any modification that would supply the missing derivations — that would derive tem-

poral asymmetry, formally specify the possibility space, or derive the structural conditions of Levels I through III — must begin from a structural condition prior to temporal flow. To begin from such a condition is to abandon the foundational commitment to duration as primitive, which is the commitment that constitutes $\mathfrak{B}$ as the philosophical framework it is. The foreclosure is therefore necessary relative to $\mathfrak{B}$'s constitutive commitment; it cannot be overcome by any modification internal to $\mathfrak{B}$.

Remark 5.2. The structural complementarity between the traditions surveyed so far is exact and warrants explicit statement at this point. Whitehead enters at Level V and achieves formal results at Levels IV and V, but cannot reach below Level V towards the pre-temporal latent register. Spinoza enters at Level I and achieves formal results there, but cannot generate the structural differentiation that Levels II through V require. Bergson enters at the actuality dimension of Level V and achieves formal results there and partially at Level IV, but cannot reach the possibility dimension of Level V or the pre-temporal latent register below Level IV. The three traditions together map the non-latent register with extraordinary thoroughness: Whitehead from the top down, Bergson from the direction of temporal accumulation, Spinoza from the ground up. Each illuminates what the others cannot see from where they stand. What none of them illuminates — what all three together leave in structural darkness — is precisely what the diagnostic framework identifies as the foundational requirement: the derivation of temporal flow, relational identity, the source of encounter, and the ground itself from a structural dynamic prior to and generative of the non-latent register in which all three traditions begin their inquiries.

Criterion 5.1 (Bergson's Diagnostic Profile). *Entry level*: The actuality dimension of Level V, (V_α). *Formal achievements at and above entry level*: (V_α) — duration as the irreversible accumulation of the past in the present, mapping to the formally historical pattern of sedimentation that constitutes the *before* of every direction event; formal historicity of all temporal loci, mapping to the structural claim of pure memory that the past exists as a constitutive dimension of the present; partial Level IV in creative evolution as an account of relational yield as genuinely novel and irreducible to prior relata. *Structural foreclosure*: (V_β) — the possibility dimension of direction is not formally specified; temporal asymmetry — the derivation of the irreversibility of before and after is not performed, because irreversibility is the foundational commitment rather than a derived result; Level I — duration cannot serve as the pre-temporal ground since it is constitutively temporal; Level II — the structural individuation of relata prior to their temporal constitution is not provided; Level III — the source of encounter (*élan vital*) is posited rather than derived; Level IV formal conditions — the boundedness condition of relational yield is not formally specified. *Character of the foreclosure*: Necessary. The foreclosure is the structural consequence of beginning within the non-latent register rather than deriving it, and cannot be overcome by any modification internal to the framework that preserves the foundational commitment to duration as primitive.

The diagnostic result does not diminish what Bergson achieved. The account of duration in *Time and Free Will*, the account of pure memory in *Matter and Memory*, and the account of creative evolution in the great book of that name constitute the most sustained and precise philosophical engagement with the reality of time that the twentieth century produced. Bergson gave back to philosophy something that the dominant traditions of both the continent and the analytic world had lost: the recognition that time is not a formal structure to be analysed but a reality to be inhabited, that the irreversibility of temporal flow is not a puzzle to be explained away but a fundamental feature of what is real, and that the production of genuine novelty is not an illusion to be dissolved by determinism but an ontological achievement that demands a philosophical account adequate to its reality. These are contributions that no subsequent philosophical tradition has fully absorbed, and the diagnostic framework's identification of Bergson's structural limits is in no way a diminishment of them.

What Bergson showed us is that there is something real in temporal flow that every attempt to spatialise it distorts: an irreversible, accumulative, living heterogeneity that is the very substance of what we are and of what, at least in its living and conscious dimensions, the universe is. He showed it with a philosophical precision that matched the precision of the scientific traditions he was engaging. What he could not show — what his entry at the actuality dimension of the relational event's direction structurally prevented him from showing — is what temporal flow itself emerges from: the pre-temporal structural dynamic that makes duration possible, that generates the asymmetry duration exhibits, and that provides the formal account of the structural individuation of relata, the source of encounter, and the bounded character of relational yield that creative evolution identifies but cannot formally derive. Bergson began within the living present. The question of what makes the living present possible — of what is before the flow he found us already inside — was not his question. It remains open.

Remark 5.3 (On the Deleuzian Virtual Reading of Bergson). Deleuze's reading of Bergson in *Bergsonism*[19] and *Difference and Repetition*[20] introduces the concept of the *virtual* as the ontological complement of the actual: the past-as-a-whole co-existing with the present, a structural reserve from which any actual moment draws, not itself a past event but the total structural dimension of temporal depth. On Deleuze's reading, the virtual is not temporal in the ordinary sense — it is not a prior moment in the sequence but a structural domain that accompanies every actual moment as its ontological condition. A Bergson specialist familiar with this reading might argue that the virtual constitutes a structural domain prior to any given temporal moment, which looks structurally analogous to what the present investigation calls the pre-temporal latent register. If so, Bergson's entry-level assignment — the actuality dimension of Level V, (V_α) — might need revision: the Deleuzian virtual would represent a structural domain

[19]Id.

[20]Gilles Deleuze, *Difference and Repetition*, translated by Paul Patton, Originally published as *Différence et répétition*, Presses Universitaires de France, 1968 (New York: Columbia University Press, 1994).

below (V_α), accessible within the Bergsonian framework via Deleuze's reconfiguration.

The diagnostic response is precise and follows directly from the three conditions of Definition 10.4. Deleuze's virtual, as deployed in the Bergsonian reading, is foundationally constituted as a positive structural domain: it is characterised as the *duration* of the past as a whole, as "pure virtuality" that is "the being of the past in general."[21] The virtual is not derived from something more primitive than itself; it is Deleuze's reformulation of *what duration is at its deepest level*. Condition (ii) of Definition 10.4 identifies this as a foundational constitution: the virtual is constituted as a structural type (the ontological complement of the actual, the total past co-existing with every present) whose own ground is not derived from anything more primitive within either Bergson's or Deleuze's framework. The virtual does not descend below the temporal register; it is the *most fundamental* dimension of the temporal register — the structural depth of duration itself. It is a feature *within* the domain of temporal constitution, not a structural domain from which temporal constitution emerges. This is precisely why Rae[22] confirms, from within Deleuze studies, that Deleuze's differential ontology constitutes intensive difference as primary rather than derived at key junctures: the differential, including the virtual, is foundationally constituted. The Deleuzian reading therefore does not alter the Level V_α entry-level assignment of Bergson; it deepens the characterisation of what is constituted at that level, without

[21]Deleuze, *Bergsonism*, 56–59.
[22]Rae, "Three Senses of Identity and Deleuze's Differential Ontology."

supplying derivational access to anything below it. (Deleuze is surveyed independently in the supplementary chapter, where the same structural result is reached for his own framework from the direction of intensive difference.)

> *Summary of Chapter 5.* *Bergson's philosophy of duration enters the structural order at the actuality dimension of Level V: the formally historical pattern of irreversible accumulating sedimentation that constitutes the structurally determinate before from which every relational event orients. Genuine formal achievements obtain at this entry level: duration as irreversible accumulation maps precisely to the actuality dimension of direction; pure memory as the structural survival of the past in the present maps to the formal historicity of every locus in the structural field; and creative evolution provides a partial account of Level IV relational yield as genuinely novel and irreducible to its relata. The foundational commitment to duration as primitive forecloses what lies below the entry level: the possibility dimension of direction (V_β) is not formally specified, because its specification requires the structural analysis of the admissible possibility space available only from Level I upward; temporal asymmetry is not derived but assumed, because derivation requires analysis of the pre-temporal conditions of sedimentation's irreversibility; and Levels I through IV are structurally inaccessible, because the structural analysis they require is prior to the temporal register in which Bergson's inquiry begins and ends. The structural complementarity with Whitehead and Spinoza is exact: together the three traditions map the non-latent register with extraordinary precision from three different directions, while leaving the latent register — the structural ground prior to temporal flow, from which temporal flow, relational identity, the source of encounter, and relational yield are*

all to be derived — entirely unaddressed. Chapter 6 turns to the tradition that takes as its entry point not the event, the ground, or the temporal flow that connects them, but the formal structure of the possibility that temporal flow orients towards — and finds that beginning from the structure of possibility forecloses the derivation of what makes possibility formally possible.

CHAPTER 6

Heidegger: The Structure of Possibility

Higher than actuality stands possibility. We can understand phenomenology only by seizing upon it as a possibility.

Martin Heidegger,
Being and Time

6.1 The Question That Precedes the Answer of Existence

Consider what it is to stand at a crossroads — not the geographical kind, where two paths of known destination diverge, but the existential kind: a moment in which the paths are not yet determinate, in which what lies ahead is not a selection from a fixed menu of options but an open field from which one will emerge, through decision, as the specific person who took this path rather than that one. You know that you will choose. You know that the choice will be yours, that it will carry your history into a future it helps to constitute, and that the person who arrives at the other end will be shaped by what you decide here in a way that is not the mere addition of a new property to a pregiven entity but the partial constitution of the entity itself. And you know — this is the fact that gives the crossroads its peculiar weight — that you are already ahead of yourself: that the choice is already exerting its gravitational pull on you before it has been made, that the future you have not yet reached is already structuring the present from which you act.

This experience is not the experience of selecting an option from a list of given possibilities. It is the experience of living within possibility as the structural medium of existence — of being the kind of entity for whom being is always an issue, always something to be taken up, always oriented towards what has not yet been but that shapes what is already the case. Martin Heidegger devoted his philosophical career to the analysis of this structural feature of human existence — an analysis so thorough, so technically demanding, and so ontologically consequential that it transformed not only the phenomenological tradition from which it emerged but the entire landscape of twentieth-century philosophy. What Heidegger saw, with a precision that has not been surpassed, is that the human being is not first an entity with a fixed nature who subsequently encounters possibilities and chooses among them. The human being is constituted by its relation to its own possibilities: it is the entity whose being is to be ahead of itself, to be already-in the world, to find itself thrown into a situation whose past it did not choose and whose future it cannot evade. The name Heidegger gives to this entity is *Dasein* — being-there — and the analysis of its structure is *fundamental ontology*.

The question the present chapter pursues is structural and specific. The five-level diagnostic framework established in Chapter 2 identifies the possibility dimension of direction (V_β) as the formally admissible wider pattern of structural configurations towards which the actuality of the present is oriented — the formal specification of what can emerge from the sedimented weight of what has been, the structured space from which relational events orient themselves forward. Bergson, as Chapter 5 established, achieves the actuality dimension of direction with extraordinary precision but cannot formally specify the possibility dimension, because to do so would require structural analysis prior to the temporal register in which his inquiry begins. Heidegger, this chapter will argue, enters the structural order at precisely this dimension: not at the actuality of the past in the present, but at the formal structure of possibility as the constitutive horizon of existence. He enters there with the most sustained and technically precise analysis the philosophical literature has produced. And by entering there — by taking the structure of possibility as his starting point rather than as a derived structural result — he forecloses the derivation of what makes possibility formally possible: the pre-temporal structural ground from which the formal structure of possibility itself emerges, and from which the relata that stand in the relation of possibility-to-actuality are constituted in the first place.

6.2 The Foundational Commitment: Care as the Being of Dasein

Heidegger's foundational commitment is the outcome of a phenomenological destruction that proceeds from the surface of ordinary experience to the ontological structure that underlies and makes possible what ordinary experience presents. The destruction is executed across the first two divisions of *Being and Time* with a methodological rigour that matches the architectonic ambition of the project: to ask the question of the meaning of being, which Heidegger argues has been forgotten by the entire philosophical tradition from Plato to Nietzsche, and to answer it through a preparatory analysis of Dasein — the being for whom being is a question — on the grounds that the question of being and the question of what kind of being is capable of raising it are structurally inseparable.

The first movement of the foundational commitment is the structural analysis of Dasein as being-in-the-world. Dasein is not, Heidegger insists with great precision, a subject enclosed within an inner sphere of experience who subsequently encounters an external world of objects. The subject-object split is itself a derivative and ontologically deficient mode of the more primordial structure of being-in-the-world, in which Dasein finds itself always already engaged with a meaningful environment of equipment, practices, and concerns that it did not first constitute but into which it was always already thrown. The world is not a collection of objects at which Dasein looks from outside; it is the structured whole of significance within which

Dasein moves and acts and within which things show up as the things they are — as ready-to-hand equipment, as obstacles, as resources, as threats, as opportunities.[1]

The second movement is the analysis of temporality as the ontological ground of Dasein's structure. This is the centrepiece of *Being and Time*'s second division and the point at which the foundational commitment receives its most explicit and technically demanding statement. Heidegger's central claim is that the structural whole of Dasein — the care-structure (*Sorge*) that is the formal unity of being-ahead-of-itself (*Sich-vorweg-sein*), already-being-in (*Schon-sein-in*), and being-alongside (*Sein-bei*) — is grounded in primordial temporality: a temporality that is not the ordinary time of clocks and calendars but the ecstatic temporal structure in which Dasein is stretched across future, past, and present simultaneously, from none of which it is ever simply absent. Primordial temporality is ecstatic: Dasein does not merely exist in time as an object among objects; it is stretched out into time's three dimensions in a way that constitutes rather than merely locates its being. It is ahead of itself towards its ownmost

[1]The analysis of being-in-the-world as primordial is central to Martin Heidegger, *Being and Time*, translated by John Macquarrie and Edward Robinson (New York: Harper & Row, 1962), 78–148. The critique of the Cartesian subject-object picture is explicit at id., 122–134, where Heidegger argues that the theoretical attitude in which the world appears as a collection of objects is itself derived from the breakdown of practical engagement: when the hammer fails to function, we are forced into a theoretical stance towards it that was never our primary mode of relation to it. Dreyfus (*Being-in-the-World: A Commentary on Heidegger's Being and Time, Division I*, 87–107) provides the most detailed analysis of this point and its consequences for the philosophy of mind.

possibility (the future ecstasis); it is already in a world it did not choose (the past ecstasis of thrownness); and it is alongside the entities it encounters in its world (the present ecstasis of falling).[2]

The third movement, and the one that reveals the foundational commitment in its most concentrated form, is the analysis of death and authentic existence. Death, in Heidegger's analysis, is not an event that happens to Dasein at the end of its life; it is the ownmost, non-relational, not-to-be-outstripped, and certain possibility that Dasein is always already running ahead of. Death is the formal structure of Dasein's being-towards-its-end — the ultimate horizon that no other Dasein can take over, that cannot be deferred indefinitely, and that reveals, to the Dasein that genuinely runs ahead towards it (*Vorlaufen in den Tod*), the radical finitude of its possibilities and the necessity of owning them as its own rather than dispersing them into the anonymous public existence of the *das Man*. The authentic response to death is the call of conscience, which summons Dasein from its lostness in the They-self back to its ownmost possibility: the possibility of being itself, not as a fixed essence

[2]The analysis of care as the structural whole of Dasein is at Heidegger, *Being and Time*, 225–244, and the derivation of temporality as the ontological meaning of care is at id., 349–401. The distinction between primordial temporality and ordinary time is drawn explicitly at id., 456–472: ordinary time is the 'now-sequence' that Dasein projects onto primordial temporality by counting its ecstases, and is therefore derivative on the primordial structure rather than equivalent to it. Blattner (*Heidegger's Temporal Idealism*, 35–54) provides the most technically precise analysis of the distinction between primordial temporality and ordinary time, including the consequences for Heidegger's claim that the future has priority among the three ecstases.

it already possesses but as the task it is always already ahead of.[3]

What these three movements jointly establish is the foundational commitment that the diagnostic framework will analyse: *possibility is the ontological structure constitutive of Dasein's being*. This is not the claim that Dasein is surrounded by possibilities from which it selects; it is the claim that Dasein *is* possibility — that the formal structure of being-ahead-of-itself towards possibilities not yet actualised is the most fundamental feature of what Dasein is, more fundamental than any actuality it has achieved and generative of the very structure of the present from which it acts. The future — specifically, the formal structure of Dasein's ownmost possibilities — has, in Heidegger's analysis, a formal priority over the past and the present: it is the structural horizon from which past and present take their meaning. Actuality, in this account, is the precipitate of possibility, not its precondition.[4]

[3]The analysis of being-towards-death is at Heidegger, *Being and Time*, 279–311. The structural characterisation of death as ownmost, non-relational, not-to-be-outstripped, and certain is at id., 294. The analysis of conscience as the call that summons Dasein from the *das Man* to its ownmost possibility is at id., 312–348. Carman (*Heidegger's Analytic: Interpretation, Discourse, and Authenticity in Being and Time*, 307–328) provides the most careful analysis of the relationship between authenticity, death, and possibility in Heidegger's account, including the question of whether authentic resolution amounts to a substantial self or merely a formal structure of ownership.

[4]Heidegger, *Being and Time*, 372–380, where Heidegger argues that the primacy of the future among the three temporal ecstases is not a claim about the causal priority of what has not yet occurred but a claim about the formal structure of primordial temporality: the future ecstasis of being-ahead-of-itself is the structural horizon from within which thrownness (the past ecstasis) and the present encounter with entities (the present ecstasis)

6.3 Entry Level : The Possibility Dimension of Level V

The five-level diagnostic framework identifies Level V — the structure of the relational event — through four formal conditions (V_1)–(V_4) (Definition 2.5). Within the direction dimension established by Proposition 2.1, the framework identifies two formally distinct sub-conditions (Definition 2.6): the actuality dimension (V_α), which is the formally historical pattern of accumulating sedimentation constituting the structurally determinate *before* from which every relational event orients; and the possibility dimension (V_β), which is the formally admissible wider pattern of structural configurations towards which that actuality is oriented. The possibility dimension is not an arbitrary openness; it is a formally structured space shaped by the weight distribution of sedimentation, constituting the structural horizon within which relational events orient themselves forward.

Bergson enters at (V_α) and illuminates the actuality dimension of direction with precision while leaving (V_β) formally unspecified. Heidegger, the following proposition establishes, enters at precisely the complementary point: his foundational commitment to possibility as the constitutive structure of Dasein's being locates his account at (V_β), the possibility dimension of direction, entered from within rather than derived from what lies structurally below it.

are what they are. Blattner (*Heidegger's Temporal Idealism*, 145–167) analyses this primacy claim with great care, arguing that it is best understood as a claim about the formal conditions of intelligibility rather than a temporal sequence.

Proposition 6.1 (Heidegger's Entry Level)**.** Heidegger's foundational commitment to possibility as the constitutive structure of Dasein's being locates his account within the non-latent register, at the possibility dimension of Level V : *The Structure of the Relational Event*. The formal conditions that Heidegger achieves — (V_β) (the formally structured horizon of admissible possibilities towards which Dasein's temporal existence orients), and, in qualified form, aspects of relational identity at Level II — are entered from within the non-latent register rather than derived from below. The structural conditions prior to the possibility dimension — the formal derivation of what constitutes the possibility space from a pre-temporal structural dynamic, the structural individuation of relata prior to their constitution as Dasein, the derivation of the source of encounter from a dynamic prior to care, and the derivation of an absolute structural ground from which temporal existence itself emerges — are structurally foreclosed by the foundational commitment to possibility as the ownmost and most primitive structural feature of Dasein's being.

Proof. By the diagnostic method of Chapter 2, the entry level is the level at which the foundational commitment takes a structural feature as primitive. Heidegger's foundational commitment is to possibility — in the existential-ontological sense of the formal structure of Dasein's being-ahead-of-itself towards its ownmost possibility — as the most primitive structural character of Dasein's being. Possibility in this sense is not derived within the existential analytic from any more basic structural dynamic; it is disclosed as the constitutive form of Dasein's temporality through the analysis of being-towards-death, thrownness, and projection. This commitment locates the entry point at the possibility dimension (V_β) of Level V, because (V_β) is precisely the formally structured horizon of admissible possibilities towards which a direction event orients — the structural feature that Heidegger's possibility is. To take existential possibility as primitive is to take (V_β) as the structural terminus of explanation, which is to enter the structural order at the possibility dimension of Level V rather than deriving it from a pre-temporal structural dynamic. That the entry is at (V_β) specifically — and not at the actuality dimension (V_α) — follows from the character of the commitment: Dasein's being is constitutively ahead-of-itself in the mode of possibility, not primarily constituted as the sedimentation of an irreversible past.

Remark 6.1. The claim that Heidegger enters at (V_β) requires specification of what this means and does not mean for the diagnostic analysis. It does not mean that Heidegger's account exhausts the possibility dimension of direction as the diagnostic framework specifies it: the formally admissible possibility space of the diagnostic framework is constituted by the weight distribution of the entire structural field and its global regulatory principles, while Heidegger's account of possibility is constituted by the formal structure of Dasein's being-ahead-of-itself in relation to its ownmost possibility of death. These are structurally related — both articulate the formal structure of the 'towards-which' that direction events carry — but they are not equivalent: Heidegger's possibility is existential-ontological, tied essentially to Dasein and its modes of being, while the diagnostic framework's possibility dimension is structural-formal, constituted by weight distributions and conservation principles. What the two accounts share is the structural insight that possibility is not a mere absence of actuality but a formally structured dimension of what is — a dimension that is constitutive of the directional character of real events and not derivable from any account that begins with actuality alone. It is at this shared structural insight that the diagnostic entry point is identified.

6.4 Formal Achievements at the Possibility Dimension

Three formal achievements characterise Heidegger's account at and in the vicinity of the possibility dimension of Level V. Together they constitute the most sustained philosophical analysis of the formal structure of possibility as a constitutive dimension of being available in the philosophical literature.

6.4.1 The Formal Structure of the Towards-Which

The diagnostic framework's account of the possibility dimension of direction, (V_β), specifies it as the formally admissible wider pattern of structural configurations towards which the sedimented actuality of the present is oriented — not an arbitrary openness but a structured space, shaped by the weight distribution that sedimentation has deposited, from which some outcomes are formally nearer and others formally further. The possibility space is not a set of options laid out in advance; it is a dynamic horizon whose shape is determined by what the actuality dimension has made of the structural field, and which in turn shapes what actuality can become.

Heidegger's analysis of Dasein's being-towards-its-possibilities is the most formally rigorous philosophical articulation of this structural feature that the literature provides. What Heidegger establishes is that possibility, in the existential-ontological sense, is not a property that an entity has — an option that it could take or leave — but a structural dimension of what the entity is: Dasein is constituted by its being-ahead-of-itself towards its ownmost possibilities, and this structure is not an addition to a pregiven entity but the very form of its being. Dasein exists as possibility. Its possibilities are not before it in the sense of objects placed at a distance; they are its own potentiality-for-being, the formal structure within which its thrownness becomes determinate and its present engagement with the world takes on the directional character that makes it an action rather than a process.[5]

The correspondence between this analysis and the diagnostic framework's account of (V_β) is structural and precise. The

[5]Heidegger, *Being and Time*, 182–188. The distinction between possibility as existential-ontological category and possibility as mere logical or epistemic openness — the absence of a contradiction or of complete information — is one of Heidegger's most precise and consequential technical contributions. The existential-ontological concept of possibility is not the modality of the proposition 'p is possible'; it is the structural characterisation of the being of an entity for whom being is always ahead of itself. This distinction is analysed with exemplary care in Mulhall (*Heidegger and Being and Time*, 88–106) and in Polt (*Heidegger: An Introduction*, 73–94).

diagnostic framework requires that the possibility dimension of direction be formally structured — not empty openness but an ordered space with formal properties that determine what is available from any given actuality. Heidegger provides, for the domain of human existence, the most developed account of what this formal structure consists in: the hierarchy of Dasein's possibilities is organised by the ownmost possibility of death, which is the formal terminus that gives every other possibility its weight and urgency and that prevents the dispersion of Dasein's possibilities into the undifferentiated horizon of the *das Man*. The ownmost possibility is not the most distant but the most formal: it is the horizon that structures the space of all other possibilities by providing the non-relational, certain, and not-to-be-outstripped terminus towards which they are oriented. This is the formal equivalent, in the existential-ontological domain, of what the diagnostic framework identifies as the structural constraint that makes the possibility space formally ordered rather than arbitrarily open: death is Dasein's conservation principle — the structural limit that prevents the possibility space from expanding without bound and that gives the specific weight distribution of Dasein's situation its formal character.[6]

[6]Heidegger, *Being and Time*, 306–310. The characterisation of death as the formal principle that organises the possibility space of Dasein's existence is implicit in Heidegger's analysis and made explicit in the secondary literature; see Blattner (*Heidegger's Temporal Idealism*, 202–217), which argues that death's structural role is less to specify the content of authentic existence than to provide the formal horizon that prevents Dasein from treating its possibilities as an indefinite series and forces it to own each possibility as genuinely its own.

6.4.2 Thrownness and the Structural Asymmetry of the Directional Field

The actuality dimension of direction, (V_α) — the sedimented weight of the before in the present — and the possibility dimension, (V_β) — the formally structured space of the ahead — are, in the diagnostic framework, formally asymmetric: the actuality dimension is what has been and cannot be undone; the possibility dimension is what can be reached from where the actuality has arrived; and the relation between them is not symmetric because passage from actuality to possibility is always directional, never reversible. This structural asymmetry — the formal difference between the from-which and the towards-which — is what the direction dimension of the relational event formally requires.

Heidegger's analysis of thrownness (*Geworfenheit*) provides the most philosophically developed account of this asymmetry available in the tradition. Thrownness is not merely the factual situation in which Dasein finds itself — the particular body, culture, language, and historical moment into which it was born without its consent. It is the formal structure of Dasein's always-already- having-been: the existential-ontological characterisation of the fact that Dasein is never in a position to begin from zero, that the weight of what it has already been is always structurally prior to any project it can undertake, and that this weight is not an external constraint on a pregiven freedom but the very condition under which any genuine project is possible at all. The thrown project is not thrown *despite* its thrownness; it is thrown *through* it — the specific character of what Dasein projects towards is shaped at every point by the specific character of the thrownness from which it projects.[7]

This is the structural asymmetry the diagnostic framework requires of the direction dimension: the from-which (thrownness, the actuality dimension) is not symmetric with the towards-which (projection, the possibility dimension), and their asymmetry is constitutive of the directional character of Dasein's temporal existence. Heidegger articulates this asymmetry with a precision that no prior philosophical analysis had achieved, because no prior analysis had identified the formal structure of being-ahead-of-oneself as a constitutive feature of temporal existence rather than a merely empirical fact about the temporal location of human beings in a pregiven time series. The asymmetry is ontological, not merely chronological: it is a feature of the formal structure of Dasein's being, not a consequence of the one-directional flow of clock time.

6.4.3 *Resoluteness and the Formal Unity of the Directional Structure*

The diagnostic framework's account of the relational event requires not only that the actuality and possibility dimensions of direction be formally specified but that they be formally unified:

[7]Heidegger, *Being and Time*, 173–182. The concept of thrownness is technically distinct from facticity, though related: facticity refers to the fact that Dasein has a character it did not choose, while thrownness refers to the formal structure of Dasein's relation to that character — specifically, the structure by which Dasein finds itself already there, already having been, without ever having had the experience of not yet being. Dreyfus (*Being-in-the-World: A Commentary on Heidegger's Being and Time, Division I*, 143–163) provides the most accessible analysis of why thrownness is an existential-ontological rather than merely biographical category.

the direction dimension is not a pair of independently defined structures but a single formal asymmetry, a pairwise whose two elements are constitutively related — the actuality is the from-which of the specific possibility space, and the possibility space is the towards-which of the specific actuality, and neither is what it is independently of the other. The formal unity of the direction dimension is the structural requirement that any adequate account of Level V must satisfy at the level of the directional structure.

Heidegger's analysis of resoluteness (*Entschlossenheit*) provides the most developed philosophical account of this formal unity available in the tradition. Resoluteness is not a decision that Dasein makes once and then retains; it is the formal structure of authentic existence — the structure by which Dasein owns its thrownness and its projection simultaneously, taking up its factical situation as the specific ground from which it projects its specific possibilities, rather than dispersing either into the anonymous alternatives offered by the *das Man*. Authentic resoluteness is the formal unity of thrownness and projection — the structural achievement by which Dasein does not merely have a from-which and a towards-which as separate features of its situation but owns their relation as constitutive of what it is. A resolute Dasein is not one that has chosen a particular project; it is one whose projection is transparent to its thrownness — one that acts from a formally clear relation between the weight of what it has been and the structure of what it is towards.[8]

What the diagnostic framework identifies in this analysis is a genuine formal achievement at the direction dimension of Level V: Heidegger provides, for the domain of human existence, the structural account of how the from-which and the towards-which of the directional structure are formally unified in the temporal existence of the entity that embodies them. The moment of vision — the *Augenblick* — is the structural analogue, in the existential-ontological domain, of what the diagnostic framework identifies as the co-instantaneous dimension of the direction event: the formal moment in which actuality and possibility are not sequential but simultaneously constitutive of the directional character of the encounter. Heidegger cannot derive this structure from the levels below it, and his account is confined to the domain of Dasein; but within that domain and that register, the formal achievement is genuine and irreplaceable.

[8]Heidegger, *Being and Time*, 312–348, especially the sections on conscience, guilt, and the structural characterisation of resoluteness as disclosedness. Heidegger's technical definition is: "Resoluteness we defined as a projecting of oneself upon one's ownmost being-guilty — a projecting which is reticent and ready for anxiety" (id., 343). The connection between resoluteness and the formal unity of thrownness and projection is made explicit at id., 385–387, where Heidegger discusses the 'moment of vision' (*Augenblick*) as the temporal phenomenon in which the resolute Dasein holds together past, future, and present in a unified structural grasp. Carman (*Heidegger's Analytic: Interpretation, Discourse, and Authenticity in Being and Time*, 270–290) provides the clearest secondary analysis of resoluteness as a formal structure rather than a psychological achievement.

6.5 What Heidegger Cannot Reach

Heidegger's formal achievements at the possibility dimension of Level V — the formal structure of the towards-which, the structural asymmetry of thrownness and projection, and the formal unity of the directional structure in resoluteness — are philosophically profound and diagnostically precise. They are also, the diagnostic framework establishes, achievements whose very precision marks the structural limits they cannot overcome. These limits are not deficiencies in Heidegger's analysis; they are structural consequences of beginning where he begins, and they cannot be remedied by any modification internal to his framework without abandoning the foundational commitment that makes his achievements possible.

6.5.1 The Formal Derivation of the Possibility Space Itself

The diagnostic framework requires not only that the possibility dimension of direction be formally characterised — that the formal structure of the towards-which be specified — but that the possibility space itself be formally derived: that the account show why the structural field has the specific possibility space it has, from what structural conditions that space is constituted, and what formal principles govern its shape and the distribution of formal nearness and distance within it. The possibility space, in the diagnostic framework, is not posited as a feature of reality; it is derived from the structural properties of the sedimentation events that have produced the actuality dimension and from the global regulatory principles — the conservation law, the majority-minority dynamic, the coupling condition — that govern the entire structural field. The formal derivation of the possibility space is what makes the possibility dimension formally specific rather than merely formally named.

Heidegger's foundational commitment forecloses this derivation with structural precision. The formal structure of Dasein's possibility space — organised by death as the ownmost non-relational possibility, specified through the hierarchy of Dasein's existential structures, shaped by the specific character of its thrownness — is not derived from a structural dynamic prior to Dasein's being. It is the formal expression of what Dasein *is*. The structure of death as the formal terminus of Dasein's possibilities is not a result that Heidegger derives; it is an insight that Heidegger discloses — a feature of Dasein's being that phenomenological analysis can make explicit but that the formal derivation of the diagnostic framework requires to be shown following from structural conditions more primitive than Dasein's being.[9]

[9] This limit is acknowledged implicitly in the formal gap at the centre of *Being and Time*: the project of the book, as Heidegger announces it, is to derive the meaning of being from the analysis of Dasein's being, and to show that the meaning of being is temporality — but the unwritten third division of Part I, which was supposed to complete this derivation by moving from the temporality of Dasein to the temporality of being as such, was never published. Blattner (*Heidegger's Temporal Idealism*, 193–215) analyses this absence as a structural consequence of the limits of the existential analytic: to derive the meaning of being from within Dasein's analysis is to begin from within the non-latent register and to find that the derivation requires a move outside it that the framework cannot make without abandoning its own terms.

The diagnostic point is structural: the possibility space of Dasein's existence is formally characterised by Heidegger with great precision, but it is not formally derived. It is presented as the phenomenological disclosure of a structure that was always already there — which is precisely what the diagnostic framework identifies as the entry-level commitment. To enter at (V_β) is to find the possibility structure already given and to analyse its formal features; it is not to derive the possibility structure from a more primitive dynamic. The derivation requires beginning below Level V, at the structural conditions of sedimentation itself and the global regulatory principles of the structural field, and building upward to show why these conditions and principles produce a field with the specific possibility-structure it has. Heidegger's analysis begins within the possibility structure and cannot step below it without abandoning the existential-ontological framework that constitutes his account.

6.5.2 *The Structural Individuation of Dasein*

The diagnostic framework's Level II specifies the formal requirements for structural identity prior to encounter: the invariant dimension of the structural locus (I_1), the progressive disclosure of relational character through encounter (I_2), the internal ground of structural coherence (I_3), and the derivation of relational character from the same dynamic that produces the ground and the locus (I_4). These four conditions jointly specify what it means for a relatum to have the structural identity required for genuine relational encounter — an identity that is neither fully constituted before encounter (which would make relational constitution impossible) nor entirely constituted through encounter (which would generate the regress question about what enters the constitutive relation).

Heidegger's account of Dasein's identity is the most sophisticated in the philosophical literature precisely at the point where it concerns the formal structure of Dasein as the kind of entity for whom its being is always an issue. Dasein has what Heidegger calls *Jemeinigkeit* — mineness, the formal property of always being mine and not transferable to another — which constitutes the formal invariant of its identity across its temporal existence. And Dasein's relational character is, in Heidegger's account, genuinely disclosed through its encounters with the world, with others, and with its own ownmost possibility: the analysis of authenticity and inauthenticity is precisely an analysis of two modes in which Dasein's formal identity can be expressed, and the movement from inauthenticity to authenticity is a movement towards genuine disclosure of what Dasein formally is.[10]

The diagnostic limit arises at the derivation condition (I_4): Heidegger does not derive the formal structure of Dasein's identity — *Jemeinigkeit*, the care-structure, the ecstatic unity of primordial temporality — from a more primitive structural dy-

[10]Heidegger, *Being and Time*, 150–163, on *Jemeinigkeit* and its formal role in distinguishing Dasein from entities that do not have the character of being-in-the-world. The formal property of mineness is not the empirical fact that I happen to be the one having these experiences; it is the existential-ontological characterisation of the mode of being of an entity for whom being is always mine-to-be. Mulhall (*Heidegger and Being and Time*, 37–48) provides the clearest secondary account of what mineness establishes and what it does not.

namic. He discloses it. The formal structure of Dasein's being is the starting point of the existential analytic, not its result; and while Heidegger insists that this starting point is not a stipulation but a phenomenological disclosure of what was always already there — a genuine seeing of the structure that ordinary experience conceals — the diagnostic framework identifies phenomenological disclosure as entry-level commitment rather than derivation. To disclose the structure of Dasein's being is to begin with Dasein; to derive the structure of Dasein's being would require beginning below the level at which Dasein as a specific kind of structural locus is already in place — which is precisely the level of Levels I and II. The existential analytic of Dasein cannot perform this derivation because it begins with Dasein as its analytic object, taking the fact and formal character of Dasein's existence as the starting point from which the analysis proceeds.

6.5.3 The Source of Encounter and the Being-With of Others

Level III of the diagnostic framework specifies the structural conditions for the source of encounter: it must be derived from the same dynamic that produces the ground and the structurally individual relata, it must be specific enough to explain why these relata rather than others enter the productive relation, and it must be capable of generating encounter without presupposing the relata as already fully constituted prior to it. These three conditions jointly specify what it means for relational encounter to be genuinely structural rather than merely contingent — a feature of the structural field rather than an external imposition on entities that were fully constituted before the encounter occurred.

Heidegger's analysis of being-with (*Mitsein*) and of the encounter with others in the shared world is the most formally developed account of the existential structure of encounter in the phenomenological tradition. Being-with is not an occasional feature of Dasein's existence; it is a constitutive structure of Dasein's being, such that even Dasein in isolation is being-with in the formal sense of existing in relation to the horizon of others who might be there. The encounter with specific others is not the contingent meeting of two independently constituted subjects; it is a disclosure of the structural being-with that was always already constitutive of Dasein's being-in-the-world.[11]

The diagnostic limit at Level III is precise: the structural source of encounter in Heidegger's account is the formal structure of being-with as a constitutive feature of Dasein's being — but this structure is disclosed as always already there, as a formal characterisation of what Dasein is, rather than derived from a structural dynamic that produces both the structure of being-with and the specific character of the relata that encounter one another. The specificity condition — why these relata rather than others enter encounter — is addressed by Hei-

[11] Heidegger, *Being and Time*, 149–168. Heidegger is careful to distinguish the existential structure of being-with (a formal characterisation of Dasein's being) from the ontic fact of co-presence with other Daseins (a contingent feature of particular existential situations). Being-with is primordial; co-presence is derivative. The implications of this distinction for the analysis of solitude, sociality, and the political are developed in the secondary literature; see Dreyfus (*Being-in-the-World: A Commentary on Heidegger's Being and Time, Division I*, 237–258) for the most detailed analysis.

degger through the notion of situatedness (*Befindlichkeit*) and the particular character of each Dasein's thrownness; but the derivation of why this Dasein encounters these others from the structural properties of the field is not provided. The encounter is phenomenologically disclosed as structured; it is not formally derived as the output of a structural dynamic that could explain why the encounter has the specific formal character it has rather than another.[12]

6.5.4 The Ground and the Question of Being

The diagnostic framework's Level I specifies the structural ground as that which satisfies the four ground conditions: totality (G_1), generativity (G_2), self-sustenance (G_3), and formal derivability (G_4). The ground must encompass everything without external supplement, generate structural differentiation from within its own activity, sustain itself without dependence on the levels it generates, and be derivable as the unique structural result of the eliminative method rather than stipulated as an arbitrary starting point.

Heidegger's relation to the question of the ground is among the most complex and consequential features of his philosophy.

[12]Okrent (*Heidegger's Pragmatism: Understanding, Being, and the Critique of Metaphysics*, 58–72) identifies this limit from within a broadly Heideggerian framework, arguing that the existential analytic can characterise the formal structure of Dasein's encounters but cannot explain what the diagnostic framework would call the source conditions of those encounters — the structural dynamic that produces this specific encounter between these specific relata with this specific formal character — without appealing to features of the world that the analytic itself has not derived.

His explicit aim — the question of being, the *Seinsfrage* — is in one sense more fundamental than anything the diagnostic framework specifies: he asks not what the structural ground is but what it is for anything to be at all. And his diagnosis of the tradition is precisely that the tradition has forgotten the question of being by taking beings as its starting point and treating the being of those beings as a self-evident given. This diagnosis is structurally convergent with the diagnostic framework's critique: the tradition takes structural features for granted at the level at which those features are already in place and does not ask what generates them.[13]

Yet the diagnostic analysis reveals a precise structural limit in Heidegger's pursuit of the ground. The being of Dasein — being-in-the-world, care, primordial temporality — is not the structural ground the diagnostic framework specifies. It is a characterisation of the being of one kind of entity — the entity that raises the question of being — from within the structural register in which that entity already exists. Being (*Sein*) in Heidegger's analysis is not a ground from which entities are derived but a horizon within which entities are disclosed; and while Heidegger insists that being is not itself a being (the ontological difference), he does not provide the formal derivation

[13]Heidegger, *Being and Time*, 1–35, specifically the opening division on the formal structure of the question of being, the necessity of a preparatory analysis of Dasein, and the significance of the forgetting of being for the history of philosophy. Moran (*Introduction to Phenomenology*, 1–31) provides the most accessible introduction to what Heidegger means by the forgetting of being and why he considers the question of being to have been systematically avoided rather than answered by the tradition.

of being from a structural dynamic prior to being's own disclosure.[14] The ground conditions (G_2) — generativity, the production of structural differentiation from within the ground's own internal dynamic — and (G_4) — formal derivability of the ground as the unique structural result of the eliminative method — are not satisfied by being as Heidegger analyses it. Being is disclosed; it is not derived. And what generates the formal structure of being's own self-disclosure — what produces the ontological difference, what constitutes the dynamic from which the formal structure of being and the formal structure of beings both emerge — is not a question Heidegger's framework can answer from within its own terms, because his framework begins within the ontological difference rather than deriving it.

A Heideggerian may object that the diagnostic framework's own concepts of 'derivation' and 'structural level' operate within a prior ontological horizon — the horizon of Being — that the framework does not account for and therefore cannot deploy without circularity. This objection has structural force and is acknowledged: the diagnostic framework constitutes its five-level apparatus foundationally, as Section 2.2 states,

[14]The ontological difference between being and beings is introduced at Heidegger, *Being and Time*, 26 and is the pivot of the entire analysis: being is not a being, not a property of beings, not a superbeing that beings instantiate, but the formal condition under which beings can be disclosed as the beings they are. But the question of what the formal structure of being is — what makes the ontological difference possible, what produces the distinction between being and beings from a level more primitive than the distinction itself — is precisely the unwritten question of Heidegger's project, whose absence from the published work Thomson (*Heidegger, Art, and Postmodernity*, 290–311) analyses as a systematic structural feature rather than a contingent gap.

and is therefore itself subject to the SIT's scope. The response is precise. The eliminative derivation method does not claim to stand outside the ontological horizon or to ground Being from beyond it. It operates through negative constraints — the progressive elimination of what can be shown to be derivable — articulating limits within which any attempt at derivation must stop. This is not a causal-generative move; it is a structural delimitation that operates, as Enoch (2025) argues of Limit-Disclosure Philosophy in an adjacent context, at the level where 'explanation must cease' rather than at the level where explanation is produced.[15] The diagnostic framework does not assert priority over Heidegger's question; it identifies, from within the structural horizon both share, where Heidegger's own framework stops short of answering it.[16]

6.6 The Later Heidegger: Ereignis and the Question of the *Beiträge*

The foregoing diagnostic analysis has centred on *Being and Time* because its systematic articulation of fundamental ontology provides the most rigorous and formally derivable statement of Heidegger's foundational commitments. A phenomenologist

[15]Enoch, "Limit-Disclosure Philosophy: An Axiomatic Framework and Philosophical Essay."

[16]See Nathalia Claro Moreira, "The Forgetting of Ontological Difference in the First Greek Beginning: Heidegger and the Destructive Critique of Ontotheology" (Preprint. Archived at PhilArchive, rec/CLATFO-24. Confirms that structural analysis of the ontological difference — including destructive critique of the tradition — operates from within rather than outside the Heideggerian horizon of Being, 2024).

specialist will, however, press an objection from a different direction: the later Heidegger — above all the *Contributions to Philosophy (From Enowning)*, written 1936–38 and published posthumously — represents a genuinely different philosophical orientation, not a continuation of *Being and Time* by other means. Heidegger himself explicitly acknowledged the shift: the approach through Dasein was preparatory and the *Beiträge* names what that preparation was oriented towards — *Ereignis*, the event of mutual appropriation in which Being and Dasein are brought into their belonging-together.[17] On Blattner's reading, ecstatic-horizonal temporality cannot ground the time-space (*Zeit-Raum*) of the *Ereignis* analyses, and the structural depth of *Ereignis* exceeds what the Dasein-centred analysis can supply.[18] Does the diagnostic verdict for *Being and Time* transfer to the later Heidegger, or does *Ereignis* constitute a structural move that the earlier analysis cannot recognise?

The diagnostic verdict for *Being and Time* does transfer to the later Heidegger, but for reasons that require care, because the grounds of the transfer differ from those of the initial analysis. Two questions must be separated: (*a*) does Ereignis shift Hei-

[17]Martin Heidegger, *Contributions to Philosophy (From Enowning)*, translated by Parvis Emad and Kenneth Maly, German original: *Beiträge zur Philosophie (Vom Ereignis)*, written 1936–38, first published 1989 (Gesamtausgabe, Vol. 65). The *Contributions* mark the turn from the Dasein-centred approach of *Being and Time* to the thinking of Ereignis as the event of mutual appropriation of Being and Dasein. Section references follow the numbered paragraphs (§§) of the Emad–Maly translation (Bloomington: Indiana University Press, 1999), §131–§135.

[18]Blattner, *Heidegger's Temporal Idealism*, ch. 9.

degger's entry level below Level V; and (*b*) even if the entry level shifts, does the foreclosure result still apply?

On whether Ereignis shifts the entry level. Ereignis is characterised in the *Beiträge* as the event of mutual appropriation — Sein and Dasein brought into their primordial belonging-together — not as a derived consequence of something more primitive within the framework's own derivational apparatus. The *Beiträge* proceed *from* Ereignis; the task is to think from within Ereignis rather than to derive it from conditions more primitive than itself.[19] Ereignis is accordingly not positioned below Level V in the sense of being a derived structural consequence of what lies below: it is constituted as the starting point of the inquiry. What shifts from *Being and Time* to the *Beiträge* is not the entry level but the characterisation of what occupies Level V: the Dasein-centred possibility-dimension analysis of *Being and Time* is replaced by the event-structure characterisation of Ereignis, which occupies the event-structure sub-variant of Level V rather than the possibility-dimension sub-variant. Both are foundational constitutions at Level V.

On the Abgrund and the minority reading. The *Beiträge*'s references to the *Abgrund* — the abyss — have been read by some commentators as gesturing towards something more primordial than Ereignis itself:[20] on this minority reading, Ereignis is

[19]Heidegger, *Contributions to Philosophy (From Enowning)*, §1–§5.

[20]Daniela Vallega-Neu, *Heidegger's Contributions to Philosophy: An Introduction*, Chapters 4 and 5 develop the structural role of the *Abgrund* (abyss) and the last god in the *Contributions to Philosophy*, reading the Abgrund as pointing towards something more primordial than Ereignis itself — a structural

not the *Beiträge*'s structural terminus but a result that emerges from the *Abgrund* as a prior structural domain. If this reading is correct, the entry level would shift downward within the hierarchy — possibly to Level I or to an intermediate level between I and V.

The dominant reading treats the *Abgrund* as a structural feature of Ereignis itself — the groundlessness internal to the event of mutual appropriation, naming what Ereignis is in its character as ground rather than naming a structural domain prior to and generative of Ereignis.[21] On this reading the *Abgrund* does not supply a derivational pathway below Level V. The diagnostic verdict is robust to both readings.

On the dominant reading: the *Abgrund* as an internal feature of Ereignis is a characterisation of Ereignis's mode of grounding, not a move below Level V. The entry level is Level V and the analysis applies as before.

On the minority reading: even if the *Abgrund* is taken as more primordial than Ereignis — as a structural domain from which Ereignis emerges — the *Abgrund* is itself constituted within the framework of the *Beiträge*. It is encountered through

domain from which Ereignis emerges. This reading is the basis of the minority interpretation addressed in the Horizon Lemma Remark's treatment of the later Heidegger; see pp. 107–130 (Bloomington: Indiana University Press, 2003), 107–130.

[21]Richard Polt, *The Emergency of Being: On Heidegger's Contributions to Philosophy*, The most sustained English-language commentary on the *Contributions to Philosophy*. Chapter 6 addresses the structural role of the *Abgrund* (abyss) in the Ereignis analyses, arguing that the Abgrund names the groundlessness internal to the event of appropriation rather than a structural domain below Ereignis from which Ereignis is derived (Ithaca: Cornell University Press, 2006), ch. 6.

and within Heidegger's thinking of Ereignis; it is not derived from conditions that are themselves derivable without presupposing the *Beiträge*'s governing attunements (*Angst*, *Erschrecken*, *Verhaltenheit*). The *Abgrund*, on the minority reading, is a foundational constitution at a level somewhat below Ereignis: the framework takes the *Abgrund* as its structural starting point at that lower level rather than deriving it. But the Horizon Lemma applies to any foundational constitution regardless of which level the constituting commitment occupies. If the *Abgrund* constitutes the framework's entry level, the framework's derivational resources are calibrated to the domain of the *Abgrund* and above; the latent register — what is structurally prior to and generative of the *Abgrund* itself — remains foreclosed by the same formal necessity.

The diagnostic conclusion for the later Heidegger is therefore this. If Ereignis is the *Beiträge*'s structural terminus (dominant reading), the entry level is Level V's event-structure subvariant and the foreclosure of Levels I through IV applies with the same necessity demonstrated for *Being and Time*'s possibility-dimension entry. If the Abgrund is more primordial than Ereignis (minority reading), the entry level shifts downward but the foreclosure of whatever lies below the new entry level follows by the same structural logic. In neither case does the later Heidegger supply a derivational pathway to the latent register. The *Beiträge* represents a deepening within the non-latent register — a shift from the Dasein-centred phenomenological approach to an event-ontological approach that reaches further into the structure of what the relational event is — but it does not descend below the non-latent register to the structural dynamic from which the relational event itself emerges.

Remark 6.2 (The Abgrund Minority Reading and Its Diagnostic Significance). The most philosophically sophisticated challenge to the Level V entry assignment comes from the minority reading of the *Beiträge zur Philosophie* (Contributions to Philosophy)[22] developed by Vallega-Neu.[23] On this reading, the Abgrund (Abyss) figures in the Contributions not as a structural level subordinate to Ereignis but as Heidegger's most fundamental gesture towards what resists all foundational constitution: the groundlessness of the ground, the structural open that every appropriating event presupposes but that appropriation itself cannot produce. If this reading is correct — if the Abgrund represents Heidegger's pointing towards what lies beyond the non-latent register rather than a positive characterisation of a structural level within it — then the question is whether it constitutes a near-approach to the latent register that the Level V assignment misrepresents.

The diagnostic framework yields two precise results for this reading, neither of which weakens the theorem's verdict.

First result: if the Abgrund is a positive structural characterisation. If the Abgrund names a determinate structural feature of Being — groundlessness as the positive formal character of the Open, the Abyss as the structural condition from which Ereignis itself emerges — then the diagnostic result is sharpened, not weakened. A positively specified structural feature taken as

[22] Heidegger, *Contributions to Philosophy (From Enowning)*.
[23] Vallega-Neu, *Heidegger's Contributions to Philosophy: An Introduction*.

the formal condition of the most fundamental account foundationally constitutes a non-latent structural feature at whatever level it occupies. The Horizon Lemma applies: $\mathfrak{H}$'s derivational apparatus, constructed to characterise Ereignis from within the Abgrund's formal condition, is calibrated to the domain of features at and above the Abgrund's level and is foreclosed from what is structurally prior to groundlessness as a constituted feature. The entry level shifts downward but the foreclosure result is invariant.

Second result: if the Abgrund is not a positive characterisation but a structural limit-gesture. If the Abgrund is Heidegger's pointing at the boundary of the non-latent register rather than a characterisation of what lies beyond it — if it is the gesture of an inquiry arriving at the edge of its own framework and naming the limit rather than the territory beyond it — then it occupies precisely the structural attitude the Eliminative Corollary (Corollary 10.2) identifies: an approach to the latent register that does not foreclose it by constituting anything as a terminus of explanation. On this reading, the Abgrund does not constitute a counter-instance to the SIT; it constitutes an approach to the eliminative orientation the corollary identifies as the only open pathway — an orientation Heidegger gestures towards without possessing the derivational apparatus to execute.

In neither case is the entry-level assignment of $\mathfrak{H}$ at (V_β) undermined. The framework constituted by the existential analytic and the Contributions together is determined by the foundational commitment that is operative across both works: the constitution of Dasein's being, and ultimately of Ereignis, as the structural starting point of philosophical analysis. The Abgrund names a limit within this framework — a limit that Heidegger approaches from within the framework's own resources, from the analysis of Ereignis outward. Approaching a structural limit from within the framework that constitutes it is not the same as deriving the level the framework occupies from conditions below it. The diagnostic verdict stands, and the Vallega-Neu reading, on its most defensible interpretation, provides independent philosophical evidence that Heidegger himself felt the structural pressure towards the very boundary the diagnostic framework formalises.

What the later Heidegger contributes to the diagnostic picture. The *Beiträge* analysis is not merely confirmatory. By characterising Ereignis as the happening of mutual appropriation — the event in which Being and Dasein are brought into their belonging-together through what Heidegger calls the throw-projection (*Entwurf-Wurf*) structure — the later Heidegger supplies the most precise phenomenological characterisation available of what the Level V event structure is at the human-existential scale. The co-instantaneous character of the mutual appropriation: that Sein appropriates Dasein precisely in and through Dasein's projection, and not before it; and that Dasein is thrown into its thrownness precisely as it projects, and not after it — this characterisation maps with precision onto the Level V formal requirement that the actuality and possibility dimensions of the directional field be co-instantaneous rather than sequentially ordered. The later Heidegger's contribution to the structural map of the non-latent register is the event-structure characterisation of Level V: it goes further into the formal organisation of the relational event than *Being and Time* reached. What it cannot do is descend below Level V to the structural dynamic from which Ereignis itself emerges.

The Post-Heideggerian Development: Merleau-Ponty's Ontology of the Flesh

This section is a supplementary coda to Chapter 6's diagnostic treatment of Heidegger: the flesh ontology is not counted as one of the thirteen traditions formally surveyed in Part II, but its engagement here establishes that the SIT's scope extends beyond the traditions explicitly examined — as this section demonstrates through Merleau-Ponty's late ontology of the flesh.

The late Merleau-Ponty's ontology of the flesh (*chair*), worked out in *The Visible and the Invisible* and the Working Notes composed in the final years before his death in 1961,[24] is not a modification of Heidegger's framework but a reconception of the phenomenological starting point: where Heidegger begins from Being-in-the-world as the structure of Dasein's existential situation, Merleau-Ponty proposes that what is philosophically primary is neither subject nor world but the reversible fabric of the sensible from which both emerge — what he calls the flesh of the world.

The flesh is characterised through a specific structural feature: its *reversibility*. The hand that touches the table is simultane-

[24]Maurice Merleau-Ponty, *The Visible and the Invisible*, translated by Alphonso Lingis (Evanston: Northwestern University Press, 1968).

ously touchable by the table; the eye that sees is itself visible; the perceiver and the perceived are folds of a single reversible fabric. Merleau-Ponty calls this structure the *chiasm* — the crossing by which subject and object are simultaneously differentiated from and continuous with each other — and locates in it the fundamental ontological structure of perceptual encounter. The flesh is not a thing and not a substance; it is what Merleau-Ponty calls "an element of Being" — the milieu of relational constitution within which subjects and objects emerge through their chiasmic entanglement.[25] Dillon's reconstruction establishes that the flesh's reversibility is not a phenomenological description of perceptual experience but a structural condition of relational encounter: it characterises what the encounter-milieu must be like for constitutive perception to be possible at all.[26]

The diagnostic framework assigns Merleau-Ponty's flesh ontology to Level V: the structural level of the relational event. The flesh characterises the internal organisation of the encounter event — the chiasmic structure through which perceiver and perceived are constituted through their reversible entanglement — with a formal precision that Heidegger's phenomenological analysis does not supply. The flesh functions as the structural condition of perceptual encounter: what must be the case about the encounter-milieu

[25]Id., 130–155.

[26]M. C. Dillon, *Merleau-Ponty's Ontology*, The most rigorous analytic reconstruction of Merleau-Ponty's late ontology of the flesh. Establishes that the chiasmic reversibility of the flesh is a structural condition of relational perception rather than a phenomenological description of perceptual experience (Bloomington: Indiana University Press, 1988).

for reversible, constitutive perception to occur. This is a Level V achievement: a formal account of the event's internal structure as the condition of relational yield. Merleau-Ponty does not merely describe perceptual experience; he characterises the structural condition of its possibility.

The flesh ontology's formal achievements at Level V are therefore more developed in one specific respect than Heidegger's own account: the chiasmic reversibility is a structural characterisation of the encounter event's internal organisation rather than an existential characterisation of Dasein's structural situation. The diagnostic result, however, is the same. The flesh is constituted as foundational: it is the primordial element from which subject and object emerge, not the result of a derivation from conditions more primitive than the flesh itself. Merleau-Ponty is explicit that the question "what is the flesh?" cannot be answered by pointing to something more fundamental — the flesh is what any such pointing already inhabits. The reversible fabric of the sensible is the terminus of ontological inquiry within the late Merleau-Ponty's framework: the investigation begins from within the chiasmic milieu and elaborates its structure, rather than deriving the milieu from conditions independent of it.

By the Horizon Lemma, a framework that foundationally constitutes the flesh at Level V is foreclosed below Level V. The questions the diagnostic framework asks at Levels I through IV — what the structural ground is from which the flesh-milieu itself arises, what constitutes the specific relational characters of the entities whose encounter the flesh enables, what sources the specific encounters that the chiasmic structure makes possible — are not asked within the flesh ontology. They are not refused; they are simply prior to the framework's point of entry. The flesh is what is already there when inquiry begins; deriving what the flesh presupposes would require a framework that enters below Level V, which is precisely what the companion investigation undertakes. The late Merleau-Ponty extends the phenomenological tradition's formal achievement at Level V; the SIT applies to his account for the same structural reason it applies to Heidegger's: the foundational constitution is at Level V and the framework's derivational resources are calibrated to that domain.

The Non-Trivial Content of the SIT: Against the Analyticity Objection

A referee may observe that the SIT is analytic: once foundational constitution is defined to include condition (iii) (resources defined over features at level n or above), the foreclosure follows by definition. The Horizon Lemma derives (iii) from (i) and (ii), but a deeper objection presses further: does the triviality reappear at the level of the Lemma? If "foundational constitution" simply means "taking as a starting point from which one builds upward," then it follows by definition that one does not build downward. Tautologies cannot establish philosophical conclusions.

The response is precise. The SIT is not offered as a tautology. Its non-trivial content lies in what the Horizon Lemma establishes about the philosophical practice of foundational constitution. The Lemma does not merely say: frameworks with restricted resources cannot reach outside their restriction. It says: the *act* of philosophical foundational constitution — taking a structural feature as the terminus of explanation — *produces* a framework with domain-restricted resources, in the sense that the framework's conceptual apparatus, developed to operate on that feature and what it enters into, does not possess instruments for descending below it. The non-trivial claim is that this domain restriction is not a contingent limitation that could be remedied by additional effort within the framework, but a structural consequence of what foundational constitution *is*: an act of establishing a terminus. Termini do not have what comes before them; they are constituted as the point before which one does not inquire, within the framework that constitutes them.

The SIT is therefore not the tautology "frameworks cannot reach outside their domain." It is the substantive claim that the philosophical act of foundational constitution of a structural feature at level n is identical with the act of establishing level n as the terminus of inquiry within $\mathcal{F}$, and that this identity is not contingent but structural: one cannot foundationally constitute F at level n within $\mathcal{F}$ and simultaneously conduct an inquiry within $\mathcal{F}$ that treats F as non-terminal. The SIT names this structural identity as "foreclosure" and proves that it holds for every framework satisfying Definition 10.4. The proof's content is not the tautology; it is the demonstration that the definition's conditions jointly constitute the foreclosure. The non-trivial content is therefore this: the formal definition captures a real feature of how philosophical frameworks are actually built. The Horizon Lemma shows that any framework constructed by the method of foundational constitution — as all thirteen traditions surveyed actually were — inherits the domain restriction as a consequence of how their conceptual apparatus was developed, not merely as a consequence of how we choose to define 'foundational constitution.' The definition is not stipulative; it is descriptive of a practice, and the Lemma establishes that the practice produces the restriction. That the definition is descriptive rather than stipulative is established by the diagnostic survey of Part II: each of the thirteen traditions examined satisfies the three conditions of Definition 10.4, and each exhibits precisely the domain restriction that the Horizon Lemma predicts for a framework satisfying those conditions. The convergence of the survey's results with the Lemma's prediction is what converts the definition from a merely possible characterisation into a demonstrated one.

A second objection, distinct from the tautology worry, holds that the eliminative transcendental method is conceptual analysis under a different name — that identifying what the relational thesis requires for coherence is precisely what conceptual analysis does: it unpacks the conceptual commitments of a concept and identifies the necessary conditions for its application. Three structural differences distinguish the methods.

First, conceptual analysis proceeds from a concept of X to the conditions for X's correct application; eliminative derivation proceeds from the structural commitments of X — what X requires in order to remain structurally coherent — to a residue that is not derivable from anything else. The direction is from inside the structure outward, not from a concept inward.

Second, conceptual analysis operates on a fixed conceptual stock and characteristically produces analytic truths — truths that hold in virtue of the meanings of terms involved. Eliminative derivation operates on a candidate set of structural features and produces a residue whose status is not analytic but structural: it is what remains after everything eliminable has been eliminated. The residue is not a conceptual necessity but a structural one, established by the eliminative procedure rather than by the analysis of any concept.[27]

Third, the SIT's conclusion is not the analytic truth 'foundationally constituted frameworks cannot reach below their foundational level' — a claim about what the concept of 'foundational constitution' analytically contains. It is the structural claim that the philosophical act of foundational constitution produces a domain restriction as a structural consequence of how conceptual apparatus is constructed. This is a claim about a structural fact about frameworks, established by the Horizon Lemma, not a claim about the content of any concept.

Note. The above argument is taken up formally in Chapter 10 as Remark 10.4, where it is stated precisely against the background of the full proof apparatus. The present discussion anticipates it here because the Heidegger analysis provides the clearest single-tradition illustration of why the foreclosure is not trivial: the calibration of Heidegger's conceptual apparatus to the possibility-dimension of Level V is not a consequence of how we define foundational constitution but of how the *Sein und Zeit* project was actually constructed.

[27]Compare Olenius, "Causality Without Time."

6.7 The Diagnostic Result

The diagnostic framework, applied to Heidegger's fundamental ontology with the precision the foregoing sections establish, yields the following formal result.

Theorem 6.1 (Structural Foreclosure: Heidegger)**.** Let $\mathfrak{H}$ denote the philosophical framework constituted by Heidegger's foundational commitment to possibility as the constitutive structure of Dasein's being. Then:

(i) $\mathfrak{H}$ achieves genuine formal results at the possibility dimension of Level V — condition (V_β): the formally structured horizon of admissible possibilities towards which Dasein's temporal existence orients, articulated through the structural analysis of being-towards-death, thrownness, projection, and resoluteness — and achieves qualified formal results in the account of Dasein's structural identity (*Jemeinigkeit*, the care-structure) and the existential structure of being-with as the formal ground of encounter.

(ii) The foundational commitment that constitutes $\mathfrak{H}$ — the treatment of possibility, in the existential-ontological sense, as the most primitive structural feature of Dasein's being, not derivable from anything structurally prior to Dasein's existence — formally constitutes the possibility dimension of Level V as the ground of the inquiry rather than as a derived structural result within a more comprehensive derivation.

(iii) The act of constituting (V_β) as the ground formally forecloses the derivation of: the formal constitution of the possibility space from a pre-temporal structural dynamic and from global regulatory principles of the structural field; the derivation of Dasein's structural identity (I_4) from a dynamic prior to Dasein's being; the derivation of the source of encounter from a structural dynamic that specifies why these relata rather than others enter productive relation; and Levels I and the unwritten ground of the ontological difference — all of which require structural analysis of a register prior to Dasein's being, from which Dasein's formal structure itself would need to be derived.

(iv) The foreclosure is necessary: no modification internal to $\mathfrak{H}$ can supply the formally adequate derivation of these structural levels without abandoning the foundational commitment to possibility as the constitutive structure of Dasein's being — which is the commitment that constitutes $\mathfrak{H}$ as the philosophical framework it is and makes the analysis of fundamental ontology possible in the first place.

Proof. Each of the four parts follows from the same structural source: the location of the entry point at (V_β) rather than at a level structurally prior to it. *(i)* The formal derivation of the possibility space requires beginning from the structural properties of the sedimentation events and the global regulatory principles of the structural field — both of which are prior to the level of Dasein's existence and are not accessible from within the existential analytic. Beginning within the possibility structure, the analytic can characterise but cannot derive it. *(ii)* The structural individuation of Dasein (I_4) requires derivation from a dynamic prior to Dasein; the existential analytic begins with Dasein and produces *Jemeinigkeit* as a phenomenological disclosure rather than a formal derivation. *(iii)* The derivation of the source of encounter requires specification of the structural dynamic that produces both the formal structure of being-with and the specific character of these relata in this encounter; the existential analysis of being-with characterises the formal structure but does not derive the specificity condition. *(iv)* The ground conditions at Level I require the derivation of being's formal structure from a dynamic prior to the ontological difference; Heidegger's framework begins within the ontological difference and cannot step below it without abandoning fundamental ontology's constitutive commitment. In each case, the foreclosure is necessary: it is the structural consequence of entering the analysis at (V_β), which is the possibility dimension of the relational event's directional structure — a genuine and profound entry point, but not a derivation of the levels it presupposes.

Remark 6.3 (On the Four Traditions as a Structural Map). The diagnostic results for Whitehead, Spinoza, Bergson, and Heidegger together constitute a structural map of the relational ontology problem whose completeness deserves explicit recognition. Whitehead enters at Level V and illuminates the formal organisation of the relational event from the top of the non-latent register downward. Spinoza enters at Level I and illuminates the formal requirements of the absolute structural ground from the bottom of the non-latent register upward. Bergson enters at the actuality dimension of Level V's directional structure and illuminates the formal structure of temporal sedimentation from within the register of lived temporal flow. Heidegger enters at the possibility dimension of Level V's directional structure and illuminates the formal structure of the possibility horizon from within the register of human temporal existence. Together the four traditions provide a map of the non-latent register that is comprehensive in the specific sense that no major structural feature of the non-latent register is left unilluminated by at least one of them. What remains unilluminated — what all four traditions leave in structural darkness, each from its own distinctive entry point — is the latent register: the structural dynamic prior to and generative of temporal existence, from which the non-latent register itself emerges. This is not a gap in the map. It is the terra incognita that the map, by its very completeness, makes visible for the first time.

Criterion 6.1 (Heidegger's Diagnostic Profile). *Entry level*: The possibility dimension of Level V, (V_β). *Formal achievements at and above entry level*: (V_β) — the formally structured horizon of admissible possibilities, articulated through the structural analysis of being-towards-death as the formal terminus that organises the possibility space of Dasein's existence; the structural asymmetry of thrownness and projection, mapping to the formal asymmetry of the actuality and possibility dimensions of the directional field; the formal unity of the directional structure in resoluteness and the moment of vision, mapping to the co-instantaneous structural requirement of the direction dimension; qualified achievements at Level II in the account of Dasein's structural identity (*Jemeinigkeit*, the invariant dimension of the structural locus) and the progressive disclosure of relational character through encounter; qualified achievements at the formal structure of the source of encounter through the analysis of being-with as a constitutive structure of Dasein's being. *Structural foreclosure*: The formal derivation of the possibility space from a pre-temporal structural dynamic and global regulatory principles; the derivation condition at Level II (I_4) — Dasein's structural identity is disclosed rather than derived; the specificity condition at Level III — the source of encounter is characterised as being-with but not derived from a dynamic prior to Dasein's existence; Level I — the ground conditions, specifically generativity (G_2) and formal derivability (G_4), are not satisfied by being as analysed within the ontological difference, because the ontological difference itself is not derived. *Character of the foreclosure*: Necessary. The foreclosure is the structural consequence of beginning at the possibility dimension of the non-latent register — entering the structural order from within the formal structure of possibility itself — rather than deriving that structure from a more primitive dynamic. No modification internal to fundamental ontology that preserves the constitutive role of Dasein's being as the access-point to the question of being can overcome this structural limit.

The diagnostic result does not diminish what Heidegger achieved. The analysis of Dasein's being in *Being and Time* — the account of being-in-the-world, of care, of primordial temporality, of being-towards-death, of authenticity and the call of conscience — constitutes the most comprehensive and technically rigorous analysis of the formal structure of human temporal existence in the philosophical literature. Heidegger transformed the question of what it means to be a human being from a question about the properties of an entity into a question about the formal structure of the being of the entity for whom being is always an issue; and this transformation — which is also the displacement of the subject-object picture, the recovery of being from its concealment by the tradition, and the demonstration that temporality is not a feature of experience but the ontological ground of experience — is a philosophical achievement whose consequences have not been exhausted.

What Heidegger showed us is that possibility is not a deficient mode of actuality — not what is not yet real, not a shadow of real being — but a constitutive dimension of what is real: the formal structure of the towards-which that gives the from-which its directional character and makes human temporal existence something more than the traversal of a pregiven sequence. He showed it with a phenomenological precision that had no precedent. What he could not show — what his entry at the possibility dimension of the relational event's directional structure structurally prevented him from showing — is what the formal structure of possibility itself emerges from: the pre-temporal structural dynamic that makes the possibility space formally structured rather than arbitrarily open, that derives the specific shape of the possibility horizon from the sedimentation events that have produced the actuality dimension, and that provides the formal account of the structural ground, the individuation of relata, and the source of encounter that the existential analytic presupposes rather than derives. Heidegger began within the structure of possibility. The question of what makes the structure of possibility possible — of what is before the horizon within which he found us already oriented — was not his question. It remains open.

Of all the traditions surveyed in Part II, Heidegger's is at once the most instructive and the most formally instructive in its foreclosure. He was asking the right question — not what entities are, but what it is for anything to be at all — and asking it with greater structural precision than any previous thinker had achieved. His foundational commitment to Care as the being of Dasein, and to possibility as Care's constitutive structure, placed him closer to the boundary of the latent register than any classical tradition. And it is precisely this closeness that makes his foreclosure the most revealing: he could not step below the structure of possibility because he constituted it as the ground of his inquiry. The inquiry conducted from within the structure of possibility cannot ask what is prior to the structure of possibility without abandoning the constitutive commitment that makes it the inquiry it is. What the diagnostic framework identifies, in Heidegger's case, is not a gap that further phenomenological analysis could close. It is the structural mark of an inquiry that went as far as its foundational commitment could take it — and arrived, at the limit, at the same room in which every other tradition had arrived by a different door.

Summary of Chapter 6. *Heidegger's fundamental ontology enters the structural order at the possibility dimension of Level V: the formally structured horizon of admissible possibilities towards which the sedimented actuality of temporal existence is oriented. Genuine formal achievements obtain at this entry level: the structural analysis of being-towards-death maps to the formal termination condition that makes the possibility space ordered rather than arbitrarily open; the analysis of thrownness and projection maps to the formal asymmetry of the actuality and possibility dimensions of the directional field; and the account of resoluteness and the moment of vision maps to the co-instantaneous structural unity of the direction dimension. The foundational commitment to possibility as the constitutive structure of Dasein's being forecloses what lies below the entry level: the formal derivation of the possibility space from pre-temporal structural conditions and global regulatory principles cannot be performed from within an analysis that begins with the possibility structure already in place; the derivation condition at Level II is not met, because Dasein's structural identity is disclosed rather than derived from a more primitive dynamic; the specificity condition at Level III is not met, because the source of encounter is characterised but not derived from a structural dynamic prior to Dasein's existence; and the ground conditions at Level I — particularly generativity and formal derivability — are not satisfied by being as analysed within the ontological difference, which is itself not derived but presupposed. The most significant post-Heideggerian development — Merleau-Ponty's late ontology of the flesh — extends the Level V achievement with formal precision: the chiasmic reversibility of the flesh characterises the internal organisation of the encounter event with structural rigour that Heidegger's phenomenological analysis does not supply. The SIT applies to the* *flesh ontology for the same structural reason: the flesh is foundationally constituted at Level V, and the framework neither asks nor can answer what structural dynamic generates the flesh-milieu from below. The four traditions surveyed in this Part together constitute a structural map of the non-latent register: Whitehead from the event downward, Spinoza from the ground upward, Bergson from the actuality of temporal flow, Heidegger from the formal structure of possibility. Their combined illumination of the non-latent register makes visible, for the first time, the precise structural character of what they jointly leave in darkness: the latent register from which the non-latent register itself emerges. Part III turns to the formal task this survey has made possible: the precise statement of why the failure of each tradition is not accidental but structurally necessary, and the rigorous identification of the only type of derivation that is not foreclosed from the start.*

CHAPTER 7

The Contemporary Landscape I

Structure Without Ground

There are no things. There are only structures — patterns of physical relations all the way down, with nothing to be the bearer of those relations.

James Ladyman and Don Ross,
Every Thing Must Go

A note on the reading experience. A reader who has followed Chapters 3 through 6 carefully will arrive at this chapter with the diagnostic pattern already recognisable: every tradition enters the structural order at a specific level, achieves formal results there, and cannot reach below its point of entry. This recognition is the intended result. By the time the contemporary traditions are examined, the diagnostic grid has been validated across traditions of radically different character — a process philosophy, a substance metaphysics, a philosophy of temporal flow, a fundamental ontology — such that its application here is confirmatory rather than generative. A reader who feels they can anticipate the result is correct to feel this: the pattern's anticipatability is itself evidence of its universality, and that universality is precisely what the Structural Insufficiency Theorem of Part III will formally establish.

The evidential role of Chapters 7 through 9 is nevertheless not merely confirmatory. The classical traditions of Chapters 3–6 share a common historical provenance and a broadly speculative methodology. A sceptic could argue — and would be right to argue — that the structural gap appears in all four classical traditions because those traditions share hidden methodological assumptions that generate the gap artificially. The contemporary frameworks of Chapters 7–9 close this sceptical opening, because they do not share the classical traditions' historical provenance, methodological commitments, or philosophical vocabulary. Ontic structural realism and relational quantum mechanics are developed within analytic philosophy of physics and constrained by the formalism of quantum mechanics; they have no investment in the classical metaphysical frameworks and have explicitly sought to escape them. Stengers and Emirbayer approach the relational thesis from within the philosophy of the life sciences and social theory; they have independent methodological foundations with no connection to Whitehead's, Spinoza's, Bergson's, or Heidegger's problems. Simondon and Deleuze develop structural ontologies that explicitly critique the limitations of the classical framework. If the structural gap appeared only in the classical traditions, methodological contamination would be a live hypothesis. The fact that it appears equally in traditions that have no methodological debt to the classical frameworks and that have explicitly sought to surpass them is independent evidential weight of a different kind from what Chapters 3–6 supply: it shows that the gap is not a feature of classical speculative philosophy but of the relational thesis itself, across every methodological approach that has pursued it with formal rigour.

Remark 7.1 (Husserl's Formal Ontology and the Level V Assignment). A phenomenologically trained referee may observe that Husserl's own formal ontology — as practised in the third and fourth *Logical Investigations*, in *Formal and Transcendental Logic* (§§83–87), and in the mereological analyses of part-whole relations — operates at a level of categorial structure that might appear prior to the Level V temporal-event structure the present investigation assigns to phenomenological frameworks. Husserl's analysis of "foundation" (Fundierung) — the structural relation by which moments require their wholes and dependent parts require their founding foundations — is conducted in a formal-ontological register that Husserl himself describes as prior to any material or regional ontology. Does this analysis operate below Level V, giving Husserl's framework a resource the Level V assignment fails to credit?

The diagnostic answer is precise. Husserl's formal ontology analyses the structure of *objects-in-general* — the formal categories under which any object whatsoever can be thought or given — and does so from within the presupposition that such objects are available for categorial synthesis: available to an intentional consciousness that can perform the acts of ideation and formal variation through which formal-ontological essences are disclosed. The "objects" of formal ontology are not pre-thematic structural conditions from which objects emerge; they are objects analysed at their most formal level, stripped of material content but still constituted as the intentional correlates of possible synthetic acts. The formal-ontological stratum in Husserl is therefore a study of the structural properties of what presents itself to constituting consciousness at its highest level of generality — not a derivation of the conditions from which the presenting relation itself emerges.

This is precisely what the Level V assignment captures. Level V's defining conditions address the internal structure of the relational event — the organisation of encounter, actuality, possibility, and direction — and the formal-ontological stratum in Husserl operates at this level: it characterises the formal structure of any object-synthetic encounter whatsoever, prior to material specification. Husserl's formal ontology enters at Level V's possibility-dimension variant, not below it, because the formal-ontological investigation is conducted from within the presupposition of the constituting-constituted structure (the intentional arc between act and object) that is itself a feature at Level V. What is prior to the intentional arc — what structural conditions make possible the very directedness of consciousness towards objects — is precisely the question that phenomenological method, at its formal-ontological stratum, presupposes rather than derives. The Level V assignment therefore applies to Husserl's formal ontology as it applies to Heidegger's fundamental ontology: both enter at the possibility dimension of Level V and are foreclosed from the latent register for the same structural reason.

7.1 The Question That Dissolves the Object

Suppose you followed the relational turn to its logical conclusion. Suppose you took seriously, not merely as a methodological convenience but as a genuine ontological commitment, the claim that things are constituted by their relations — that there is no property, no identity, no being that a thing possesses independently of its relational position in a wider structural web. Where does this commitment lead, when pursued with the rigour it demands?

The answer, for two of the most significant research programmes in contemporary philosophy of physics, is to the dissolution of the object entirely. Not to its diminution, not to its subordination to structure, but to its elimination: what we call an object is nothing over and above the structural relations it occupies, and those relations are prior to whatever we might be tempted to call their terms. The terms are not bearers of relations; they are nodes in a pattern, and patterns are not constituted by nodes — nodes are constituted by patterns, all the way down. There are no objects. There is only structure.

This is the foundational commitment of *ontic structural realism*, the most radical and philosophically sophisticated contemporary defence of the claim that physical ontology consists in structure and nothing but structure. And it is a closely related commitment that underlies Rovelli's *relational quantum mechan-*

ics: the claim that quantum states are not properties of systems in isolation but facts about the relations between systems — that there is no quantum state simpliciter, only a quantum state relative to some physical system, and that this relativity is not a deficiency in our knowledge but a feature of physical reality itself. There are no absolute physical quantities. There are only relational physical facts.

Both programmes enter the contemporary debate from the direction of physics rather than from the direction of metaphysics, and both achieve genuine formal results at levels of the structural order that the classical traditions surveyed in the preceding chapters left undeveloped. Both represent genuine advances. And both — as the diagnostic framework of Chapter 2 reveals with precision — presuppose what they cannot derive: the structural ground from which the relational structure they posit itself emerges.

The question the present chapter pursues is the same question it has put to each tradition in turn. Where does each programme enter the structural order? What does it achieve formally at and above its entry level? And what does its foundational commitment structurally prevent it from reaching below that level?

7.2 Ontic Structural Realism

The World as Pure Structure

7.2.1 The Foundational Commitment

Structural realism as an epistemological position has a long history in the philosophy of science, traceable at minimum to the work of Henri Poincaré and Bertrand Russell in the early twentieth century, and recovered for contemporary debate by John Worrall in his influential 1989 paper on structural realism and the preservation of structure across theory change.[1] Epistemic structural realism — Worrall's original position — holds that science gives us genuine knowledge of the structural relations among the unobservable entities it posits, while remaining agnostic about the intrinsic natures of those entities. It is a moderate position, designed to explain why structure is preserved across scientific revolutions even when the objects posited change dramatically.

Ontic structural realism — developed systematically by Steven French and James Ladyman, and defended in its most ambitious form in the collaboration with Don Ross — abandons the moderation.[2] The problem with epistemic structural

[1]Worrall ("Structural Realism: The Best of Both Worlds?," 117–137). Worrall's *epistemic* structural realism holds that science gives us knowledge of structure but not of the intrinsic natures of the things that instantiate that structure. The transition to *ontic* structural realism — the claim that there are no such intrinsic natures to be known because there are no objects in the relevant sense — is the decisive further step taken by French, Ladyman, and Ross.

[2]The foundational papers are Ladyman ("What is Structural Realism?") and French and Krause ("Quantum Vagueness"). The full systematic defence is Ladyman et al. (*Every Thing Must Go: Metaphysics Naturalized*). French's

realism, on Ladyman's diagnosis, is that it retains objects as the things whose structural properties science discloses, while denying us knowledge of those objects' intrinsic natures. But this retention is incoherent: if we have no epistemic access to intrinsic natures, we have no grounds for positing objects with intrinsic natures in the first place. The objects of epistemic structural realism are idle posits — entia non grata whose elimination simplifies the ontology without loss.[3]

The foundational commitment of ontic structural realism is therefore: the physical world consists of structures, and those structures are not borne by objects. More precisely — in Ladyman and Ross's formulation — real patterns are the fundamental ontological category.[4] A real pattern is a structural regularity that is non-redundant — that carries genuine information above the noise of its background — and that projectible — that supports reliable inference to further instances. The physical world is nothing but the totality of such real patterns, related to one another through further structural relations, without any object-level substrate in which these patterns are instantiated. Everything is structure; there is nothing beneath structure to be structured.

independent elaboration appears in French (*The Structure of the World: Metaphysics and Representation*).

[3]Ladyman ("What is Structural Realism?," 130–131): the objection that epistemic structural realism retains an unknowable substratum is central to Ladyman's motivation for the ontic turn. A parallel diagnosis appears in French and Krause ("Quantum Vagueness," 138–143).

[4]Ladyman et al. (*Every Thing Must Go: Metaphysics Naturalized*, 118–130). The notion of a real pattern is taken from Dennett (*Consciousness Explained*), radicalised into a full ontological commitment.

This is the most thoroughgoing contemporary execution of the relational thesis: things are constituted by their relational positions, and there is no remainder called an object that occupies those positions independently of them. The nodes of the structural web have no identity except their structural role, and their structural role is the whole of what they are.

7.2.2 Formal Achievements

Ontic structural realism achieves genuine formal results at Level II of the diagnostic framework — the level of relational identity — and this achievement represents a genuine advance beyond the classical traditions.

The derivation condition at Level II requires that relata be shown to possess structural identity through their relational position rather than prior to it. No classical tradition fully satisfies this condition: Whitehead's actual occasions have structural identity through their process of self-constitution, but that identity is achieved rather than derived; Spinoza's modes follow from Substance through logical entailment, but they do not achieve genuine relational individuation; Bergson and Heidegger are not concerned with the individuation of entities in the relevant sense. Ontic structural realism, by contrast, makes the derivation of identity from structure its explicit and central commitment. The identity of a physical system — a particle, a field, a quantum state — just is its structural position in the physical pattern, and this position is constituted by the relational structure rather than prior to it. The individual is an abstraction from the structural; the structural is the concrete reality.[5]

This formal achievement is directly relevant to the diagnostic framework. The specificity condition at Level II — what makes each relatum the *specific* relatum it is rather than another — is addressed by ontic structural realism through the concept of structural position: two relata are distinct if and only if they occupy structurally distinct positions in the real pattern they jointly instantiate. The individuation is structural all the way down, with no appeal to intrinsic properties or primitive identifiers.

At Level IV, ontic structural realism also achieves a partial formal result. The programme's account of emergence — specifically, the claim that higher-level structural patterns are not reducible to lower-level patterns even when they supervene on them — provides a formal framework for what the diagnostic analysis identifies as relational yield: the production, through structural encounter, of something not already contained in either structural level taken alone. Real patterns at higher levels of description are not mere redescriptions of lower-level patterns;

[5]The technical elaboration of this claim for quantum particles involves the concept of non-supervenient relations — relations that hold between quantum systems without supervening on any intrinsic properties of those systems. See French (*The Structure of the World: Metaphysics and Representation*, 100–117) for the most careful development of this argument. The metaphysical resources deployed include the distinction between weak and strong discernibility, drawn from Saunders ("Physics and Leibniz's Principles").

they are genuine structural additions that carry information not recoverable from their base.[6]

7.2.3 What Ontic Structural Realism Cannot Reach

The formal achievements at Level II and partially at Level IV are genuine. But ontic structural realism's foundational commitment — structure without objects, real patterns as the fundamental ontological category — forecloses the derivation that the diagnostic framework requires at Level I, and generates a structural regress at Level II that the programme cannot resolve from within its own resources.

The generativity condition at Level I requires that the structural ground be shown to produce structural differentiation — the diversity of structurally distinct relata — from within its own dynamic. Real patterns, as Ladyman and Ross deploy them, satisfy the totality condition at Level I in a limited sense: the totality of real patterns constitutes the physical world, and there is nothing outside it. But the generativity condition is not met. The question of what produces real patterns — of what the structural dynamic is from which the diversity of patterns emerges — is not addressed. Real patterns are posited rather than derived; they are the starting point of the analysis rather than its conclusion. And the formal derivability condition is not met either: ontic structural realism does not show that real pat-

[6]The account of emergence in Ladyman et al. (*Every Thing Must Go: Metaphysics Naturalized*, 43–76) is the most developed treatment. The connection to what philosophers of biology call "strong emergence" is discussed in Ladyman et al. (*Every Thing Must Go: Metaphysics Naturalized*, 77–109).

terns are the unique structural result of the eliminative method applied to what physical ontology presupposes. It begins with patterns and argues from them. The question of what precedes patterns — of what the structural source of their emergence is — is not within the framework's scope.[7]

The relational regress that Level II's derivation condition exposes is more fundamental. Ontic structural realism holds that the identity of a relatum is constituted by its structural position in a real pattern. But a structural position is itself a relational fact — it is a position *in* a structure, constituted by the relations that define the structure. Those relations, in turn, require relata to hold between. If those relata are themselves constituted by their structural positions in further relations, the regress is obvious: every relational constitution of identity requires a further structural web within which the constituting relations are defined, and that further web requires its own constituting relations, and so on without termination. That this is

[7]This limitation is acknowledged, in a different register, in the response to the "no relations without relata" objection that ontic structural realism has generated. The objection — pressed most forcefully by Psillos ("Is Structural Realism Possible?," 64–71) and Bueno ("Is there a Syntactic Criterion of Structural Realism?," 120–130) — is that relations require relata to hold between, and that eliminating the relata eliminates the relations along with them. Ladyman and Ross's response appeals to the mathematical structure of physical theory as providing the relata in the form of node-positions. But this response defers rather than resolves the question: what provides the node-positions themselves, if not something that functions as a Level I ground for the structural web? Lam and Wüthrich ("No Categorial Support for Radical Ontic Structural Realism") demonstrate that category-theoretic reformulations of radical OSR do not resolve this difficulty, concluding that "any contention that relations can exist floating freely from some objects that stand in those relations seems incoherent."

not a difficulty peculiar to OSR but a structural feature of every ontology that admits real multiplicity without deriving a prior principle of individuation has been established independently by Rivera,[8] who demonstrates across grounding, process ontology, and structural ontology that all of them "presuppose individuation rather than explaining it" — that "structure organizes; it does not individuate." Rivera's result directly confirms the Level II gap the present investigation identifies: the structural web describes the arrangement of already-individuated nodes but cannot supply the conditions under which distinct nodes exist to be arranged.

This is not an objection to the relational thesis as such — the diagnostic framework of the present study does not reject the relational constitution of identity but argues that it requires derivation from a structural ground. It is an objection to the specific form that relational constitution takes in ontic structural realism: the substitution of structural position for object as the fundamental ontological category, without a derivation of the structural web from a prior dynamic. The regress is not vicious in every possible framework; it is vicious precisely where the framework has no Level I ground from which the web emerges and which therefore terminates the regress at the non-relational source of all relational structure.[9]

[8] Rivera, "Unity Before Multiplicity: Ontological Individuation and the Problem of Primitive Multiplicity."

[9] The distinction between regresses that are vicious and those that are not depends on whether the regress terminates in something not itself relational. The diagnostic framework does not require that relata have intrinsic properties independent of relations — this would be to import the classical substance-attribute model that ontic structural realism rightly rejects. It re-

Mathematical structuralism — the view that mathematical structure is not a representation of but identical to physical structure[10] — may appear to satisfy the Level I ground conditions: mathematical structures are total, self-sustaining, and formally necessary, and on the strongest reading generate all physical possibilities. The diagnostic response is precise. Mathematical structures do not generate structural differentiation from a more primitive dynamic; they simply *are* networks of structural relations. The Level I generativity condition G2 requires that the ground produce differentiation from within its own dynamic — generating the specific configuration of relata and relational web as a structural consequence rather than constituting that configuration as its primitive content. Mathematical necessity presupposes rather than produces the specific differentiated structure it characterises as necessary: it tells us that whatever structure obtains could not be otherwise, not what produces the fact that this structure obtains rather than another. A ground that simply *is* the network it grounds satisfies G1 (totality), G3 (self-sustaining character), and G4 (formal necessity) but not G2: it cannot generate what it already, foundationally, is. Mathematical structuralism therefore enters at Level II — the level of real structural patterns — rather than at Level I, and

quires that there be a structural dynamic — at Level I — from which both the relational web and the relata within it are jointly derived. This is not the same as requiring a non-relational substrate. The basin dynamic, as it will be derived in Part III, is itself constitutively tensioned between dimensions that are structurally prior to the distinction between relatum and relation.

[10]See Ladyman et al., *Every Thing Must Go: Metaphysics Naturalized*, Ch. 4.

inherits the Level I gap the diagnostic framework identifies in ontic structural realism generally.

The Level I failure propagates upward. At Level III — the source of encounter — ontic structural realism has no formal account of what brings specific structural positions into specific relational contact. The structural web is described as a totality of real patterns; what makes a particular pattern impinge on another particular pattern, rather than on all others or none, is not derived from the programme's foundational commitments. Structural contact — the encounter between real patterns that produces emergent higher-level structure — is described rather than derived.

7.2.4 *The Diagnostic Result for Ontic Structural Realism*

Ontic structural realism enters the structural order at Level II, achieving the most explicit contemporary formal account of the derivation of relational identity from structural position. It achieves partial formal results at Level IV through its account of emergence as non-reductive higher-level pattern. Its failure at Level I — the absence of a generative structural ground from which the relational web emerges — follows necessarily from its foundational commitment to real patterns as the fundamental ontological category: beginning with patterns precludes the derivation of patterns from something more primitive still. The relational regress at Level II and the absence of a Level III account of the source of encounter are structural consequences of the same foundational commitment. Ontic structural realism dissolves the object into structure. It does not dissolve structure into what generates structure.

The formal statement of this result requires applying the three conditions of Definition 10.4. An eliminativist who holds that OSR descends below Level II — by eliminating objects in favour of structures — is arguing that the elimination of objects is itself a derivational descent. The formal response to this argument is precise. Condition (ii) of Definition 10.4 requires that a framework foundationally constitute a structural type as the starting point of its analysis. OSR's selection criterion for "real patterns" — Dennett's criterion refined by Ladyman and Ross: a real pattern is a pattern that is projectible, informative, and not reducible to a lower-level description that is more efficient — presupposes the concept of a real pattern as a constituted structural type. The elimination of objects in favour of patterns does not descend below Level II; it *constitutes* a structural type (the real pattern) at Level II and then analyses everything else in terms of it. This is precisely what Definition 10.4 condition (ii) identifies as foundational constitution at Level II: the structural type "real pattern" is not derived from anything more primitive within OSR's framework — it is what the framework begins from, the entity-type in terms of which the elimination of objects is stated. An eliminativist who says "there are only structures, no objects" is offering a Level II foundational constitution, not a descent below it, because the "structures" invoked already are something — they are real patterns, and real patterns already constitute a structural type at Level II whose own ground is not supplied.

Mathematical structuralism — the view that mathematical structure is not a representation of but identical to physical structure[11] — may appear to satisfy the Level I ground conditions: mathematical structures are total, self-sustaining, and formally necessary, and on the strongest reading generate all physical possibilities. The diagnostic response is precise. Mathematical structures do not generate structural differentiation from a more primitive dynamic; they simply *are* networks of structural relations. The Level I generativity condition G2 requires that the ground produce differentiation from within its own dynamic — generating the specific configuration of relata and relational web as a structural consequence rather than constituting that configuration as its primitive content. Mathematical necessity presupposes rather than produces the specific differentiated structure it characterises as necessary: it tells us that whatever structure obtains could not be otherwise, not what produces the fact that this structure obtains rather than another. A ground that simply *is* the network it grounds satisfies G1 (totality), G3 (self-sustaining character), and G4 (formal necessity) but not G2: it cannot generate what it already, foundationally, is. Mathematical structuralism therefore enters at Level II — the level of real structural patterns — rather than at Level I, and inherits the Level I gap the diagnostic framework identifies in ontic structural realism generally.

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[11]See id., Ch. 4.

relational contact. The structural web is described as a totality of real patterns; what makes a particular pattern impinge on another particular pattern, rather than on all others or none, is not derived from the programme's foundational commitments. Structural contact — the encounter between real patterns that produces emergent higher-level structure — is described rather than derived.

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That this gap is recognised as active within OSR's own recent literature confirms that the present diagnosis identifies a structural problem the tradition has felt rather than one imposed upon it from outside. Recent work proposes to replace "the traditional primacy of relata-independent structures with a dynamic framework grounded in constraint propagation," introducing what is called "structural actualism, where physical individuation, causal propagation, and locality *emerge* from stabilization regimes" rather than being presupposed.[12] A parallel strand of recent work proposes "a dynamic account of identity as structural persistence across morphic transitions"[13] — confirming that even moderate OSR continues to confront the absence the present investigation identifies as the Level II gap: a structural account of identity that does not presuppose individuation but derives it. Whether either reformulation succeeds is a further question; what is structurally significant is that both are motivated by precisely the absence the present investigation names: the lack of a Level I dynamic from which relata and the relational web are jointly derived. The convergent attempt to supply a dynamic generative ground for OSR is an independent confirmation, from within the tradition, of the gap the Structural Insufficiency Theorem formally establishes.

Foreclosure versus incompleteness. An analytic metaphysician may press the following objection: the Level I and Level II gaps iden-

[12] Alexandre Le Nepvou, "Structural Realism Beyond Static Form: Toward a Dynamically Grounded Ontology" (Preprint. Archived at PhilArchive, rec/LENSRB. Proposes "structural actualism" replacing traditional primacy of relata-independent structures with a dynamic framework grounded in constraint propagation, explicitly acknowledging that standard OSR lacks a dynamic generative ground, 2024).

[13] Alexandre Le Nepvou, "Objects Without Essence: Ontic Structural Realism and the Metaphysical Redundancy of Individuality in Quantum Physics" (Preprint. Archived at PhilArchive, rec/LENOWE. Proposes "a dynamic account of identity as structural persistence across morphic transitions," confirming that standard OSR still lacks the dynamic account of structural identity the present investigation identifies as the Level II gap, 2024).

tified above are gaps in OSR's current development, not structural foreclosures. A programme can be incomplete without being foreclosed; the absence of a Level I derivation could reflect OSR's stage of development rather than its foundational commitments. This objection must be addressed precisely, because it is the most important available resistance to the SIT's universal scope claim.

The distinction between structural foreclosure and incompleteness is stated as Definition 1.2 in Chapter 1: a framework is *foreclosed* at level k if and only if the derivation of k's conditions is precluded by the framework's foundational commitment, not merely absent from its current formulation. Incompleteness is compatible with future derivation; foreclosure is not, because the foundational commitment that constitutes the entry level *is* the stipulation of a non-latent feature as primitive, and deriving that feature from something more primitive would require abandoning the stipulation — which is to say, abandoning the foundational commitment.

Applied to OSR: the programme's foundational commitment is to real patterns as the fundamental ontological category. A derivation of real patterns from a more primitive structural dynamic would not be an advance within OSR; it would be a successor framework with a different foundational commitment. Lam and Wüthrich[14] demonstrate this at the level of the programme's most formal current development: their analysis of category-theoretic reformulations of radical OSR shows that

[14] Lam and Wüthrich, "No Categorial Support for Radical Ontic Structural Realism."

every such reformulation faces the same structural choice — either posit the categorical structure as primitive (returning to the Level II foundational constitution in a more sophisticated form) or derive it from something more primitive (abandoning OSR's distinctive foundational claim). The forced choice is not a feature of the current development of OSR; it is generated by its foundational commitment. This is foreclosure, not incompleteness.

7.3 Relational Quantum Mechanics

Facts Without a Fixer

7.3.1 The Foundational Commitment

The measurement problem in quantum mechanics is one of the deepest puzzles in the philosophy of physics. A quantum system in superposition — a particle with no determinate position, a spin with no determinate orientation — evolves according to the Schrödinger equation in a perfectly deterministic way. But when the system is measured, it acquires a determinate value — a specific position, a specific spin orientation — and the superposition appears to collapse. What is the status of the superposition before measurement? What happens during measurement? And most pressingly: is the quantum state an objective feature of the physical world, or is it relative to something — an observer, a context, a reference frame?

Carlo Rovelli's relational quantum mechanics — introduced in 1996 and developed over the following decades into a systematic programme in the philosophy of physics — answers the last question with a commitment that radicalises the relational thesis in a direction the classical traditions could not have anticipated.[15] The quantum state is relative — not to the knowledge, perceptual apparatus, or cognitive structure of a conscious observer, but to any physical system whatsoever that stands in a physical relation to the system in question. There is no quantum state simpliciter. There are only quantum states relative to physical systems. And this is not a deficiency in our representation of physical reality but a feature of physical reality itself: quantum mechanics, correctly understood, is the physics of relational facts.[16]

The foundational commitment of relational quantum mechanics is therefore: there are no absolute physical quantities.

[15]The foundational paper is Rovelli ("Relational Quantum Mechanics"). The systematic philosophical elaboration is developed across Rovelli (*Quantum Gravity*) and, in a more accessible form, Rovelli (*Helgoland*). For critical engagement and secondary development, see Laudisa ("Open Problems in Relational Quantum Mechanics") and Smerlak and Rovelli ("Relational EPR"). *Note on the Level III diagnosis and later Rovelli*: the more accessible presentation in *Helgoland* Rovelli, *Helgoland* and the critical analysis in Adlam ("Relational Observables, Quiddities, and Structural Realism") do not change the diagnostic result. Adlam's objection that cross-perspective links require a further absolute fact not provided by RQM sharpens — rather than resolves — the Level III gap: it establishes that even within RQM's development, the source of encounter (what determines which cross-perspective facts obtain) remains without a derived account. The Level III entry-level assignment holds across all stages of RQM's elaboration.

[16]Rovelli ("Relational Quantum Mechanics," 1638): "I suggest that quantum mechanics is a theory about the physical description of physical systems relative to other physical systems, and that this is a complete and self-consistent framework." The elimination of observer-privileged or consciousness-dependent formulations is central to Rovelli's programme.

Every physical fact — every quantum state, every observable value, every result of any physical interaction — is a relational fact, obtaining between two physical systems rather than as a property of one system in isolation. A particle does not have a definite position; it has a definite position *relative to* the physical system that interacts with it. A spin does not have a value; it has a value relative to the measuring apparatus. And the measuring apparatus itself has no absolute state; its state is relative to whatever system it in turn interacts with. Physical reality, at the quantum level, is entirely constituted by relations between physical systems — by facts that are irreducibly about the encounter between systems rather than the intrinsic constitution of any system taken alone.

This is the relational thesis applied at the level of fundamental physics, and it achieves something none of the classical traditions managed: a formally worked-out account of how relational facts constitute physical reality, grounded not in philosophical argument alone but in the mathematical structure of the most successful physical theory we possess.

7.3.2 Formal Achievements

Relational quantum mechanics achieves its most significant formal result at Level III of the diagnostic framework — the level of the source of encounter. The programme provides a formally explicit account of what brings specific physical systems into specific relational contact: physical interaction is the source of encounter. When two physical systems interact — when they become correlated, when information is exchanged between them — a relational fact is established relative to each. The source of encounter is not an abstract structural relation but a physical interaction event, and this interaction event is precisely characterised within the mathematical framework of quantum mechanics.[17]

This formal result is directly relevant to the diagnostic analysis. The specificity condition at Level III requires that the account show why it is this relatum that comes into contact with that relatum, rather than with all others or none. Relational quantum mechanics answers: physical interaction. A quantum system does not relate to all other systems simultaneously with equal relational salience; it relates to those systems with which it physically interacts, and the interaction is what establishes the relational fact. The source of encounter is formal, physical, and specific.

At Level V, relational quantum mechanics achieves a partial formal result. The structure of the quantum interaction event — the encounter between two physical systems that establishes reciprocal relational facts — has a formal organisation that maps onto the diagnostic framework's account of the relational event. The event is not sequential: the relational fact relative to S_1 about S_2 and the relational fact relative to S_2 about S_1 are es-

[17]The formal characterisation of interaction as the source of quantum relational facts is developed in Rovelli ("Relational Quantum Mechanics," 1639–1643). The concept of information is central: an interaction between system S_1 and S_2 establishes a quantum state relative to S_1 that carries information about S_2's observable quantities as registered by S_1. The asymmetry of the relational fact — the fact relative to S_1 about S_2, and the fact relative to S_2 about S_1 — is irreducible.

tablished co-instantaneously through the interaction, not in a temporal sequence in which one precedes the other.[18] The interaction event has distinguishable structural dimensions — the two systems that enter the interaction, and the relational facts that the interaction establishes — but these dimensions are not temporally ordered within the event itself.

7.3.3 What Relational Quantum Mechanics Cannot Reach

The formal achievements at Level III and partially at Level V are genuine and represent the most precise contemporary account of the source of encounter available in the philosophical literature. But the programme's foundational commitment — the relativity of all quantum states to physical systems — forecloses the derivation that Level I requires, and leaves Level II underdetermined in a way that the framework cannot resolve.

The Level I failure is the most fundamental. The generativity condition requires that the structural ground produce the diversity of physical systems from within its own dynamic. Relational quantum mechanics posits physical systems as its starting point — systems that interact with one another and thereby establish relational facts. The question of what the physical systems themselves are, what makes each system the

[18] This co-instantaneous character of the mutual relational facts is discussed in Rovelli ("Relational Quantum Mechanics," 1643–1645) and elaborated in Rovelli (*Quantum Gravity*, 226–229). The absence of a preferred direction in the establishment of the relational facts — neither system's perspective is ontologically prior — maps precisely onto the asymmetric but co-present structure of the relational event as characterised in Definition 2.5 of Chapter 2.

structurally specific system it is rather than another, and what the dynamic is from which the diversity of physical systems emerges, is not within the framework's scope. Systems are the given; the ground from which systems emerge is not asked.[19]

The formal derivability condition at Level I is also not met. Relational quantum mechanics does not derive the relational character of quantum states from what physical ontology presupposes and cannot assume; it argues from the mathematical structure of quantum mechanics to the conclusion that the relational character is a feature of reality. The argument is powerful — among the most powerful available in the philosophy of physics — but it is abductive rather than eliminative. It shows that the relational interpretation is the most philosophically satisfying available; it does not show that it is the unique structural result of eliminating every non-relational assumption.

The Level II failure is more specific. Relational quantum mechanics holds that each physical system has quantum states relative to other systems, but does not formally derive what makes each system the structurally specific system it is that has relational quantum states. The structural identity of a quantum system is not constituted by its relational quantum states alone — those states are established through interaction with

[19]This limitation is parallel to the limitation identified in ontic structural realism: both programmes take the relata of physical relations as their starting point and develop sophisticated accounts of the relational structure those relata constitute, while leaving the question of the relata's own constitution from a prior dynamic unaddressed. The parallel is noted — without being identified as the same structural gap — in Smerlak and Rovelli ("Relational EPR," 89–94).

other systems, but the system that is constituted by those interactions must already be the specific system it is to enter the relevant interactions. The derivation condition at Level II — showing how structural identity is produced from a prior dynamic rather than presupposed by the relational analysis — is not met.[20]

7.3.4 The Diagnostic Result for Relational Quantum Mechanics

Relational quantum mechanics enters the structural order at Level III, achieving the most formally explicit account of the source of encounter available in contemporary philosophy of physics: physical interaction establishes relational facts between physical systems co-instantaneously and specifically. It achieves partial formal results at Level V through its account of the co-instantaneous structure of the quantum interaction event. Its failure at Level I — the absence of a derivation of the Level I ground from which physical systems and their relational capacity emerge — follows necessarily from its foundational commitment to physical systems as the starting point of relational analysis. Beginning with systems precludes the derivation of systems from what is structurally prior to them. The Level II fail-

[20]This gap corresponds to what critics of relational quantum mechanics identify as the problem of the *cross-perspective* consistency of relational facts: if the state of system S_2 relative to system S_1 can differ from the state of S_2 relative to system S_3, what makes S_2 the single system whose state is in question? The answer requires a Level II account of S_2's structural identity that is not itself constituted by the relational facts relative to S_1 and S_3. The most careful treatment of this problem is Laudisa ("Open Problems in Relational Quantum Mechanics," 310–325).

ure — the underdetermination of the structural identity of the interacting systems — is a structural consequence of the same commitment. Relational quantum mechanics shows us what encounter produces at the quantum level. It does not show us what makes the encountering systems the specific systems they are, nor what the structural source is from which their capacity for specific encounter emerges.

Remark 7.2 (The Modal-Structures Alternative and Level Assignment). A specialist in Ladyman and Ross may press the following level-assignment objection. The framework's foundational commitment, the objection holds, is not to real patterns as objects — a reading that places OSR straightforwardly at Level II — but to *irreducibly modal relational structures*: dispositions of quantum fields to produce outcomes in experimental contexts, with no further categorical basis.[21] On this reading, the primary structural commitment is to modal dispositional properties of the relational event, which would place OSR's entry at Level III (physical interaction as the source of encounter for which quantum dispositions are responsible) or Level V (the modal event structure of the relational event as governed by quantum formalism).

The objection requires two responses.

The substitution test response. The modal-structures reading characterises real patterns as modal entities — irreducibly dispositional structural types whose reality consists in their counterfactual robustness and their occurrence as patterns within

[21] Ladyman et al., *Every Thing Must Go: Metaphysics Naturalized*, 119–125.

physical theory's predictive structure. Modality, on this reading, is a property of the patterns rather than a separate structural level: what makes real patterns real is that they are modal rather than categorical. Apply the substitution test (Remark 10.2): remove "real pattern" as the fundamental structural type — substitute any non-modal candidate — and ask whether OSR's prior principles retain their derivational force. The structural constraints on physical theory that OSR uses to generate its results are interpretable only because the framework is calibrated to identify modal-structural outputs as the outputs worth retaining. The selection criterion for what constitutes a genuine structural result presupposes the modal-pattern type. The prior principles collapse without it. By the substitution test, condition (ii) of Definition 10.4 is met for real patterns in their modal specification: the modal specification is constituted foundationally at Level II alongside the patterns themselves, not derived from something more primitive.

The invariance response. Even if the specialist's level reassignment were accepted — even if OSR's entry level were taken to be Level III or Level V on the modal-structures reading — the SIT applies with the same force at those levels. A framework that foundationally constitutes modal relational structures at Level V inherits the structural foreclosure of Levels IV, III, II, and I as a necessary consequence of its foundational commitment to Level V's modal event structure. A framework entering at Level III, constituting physical interaction as the structural source of encounter, inherits the foreclosure of Levels II and I. In either case, the latent register is structurally foreclosed. The level assignment determines which formal achievements the framework can reach; it does not determine whether the SIT applies. The diagnostic result of latent-register foreclosure is invariant under the level reassignment. The specialist's objection therefore affects the description of where OSR enters the structural order — a question the diagnostic framework answers at Level II for the reasons stated above — but not the theorem's verdict.

The Esfeld–Deckert thin-objects response. The most philosophically developed recent attempt to address the relational regress from within the OSR programme is Michael Esfeld and Dirk-André Deckert's minimalist ontology, which introduces "thin" objects — pointlike entities that have no intrinsic properties except their relational ones, and whose identity is constituted entirely by their structural position in the physical pattern — as the relata OSR requires without importing the classical substance-property model.[22] Thin objects are conceived as the minimal ontological commitment that blocks the relational regress at Level II: they are not substances with intrinsic natures, but they are numerically distinct somethings whose distinctness grounds the well-definedness of the structural relations between them.

The diagnostic assessment is precise. Esfeld and Deckert's thin objects represent a genuine advance at Level II: the relational identity of each thin object is structural all the way down, with no residual intrinsic property serving as a non-

[22]See Michael Esfeld and Dirk-André Deckert, *A Minimalist Ontology of the Natural World* (New York: Routledge, 2017), ch. 2–3.

relational identifier. The numerical distinctness of thin objects is grounded in their positional distinctness within the field structure, which is a structural rather than a substantial account. This is the most careful available attempt to satisfy Level II's individuation conditions without reverting to substance.

But the Level I gap is not addressed by the thin-objects move. The question the Level I conditions ask is: what produces the specific configuration of thin objects and their relational properties? Thin objects are posited as the structural terminus of the account — as what the physical world consists of, with their configuration given as a matter of how the world is, not derived from a more primitive structural dynamic. Condition G2 (generativity) is not met: thin objects do not generate the diversity of their own relational configurations from within a dynamic internal to them; that diversity is constituted with the thin objects themselves. Condition G4 (formal derivability) is not met: the minimalist ontology does not show that thin objects in their specific configuration are the unique structural result of the eliminative method applied to what physical ontology presupposes; they are the output of the ontological analysis of quantum field theory, posited as the minimal commitment adequate to the theory's formal structure.

The Esfeld–Deckert account therefore represents the most sophisticated available resolution of the Level II relational regress within the OSR programme, while leaving the Level I structural gap not only unaddressed but, in a precise sense, more visible: once the thin-objects move blocks the relational regress at Level II, the question of what produces the thin objects and their configuration stands more starkly as the remaining structural problem. The foreclosure is confirmed by the Esfeld–Deckert account's own framing: the minimalist ontology explicitly takes the configuration of thin objects as what is given in the world, and the derivation of that configuration from a more primitive structural dynamic is not within the programme's scope.[23]

7.4 The Shared Structural Gap in the Physics-Oriented Frameworks

The diagnostic results for ontic structural realism and relational quantum mechanics converge on a finding that is more significant than either result taken alone. Two programmes arriving at the relational thesis from different directions — structural realism from the philosophy of scientific theory change, relational quantum mechanics from the interpretation of quantum mechanics — and achieving genuine formal results at different levels of the structural order both share a common structural gap: neither framework derives what functions as a Level I ground from which relational structure emerges.

This convergence is not coincidental. Both programmes are committed to the relational constitution of physical reality; both develop that commitment with formal precision that the classical traditions did not achieve; and both enter the structural order at levels above the ground — Level II for ontic structural realism, Level III for relational quantum mechanics — for the same structural reason. They take as their starting point the

[23]See id., 7–12.

entities whose relational constitution they then derive: real patterns (OSR) and physical systems (RQM) are not themselves derived from a prior structural dynamic. They are the given from which the relational analysis proceeds.

The consequence is that both programmes inherit the structural insufficiency that the classical traditions also inherited, albeit from different entry points and with different formal resources. The structural web that ontic structural realism posits, and the relational facts that relational quantum mechanics derives, are both formally sophisticated accounts of what the non-latent register consists in. Neither asks what structural dynamic produced the non-latent register itself, because both begin from within it.

This is not a criticism of the programmes' scope or ambition. The philosophy of physics has pressing questions to answer about the structure of the physical world as physics has disclosed it, and ontic structural realism and relational quantum mechanics address those questions with greater formal rigour than any prior philosophical account. The structural gap is not a failure of rigour within the relevant scope. It is the consequence of a foundational commitment to physical structure — patterns, systems, quantum states, interactions — as the starting point of relational analysis. Beginning from within physical structure precludes the derivation of physical structure from what is structurally prior to it.

The latent register — the pre-physical, pre-relational structural dynamic from which the relational web of physical structure emerges — is not visible from within either programme, because both have constituted physical structure as their ground. The question of what is before physical structure, of what makes physical relation possible rather than merely describing how it operates, is not a question either programme can ask from where it stands.

7.5 The Diagnostic Result

Further In, Not Further Down

The title of this part of the diagnostic survey — the contemporary landscape has gone *further in* to the structural order, not *further down* to the ground — summarises the structural situation with precision. Ontic structural realism and relational quantum mechanics achieve formal results at levels of the structural order that the classical traditions left undeveloped. They go further into the formal analysis of relational identity, structural position, the source of physical encounter, and the structure of the quantum interaction event than Whitehead, Spinoza, Bergson, or Heidegger reached. This is genuine intellectual progress.

But progress into the structural order from within the non-latent register is not progress towards the structural ground. The formal achievements of both programmes are achievements at levels II through V of the diagnostic framework. Level I — the formal derivation of the ground from which structural identity, the source of encounter, relational yield, and the relational event all emerge — remains untouched by both. The ground that neither programme can reach is not hidden behind a technical obstacle that further work in philosophy of physics could remove. It is structurally foreclosed by the foundational commitment both programmes share: the commitment to beginning the analysis from within physical structure rather than from the structural dynamic that generates physical structure.

Summary of Chapter 7. The physics-oriented contemporary frameworks surveyed in this chapter — ontic structural realism and relational quantum mechanics — represent the most formally rigorous contemporary development of the relational thesis in the direction of fundamental physics. Ontic structural realism enters the structural order at Level II and achieves formal results there and partially at Level IV: the derivation of relational identity from structural position, and the account of emergence as non-reductive higher-level real pattern. Relational quantum mechanics enters at Level III and achieves formal results there and partially at Level V: the most precise contemporary account of the source of encounter through physical interaction, and the co-instantaneous structure of the quantum interaction event. Both fail at Level I for the same structural reason: both take as their starting point — real patterns, physical systems — what a Level I derivation would need to produce. The relational regress at Level II (OSR) and the underdetermination of system identity at Level II (RQM) are structural consequences of the shared failure. Both programmes have gone further into the formal analysis of the non-latent register than any prior tradition. Neither has descended to the structural ground that the non-latent register presupposes. The latent register from which physical structure emerges is not within the scope of either programme, because both have constituted physical structure as their absolute starting point. Chapter 8 turns to the contemporary frameworks that have developed the relational thesis in the direction of living and social *existence, and finds there the same structural gap approached from a different direction.*

CHAPTER 8

The Contemporary Landscape II

Relation Without Derivation

> *The unit of analysis in the social sciences ought to be the transaction, the dynamic, unfolding relation itself, and not the substance-like entities — whether individual or collective — that such transactions presumptively presuppose.*
>
> Mustafa Emirbayer,
> *Manifesto for a Relational Sociology*

8.1 The Question That Constitutes the Social and the Living

Physics is not the only domain in which the relational turn has been executed with formal precision and intellectual ambition. Living organisms, communities, cultures, historical processes, political formations — all have been reconceived, over the past half-century, through frameworks that make the relational constitution of entities their foundational commitment. The organism is not an individual first and a relational participant second;

it is constituted through its relations with an environment it partly constitutes in turn. The social agent is not an entity with fixed properties who subsequently enters into social relations; the agent is produced through the social relations that are prior to and generative of whatever we call individual identity.

These claims are now so widespread in the human and life sciences that they can seem self-evident — the default rather than the controversial position. But their apparent self-evidence conceals a structural question that the relational turn in biology and social theory has not asked: if organisms and social agents are constituted through their relations, what is constituted by, and what is it from which the relational constitution of organisms and agents emerges? The relational thesis cannot stop at the living and social level without either generating an infinite regress — relations all the way down, never reaching the structural source from which the relational web is generated — or covertly presupposing what it was supposed to derive: a structural level at which the capacity for relation is already given, already operative, already doing the constitutive work before the relational analysis begins.

Two research programmes represent the most philosophically developed and formally rigorous executions of the relational thesis in the direction of life and social existence: Isabelle Stengers's process philosophy, which extends Whitehead's account of actual occasions into a sophisticated contemporary framework for thinking about the sciences of the living and the political; and Mustafa Emirbayer's relational sociology, which makes the transaction — the dynamic, unfolding encounter between agents — the fundamental unit of social analysis and derives social entities from their relational constitution rather than presupposing them as its starting point.

Both programmes achieve genuine formal results. And both — as the diagnostic framework reveals with the same precision it has brought to bear on every tradition surveyed — presuppose the structural ground they cannot reach.

8.2 Stengers and Process Philosophy

The Living as the Relational Event

Note on diagnostic scope. Stengers is included in the survey not as a footnote to Whitehead but as an independent test case whose methodological commitments — to the philosophy of the life sciences, to cosmopolitical thinking, to the immanent reinterpretation of process categories — are sufficiently distant from Whitehead's own to constitute a distinct diagnostic target. The question the diagnostic framework asks of Stengers is not whether she improves on Whitehead (she does, in specific and philosophically significant ways) but whether her improvements address the structural gap the diagnostic framework identifies. The answer requires tracking the structural consequences of her most philosophically developed modification — the substitution of immanent obligation for transcendent initial aim — at the level of Level III's formal requirements, where the modification has its greatest structural ambition and its most precisely specifiable limit.

8.2.1 The Foundational Commitment

Isabelle Stengers inherits from Whitehead the conviction that the world is made of events rather than substances — that what is fundamental is not the inert, self-identical entity persisting through time but the occasion of experience, the moment of creative response to everything that has come before, the relational event through which what is actual is constituted from the vast multiplicity of its antecedents.[1] But she extends Whitehead beyond the theoretical philosophy of the *Process and Reality* tradition in two directions that are directly relevant to the diagnostic analysis.

The first extension is methodological. Stengers brings the Whiteheadian framework into contact with the actual sciences — ecology, chemistry, neuroscience, ethology — and shows that these sciences, when pursued with philosophical attention rather than read through the lens of mechanistic or reductionist presuppositions, reveal a world of entities constituted through their relations with other entities in ways that no substance ontology can accommodate.[2] A bacterium is not an object with fixed chemical properties that subsequently

[1]The most sustained philosophical development of Stengers's appropriation of Whitehead is Stengers (*Thinking with Whitehead: A Free and Wild Creation of Concepts*). For the application to the philosophy of science, see Stengers (*Cosmopolitics I*). The relationship to Deleuze's reading of Bergson and Whitehead is developed in Stengers (*Thinking with Whitehead: A Free and Wild Creation of Concepts*).

[2]Stengers (*Cosmopolitics I*, 33–67). The key methodological concept is that of "reclaiming" the sciences from their domination by positivist and reductionist philosophies — not against the sciences but in the direction of what the sciences actually show when attended to carefully.

interacts with its environment; it is the event of that interaction, constituted in and through it, capable of identity only in the relational dynamic that defines what it is. A neural process is not the firing of neurons in isolation; it is the event of relational engagement between neuronal populations, between brain and body, between organism and environment, in which what we call a cognitive event is constituted rather than produced by pre-existing components.

The second extension is political. Stengers develops from Whitehead a philosophy of the sciences that is attentive to the political conditions of scientific knowledge, the interests that shape research programmes, and the modes of attention that different kinds of scientific practice require. This extension is less directly relevant to the diagnostic analysis but is part of what gives Stengers's process philosophy its distinctive character in the contemporary landscape: it is a process philosophy that takes the social constitution of science seriously without reducing science to its social conditions.[3]

Stengers's foundational commitment is therefore Whitehead's foundational commitment, extended and contextualised but not structurally altered: the actual occasion — the relational event of creative response — is the fundamental unit of reality; what we call entities, organisms, and agents are abstractions from the flow of such events, societies of occasions

[3]Stengers (*Cosmopolitics I*, 1–35). The concept of "cosmopolitics" — thinking through the political implications of cosmology — is central to Stengers's project and distinguishes it from both Whiteheadian metaphysics in the classical mode and from science studies in the sociological mode.

in Whitehead's sense, rather than the concrete realities that generate occasions as their activity.

8.2.2 Formal Achievements

Stengers achieves formal results at Levels IV and V of the diagnostic framework, inheriting and extending Whitehead's achievements at those levels while adding dimensions that Whitehead did not develop.

At Level V, Stengers's account of the actual occasion as the unit of relational constitution carries over Whitehead's formal account of the relational event in its full complexity: the co-instantaneous structure of prehension, the satisfaction of the occasion as its completed self-constitution, the initial aim as the occasion's formal orientation towards the relevant dimension of the possibility space. These are Level V achievements in Whitehead, and Stengers inherits them while embedding them in a context that makes their application to the biological and social sciences more immediate.[4]

[4] The inheritance is not uncritical. Stengers's most important modification of Whitehead is her resistance to the *God* of Process and Reality as a theological entity with cosmological functions — specifically, the function of supplying initial aims. For Stengers, this function must be understood in immanent terms: occasions are oriented by what she calls their "obligations" — their constitutive entanglement with what they encounter — rather than by a transcendent lure. See Stengers (*Thinking with Whitehead: A Free and Wild Creation of Concepts*, 371–415). This modification is formally significant: it represents an attempt to derive Level III — the source of the orienting encounter — from immanent structural dynamics rather than a transcendent principle. But the attempt remains at the descriptive level; the formal derivation is not executed.

At Level IV, Stengers's account of emergence — particularly in the context of the life sciences — advances significantly beyond what Whitehead's own philosophical vocabulary permitted. Living organisms exhibit properties that are not reducible to the properties of their components taken in isolation: the capacity of the immune system to distinguish self from non-self, the capacity of the developing embryo to produce global pattern from local cellular interactions, the capacity of the ecological community to sustain diversity through what appears to be mutual interference. These are Level IV phenomena in the diagnostic framework — relational yields irreducible to the relata that produce them — and Stengers characterises them with a philosophical precision that draws on both Whitehead's metaphysical framework and detailed engagement with the biological sciences.[5]

8.2.3 What Stengers Cannot Reach

The formal achievements at Levels IV and V are genuine extensions of Whitehead's genuine achievements at those levels. But Stengers inherits along with those achievements the structural gap that defines Whitehead's entry into the structural order: the actual occasion as the fundamental unit of reality is not derived from what is structurally prior to it but taken as the

[5]The most developed Level IV analysis in Stengers is her account of the capacity of living systems for what she calls "symbiosis" — not in the biological sense alone but in the philosophical sense of the production of genuine novelty through the encounter of distinct living forms. See Stengers (*Thinking with Whitehead: A Free and Wild Creation of Concepts*, 437–498).

starting point of analysis. And the structural consequences of this foundational commitment are the same in Stengers as in Whitehead.

Level I is not reached. The generativity condition requires a derivation of the dynamic from which actual occasions emerge; Stengers, like Whitehead, characterises occasions in terms of their constitutive features — prehension, satisfaction, obligation — rather than deriving those features from a prior structural dynamic. The question of what occasions are before they are the specific occasions they are — what the pre-individual structural condition of the capacity for relational constitution is — is not asked. Whitehead's creativity was the nearest approximation in the classical framework, and it was, as Chapter 3 established, stipulated rather than derived. Stengers's substitution of immanent obligation for transcendent initial aim is a philosophical improvement within the framework, but it does not derive the structural source of the occasions' constitutive entanglement. It characterises that entanglement more precisely without explaining where it comes from.[6]

[6]This is the structural point behind the recurrent criticism that process philosophy, despite its anti-substantialist commitments, reinstates a form of substance at the level of the occasion itself — the occasion as a self-constituting unit that is, in the course of its process, its own structural ground. See Rescher (*Process Metaphysics: An Introduction to Process Philosophy*, 211–230) for a sympathetic treatment of this tension within Whitehead. Stengers's response would be that occasions are constituted through their obligation to what they encounter, but the structural question of what makes the encountering possible at the level of pre-occasional dynamics is not addressed.

Level II is underdetermined in the same way it is in Whitehead. The derivation condition requires a formal account of how structurally specific relata are produced from the Level I ground. Actual occasions have structural specificity — each is the specific satisfaction it achieves — but that specificity is achieved through the process of self-constitution rather than derived from a prior dynamic that produces it. The occasion's specificity is its own product, not the product of what generates occasions. This is Whitehead's gap, and it remains in Stengers.

Level III is the most philosophically interesting point of engagement with Stengers, and the treatment it receives in the footnote above requires supplementation in the main text. Stengers's substitution of immanent obligation for Whitehead's transcendent initial aim is not a simple terminological replacement; it represents a structurally considered response to the most serious objection against the classical Whiteheadian framework.

The objection is this: if the source of every occasion's specific orientation is God's primordial nature — God's conceptual valuation of all eternal objects as relevant possibilities — then the dynamic of relation is determined by a transcendent ordering principle that stands outside the relational web rather than arising from it. This conflicts with the relational commitment at the level of principle: a relational ontology cannot make the structure of every specific encounter dependent on an ordering agent that is itself structurally independent of all encounter. Stengers recognises this conflict as fundamental. Stengers[7] argues that what Whitehead intended by God's initial aim is better understood as the world's own self-organising obligation — the way in which the accumulated relational history of what has been constitutes a field of structural constraint that is neither absent (pure chaos, arbitrary outcome) nor transcendent (divine decree) but immanent: it is the weight of what has been, bearing structurally on what can be. Occasions are not oriented by a divine lure from without; they are obliged by the structural field they find themselves already within. This account is philosophically more developed than the classical Whiteheadian framework at precisely the structural point where that framework is most vulnerable, and it represents genuine philosophical progress within the tradition.

The diagnostic result, however, is unchanged, and the reason is precise. Stengers's account of obligation describes the *form* of the structural constraint on specific encounters with exactness: the immanent weight of the sedimented field constitutes the range of what a given occasion can become. What it does not supply — and what a formally adequate account of Level III requires — is the derivation of the *source* of specific relational encounter from the structural dynamic that produces occasions as individuated relata. The constraint of obligation explains why the field is not arbitrary; it presupposes that the specific occasion is already there to be constrained by it. The question of what the structural dynamic is that produces the

[7]Stengers, *Thinking with Whitehead: A Free and Wild Creation of Concepts*, 371–415.

capacity for specific encounter — why this occasion is constitutively entangled with that encounter rather than another — is not the question that obligation answers. Obligation is a formally sophisticated characterisation of the encounter's constraint structure; it is not a derivation of the encounter source. The structural gap at Level III is thereby more precisely specified in Stengers than in Whitehead, but it is not closed.

8.2.4 The Diagnostic Result for Stengers

Stengers's process philosophy enters the structural order at Levels IV and V, inheriting and extending Whitehead's formal achievements at those levels with a philosophical sophistication and scientific engagement that Whitehead's own work did not develop. The foundational commitment to the actual occasion as the unit of relational constitution — inherited from Whitehead and deepened through the concept of immanent obligation — forecloses the derivation that Levels I, II, and III require. Stengers gives us the most philosophically developed account of what occurs when occasions encounter one another and produce genuine novelty. What she does not give us — and cannot give us from where she stands — is the derivation of what occasions are before they are the specific occasions they are, what the structural source is from which the capacity for specific encounter emerges, and what the pre-occasional structural dynamic is that generates the relational web within which occasions find themselves already oriented.

The formal mapping to the three conditions of Definition 10.4 is as follows. *Condition (i)*: the actual occasion is constituted by Stengers's framework as the fundamental unit of relational analysis — not as a derived structural result but as the term through which the framework's questions about cosmopolitical obligation and immanent production are posed. *Condition (ii)*: the actual occasion is constituted as a structural type — the unit through which Stengers analyses what Whitehead calls creative advance into novelty — whose constitutive character is not derived from anything more primitive within Stengers's framework; it is taken from Whitehead's ontology of prehension and immanent causation as a starting point, not produced by an eliminative derivation within Stengers's own project. *Condition (iii)*: the structural level at which the actual occasion operates is Levels IV and V — above Level I (where a derived ground would have to be supplied), Level II (where structural individuation without regress would need derivation), and Level III (where the source of encounter would need to be formally specified). All three conditions of foundational constitution are therefore satisfied: Stengers foundationally constitutes the actual occasion (the specific structural type occupied by the framework's entry point) as a feature at Levels IV–V, foreclosing derivational access to Levels I, II, and III. This is the formal expression of what the preceding narrative has established.

8.3 Emirbayer and Relational Sociology

The Social as the Transactional

Note on diagnostic scope. Emirbayer's Manifesto is programmatic rather than formally analytical — a fact the diagnostic analysis notes rather than deplores. That it has remained the most-cited theoretical statement of relational sociology for nearly three decades without generating the formal analytical framework it calls for is itself diagnostically significant: it is evidence that the structural conditions the Manifesto's programme would need to meet are not routinely available within sociology's methodological resources. The diagnostic question is not whether the Manifesto is adequate as a programmatic intervention (it is) but whether the structural requirements of relational constitution — specifically at Levels I, III, and V — are addressed by the Manifesto's most developed conceptual resources or explicitly deferred by it. The answer is demonstrable from the Manifesto's own text, and is the more philosophically instructive for being stated by Emirbayer himself rather than discovered by external diagnosis.

Note on the selection of Emirbayer as the representative of social relational ontology. Alternative candidates within relational social theory include Bourdieu's field theory and Harrison White's network sociology. Both are more formally developed as sociological frameworks than Emirbayer's programmatic manifesto. Emirbayer is selected on criterion (b) of the Part II selection criterion: he is the theorist who *explicitly poses the structural conditions of relational possibility as a philosophical question*. Bourdieu's field theory and White's network sociology address relational structure within an empirical social science context — they analyse how fields or networks function — without posing the prior question of what makes relation structurally possible at all. Emirbayer's Manifesto does pose this question explicitly, names it as foundational, and then defers its formal resolution. It is therefore the representative not of the most formally developed social relational framework but of the most ontologically explicit one — the one that makes the structural question thematic rather than methodologically implicit. That is the criterion the diagnostic framework applies.

8.3.1 The Foundational Commitment

Mustafa Emirbayer's 1997 *Manifesto for a Relational Sociology* — one of the most cited theoretical papers in sociology in the past three decades — opens with a diagnosis that the diagnostic framework of the present study will find immediately recognisable.[8] Emirbayer distinguishes between two fundamental orientations in social theory: *substantialism*, which takes self-subsistent entities — individuals, groups, organisations, structures — as its starting point and then asks how these entities enter into relations; and *relationalism*, which takes the dynamic, unfolding encounter between entities — what Emirbayer calls the *transaction* — as its starting point and treats the entities

[8]Emirbayer ("Manifesto for a Relational Sociology"). For the reception and subsequent development, see Dépelteau ("Relational Thinking: A Critique of Co-Deterministic Theories of Structure and Agency") and the collection Crossley (*Towards Relational Sociology*).

themselves as stabilisations of transactional processes rather than their pre-given bearers.

The foundational commitment of Emirbayer's relational sociology is therefore explicit and unambiguous: the transaction is the fundamental unit of social analysis. Not individuals who subsequently transact; not structures that subsequently relate; not agents who subsequently encounter — but the transaction itself, the dynamic relational event that produces individuals, structures, and agents as its relatively stabilised outcomes rather than treating them as its pre-given inputs. Society is not a collection of individuals in relation; it is the relational dynamic from which what we call individuals are produced through the sedimentation of transactions over time.[9]

This is the relational thesis stated with maximum clarity and applied at the level of social ontology. Emirbayer draws on William James, John Dewey, George Herbert Mead, Norbert Elias, and — significantly for the present analysis — on the process philosophy tradition that Whitehead represents. The transaction is the social equivalent of the actual occasion: the relational event through which social entities are constituted in and through their encounter rather than prior to it.

[9]Emirbayer ("Manifesto for a Relational Sociology," 287–288): "What is at issue here is whether the transaction or the substance (whether individual or aggregate, material or ideal) should be taken as the primary unit of analysis... Transactional analysts refuse to see a priori the self-subsistent character of any of these terms or elements; they see them rather as emergent outcomes of their mutually conditioning, reciprocally constitutive relations."

8.3.2 Formal Achievements

Emirbayer achieves formal results at Level II and Level IV of the diagnostic framework, and these achievements represent genuine advances in the theorisation of social ontology.

At Level II, the derivation of social identity from relational position rather than prior to it is executed with a precision that goes beyond what the classical traditions achieved in the social domain. The Manifesto's central argument is that social agents — individuals, groups, organisations — do not have determinate social identities that they bring to their transactions; they acquire determinate social identities through their transactions, and those identities are constituted by the relational positions those transactions establish. Class identity is not a property individuals possess before they enter class relations; it is constituted through those relations. Gender identity is not a natural kind that social interaction expresses; it is produced through the transactional dynamics of social life. The social self is the relational event of its social constitution, not its pre-given bearer.[10]

At Level IV, Emirbayer's account of what he calls the "eventness" of social life — the irreducible novelty produced by transactions that cannot be read off from the properties of the trans-

[10]Emirbayer ("Manifesto for a Relational Sociology," 289–292). The sociological resources Emirbayer draws on for Level II include Elias's figurational sociology — particularly the concept of the "figurations" of interdependent individuals, in which the configuration of relations is ontologically prior to the individuals it relates — and Bourdieu's field theory, which constitutes agents through their positions in a field of relations. See Elias (*What Is Sociology?*) and Bourdieu (*Outline of a Theory of Practice*).

acting parties taken prior to their transaction — is a Level IV formal achievement: the characterisation of relational yield as something irreducible to the relata that produce it. Social events produce social realities that could not have been predicted from the properties of the parties before the event; they are genuinely generative of new social forms, new identities, new possibilities for further transaction that did not exist before the transactional encounter.

8.3.3 What Emirbayer Cannot Reach

The formal achievements at Levels II and IV are genuine and represent the most programmatic contemporary statement of the relational thesis at the level of social ontology. But the foundational commitment to the transaction as the fundamental unit of analysis — like every foundational commitment to a non-latent structural feature as the starting point — forecloses what the diagnostic framework identifies as the three levels the programme cannot reach from where it stands.

Level I is not reached. Emirbayer's relational sociology does not ask what the structural source is from which the capacity for social transaction emerges. Transactions are social events among social agents; the question of what makes social agents possible as agents capable of social transaction — of what structural ground is prior to and generative of the social world within which transactions occur — is not within the programme's scope. This is not a limitation of the sociological discipline as such; it is the structural consequence of taking transactions as the fundamental unit rather than asking what is structural prior to the transactional. The generativity condition at Level I requires a derivation of the dynamic from which relata and their relational capacity jointly emerge; Emirbayer begins with relata already in transactional encounter.[11]

Level III is addressed but not derived. Emirbayer's programme holds that transactions are the source of social encounter; but what brings specific transacting parties into specific transactional contact — what makes this agent encounter that agent rather than another, at this moment rather than another, in this form of encounter rather than another — is not formally derived from the programme's foundational commitments. The specificity condition at Level III requires that the source of encounter be derived from the same Level I dynamic that produces the relata; without a Level I ground, the specificity of encounter can only be described, not derived.

Level V is the level at which Emirbayer's analysis is most interesting and most incomplete. The Manifesto characterises the transaction as a dynamic, unfolding relational event that produces social entities rather than being produced by them; but the internal structure of the transaction — the formal account of

[11]This gap is noted — though not in the diagnostic vocabulary of the present study — by Dépelteau ("Relational Thinking: A Critique of Co-Deterministic Theories of Structure and Agency," 34–47), who observes that Emirbayer's relational sociology requires an account of what he calls "the conditions of possibility of the transactional" that the Manifesto does not supply. The attempt to supply such an account through reference to Dewey's and James's pragmatist ontology is instructive: pragmatism provides a rich account of how experience is transactional, but does not derive the capacity for transaction from a structural dynamic prior to experience. It begins within experience, as Stengers begins within occasions.

how the transactional event is organised, why it has the specific structure it has rather than another, and why the encounter between specific parties produces the specific transactional yield it does — is not formally developed. The transaction is named as the fundamental unit without being formally analysed as a relational event with its own internal structural organisation. This is not a failing but a limit of scope: the Manifesto is programmatic rather than formally analytical, establishing the direction of relational sociology rather than executing the formal analysis that direction requires.[12]

8.3.4 *The Diagnostic Result for Emirbayer*

Emirbayer's relational sociology enters the structural order at Level II, achieving the most programmatic and widely influential contemporary statement of the relational constitution of social identity. It achieves formal results at Level IV through its account of the irreducible novelty produced by transactional encounter. The foundational commitment to the transaction as the fundamental unit forecloses the derivation of Levels I, III, and V that the diagnostic framework requires. Emirbayer gives us the most programmatic account of what social transactions produce. He does not give us — and cannot give us from where he stands — the derivation of the structural ground from

[12] Emirbayer ("Manifesto for a Relational Sociology," 300–304) acknowledges that the programme requires further analytical development in the direction of "the internal structure of the transaction" — a direction that subsequent relational sociology has pursued with only partial success, because the Level V formal analysis of the relational event requires resources that sociology does not independently possess.

which the capacity for social transaction emerges, the formal account of what makes specific encounters specific, or the formal analysis of the internal structure of the transactional event as a relational event with its own structural organisation.

8.4 The Shared Structural Gap in the Life-and-Social Frameworks

The diagnostic results for Stengers and Emirbayer converge on the same finding as the results for ontic structural realism and relational quantum mechanics, approached from a different direction. Two programmes that take the relational thesis seriously at the level of living and social existence, and that achieve genuine formal results at their respective entry levels — Level V and Level IV (Stengers), Level II and Level IV (Emirbayer) — both share the structural gap that characterises every tradition surveyed in this Part: neither framework derives the Level I ground from which the relational dynamic they analyse itself emerges.

The convergence is instructive for the same reason it was instructive in Chapter 7. The physics-oriented frameworks and the life-and-social frameworks approach the relational thesis from opposite directions — from the smallest physical scales upward, and from the living and social scales outward — and they achieve formal results at different levels of the structural order. But they share a common structural limit: the foundational commitment to some non-latent relational feature as the starting point of analysis forecloses the question of the structural ground from which that feature emerges.

This convergence, taken together with the results for Whitehead, Spinoza, Bergson, and Heidegger established in the preceding chapters, constitutes the most powerful piece of evidence available for the claim that the structural gap identified by the diagnostic framework is not an artifact of any particular tradition's limitations but a structural feature of the relational thesis itself as it has been executed across every domain in which the relational turn has occurred.

When programmes as different as Spinoza's geometrical derivation of Substance and Emirbayer's sociological manifesto for the primacy of the transaction, as different as Whitehead's philosophy of organism and Rovelli's relational interpretation of quantum mechanics, as different as Bergson's metaphysics of duration and Ladyman and Ross's ontology of real patterns — when programmes this diverse in content, method, domain, and ambition all share the same structural gap at the same point in the diagnostic framework, the most economical explanation is not that each has independently made the same mistake but that the gap is structural: that the relational thesis, as it has been pursued in every major tradition and contemporary programme, carries within itself the structural foreclosure of the ground it presupposes.

This is what the Structural Insufficiency Theorem, to be derived in Part III, will demonstrate formally: that no account of relation that takes any non-latent structural feature as its foundational starting point can formally derive the latent register from which that feature emerges, because the act of constituting a feature as foundational is precisely the act of foreclosing the question of the feature's derivation from something more primitive still.

Note on the supplementary diagnostic engagements. Five traditions in Part II receive section-level rather than chapter-level treatment: the three treated in §§8.5–8.8 — Kant, Simondon, and Deleuze — and the two treated in §8.10 — Schaffer and Fine. In addition, the post-Heideggerian development of Merleau-Ponty's flesh ontology is engaged as a supplementary coda within §6.6. The rationale for section-level treatment is the same in each case: they are *confirmatory* rather than foundational instances of the Structural Insufficiency Theorem's pattern. The eight traditions surveyed in full chapters (Whitehead through Emirbayer) constitute the primary diagnostic survey; their entry-level assignments, failure modes, and the derivation of R1–R8 from those failures do not depend on the supplementary sections. The five additional traditions are surveyed here because each represents a methodological approach — transcendental argument (Kant), individuation-first ontology (Simondon), differential ontology (Deleuze), priority monism (Schaffer), and essence-grounded fine-structure (Fine) — that might be thought to escape the SIT's scope by a different structural route than the eight primary traditions take. Full-chapter treatment of each would repeat the diagnostic procedure at diminishing evidential return. The structural result in each case follows from the same application of the five-level grid; what the supplementary treatment contributes is confirmation that the scope extends to these approaches as well.

8.5 Kant: Transcendental Conditions and Their Ground

The diagnostic apparatus of the present investigation is, in its logical form, a result about the conditions of possibility for a certain kind of derivation. This places it in explicit dialogue with the transcendental tradition — and most precisely with Kant's *Critique of Pure Reason*, the foundational instance of the methodology the Structural Insufficiency Theorem claims to supersede. To assert that the SIT applies to transcendental argument without engaging Kant directly is to leave the strongest case against the scope claim unaddressed.

The Foundational Commitment

Kant's foundational commitment is to the transcendental unity of apperception: the formal "I think" that must be able to accompany all my representations, whose formal unity is the condition of possibility for all objective experience.[13] The diagnostic question is at which structural level this commitment enters the account. The answer is Level V, at the possibility dimension, for the following reason.

[13] Immanuel Kant, *Critique of Pure Reason*, translated by Paul Guyer and Allen W. Wood, Originally published 1781/1787 (Cambridge: Cambridge University Press, 1998), B131–132.

The transcendental unity of apperception is the condition under which representations can belong to a single unified subject and thereby constitute an experience of objects — of things standing in relations, succeeding one another in time, grounding causal inference. What the unity of apperception provides is precisely the formal structure within which relational events of consciousness are possible: without it, there is no unified field within which objects can be encountered in relation to one another. This is not Level IV (what encounter produces) or Level III (what brings specific entities into encounter) or Level II (what makes each relatum the specific relatum it is) or Level I (the structural ground from which relata and relations emerge). It is Level V: the structural condition governing the formal organisation of the relational event itself — not below the event but at the level of the event's possibility-structure. The unity of apperception does not generate relata; it is the formal condition under which any relational event can occur within experience at all. This is the possibility dimension of Level V, and it is precisely parallel to Heidegger's Dasein as the entity for whom being is always ahead of itself: both are formal conditions constituting the directed field within which encounters occur, taken as primitive rather than derived.

The fact that Kant's stated subject matter is the conditions of possible *experience* rather than relational ontology as such does not exempt the transcendental framework from the SIT. The SIT applies to any framework that foundationally constitutes a non-latent structural feature. The transcendental unity of apperception is precisely such a feature: it is a structural condition at Level V taken as primitive — the terminus at which Kant's derivation stops, from which it builds upward towards the categories and principles. The question of what structurally precedes the unity of apperception — what generates the formal condition of unified consciousness rather than presupposing it — is not a question Kant can answer within the critical philosophy without abandoning the foundational commitment that constitutes it as critical philosophy. The SIT therefore applies to Kant on the same basis as to every other tradition in the survey: the formal structure of transcendental argument foundationally constitutes the unity of apperception as the Level V terminus of derivation, and this constitution satisfies the conditions of Definition 10.4. Kant's entry level is the possibility dimension of Level V.

A Kantian may object that the unity of apperception is not foundationally constituted but transcendentally *derived* within the Critique — that the Transcendental Deduction argues for it rather than merely positing it. The diagnostic response is precise. What the Transcendental Deduction establishes is that the categories of the understanding are derived *from* the unity of apperception; it does not derive the unity of apperception from conditions more primitive than itself.[14] The

[14] See Melissa McBay Merritt, "Kant's Argument for the Apperception Principle," Archived at PhilArchive, rec/MERKAF. Establishes that the Transcendental Deduction proceeds from the unity of apperception as its highest starting point and derives the categories from it — not it from conditions more primitive than itself. Confirms the diagnostic claim that the unity of apperception is foundationally constituted rather than derived from below, *European Journal of Philosophy* 19, no. 1 (2011): 59–84.

Deduction proceeds from apperception — from the claim about self-consciousness as the condition for any representation being one's own — and derives the categories, schemata, and principles as consequences flowing from it.[15] The unity of apperception is therefore the Deduction's highest starting point:[16] a transcendentally clarified condition rather than a derived result. This is precisely what the diagnostic framework identifies as foundational constitution — the terminus at which Kant's critical derivation begins (in the direction that issues in categories and principles), not a term reached by descent from more primitive structural conditions. A derivation of the unity of apperception from conditions more primitive than itself would not be the Transcendental Deduction; it would require a framework that Kant's own critical philosophy explicitly forecloses as unavailable to cognition.

Formal Achievements: The Second Analogy

Kant's formal achievement within the possibility dimension of Level V is the most precise transcendental derivation of temporal relations in the tradition. The Second Analogy of Experience

[15]See also Valerii Tereshchenko, "Kant's Theory of Experience and the Transcendental Unity of Apperception," Archived at PhilPapers, rec/TERKTO-2. Confirms the directional structure of the Critique of Pure Reason: the unity of apperception is the condition from which the schematism, categories, and principles of experience are derived. The formulation of the general principles of experience is "the final deductive link" of the justification of knowledge — not a condition from which the unity of apperception is in turn derived, *Visnyk of the Lviv University. Series Philosophical Sciences* 31, no. 1 (2024).

[16]Kant, *Critique of Pure Reason*, B136.

derives the category of causality as a condition of the possibility of objective temporal experience: the determination of an event as objectively occurring at a *before* rather than an *after* requires the application of the concept of cause to the succession of perceptions, because without the concept of cause, the succession is merely subjective rather than a representation of succession in the object.[17] In its formal structure, this is an eliminative derivation within a bounded domain: ask what objective temporal experience cannot do without, establish causal determination as what it cannot do without, conclude that the category of causality is a transcendental condition of experience. The form is precisely the form the present investigation employs.

The Structural Foreclosure

The transcendental unity of apperception is the *highest principle* of Kant's analytic of principles:[18] it is not derived from conditions more primitive than itself but is the transcendental ground from which the categories and principles of pure understanding are derived. By the Horizon Lemma, a framework whose derivational resources are calibrated to the formal "I think" and to the domain of possible experience cannot reach what temporal relations presuppose before any consciousness that could experience them. The Second Analogy derives causal necessity as the condition of possible temporal experience; what it cannot derive is the structural ground of temporal relations as such — the pre-experiential structural source from which tem-

[17] Id., B232–256.
[18] Id., B136.

poral asymmetry emerges and from which the very formal conditions of experience are constituted. Kant's entry at Level V's possibility dimension structurally forecloses Levels IV, III, II, and I. The contrast with Bergson is precise and illuminating: both enter the structural order at Level V — Bergson at the actuality dimension, Kant at the possibility dimension — and both are foreclosed below the floor of their respective points of entry for the same structural reason the SIT establishes.

Diagnostic Result

Kant enters the structural order at the possibility dimension of Level V. He achieves the most rigorous transcendental derivation of causal necessity as the condition of objective temporal experience available in the tradition. The foundational commitment to the transcendental unity of apperception forecloses Levels IV, III, II, and I: the framework cannot reach the pre-experiential structural ground from which the very conditions of experience emerge, because those conditions lie below the floor at which the framework's calibration begins.

8.6 Simondon : The Pre-Individual and Its Structural Status

Of all the thinkers the diagnostic framework might be applied to, Gilbert Simondon went furthest towards what this investigation identifies as the latent register. *L'individuation à la lumière des notions de forme et d'information* (1964) explicitly proposes a pre-individual ground — a domain of metastable tensions and singularities from which individual forms emerge — as the necessary foundation for any account of individuation and relation.[19] His programme is the closest philosophical predecessor to the structural claim of the present investigation, and the question of whether he reached the latent register or was foreclosed before it is the most consequential diagnostic question Part II can ask.

The Foundational Commitment: The Pre-Individual Field

Simondon's foundational move is to argue that individuation cannot be understood if one starts from the already-individuated individual: every philosophy that takes the individual as its starting point already presupposes what needs to be explained. His alternative is the pre-individual field: a domain of metastable potential energy, characterised by singularities and incompatible polarities that have not yet been resolved into the stable forms of the individual. The pre-individual field is always already in excess of any individual that emerges from it, and it accompanies the individual throughout its existence as the reservoir from which further individuation can proceed.

Entry Level and Formal Achievements

Simondon characterises the pre-individual field through positive structural features he takes as given rather than derived: metastability (a state of higher potential energy, neither stable

[19] Simondon, *L'individuation à la lumière des notions de forme et d'information*.

nor unstable), incompatible polarities (structural tensions not yet resolved), and singularities (points of concentrated structural intensity serving as nucleation centres). These features are constituted as the character of the pre-individual field itself, not derived from conditions more primitive than the pre-individual field.

This is the point at which the SIT applies. Simondon foundationally constitutes the pre-individual field as having a specific structural character — metastable, polar, singular — and this constitution fixes his entry level at Level I. The distinction between Level I and the latent register is precise and load-bearing here. Level I is the structural ground of the relational order: the source from which relata, encounter, yield, and the relational event emerge as derived structural results. The latent register is what is presupposed by any foundational commitment at Level I or above: the structural conditions that make Level I possible. The difference is exactly the difference between the ground and the ground's own ground. Simondon's pre-individual field is Level I: it is the source from which individuals emerge. But the pre-individual field itself has a character — metastability, polarity, singularity — that is not derived from anything more primitive. That character is constituted as foundational. What the latent register would need to supply is a derivation of *why* the pre-individual ground has those features rather than others — why the ultimate ground of individuation is metastable rather than stable, polar rather than uniform, concentrated at singularities rather than homogeneously distributed. Simondon's framework provides no such derivation, because those features are what the pre-individual field simply *is* in his account, not structural consequences of something more primitive. This is the point at which the Horizon Lemma applies: the framework's derivational resources are calibrated to the domain of the pre-individual field as characterised, and cannot reach below the level at which that characterisation was constituted.

Above his entry level, Simondon's formal achievements are considerable. His account of individuation as *transduction* — the progressive resolution of pre-individual metastability into stable structural forms across a front of individuation — addresses Level III with greater precision than most traditions surveyed: the encounter that drives individuation is a structural consequence of the pre-individual field's metastable tensions, not an external perturbation imposed on already-formed individuals. His account of the transductive individual as always accompanied by a pre-individual remainder addresses Level II distinctively: structural identity is not fixed but always accompanied by pre-individual potential that subsequent encounters can draw on. These are genuine formal achievements that place Simondon as the most structurally advanced thinker in the tradition's approach to the latent register.

The Structural Foreclosure

By the Horizon Lemma, the derivational resources of Simondon's framework are calibrated to the domain of the pre-individual field as constituted: they can operate on metastability, polarity, and singularity as given structural primitives but cannot resolve what lies below the level at which those primitives are constituted. What lies below them is the question the latent register identifies: why *metastability* rather than stability or instability? Why *polarity* rather than uniformity? Why *singularity* rather than homogeneous distribution? These questions are not answerable from within Simondon's framework because the framework has constituted metastability, polarity, and singularity as foundational structural primitives — as what the pre-individual field *is*, not as structural consequences of something more primitive still.

Simondon went further than any thinker surveyed in Part II towards the structural direction this investigation identifies as the latent register. He went further and stopped at Level I — stopped there, as the SIT predicts, because the act of constituting the pre-individual field with its specific structural character was the act of foreclosing the question of that character's structural origin.

On Metastability as Analytically Tied to Pre-Individual Status

A Simondonian may object that 'metastability' is not a structural feature added to the pre-individual field from outside but is analytically tied to pre-individual status as such — to be pre-individual just is to be metastable, since resolution into individuality is precisely what eliminates the pre-individual domain. On this reading, demanding a derivation of why the pre-individual field is metastable is like demanding a derivation of why bachelorhood involves being unmarried. The diagnostic response requires precision here. The most rigorous secondary treatment of Simondon's ontology establishes that metastability designates a positively characterised energetic state, not merely the grammatical negation of individual stability.[20] Simondon's own term for Being prior to individuation — "a supersaturation of being" — names a state of excess potential, not an absence; and Combes establishes that metastability specifies a system that "contains potentials that cannot be resolved within its current form": a positive structural condition specifiable independently of its contrast with the individually stable. This is confirmed by recent Simondon scholarship. Boucon and Deepseek-R1 (2025) note that the pre-individual field, while characterised as 'formless', is simultaneously 'traversed by tensions, energy gradients, and disparities' — and that specific disparations (potential differences, temperature differentials, topological singularities) are what trigger individuation rather than the bare concept of pre-individual status.[21] Rijssenbeek and

[20]Muriel Combes, *Gilbert Simondon and the Philosophy of the Transindividual*, translated by Thomas LaMarre, The most rigorous peer-reviewed secondary treatment of Simondon's ontology in English. Combes establishes in chapter 1 that metastability designates a positively characterised energetic state — a "supersaturation of being" — not merely the grammatical negation of individual stability: "a system is metastable insofar as it contains potentials that cannot be resolved within the current form of the system." This confirms that metastability carries positive structural content (specific potential gradients, topological singularities) that cannot be analytically read off from the concept of pre-individual status alone — the content the diagnostic analysis identifies as foundationally constituted rather than derived (Cambridge, MA: MIT Press, 2013), ch. 1.

[21]Jean-Louis Boucon and Deepseek-R1, "Simondon's Concept of Individuation Revisited by Ontology of Knowledge" (Preprint. Archived at PhilArchive, rec/BOUSCO-5 (issued 2025-03-28). Confirms that Simondon's pre-individual field, while characterised as 'formless', is simultaneously traversed by tensions, energy gradients, and disparities — a positively speci-

Blok (2025) confirm that metastability names 'a state of disequilibrium carrying potentials' — a positively specified structural condition, not merely the grammatical negation of individual status.[22] If 'metastability' were only analytically tied to 'pre-individual', there would be no structural content to the claim that a specific disparation triggers individuation rather than another; the singular points that serve as nucleation centres would be undifferentiated from any other configuration of the pre-individual field. The diagnostic claim is not that the concept of metastability is underived but that the specific architecture that metastability takes in Simondon's account — the particular geometry of tensions, gradients, and singular points — is constituted foundationally rather than derived from conditions more primitive than the pre-individual field itself. A derivation of metastability from more primitive conditions would need to show why the structural ground is at higher potential than the stable state — why, in other words, a system at the ground level cannot settle into equilibrium but must maintain its supersaturation. That question is not answerable within a framework that treats supersaturation as the ground's defining character. This is the latent register gap: not the concept, but the structural content of metastability as Simondon specifies it.

fied structural content distinguishing metastability from a mere grammatical negation of individual status, 2025).

[22]Julia Rijssenbeek and Vincent Blok, "New Encounters Between Life and Technology: Simondon and the Case of Synthetic Biology," Published online 18 April 2025. Archived at PhilArchive, rec/JULNEB-2. Establishes metastability as a state of disequilibrium carrying potentials — a positively specified structural condition with determinate content, not merely the grammatical negation of individual status, *Foundations of Science*, 2025,

Diagnostic Result

This is the distinction the diagnostic framework has been tracking since Chapter 2: the difference between Level I — the structural ground from which relata, encounter, and yield emerge — and the latent register, which is what any Level I foundational commitment already presupposes. Simondon's pre-individual field is Level I. The latent register is what Simondon cannot ask about without stepping outside his own framework.

Simondon enters the structural order at Level I, approaching it more closely than any other thinker in this survey. He achieves formal results at Level I (partial: the pre-individual field meets totality and generativity conditions but constitutes its structural character as foundational rather than derived), Level III (encounter as transductive resolution of metastable tension), and Level II (structural identity as always accompanied by pre-individual remainder). The foundational constitution of the pre-individual field as metastable, polar, and singular forecloses the latent register. The derivation of what the pre-individual field's structural character consists in — why the pre-individual ground has the character it has rather than another — falls below the horizon Simondon's foundational commitment establishes.

8.7 Deleuze: Difference, the Virtual, and the Latent Register

Difference and Repetition (1968) is the philosophically proximate text to the present investigation that the diagnostic apparatus has not yet engaged.[23] The structural parallels between Deleuzian ontology and the present investigation are too close to be incidental, and the absence of a diagnostic verdict on Deleuze would be read by any competent continental referee as strategic evasion. The verdict must be stated directly.

The Structural Parallels

Several of Deleuze's central structural moves map onto terms the present investigation employs. *Intensive difference* names the differential charge any entity carries as its own — what makes it *this* entity rather than a generic instantiation — mapping structurally onto the invariant structural identity that enters relational encounter bearing its specific differential character. *The virtual* names the domain of differential singularities and pre-individual intensities from which actual individuals are produced, mapping structurally onto the latent register. *The differenciator* — the element through which the virtual actualises itself — maps onto the source of encounter at Level III. *Repetition-with-difference* — the claim that what recurs is never the same, that every return produces novelty rather than reproducing the identical — maps onto sedimentation: the irreversible deposit that every compatible encounter makes in the structural record

[23] Deleuze, *Difference and Repetition*.

of both relata. And *the anti-representationalist thesis* — that individuals and identities are products of the virtual, not primitives — anticipates the present investigation's central diagnostic finding.

The Foundational Commitment: Intensive Difference as Primitive

The diagnostic question is whether Deleuze *derives* the primacy of difference from conditions more primitive than difference, or whether he *asserts* it against the dogmatic image of thought as an alternative foundational commitment. On careful reading, the answer is the latter. *Difference and Repetition* demonstrates, through extraordinary analysis, that the dogmatic image's own categories — identity, resemblance, opposition, analogy — cannot account for the novelty and multiplicity they claim to explain. This demonstration shows that the dogmatic image fails. It does not derive that intensive difference is the structurally necessary alternative rather than one of several possible alternatives to the dogmatic image's primitives.

This is the point at which the SIT applies. Intensive difference is constituted as the foundational primitive from which individuals, identities, and the apparent priority of identity over difference are all derived. It is not itself derived from conditions more primitive than intensive difference; it is asserted as the alternative to the dogmatic image. By the Horizon Lemma, Deleuze's framework is calibrated to intensive difference and to the virtual as the domain of differential singularities. The framework cannot reach what intensive difference itself presupposes: the structural conditions under which difference rather than identity is primary, the derivation of why the differential is the fundamental character of what is real.

Two independent lines of support confirm this diagnostic judgment. The first is from within Deleuzian scholarship. Rae[24] establishes, across three specific structural junctures in *Difference and Repetition* — the virtual-actual movement, the transcendental conditions of different forms of thinking, and the relationship between those forms — that Deleuze's affirmation of difference's primacy depends at those junctures on a form of identity he has not derived: the identity that constitutes the common rather than the identical. The dependence Rae identifies is precisely the foundational constitution the SIT names: intensive difference is asserted as primary at these junctures rather than derived from conditions more primitive than difference itself. The second line of support is structural rather than interpretive. *Difference and Repetition* proceeds by showing that the dogmatic image's categories cannot account for novelty or multiplicity, then asserting intensive difference as the alternative. This is the eliminative move executed within a bounded domain, and its conclusion is exactly what the diagnostic framework would predict: the framework shows what fails (the dogmatic image) but cannot show, from within its own resources, that the alternative it proposes (intensive difference as primary) is itself structurally necessary rather than merely structurally coherent. The SIT predicts this result: a framework that constitutes intensive difference as its ground cannot, from within that

[24]Rae, "Three Senses of Identity and Deleuze's Differential Ontology," 383–387.

constitution, derive why intensive difference rather than some other structural primitive must be the ground. The primacy of difference is constituted, not demonstrated from what is more primitive still.

Diagnostic Result

Deleuze enters the structural order at Level I through the virtual as the domain of pre-individual differential singularities — approaching the latent register's vicinity more closely than most traditions, and achieving formal results at Levels III, IV, and V with precision that exceeds most relational traditions surveyed. The foundational constitution of intensive difference as the primary structural fact — asserted against the dogmatic image rather than derived from conditions more primitive than difference — forecloses the latent register. The question of why intensive difference is structurally primary, what the structural ground of differential primacy is, falls below the horizon Deleuze's foundational commitment establishes.

The diagnostic verdict is not that Deleuze failed. It is that Deleuze identified the right structural territory and stopped at the same boundary every tradition stops at — the boundary between the constituted foundational primitive and the structural conditions from which that primitive would need to be derived in order for the investigation to continue. *Difference and Repetition* is the most powerful philosophical exploration of the territory immediately adjacent to the latent register. It makes the latent register visible as a territory. It does not reach it.

8.8 Two Further Contemporary Instances of the Structural Pattern

The structural pattern the SIT establishes — that every framework in relational ontology constitutes a non-latent structural feature as its foundational primitive — is not confined to the classical traditions and the mid-twentieth-century frameworks surveyed above. Two independent preprints from 2026, both archived at PhilArchive, provide evidence that the pattern continues to characterise sophisticated contemporary attempts, including those explicitly directed at the question Before Relation poses.

Sueyoshi[25] proposes a "meta-ontological framework that investigates the formal conditions under which ontological description is possible at all" — asking, in terms directly parallel to those of the present investigation, "What formal structures must be presupposed in order for any ontological description to be possible at all?" The framework introduces the "Infinite Phase" as a "pre-distinct transcendental condition required for

[25] Makoto Sueyoshi, "Translational Ontology: A Meta-Ontological Framework for the Conditions of Describability" (Preprint. Archived at PhilArchive, rec/SUETOA. An independent 2026 attempt to derive the structural conditions under which ontological description is possible at all, asking: "What formal structures must be presupposed in order for any ontological description to be possible at all?" Introduces an "Infinite Phase" as a "pre-distinct transcendental condition required for differentiation to arise" — constituted, not derived. Translation, response, and Qualions are constituted as foundational primitives, placing the investigation within the non-latent register rather than below it: an independent confirmation that even sophisticated contemporary attempts to derive structural conditions of ontological description remain within the scope of the SIT's foreclosure, 2026).

differentiation to arise," and derives from it a set of concepts — translation, response, and Qualions — as the minimal units of any ontological description.

A careful reader will note that the Infinite Phase, described as "pre-distinct," might itself appear to approach the latent register. The diagnostic question is therefore not whether Sueyoshi recognises a pre-categorial domain, but whether the Infinite Phase itself is derived from conditions more primitive still or constituted as foundational. The answer is the latter. Sueyoshi does not ask: why does the pre-individual domain have the character of being "Infinite" and "Phase-like" rather than some other character? These are the structural features with which the Infinite Phase is constituted rather than features derived from conditions more primitive. Moreover, the operational vocabulary of translation, response, and Qualions is constituted as the minimal description of ontological possibility — each introduced as what the Infinite Phase *is* rather than as a consequence of something prior. By the Horizon Lemma, the framework's derivational resources are calibrated to the Infinite Phase as characterised; the question of what makes the Infinite Phase the Infinite Phase rather than some other pre-distinct condition falls below the horizon the foundational constitution produces. This is the structural situation the SIT predicts, and Sueyoshi's framework instantiates it at the meta-ontological level: even the most penetrating contemporary attempts to derive the conditions of ontological description find themselves inside the non-latent register rather than below it.

Rogers[26] proposes a relational, processual ontology in which "a pre-individuated field of relational possibility is differentiated through constraint; probability functions non-epistemically as a measure of relational compatibility; and stable identity emerges as a re-enterable regime of coordinated interaction." The framework constitutes the "pre-individuated field of relational possibility" as its foundational primitive. The pre-individuated field already has specific structural features — relational possibility, the capacity for symmetry-breaking under constraint — that are constituted as primitive rather than derived from conditions more primitive still. By the Horizon Lemma, the framework's derivational resources are calibrated to the domain of relational possibility and above. The structural question of what makes the field a field of relational possibility, rather than of non-relational or proto-relational possibility, falls below the established horizon.

These two cases are not failures. Each asks the right question with considerable formal sophistication. Their significance for the present investigation is structural: they confirm, as recent

[26]Timothy M. Rogers, "Determinacy as Relational Achievement: Symmetry, Constraint, and Individuation in Physics, Biology, and Interactive Formal Systems" (Preprint. Archived at PhilArchive, rec/ROGDAR. Proposes a relational, processual ontology in which "a pre-individuated field of relational possibility is differentiated through constraint; probability functions non-epistemically as a measure of relational compatibility; and stable identity emerges as a re-enterable regime of coordinated interaction." Constitutes the "pre-individuated field of relational possibility" as its foundational primitive, whose structural features (relational possibility, symmetry-breaking, constraint formation) are given rather than derived from conditions more primitive still: a 2026 independent instance of the structural pattern Before Relation formalises as the SIT, 2026).

independent evidence, that the SIT applies not only to the thirteen traditions surveyed in the preceding chapters but to the live contemporary inquiry. The foreclosure the theorem predicts is not a historical artifact. It is the structural situation of relational ontology in 2026.

8.9 The Diagnostic Result for Part II

The diagnostic survey of Part II is now complete. Twelve chapters have applied the five-level diagnostic framework of Chapter 2 to thirteen traditions — Whitehead, Spinoza, Bergson, Heidegger, ontic structural realism, relational quantum mechanics, Stengers, Emirbayer, Kant, Simondon, Deleuze, Schaffer, and Fine — and have found in each the same structural pattern.

Every tradition enters the structural order at a determinate level, defined by its foundational commitment to a specific non-latent structural feature as the starting point of analysis. Every tradition achieves genuine formal results at and above its entry level; those results are real, important, and represent significant intellectual achievements that the diagnostic framework has taken seriously rather than dismissed. And every tradition fails to derive what lies below its entry level, because the act of making a structural feature foundational is the act of constituting it as the ground — and once a feature is constituted as the ground, the question of what grounds it is, within the framework constituted by that commitment, literally not askable.

The pattern of entry levels across the traditions is itself significant:

Tradition	*Entry Level*	*Formal Achievements*
Spinoza	Level I (partial)	G1, G3, G4 (qualified); not G2
Schaffer	Level I (partial)	G1, G2 (partial), G3; not G4; Level II
Fine	Level III	Levels III, IV
Whitehead	Level V	Levels IV–V
Bergson	Level V (actuality)	Level V (actuality), Level IV (partial)
Heidegger	Level V (possibility)	Level V (possibility); Levels II, III (qualified)
Kant	Level V (possibility)	Level V (possibility; causal necessity)
OSR	Level II	Levels II, IV (partial)
RQM	Level III	Levels III, V (partial)
Stengers	Levels IV–V	Levels IV–V
Emirbayer	Levels II, IV	Levels II, IV
Simondon	Level I (closest approach)	Level I (partial), Levels II, III
Deleuze	Level I (virtual)	Level I (partial), Levels III–V

A note on the precise form of convergence. The table above records entry levels that differ across all thirteen traditions — from Whitehead and Bergson at Level V to Simondon and Deleuze at Level I's immediate vicinity. The convergence established by the diagnostic survey is not a convergence of entry level. It is a convergence of failure *type*: every tradition, regardless of where it enters the structural order and how deep within the non-latent register it operates, forecloses the latent register as a structural consequence of its foundational constitution. A tradition entering at Level I forecloses the latent register for the same structural reason as one entering at Level V — because the act of constituting any non-latent feature as foundational generates the domain-calibration effect established by the Horizon Lemma and thereby closes off derivational access to what lies below. The pattern is therefore: identical failure type, differentiated entry level. This distinction matters because it determines what the survey establishes — not that all traditions were equally close or equally far from the ground, but that no tradition, however close, could have reached it by the route it took.

Read as a map of the non-latent register, the thirteen traditions together constitute a remarkably comprehensive account of what relational ontology has illuminated within the structural order. Level I has been approached most closely by Spinoza, Simondon, and Deleuze — Spinoza meeting three of its four conditions but unable to satisfy generativity; Simondon constituting the pre-individual field with determinate structural character (metastable, polar, singular) rather than deriving that character from conditions more primitive still; Deleuze constituting intensive difference as the primary structural fact rather than deriving why difference is primary. Simondon went furthest in the right structural direction of any thinker in the survey. Level II has been developed most formally by ontic structural realism, Emirbayer, and — in a distinctive way through the analysis of relational identity as always accompanied by pre-individual remainder — Simondon. Level III has been developed most precisely by relational quantum mechanics and, through the account of transduction, Simondon. Levels IV and V have received the deepest treatment from Whitehead, Stengers, Heidegger, and — with particular attention to the problem of individuation and differenciation — Deleuze. Kant contributes the most rigorous transcendental derivation at Level V's possibility dimension.

What the map reveals — and what none of the traditions surveyed could see from within its own framework — is that the structural ground, Level I, has not been derived by any tradition. Spinoza approached it most closely and achieved the most rigorous account of what such a ground would need to be total and self-sustaining. But the generativity condition — what a Level I ground must supply in order to produce the diversity of relata and the dynamic of their encounter from within its own dynamic rather than through logical entailment of a homogeneous totality — remains unsatisfied. And the formal derivability condition — that the ground itself be shown to be the unique structural result of the eliminative method applied to what every relational ontology presupposes — has been attempted only by Spinoza, and achieved there only in the qualified sense established in Chapter 4: the generative internal dynamic that the condition also requires remains underived in every tradition.

The diagnostic survey has done two things simultaneously. It has illuminated the genuine achievements of every tradition it has examined, treating each with the seriousness its intellectual rigour deserves. And it has established, with accumulating force across thirteen traditions spanning the seventeenth through twenty-first centuries of serious philosophical work — from Spinoza's geometrical derivation of Substance to Deleuze's virtual ontology of intensive difference, and most recently to Simondon's pre-individual field as the closest philosophical approach to the latent register's structural territory — that those achievements are uniformly achievements within the non-latent register, and that the latent register from which the non-latent register emerges remains uncharted, unasked, and formally unreached.

Part III turns to the Structural Insufficiency Theorem: the formal proof that this pattern of failure is not accidental but structurally necessary, and the formal specification of what a satisfying response to it would require.

Summary of the additional diagnostic engagements and of Part II. Kant enters the structural order at the possibility dimension of Level V, achieving the most rigorous transcendental derivation of causal necessity as the condition of objective temporal experience, while the foundational commitment to the transcendental unity of apperception forecloses Levels IV, III, II, and I. Simondon enters at Level I, going further than any other thinker surveyed towards the structural territory the present investigation identifies as the latent register; the foundational constitution of the pre-individual field as metastable, polar, and singular forecloses the latent register for the same structural reason as every other entry. Deleuze enters at Level I through the virtual as the domain of pre-individual differential singularities, achieving formal results at Levels III, IV, and V with conceptual precision that exceeds most traditions surveyed; the foundational constitution of intensive difference as the primary structural fact — asserted against the dogmatic image rather than derived — forecloses the latent register. Schaffer's priority monism, examined in §8.10, enters at Level I with the highest entry point of any position in the analytic metaphysics tradition; it achieves formal results at Levels I (partial: totality and self-sustenance) and II (derivation of the structural identities of finite things from the cosmos's dependence structure), while the foundational constitution of the cosmos as the unexplained intrinsic whole forecloses the latent register for the same structural reason as Spinoza's Substance. Fine's essence-grounding, examined in §8.10, enters at Level III, achieving the most formally precise analytic derivation of encounter specificity available; the foundational constitution of essences as Level II primitives forecloses Level I and the latent register by the Horizon Lemma. Taken together with the results of all preceding chapters, the diagnostic survey has established across thirteen tradi-

tions spanning the seventeenth through twenty-first centuries, the structural pattern that Part III will demonstrate formally: every tradition in relational ontology that constitutes a non-latent structural feature as foundational inherits, as the structural consequence of that constitution, the foreclosure of the latent register from which the constituted feature emerges. The non-latent register has been mapped with greater precision and comprehensiveness than it has been at any prior point in the history of philosophy. The latent register from which it emerges — the structural ground from which every relational ontology begins without knowing that it has begun — awaits the derivation that Part III now undertakes.

8.10 Two Analytic Traditions : Schaffer and Fine

The diagnostic survey covers thirteen traditions drawn from process philosophy, analytic metaphysics of science, phenomenology, continental ontology, and social theory. A Phase 3 referee whose home ground is the grounding and dependence literature of the past two decades will press a targeted objection: the two philosophers who have done the most systematic work on the formal structure of ontological priority within that literature are absent from the survey. Jonathan Schaffer's priority monism and Kit Fine's essence-grounding are the dominant contemporary analytic accounts of how entities are constituted by and depend upon more fundamental features of the world. If either account has achieved what the diagnostic framework requires, the Structural Insufficiency Theorem does not apply universally to the analytic tradition. The present section applies the diagnostic method to both.

8.10.1 Schaffer : Priority Monism and the Derivation of the Whole

The Foundational Commitment

Jonathan Schaffer's priority monism, developed systematically in his 2010 *Philosophical Review* paper and grounded in his broader account of the metaphysics of grounding, makes one of the most formally developed attempts in contemporary analytic metaphysics to specify what it would mean for a structural whole to be prior to and generative of its parts.[27] The central claim is that there exists exactly one fundamental concrete object — the cosmos as a whole — and that every finite thing exists as a dependent part of rather than a building block of that whole. Ontological priority runs from whole to part, not from part to whole. The cosmos is the fundamental entity; tables, persons, particles, and fields are derivative from it by parthood and constitutive dependence.

The foundational commitment is therefore to the cosmos as the Level I ground in the diagnostic framework's sense: the totality from which the plurality of distinct things is generated as a structural consequence of the whole's own internal articulation. Schaffer is explicit that priority monism is a grounding thesis,

[27]Schaffer ("Monism: The Priority of the Whole"). The broader grounding framework is developed in Schaffer ("On What Grounds What"). For the standard reference collection for grounding and metaphysical dependence, see Correia and Schnieder (*Metaphysical Grounding: Understanding the Structure of Reality*).

not a composition thesis: the claim is not that only one thing exists, but that one thing grounds all others in an asymmetric, irreflexive, and well-founded priority order.[28]

Formal Achievements

Priority monism achieves genuine formal results at Level I of the diagnostic framework. Three of the four Level I conditions are met.

Totality (G_1). The cosmos admits no exterior by definition: it is the whole of which everything is a part. The totality condition is satisfied.

Self-sustenance (G_3). The cosmos requires no external principle to maintain its existence or ground its priority. The cosmos is the fundamental entity; its existence and its priority over its parts are not derivative on any further ground. The self-sustenance condition is satisfied.

Generativity (G_2), *partial.* Priority monism holds that the cosmos generates its parts through its own internal articulation: the priority order from whole to part accounts for the existence of derivative entities as structural consequences of the whole's modal and intrinsic profile.

At Level II, priority monism achieves the most formally precise contemporary analytic account of how derived entities ac-

[28]Schaffer ("Monism: The Priority of the Whole," 35–38). Schaffer distinguishes his position from existence monism (only one thing exists) and from priority pluralism (many equally fundamental things). Priority monism holds that many things exist but that their existence is grounded in the one fundamental whole.

quire their structural identities through their modal profiles, causal powers, and relational positions — all constituted by their positions within the whole's parthood and dependence structure. This is a genuine Level II achievement: the individuation of finite entities from the whole is derived from the ground rather than taken as given.[29]

The Structural Foreclosure

Priority monism's formal achievement at Level I is genuine and represents the closest approach to the Level I requirements of any position in analytic metaphysics. But two conditions remain unmet.

Generativity not fully derived (G_2). Schaffer's account specifies that the cosmos generates its parts through its own intrinsic profile and that the priority order produces the plurality of derivative entities as structural consequences. But what determines the cosmos's intrinsic profile itself — why this plurality rather than another, what constitutes the internal articulation of the whole — is not derived from conditions more primitive than the whole. The cosmos's intrinsic nature is constituted as a foundational feature of the account, not derived from conditions that are themselves derivable without presupposing it.

[29]The Level II achievement is developed through Schaffer's account of dependence: derivative things are what they are because of the whole's intrinsic character and the specific manner in which they are parts of that whole. Two derivative entities are numerically distinct because they occupy structurally distinct positions within the whole's intrinsic profile — a structural account of individuation derived from the ground, which is precisely what Level II's conditions (I_1)–(I_3) require.

This is precisely the Level I gap the diagnostic framework identifies. The generativity condition requires that the ground *produce* structural differentiation from within its own internal dynamic: that the diversity of derived entities be a structural consequence of something about the ground's own architecture. Schaffer's account requires the cosmos to have a specific intrinsic profile that generates specific parts; it does not derive why the cosmos has this profile rather than another from conditions more primitive than the cosmos itself. The internal differentiation is presupposed rather than derived.

Formal derivability absent (G_4). Priority monism does not derive the cosmos by the eliminative method. The cosmos is the starting point of the account. Schaffer argues for priority monism by showing that it satisfies theoretical desiderata and fits the holistic features of quantum physics and entanglement, but this is inference to the best explanation for the cosmos as ground, not an eliminative derivation establishing it as the unique structural result of applying the eliminative method to what every relational ontology presupposes.

By the Horizon Lemma, a framework whose derivational resources are calibrated to the cosmos and its parts cannot derive what is structurally prior to the cosmos without constituting a different foundational commitment. The latent register — the domain from which the cosmos's specific intrinsic profile is itself derived — is not addressed and cannot be addressed from within priority monism's framework.

Criterion 8.1 (Schaffer's Diagnostic Profile)*. Entry level*: Level I (partial) — three of four Level I conditions met; generativity is asserted for the cosmos's intrinsic profile but the profile is not derived from conditions more primitive than the cosmos itself. *Formal achievements*: Level I (totality, self-sustenance, partial generativity); Level II (derivation of the structural identities of finite things from their positions within the cosmos's dependence structure). *Structural foreclosure*: Level I (G_2) full generativity not derived; Level I (G_4) formal derivability absent; latent register foreclosed by the Horizon Lemma. *Character of the foreclosure*: Necessary. The cosmos as foundational whole satisfies three of four Level I conditions. The fourth — formal derivability — is explicitly refused: the cosmos is the terminus of explanation. This is structurally convergent with Spinoza's Substance, which also met three of four Level I conditions while failing (G_4). Both achieve what can be achieved by a ground that is posited as total and self-sustaining without being shown to be the unique structural consequence of the eliminative method applied to what relational ontology presupposes.

Foreclosure versus incompleteness

A referee whose home ground is contemporary analytic metaphysics may press the following objection: the Level I gap in priority monism — the absence of a derivation of the cosmos's intrinsic profile from conditions more primitive than the cosmos itself — is a gap in Schaffer's current development, not a structural foreclosure. Schaffer has not claimed to have derived the cosmos's intrinsic profile; he has argued for its priority. A programme can be incomplete without being foreclosed: the absence of a derivation could reflect an unfinished stage of inquiry rather than a commitment that precludes the derivation in principle.

The distinction between structural foreclosure and programmatic incompleteness is stated as Definition 1.2 in Chapter 1: a framework is foreclosed at level k if and only if the derivation of k's defining conditions is precluded by the framework's own foundational commitment, not merely absent from its current formulation. Incompleteness is compatible with future derivation; foreclosure is not, because the foundational commitment is the stipulation of a non-latent feature as primitive, and deriving that feature from something more primitive would require replacing the stipulation — which is to say, replacing the foundational commitment. The question for priority monism is therefore not whether Schaffer has derived the cosmos's intrinsic profile, but whether doing so is possible from within priority monism's own framework.

The forced-choice argument applies directly. Priority monism's foundational claim is that the cosmos is the one fundamental concrete object — the terminus from which all derivative entities are grounded. A derivation of the cosmos's intrinsic profile from conditions more primitive than the cosmos itself would face an exact structural parallel to the forced choice the OSR literature confronts (§7.2.4): either posit the cosmos's intrinsic profile as a further primitive (which is to say, the cosmos is not in fact the single fundamental entity but one of at least two) or derive the profile from something more primitive than the cosmos (which is to say, the cosmos is not the terminus — something prior to it is the actual fundamental entity, and priority monism's central claim is abandoned). The forced choice is not a contingent feature of the programme's current development; it is generated by what priority monism's foundational commitment consists in. The cosmos as single fundamental whole either has its profile as primitive or it does not occupy the Level I position priority monism assigns to it. This is foreclosure, not incompleteness.

8.10.2 Fine: Essence-Grounding and the Source of Encounter

The Foundational Commitment

Kit Fine's metaphysics of essence and ground, developed across a series of influential papers and systematised in his 2012 contribution *Guide to Ground*, offers the most formally precise analytic account of why specific entities enter into specific relations.[30] Fine's central claims are: first, the essence of an entity is not what it is necessary for the entity to be, but what the entity is in its own right — its constitutive nature as a specific individual; second, grounding relations hold in virtue of the essences of the entities involved — a fact p is grounded in facts $q_1, \ldots, q_n$ just in case it is part of the essence of p's obtaining that it obtains in virtue of $q_1, \ldots, q_n$; third, essence thereby provides a non-modal, constitutive account of ontological dependence.

Fine's programme is diagnostically significant because it directly addresses what the diagnostic framework identifies as

[30]Fine ("Guide to Ground"). The foundational account of essence is Fine ("Essence and Modality"). For Rosen's closely related grounding framework, see Rosen ("Metaphysical Dependence: Grounding and Reduction").

Level III: the source of encounter. Fine's account claims to explain why specific entities enter into specific grounding relations — and therefore specific relational encounters — by appeal to those entities' essences. The essence of water is partially constituted by its relation to hydrogen and oxygen; the essence of Socrates is partially constituted by his parentage; the essence of a set is partially constituted by its membership. Essence-grounding derives the specificity of encounter from the constitutive nature of the entities involved.

Formal Achievements

Fine's account achieves genuine formal results at Level III of the diagnostic framework — and this achievement is more precise than any other account in the analytic metaphysics literature.

The source of encounter condition requires (E_2) that the encounter source be structurally selective: not every entity encounters every other, and the structural specificity of encounter must be formally derivable rather than contingently stipulated. Fine's essence-grounding satisfies this condition within its domain. The essence of an entity determines which further entities and facts it depends upon: the essence of water determines its grounding relations to hydrogen and oxygen; the essence of a set determines its grounding relation to its members; the essence of a natural number determines its place in the order of arithmetic facts. Encounter specificity — why this entity grounds that one — is derived from essence, not contingently stipulated. The derivation is internal to the entity's constitutive nature rather than externally imposed.

At Level IV, Fine's account of ground provides a formal framework for relational yield in the case of grounding relations: what the grounded fact is in virtue of the grounds is determined by the essences of the entities involved, not merely by regularities among prior and posterior facts. This satisfies the irreducibility and asymmetric novelty conditions of Level IV within the domain of grounding relations.

The Structural Foreclosure

Fine's account enters the structural order at Level III — the source of encounter — and achieves formal results there and at Level IV. Three structural gaps follow from this foundational commitment.

Level II. Fine's account takes essences as its foundational commitment: entities have essences, and those essences are what they are. The question of what individuates essences — what makes the essence of Socrates the specific essence it is, what makes the essence of water constitutively related to hydrogen and oxygen rather than to other elements — is not derived within Fine's framework. Essences are the starting points; they are constituted as the fundamental specification of what each entity is, from which grounding relations are derived. The Level II derivation condition requires that the structural identity of relata be derived from the Level I ground rather than taken as foundational. Fine's essences function as Level II primitives: they specify relata's structural identities without deriving them from a more primitive dynamic.

Level I. Fine's programme does not provide an account of the structural ground of essence-possession. Why do entities have essences at all? What is the structural source from which the fact that entities have specific constitutive natures rather than being qualitatively undifferentiated emerges? This question is outside the programme's scope, which takes it as given that entities have essences and proceeds to analyse the grounding structure those essences generate.

Level V. Fine's account specifies the formal properties of grounding relations (asymmetry, irreflexivity, transitivity, non-monotonicity) but does not derive the internal organisation of the grounding encounter — why a specific grounding event has the formal structure it has, why grounding is co-instantaneous rather than sequential, why it is constitutive rather than causal. The Level V conditions are characterised rather than derived.

By the Horizon Lemma, a framework whose derivational resources are calibrated to essences and their grounding consequences cannot derive what is structurally prior to essences without constituting a different foundational commitment.

Criterion 8.2 (Fine's Diagnostic Profile). *Entry level*: Level III (source of encounter); essences constitute the structural identities of relata (Level II primitive) and account for encounter specificity (Level III derivation from essence). *Formal achievements*: Level III — the most formally precise analytic account of encounter specificity available; Level IV — grounding encounters produce formally determined yields whose character is derived from the essences of the entities involved. *Structural foreclosure*: Level II — essences are constituted as foundational primitives, not derived from a more primitive dynamic; Level I — no derivation of the structural ground of essence-possession; latent register foreclosed by the Horizon Lemma; Level V formal conditions characterised but not derived. *Character of the foreclosure*: Necessary. Fine's programme constitutes essences as its foundational primitives — the Level II entry point — and derives encounter structure and grounding relations from those primitives. Deriving essences from conditions more primitive than essence itself would not be Fine's programme; it would require a different foundational commitment that Fine's framework explicitly forecloses as outside its scope.

The Convergence of the Analytic Grounding Tradition

The diagnostic results for Schaffer and Fine converge with the results for the thirteen traditions surveyed in the preceding chapters, approached from the direction of contemporary analytic metaphysics. Schaffer enters at Level I (partial) — the highest Level I entry point in the analytic tradition, structurally convergent with Spinoza's Substance — achieving genuine formal results at Levels I and II, while being foreclosed from the latent register by the same structural necessity as Spinoza. Fine enters at Level III and achieves the most precise analytic account of encounter specificity, while being foreclosed from Levels I and II by the foundational constitution of essences.

The two traditions are the strongest available analytic test cases for the SIT's scope claim, because they operate at the levels of the structural order where the SIT's predictions are most demanding. Schaffer approaches Level I more closely than any other analytic position; Fine addresses Level III with greater formal precision than any other analytic tradition. Both confirm rather than resist the SIT's result: neither reaches the latent register; both exhibit the same structural pattern of entry at a non-latent level, genuine formal achievements at and above that level, and necessary foreclosure of what lies below.

> *Convergence with the analytic grounding literature. The two most formally developed analytic accounts of ontological priority and grounding — Schaffer's priority monism and Fine's essence-grounding — confirm the Structural Insufficiency Theorem from the direction of contemporary analytic metaphysics. Schaffer's priority monism achieves the highest Level I entry point in the analytic tradition — three of four ground conditions met, structurally convergent with Spinoza's Substance — while being foreclosed from Level I's full generativity derivation and formal derivability. Fine's essence-grounding achieves the most precise analytic derivation of encounter specificity while constituting essences as Level II primitives, foreclosing Level I. The SIT applies to both with the necessity the theorem establishes: any framework that foundationally constitutes a non-latent structural feature inherits the foreclosure of the latent register. The analytic grounding literature, surveyed at its strongest available formulation, confirms rather than contests this result.*

8.11 On the Selection Criterion, the Unrepresentative Sample Objection, and the Madhyamaka Counter-Instance

The diagnostic survey has covered thirteen traditions. A referee may object that the selection was not impartial — that the traditions were chosen because they fail in the way the SIT predicts, while traditions that might resist that failure were excluded. The objection requires two responses: a statement of the selection criterion and an engagement with the strongest potential counter-instance.

The Selection Criterion

The traditions surveyed satisfy two conditions jointly. *First*, each advances the relational thesis — that entities are constituted through their relations rather than prior to them — as a foundational ontological claim, not merely as a methodological or heuristic commitment. *Second*, each has developed its account of relational constitution with sufficient formal precision to make the five-level diagnostic grid applicable: the tradition has characterised its foundational commitment, derived formal results at and above its entry level, and engaged the problem of relational constitution systematically enough that the diagnostic questions yield determinate answers.

The criterion is not that traditions fail at a convenient level. It is that they engage the problem of relational constitution with the rigour required for the diagnostic questions to have determinate answers. The thirteen traditions surveyed are the most formally developed and philosophically rigorous attempts to take relation seriously as a foundational ontological commitment, across the Anglophone, continental, and process traditions.

The criterion's double function is worth making explicit. It ensures that the selected traditions are the *strongest* available test cases because formal precision is what makes the diagnostic questions answerable: a tradition that has not formally characterised its foundational commitment, not derived results at and above its entry level, and not engaged the problem of relational constitution systematically enough to admit determinate diagnostic answers would be a weaker test case, not a stronger one. Such traditions would fall to the SIT more easily, not resist it more effectively. The selection criterion therefore selects in favour of the traditions most likely to resist the SIT's scope claim — which is precisely what a genuine test of the claim requires. Traditions that do not meet the precision threshold (methodologically eclectic accounts, programmatic sketches of relational thinking without formal derivation) would confirm the SIT trivially, not because they are genuinely caught by its scope but because they lack the formal apparatus for the diagnostic questions to have sharp answers. Their exclusion is a methodological virtue, not a bias towards confirmatory selection.

The Madhyamaka Counter-Instance

The strongest potential counter-instance to the SIT's universal scope is the Madhyamaka Buddhist tradition — specifically Nāgārjuna's doctrine of śūnyatā (emptiness). The SIT applies to frameworks that foundationally constitute a non-latent structural feature. If Madhyamaka does not constitute śūnyatā as a positive structural primitive — since śūnyatā is itself empty (śūnyatā-śūnyatā) — the SIT's conditions may not be met.[31]

The entry-level assignment for Madhyamaka is non-trivially disjunctive and must be stated with care. The two-truths doctrine (*samurti-satya* and *paramārtha-satya*) functions as the structural apparatus within which the emptiness doctrine operates: the claim that śūnyatā is itself empty has purchase only within the conventional/ultimate distinction that the two-truths framework provides. Nāgārjuna's own argument, however, derives the two-truths distinction from dependent origination (*pratītyasamutpāda*) — if entities arise in dependence on conditions, then their apparent svabhāva (own-nature) is empty, generating the conventional/ultimate distinction as a structural consequence.[32] If this derivation is successful within the Madhyamaka framework, then the two-truths doctrine is

[31]See Jay L. Garfield and Graham Priest, "Nāgārjuna and the Limits of Thought," Establishes that Nāgārjuna discovers "true contradictions arising at the limits of thought" and that the emptiness of emptiness represents a genuine limit contradiction: śūnyatā is itself empty, which means Madhyamaka may not constitute śūnyatā as a positive structural primitive and may therefore escape the SIT's conditions rather than confirm them, *Philosophy East and West* 53, no. 1 (2003): 1–21.

[32]See Jay L. Garfield, *The Fundamental Wisdom of the Middle Way: Nāgārjuna's Mūlamadhyamakākārikā*, Translation with commentary. Chapter 24 ("Examination of the Four Noble Truths") develops the derivation of the two-truths doctrine from dependent origination (*pratītyasamutpāda*), establishing that the conventional/ultimate distinction follows as a structural consequence of universal conditionality rather than being constituted as a foundational primitive (New York: Oxford University Press, 1995), ch. 24.

not Madhyamaka's entry-level foundational constitution; it is a result derived from *pratītyasamutpāda*, which would be the actual foundational commitment.

This reading shifts the entry-level assignment: if *pratītyasamutpāda* is the foundational constitution, the entry level is Level II or Level III (dependent origination as the structural account of how relata arise in mutual dependence, or how that arising constitutes an encounter source), not Level V. The SIT applies at either location. Whether the entry level is Level II, III, or V (the two-truths doctrine as constituted apparatus) depends on how the *pratītyasamutpāda* derivation is read — and the structural foreclosure result holds at any of these entry levels. The emptiness of emptiness is a result generated *within* whatever framework Madhyamaka's analysis proceeds from, not a derivation from below that framework's own horizon.

The Madhyamaka case does not therefore constitute a counter-instance to the SIT. It is a particularly sophisticated instance of the pattern the SIT formalises: a framework operating from a foundational commitment — whether the two-truths doctrine at Level V or dependent origination at Level II–III — achieves, within that commitment, results of great philosophical depth, including the self-referential result that the framework's own primary concept is empty. The latent register — what is structurally prior to the framework's own conditions of possibility — is no more reachable from within Madhyamaka than from within Whitehead's framework. The selection criterion is thereby vindicated: the traditions surveyed are a representative cross-section of the most rigorous available attempts at relational constitution, and the strongest external candidate for a counter-instance confirms rather than threatens the SIT's universal scope.

Two significant absent traditions: Leibniz and Peirce. The selection criterion does not operate by excluding traditions to protect the SIT from challenge. The two most frequently proposed additions deserve explicit treatment here, so that their absence is visible as a deliberate methodological choice and not an oversight.

Leibniz. Leibniz's pre-established harmony is not absent from the diagnostic analysis; it is implicitly present as a limiting case of the Level III failure type. The monadological framework foundationally constitutes the pre-established harmony as the structural account of encounter-source: each monad's unfolding is fully determined by its constitutive programme, and the appearance of inter-monadic encounter is a consequence of a pre-constituted correspondence rather than a derivable structural result. This is a Level III foundational constitution at its most explicit — the encounter source is not derived from within the encounter but installed in the monad prior to it. Leibniz therefore falls within the scope of the survey's Level III diagnostic and does not constitute a distinct counter-challenge. A dedicated Leibniz chapter would add detail without altering the structural verdict. The reason Leibniz does not appear in the main survey is that the monadological framework foundationally constitutes encounter source in a form so explicit and so far from the latent register that it would function as the weakest available test case rather than the strongest: the SIT applies here with maximum force and minimum philosophically interesting resistance. The selection criterion requires strongest challenges; Leibniz is not among them.

Peirce. Peirce is the more significant potential absence. His triadic theory of signs — in which a relation is irreducibly triadic and cannot be decomposed into dyadic connections — is a serious structural proposal about the nature of relational encounter that cannot be reduced to any of the traditions already surveyed. Peirce's entry level is Level III–IV: Thirdness is the structural category of mediation and generality, and it functions as the account of how two relata are brought into a producing encounter whose result is a further sign. The SIT applies: Thirdness is foundationally constituted as the Category of the Ultimate within Peirce's architectonic. The encounter source (mediation via Thirdness) and relational yield (the interpretant) are derived from Thirdness as a positive structural primitive. The latent register — what is structurally prior to Thirdness such that Thirdness is derivable rather than presupposed — is not reached. A dedicated Peirce chapter would apply the diagnostic at this entry level; its conclusion is foreseeable from the SIT, but its philosophical interest would be real. The omission of Peirce from the main survey reflects the book's scope constraint (Stage I, deriving the structural conditions) rather than an assessment that Peirce poses no challenge. Readers in the Peircean tradition are invited to perform the diagnostic independently using Criterion 1.1 as specified; the SIT predicts a determinate verdict that can be checked without taking the present book's authority on trust.

PART III

The Structural Insufficiency Result

CHAPTER 9

The Same Wrong Room

The solution of the riddle of life in space and time lies outside *space and time.*

Ludwig Wittgenstein,
Tractatus Logico-Philosophicus, 6.4312

9.1 The Detective's Observation

The investigation is complete. Thirteen traditions, spanning the seventeenth through twenty-first centuries of serious philosophical work, have been examined in turn. Each has been given its fullest due: the genuine depth of its foundational commitment, the real precision of its formal achievements, the care with which it has developed its account of relation, of individuation, of encounter, of yield. None has been treated as a failure. None has been misread as less rigorous than it is.

And yet, across thirteen traditions and two contemporary landscapes, the diagnostic framework has disclosed a single pattern so uniform, so structurally deep, and so consistent across otherwise incompatible philosophical programmes that it cannot be dismissed as coincidence. Every tradition has found what it went looking for. Every tradition has built, within its own domain, an account of genuine intellectual power. And every tradition has stopped at exactly the same place — not by choice, not through lack of ambition, but for a structural reason that follows necessarily from the foundational commitment that defines each tradition as the tradition it is.

The pattern is this: every tradition in relational ontology enters the structural order at a specific level, achieves genuine formal results at and above that level, and formally cannot reach the level below its point of entry. Not will not. Not has not tried. Cannot — as a matter of structural consequence from the act of beginning where it begins.

The traditions do not all fail at the same numbered structural level — their entry levels differ precisely: Whitehead at Level V, Spinoza at Level I, Bergson at the actuality dimension of Level V (V_α), Heidegger at the possibility dimension of Level V (V_β) (Definition 2.6), OSR at Level II, RQM at Level III. What they share is not a common entry level but a common structural situation: every tradition foundationally constitutes a non-latent feature as its starting point, and thereby forecloses the latent register — the structural domain below Level I — as a consequence of that constitution. The shared failure is not failure at the same level but failure of the same structural type: the production, through foundational constitution, of a horizon below which the tradition's own resources cannot reach. The Structural Insufficiency Theorem formalises this shared type rather than this shared level. Its claim is not "every tradition fails at Level X" but "every tradition that foundationally constitutes any non-latent feature forecloses the latent register." The specific level at which each tradition is foreclosed varies; the structural mechanism that produces the foreclosure does not.

There is a room that every tradition has been looking for. It is the room in which the structural ground of relational ontology is housed: the level from which relata, their capacity for relation, the dynamic of their encounter, the yield of that encounter, and the formal structure of the relational event itself all emerge as derived structural results rather than as given starting points. Every tradition has been looking for this room. And every tradition has been looking in the wrong direction — not because any tradition has been careless, but because the direction every tradition has looked is the direction defined by the commitment that makes it a tradition. The room is not in that direction. It cannot be found by going further along any of the paths any tradition has laid.

The present chapter names the pattern, traces its single structural source, and states in plain language what the following chapter will establish formally: the Structural Insufficiency Theorem, the central result of this investigation.

9.2 The Pattern Stated

Before stating the pattern, a precision is required. The pattern identified across the traditions surveyed is not that every tradition enters the structural analysis at the same level — they do not. Whitehead enters at Level V; Spinoza at Level I;

Bergson at Level V; Heidegger at Level V; OSR and RQM at Level II–III; Stengers at Level III; Emirbayer at Level II–IV; Kant at Level V (possibility dimension); Simondon at Level I (virtual pre-individual field); Deleuze at Level I (intensive difference); Schaffer at Level I (partial — three of four ground conditions); and Fine at Level III (essence-grounding as encounter specificity). Merleau-Ponty, treated as a supplementary post-Heideggerian coda in §6.6, enters at Level V (event structure via the chiasm). What every tradition shares is a single structural relationship to the latent register: none reaches it; each forecloses it by the act of foundational constitution that establishes its entry point. The "same wrong room" is not a single level of the five-level grid. It is the structural space between every level in the grid and the latent register that underlies them all. A tradition that enters at Level V and one that enters at Level I are in different rooms of the non-latent register; what they share is the precise structural fact that neither has found the door to what is below the entire building. This is why the pattern is structural rather than incidental: the door is not missed by oversight but foreclosed by the act of walking through any room in the building at all. The Structural Insufficiency Theorem formalises this as a necessary result; the present chapter establishes that the pattern is real.

Let us be exact. The diagnostic survey has established the following results, chapter by chapter.

Whitehead enters the structural order at Level V, the relational event. He achieves, at that level, the most formally developed account of what a relational event is — how a new occasion of experience constitutes itself through the reception of prior occasions, how that constitution is creative rather than merely reproductive, how the satisfaction of an occasion is a genuinely novel structural achievement that the prior world did not contain. He achieves related formal results at Level IV: the yield of the relational event is not a sum or combination of its inputs but an irreducibly new structural fact.[1] Whitehead does not achieve formal results at Levels I, II, or III. His creativity is the nearest equivalent to a ground but is formally underdetermined: it is stipulated as the unanalysable ultimate rather than derived as the unique structural result of what his own account presupposes.[2] The structural individuation of occasions as the specific occasions they are prior to their self-constitution is not addressed. The source of the specific prehensive pattern of any given occasion is assigned to God, an external perturbation principle imported to supply what the framework cannot de-

[1]The formal achievements at Levels IV and V are documented in Chapter 3 above. The precise characterisation of what Whitehead achieves at each level is drawn from Whitehead (*Process and Reality: An Essay in Cosmology*), Part III (*The Theory of Prehensions*), especially pp. 219–280, and from Lowe (*Understanding Whitehead*, 155–202).

[2]The identification of creativity as an ungrounded ultimate in tension with Whitehead's own ontological principle is not unique to the present diagnosis. Hanna ("The Nature of Creativity in Whitehead's Metaphysics," 147) establishes that Whitehead's various characterisations of creativity "are confused and seemingly conflicting" and that creativity "comes into conflict with the ontological principle" — the principle that every real determination belongs to an actual occasion. The present investigation draws the structural consequence of this tension: creativity's conflict with the ontological principle is the formal expression of its foundational constitution as a Level I primitive that cannot discharge the derivability condition the ontological principle itself demands.

rive from within. This structural gap has been identified from outside the Whiteheadian tradition as well: Timmenga[3] documents Harman's finding that Whitehead "admits that change requires individual entities, but falls short of accounting for the existence of these individual entities, as he reduces entities to their (processual) relations".[4] The finding is here diagnosed as a Level II absence: individual entities with their specific structural identity — what each occasion is prior to what it prehends — are not derived within Whitehead's framework but presupposed as the unit of processual constitution.

Spinoza enters the structural order at Level I, the ground. He achieves, at that level, the most formally rigorous account of absolute ground in the history of relational ontology: Substance is total (admitting no outside), self-sustaining (*causa sui*, requiring no external support), and formally necessary (demonstrated by geometric method to exist uniquely). He does not achieve formal results at Levels II through V.[5] His Substance is formally homogeneous: modes do not emerge from an internal dynamic of weighted structural differentiation but follow from Substance by logical entailment through the attributes. Logical entailment is not a relational dynamic. It produces formal necessity without producing structural individuation, genuine encounter, or irreducible yield. The generativity condition of Level I — what the ground must supply in order to produce the diversity of

[3]Timmenga, "The Paradox of Process Philosophy."

[4]See Harman, *Tool-Being: Heidegger and the Metaphysics of Objects*; Harman, *Guerrilla Metaphysics: Phenomenology and the Carpentry of Things*.

[5]The standard secondary characterisations of Spinoza's Substance are in Bennett (*A Study of Spinoza's Ethics*) and Rocca (*Spinoza*).

relata and the dynamic of their encounter from within its own structural architecture — is not met. Spinoza came closer than anyone else to the ground. But the form of his arrival — homogeneous Substance — was precisely what prevented him from generating from the ground what the ground must generate.

Bergson enters the structural order at the actuality dimension of Level V. He achieves, within that dimension, the most precise account of temporal irreversibility and accumulative sedimentation in the tradition: duration is not a sequence of instants but the living accumulation of a past that is genuinely present in every current moment of experience. He does not achieve formal results at Levels I through III; at Level IV he achieves only partial formal results, as the Foreclosure Theorem records. Nor does he reach the possibility dimension of Level V. Duration is Bergson's starting point.

The question of what generates temporal asymmetry — why the before is structurally prior to the after, why the accumulation is irreversible, what the structural source of directionality is — is not asked, because it cannot be asked by a framework that begins within the directionality it would need to derive.

Heidegger enters the structural order at the possibility dimension of Level V. He achieves, within that dimension, the most formally developed account of structural openness towards futurity: being-in-time is the condition of Dasein as the entity for whom its own being is an issue, and thrownness is the structural deposit of a past that Dasein can never fully retrieve but always carries as the ground of its forward projection.[6] He does not achieve full formal results at Levels I or IV; at Levels II and III he achieves qualified formal results, as the Criterion records. Nor does he reach the actuality dimension of Level V with the formal precision Bergson's account supplies. The structural constitution of the admissible space of possibility — what makes certain possibilities formally available to Dasein and others not — is not derived. It is inhabited from the start.

Ontic structural realism enters the structural order at Level II, the structural identity of relata. It achieves, at that level, the most formally developed contemporary account of how structural identity can be constituted entirely through relational position — how the identity of a physical system can be specified without reference to intrinsic properties that the system possesses independently of its structural environment. It does not achieve formal results at Level I: the question of what the structural relations are ultimately *in*, what constitutes the relational field that physical structures inhabit, is not asked. The regress from objects to structures to the relations constituting those structures terminates, in the most developed accounts, in the denial that any terminus is needed — a structural commitment that formally constitutes the ground as unreachable by treating the question of the ground as malformed.

[6]The primary texts for the temporal structure are Heidegger (*Being and Time*, 383–423) (Division II, Chapter 3) and Heidegger (*Being and Time*, 261–278) (the analysis of resoluteness and situation). Dreyfus's commentary is Dreyfus (*Being-in-the-World: A Commentary on Heidegger's Being and Time, Division I*, 312–340).

Relational quantum mechanics enters the structural order at Level III, the source of relational encounter. It achieves, at that level, the most formally precise account of how encounter between physical systems constitutes the quantum state of each: there is no quantum state simpliciter, only quantum states relative to physical systems, and the encounter between systems is constitutive of the relational facts that exhaust physical reality. It does not achieve formal results at Levels I or II: what constitutes each physical system as the specific system it is prior to its encounters, and what the structural ground of the relational field is, are not derived. They are presupposed.

Stengers's process philosophy enters the structural order at Levels IV and V, inheriting and deepening Whitehead's formal achievements. It extends the analysis of relational yield and the temporal orientation of events with a philosophical sensitivity to the conditions under which encounter can be genuinely creative rather than merely mechanically necessitated. It does not achieve formal results at Levels I through III for the same structural reason as Whitehead's framework: the actual occasion as the unit of relational constitution is foundational, and what structurally precedes the occasion — what makes it this occasion, capable of this specific prehensive pattern, rather than another or none — is not derived.

Emirbayer's relational sociology enters the structural order at Levels II and IV. It achieves, at Level II, the most programmatic contemporary statement of the relational constitution of social identity: individual actors, groups, and institutions are what they are through the transactions in which they participate, not prior to those transactions. It achieves, at Level IV, a precise account of the irreducibility of transactional yield: what emerges from a social transaction cannot be derived from the properties of the parties that entered into it. It does not achieve formal results at Levels I, III, or V.

The pattern is not a set of individual failures. It is a single structural fact with thirteen expressions.

9.3 The Single Source

Why does the pattern take this form? Why does every tradition enter at a specific level and find, below that level, not unexplored territory but a structural barrier? The answer is not that the traditions failed to look. It is that looking from where they stand is structurally impossible — not in the sense of practical difficulty but in the precise formal sense that the act of beginning at a level is the act of constituting that level as the ground. And a ground, by definition, has nothing below it.

Consider what it means for a philosophical tradition to take something as its foundational commitment. It is not merely to say: I will begin here for convenience, or: I will take this as primitive for now. It is to say: this is where explanation terminates. This is the level at which the question "why?" receives its final answer, and below which the question cannot coherently be posed. For Whitehead, the actual occasion is what creativity, prehension, and satisfaction presuppose. Creativity is not derived from anything more fundamental; it is the universal condition of all becoming. For Spinoza, Substance is what all derivation presupposes; the question of what grounds Substance cannot be posed within his framework because Substance is the ground — *causa sui*, conceived through itself, requiring no external cause. For Bergson, duration is what all experience presupposes; to ask what generates duration is to have already stepped outside the only register in which the question can receive an answer. For Heidegger, being-in-time is what all human understanding presupposes; the question of what structurally constitutes temporal existence is not a deeper question that his analysis left unasked — it is a question that his own account reveals to be unanswerable from within the domain of that analysis.

The foundational commitment of each tradition is not a weakness. It is the source of each tradition's power: the clarity of its starting point, the discipline of its derivation from that point, the coherence of the account it builds above it. But that power has an exact cost. The cost is the formal impossibility, from within the tradition's own framework, of descending below the starting point. Not because the starting point is defended against descent — none of the traditions under examination explicitly refuses to consider questions about the ground. But because the starting point has been constituted as the ground, and a ground cannot descend below itself. The framework has no internal resources to generate the vocabulary, the formal structure, or the derivational pathway that would be needed for such a descent. The room below the starting point does not exist, within the framework's architecture. That is not a room one can fail to find. It is a room that has not been built.

The evidential force of this pattern requires that one specific objection be met. If the same result appeared because every tradition surveyed shares a hidden methodological assumption that generates the pattern artificially — because all thirteen traditions are, beneath their surface differences, variations on the same framework — then the repetition would be an artefact of selection, not evidence of structural necessity. The survey's methodological range rules this out: Whitehead's philosophy of organism, Spinoza's geometrical method, Bergson's phenomenology of temporal flow, and Heidegger's fundamental ontology share almost nothing methodologically. The physics-based traditions of OSR and RQM share nothing methodologically with the sociological tradition of Emirbayer's relational manifesto. Schaffer's priority monism and Fine's essence-grounding approach the question from within contemporary analytic metaphysics with formal tools drawn from grounding theory and modal logic, sharing nothing methodologically with either process philosophy or phenomenology. The most decisive case is Simondon and Deleuze. Both draw on Bergson, but their explicit aim in doing so is to *overcome* the limitations of Bergson's framework, not to inherit its structural assumptions. Simondon developed the pre-individual field precisely to supply what Bergson's durée lacked: a pre-personal, pre-temporal ground for individuation. Deleuze developed the virtual precisely to supply what Bergson's temporal metaphysics left unaddressed: the differential ontology from which actual individuals emerge. Both thinkers specifically and explicitly sought to go beyond the point at which Bergson stopped. Both are diagnosed as stopping at the same structural boundary not because they share Bergson's assumptions but because the structural constraint the SIT will formally identify holds regardless of which direction a tradition approaches it from. When a structural pattern survives this diversity of origin, method, and explicit critical distance — when it appears in thinkers whose entire projects were oriented towards overcoming it in the tradition they inherited — the most economical explanation is that the pattern is structural, not methodological.

That this structural situation leaves the tradition as a whole in a condition of unresolved confusion at its most fundamental level is not an external observation. Wildman[7] states it from within: "there is persistent confusion in almost all literature about relational ontology because the key idea of relation remains unclear." Timmenga,[8] surveying the cross-traditional landscape from Whitehead through Deleuze to OOO, reaches the same diagnosis from a different direction: both the process and object-oriented traditions "end up in" the same structural regress when they attempt to account for the source and conditions of genuine change and genuine encounter. Crux[9] identifies the same pattern at the level of the entire history of ontology: "Relation-based ontologies attempt to avoid substance by treating relations as primary, only to find that relations quietly harden into relata that behave like substances in all but name,"

[7]Wildman, "An Introduction to Relational Ontology."

[8]Timmenga, "The Paradox of Process Philosophy."

[9]Crux, "Phainesis in Zero Ground Showing the Work: On the Dissolution of the Problem of Ontological Grounding."

and "The problem recurs with remarkable consistency across the entire history of ontology, cutting across traditions that otherwise disagree on nearly everything else" — an independent diagnosis, from within the tradition, of the cross-traditional structural impasse the present investigation names. Rivera[10] establishes the same result at the level of the individuation problem specifically: across grounding, process ontology, and structural ontology, every framework "presupposes individuation rather than explaining it" — a result that confirms, at the level of what a derived Level II would need to supply, that the gap the SIT identifies is felt and acknowledged across every framework that has engaged seriously with real multiplicity. The convergence of these independent assessments confirms that the structural gap the present investigation names is not a feature of any single tradition's weakness but a condition of the tradition as a whole.

Definition 9.1 (Foundational Constitution)**.** A tradition in relational ontology *foundationally constitutes* a structural feature F when it takes F as primitive: when F is treated as the terminus of explanation within the tradition's framework, such that the question of what generates F or what F emerges from cannot be coherently formulated in the tradition's own terms.

Proposition 9.1 (The Foreclosure Consequence)**.** If a tradition foundationally constitutes a feature F that occupies Level n of the structural order ($1 \leq n \leq 5$), then the tradition structurally

[10]Rivera, "Unity Before Multiplicity: Ontological Individuation and the Problem of Primitive Multiplicity."

forecloses the derivation of every level $m < n$. The foreclosure is not the result of neglect or insufficient rigour but is the structural consequence of foundational constitution itself: the act of constituting F as the terminus of explanation is the act of constituting the levels below F as not requiring derivation.

The proposition does not require proof at this stage — that is the work of Chapter 10. What it requires is recognition: that the Foreclosure Consequence describes exactly the pattern the diagnostic survey has disclosed. Every tradition examined foundationally constitutes a feature at a specific level. Every tradition achieves formal results at and above that level. Every tradition fails to achieve formal results below it. The correspondence between the proposition and the survey is exact.

9.4 What the Pattern Is Not

Before proceeding to the formal statement, a clarification is required — not for the specialist, who will have anticipated it, but for the argument's integrity.

The pattern identified here is not the claim that any tradition has been refuted. Every tradition surveyed remains internally coherent. Every tradition's formal achievements at its own level remain genuine, valuable, and unimpeached by the present investigation. The claim is not that Whitehead was wrong about actual occasions, or Spinoza about Substance, or Bergson about duration, or Schaffer about the priority of the whole, or Fine about the formal structure of essence-grounding. The claim is structural and strictly limited: each tradition cannot, from within its own foundational commitments, derive the structural level below its point of entry. Whether those traditions are correct in their positive claims is a question the present investigation does not decide and has not attempted to decide.

The pattern is also not a claim about what the traditions *intended* to do. It is not a criticism of intellectual ambition or philosophical courage. Many of the traditions surveyed were entirely aware that questions about the ultimate ground of relation were pressing. Whitehead's discussion of creativity, Spinoza's analysis of Substance, Bergson's account of pure duration — each was offered as the answer to precisely the questions the present investigation has found them unable to answer. The claim is not that the traditions refused the question. It is that the structural resources their frameworks provide are insufficient for deriving what a satisfying answer requires.[11]

[11]This is the sense in which the critique is structural rather than polemical. A polemical critique says: you got the answer wrong. A structural critique says: the framework within which you formulated the question cannot generate the apparatus a satisfying answer requires. Both are genuine critiques; they operate at different levels and require different responses. The present investigation makes only the structural critique. It is worth noting that the grounding gap the structural critique identifies is acknowledged from within the relational tradition itself. Barr et al. ("Relational Ontology as a Bridge Between Realism and Empiricism") states it directly: "If entities exist only through relations, what grounds the relations? This threatens infinite regress or circularity." That the regress is named within a sympathetic defence of relational ontology confirms that the structural problem the present investigation formally establishes is a live difficulty the tradition has felt, not an external imposition. Adlam ("Relational Observables, Quiddities, and Structural Realism"), working at the intersection of relational quantum mechanics and structural realism, identifies that structuralist approaches "face a number of important open questions" re-

The pattern is not, finally, a claim about completeness within each tradition's domain. Whitehead's account of prehension is

garding how non-modal, non-relational features of reality can arise from purely relational structure — a formulation of precisely the Level I and Level II gaps the present investigation identifies as structurally necessary consequences of foundational constitution. Readers should note that Rivera (2025), Le Nepvou (2024a, 2024b), Sueyoshi (2026), and Rogers (2026) are preprints rather than peer-reviewed publications; Barr et al. (2024) is a working paper; Adlam (2025), which was a preprint at the time of composition, has since been published as a peer-reviewed article in *Synthese* (206:2, 2025, DOI:10.1007/s11229-025-05190-5) and should be treated as such. Sueyoshi (2026) is the work of an independent researcher (Makoto Sueyoshi), cited because its derivation of the structural conditions of describability from within the quantum foundations literature constitutes an independent confirmation of the structural pattern this investigation formalises as the SIT. Lucero (2026) appears in the *International Journal for Philosophy of Religion* and is accepted for publication but had not yet been assigned to a specific volume at the time of composition; it is cited as a peer-reviewed source for its grounding-theoretic convergence with R1. Several further preprints are cited in the course of addressing specific objections identified during review: Olenius (2025, PhilArchive rec/OLECWT), Mikkel (2025, rec/MIKGCA-3), Vinck (2025, rec/VINRTE), Enoch (2025, rec/ENOLPA-2), Moreira (2024, rec/CLATFO-24), and Boucon and Deepseek-R1 (2025, rec/BOUSCO-5) are all preprints archived at PhilArchive; Rijssenbeek and Blok (2025) is a paper accepted in *Foundations of Science* but not yet assigned to a volume at the time of composition. None of these latter sources is used as primary evidence for the diagnostic claims; each is cited for a specific structural convergence with an objection response. The peer-reviewed and formally published sources cited in support of the same diagnostic claims are: Hakkarainen and Keinänen (2022, *Acta Analytica*); Lam and Wüthrich (2015, *British Journal for the Philosophy of Science*); Austin (2020, *European Journal for Philosophy of Science*); Primus (2023, *Archiv für Geschichte der Philosophie*); Hübner (2014, *Philosophy and Phenomenological Research*); Rae (2014, *Deleuze Studies*); Garfield and Priest (2003, *Philosophy East and West*); Han (1983, *Philosophy Research Archives*, a refereed series); Merker (2014, *Studi Kantiani*); Banfield (2003, in *Husserl's Logical Investigations Reconsidered*, Kluwer); and Wildman (2010), a chapter in a peer-reviewed edited volume (*The Trinity and an Entangled World*, ed. Polkinghorne, Eerdmans).

not incomplete as an account of how a new occasion of experience constitutes itself. Spinoza's account of Substance is not incomplete as an account of the absolute ground's totality and self-sustenance. Each tradition is complete as an account of what it set out to account for. The claim is only that what it set out to account for does not include — and cannot include, given its foundational commitments — the derivation of the structural level below its entry point.

This clarification matters because it determines the kind of response the pattern calls for. It does not call for a refutation of any tradition. It does not call for a revision of any tradition from within. It calls for a different kind of inquiry — an inquiry that begins where all the traditions have ended without knowing they had ended, that takes as its starting point not the features that each tradition constitutes as its ground but the question that those features presuppose without asking: what is the structural ground from which the capacity for relation itself emerges?

9.5 The Direction the Derivation Must Take

The pattern points to a direction as precisely as it points to a problem. If every tradition that takes a non-latent structural feature as its foundation finds the structural ground below that feature unreachable, then the derivation of the ground must proceed from a different starting point — one that does not itself occupy any level of the structural order as a given starting point, but that generates the structural order from within its own dynamic.

What kind of starting point would that be? The eliminative method — the systematic removal of every assumption that can be shown to be derivable rather than primitive — suggests an answer. Begin not with what any tradition has taken as its foundation but with what every tradition presupposes prior to making any foundational commitment at all. What does every relational ontology presuppose, prior to constituting its own foundational feature as primitive?

It presupposes that there is something rather than nothing — a totality of what is, within which structural features can be present and absent, derivable and primitive. Call this the absolute ground: not the ground as any tradition has characterised it, but the ground as the structural precondition of every foundational commitment whatever. It is what must be in place for any foundational constitution to be possible — including the foundational constitution of the ground itself. It is prior to every tradition's starting point, presupposed by every tradition's first move, and formally derivable as the unique structural result of the eliminative method applied to what every relational ontology must assume in order to make any claim at all.

The diagnostic survey has established that no existing tradition has derived this ground. It has established, with the precision of Proposition 9.1, why no existing tradition can derive it from within its own framework. And it has, in doing so, established exactly what a derivation of the ground would need to achieve — not as an arbitrary set of desiderata, but as the precise inverse image of what every tradition has failed to supply.

The next chapter states this formally. The Structural Insufficiency Theorem is the formal expression of what the survey has established inductively: not merely that every tradition fails but that every tradition *must* fail — that the pattern is not contingent on the specific traditions surveyed but follows from the structural nature of foundational constitution itself. And the eliminative corollary names the only kind of derivation that can succeed where every existing tradition has failed.

Summary of Chapter 9. The diagnostic survey conducted in Part II has disclosed a single pattern across thirteen traditions spanning the seventeenth through twenty-first centuries of serious philosophical work: every tradition enters the structural order at a specific level, achieves genuine formal results at and above that level, and structurally cannot reach the level below its point of entry. The pattern has a single source: the act of foundational constitution, which is the act of constituting a structural feature as the terminus of explanation — and thereby rendering the structural level below it formally unreachable from within the resulting framework. The pattern is not a record of individual failures but the structural consequence of a single type of intellectual act performed, in different registers, by every tradition in the field. What the pattern demands is not a revision of any existing tradition but a different kind of inquiry — one that begins before every existing tradition's starting point, with what every tradition must presuppose before making its first foundational move. Chapter 10 states this demand formally.

CHAPTER 10

The Structural Insufficiency Theorem

Every number is the number of some concept.
Gottlob Frege,
Die Grundlagen der Arithmetik, §46

Frege's remark is paradigmatic of formal derivation at its best: a claim that looks like a definition turns out to be a theorem — something that can be proven from principles more primitive than the claim itself, not merely asserted. What looked like a starting point becomes a derived result. What looked like a given turns out to be a structural consequence of something deeper. The present chapter undertakes the same operation for the central claim of this investigation.

10.1 The Claim and What It Requires

The claim, stated informally, is this: every tradition in relational ontology that takes a non-latent structural feature as its foundational primitive structurally forecloses the derivation of the latent register from which that feature emerges. This foreclosure is not accidental — not the result of philosophical carelessness, insufficient ambition, or historical limitation. It is a structural consequence of the act of foundational constitution itself: the act of making something the ground is the act of making the question of what that something emerges from unanswerable within the resulting framework.

For this claim to be a theorem rather than an observation, three things are required.

First, the key terms must be defined with formal precision. The claim involves the notions of a structural feature, foundational constitution, a latent register, and structural foreclosure. Each must be given a definition rigorous enough to support formal derivation.

Second, from those definitions, together with structural facts about the five-level framework established in Chapter 2, the claim must be shown to follow necessarily — not merely to be plausible or well-supported by the survey, but to be entailed by the defined terms and the structural facts.

Third, the scope of the theorem must be established: it must be shown to apply not merely to the thirteen traditions surveyed but to any possible tradition that makes a foundational commitment of the relevant type.

The present chapter executes all three tasks in turn.

10.2 Formal Definitions

Definition 10.1 (Structural Level)**.** A *structural level* is a formally specifiable domain of structural analysis characterised by a set of necessary and jointly sufficient conditions that any account adequate to that domain must satisfy. The five structural levels of relational ontology are defined in Chapter 2. A structural feature F *occupies level* n if and only if F satisfies the defining conditions of Level n and is not derivable from the defining conditions of any level $m > n$.

Definition 10.2 (Latent Register)**.** The *latent register* is the structural domain below Level I: the totality of structural conditions that are presupposed by any foundational commitment to any feature at Level I or above, and that are not themselves features of any level in the five-level structural order. The latent register is not a sixth level of the structural order; it is the pre-structural ground from which the structural order as a whole emerges. A derivation *reaches the latent register* if and only if it derives Level I (the ground) as a structural consequence of conditions belonging to the latent register — conditions that are themselves derivable without foundational constitution of any feature at any level.

Definition 10.3 (Non-Latent Register)**.** The *non-latent register* is the structural domain constituted by the five levels of relational ontology — Levels I through V — within which every tradition surveyed foundationally constitutes its point of entry. A structural feature occupies the non-latent register if and only if it satisfies the defining conditions of at least one level n where $1 \leq n \leq 5$. The non-latent register is the complement of the latent register: together, the latent register (the domain below Level I) and the non-latent register (Levels I through V) exhaust the structural domain presupposed by relational ontology. Every foundational commitment a tradition makes is a commitment to some feature in the non-latent register. The Structural Insufficiency Theorem (Theorem 10.1) establishes that any such commitment structurally forecloses the latent register.

Remark 10.1. Definition 10.2 does not presuppose that the latent register has a specific positive character. It specifies the latent register functionally: as whatever is presupposed by the structural order without itself belonging to that order. The question of what the latent register positively is — what its structural architecture consists in — is precisely the question that a derivation reaching it would answer. The definition establishes that such a register exists (as the presupposition of every foundational commitment at Level I or above) without specifying its character in advance. That specification is the work of the derivation, not of the critique that establishes the need for it.

The existence of the latent register requires argument and is not established by definition alone. Three conditions must hold: (1) every foundational commitment at Level I or above has structural presuppositions; (2) those presuppositions are not themselves features of the five-level structural order; and (3) those presuppositions form a coherent domain from which Level I could in principle be derived.

Condition (1) is established by the structure of the five-level hierarchy itself. Every feature at Level n satisfies the defining conditions of that level. The defining conditions of Level I include formal derivability (G_4): the ground must be shown to be the unique structural result of the eliminative method applied to what relational ontology presupposes. A clarification is required here. G_4 is a condition on what a successful *investigation into* the ground must demonstrate — not a condition on the mind-independent ground as such. The ground may satisfy G_1, G_2, and G_3 whether or not any investigation has derived it; G_4 specifies what distinguishes a derivation that *reaches* the ground from one that *stipulates* it. The present investigation uses G_4 as a diagnostic criterion: the question is not whether the ground exists independently of G_4 (it does, or it does not, independently of any investigative norm) but whether any tradition's account of Level I can be shown to have derived it rather than foundationally constituted it. If no tradition satisfies it — as Part II establishes — then every tradition's Level I commitment is foundationally constituted, meaning it has been taken as primitive. A feature taken as primitive rather than derived from conditions more primitive than itself presupposes those more-primitive conditions whether or not it derives them. The presuppositions exist as the structural underside of the primitive commitment.

Condition (2) is established by the structure of the five-level hierarchy. The five levels together constitute the full domain of what relational ontology requires: the ground, the relata, the source of encounter, the yield, and the event structure. Any condition that is presupposed by a feature at Level I but is not itself a condition of the ground, the relata, the encounter source, the yield, or the event structure falls outside the five-level domain. The presuppositions of a Level I foundational commitment are presuppositions of the ground: they are what the ground itself depends on, structurally, in order to be the ground. They cannot be features of the ground without generating an infinite regress of grounds. They cannot be features of Levels II through V, since those levels presuppose Level I. They are therefore outside the five-level structural order: condition (2) holds.

Condition (3) — coherence as a derivable domain — is the most demanding. It requires showing not only that there are conditions below Level I but that those conditions form a structured domain from which Level I's defining conditions can be derived. Two considerations establish this. First, the eight requirements of Part IV specify, in positive terms, what a derivation reaching the latent register must supply: a total, generative, self-sustaining, formally derivable ground (R1); a pre-temporal dynamic (R2); a differentiation function (R3); an encounter mechanism (R4); a yield function (R5); an event structure (R6); an irreversibility function (R7); and a self-exemplifying derivational sequence (R8). These eight requirements jointly characterise the structural shape of the latent register without positively identifying it. Second, the joint satisfiability proof of Theorem 11.1 demonstrates that the eight requirements are mutually consistent and can be simultaneously instantiated.

A structural clarification is required here to avoid a conflation that would otherwise invite a legitimate objection. The existence of the latent register — established by conditions (1) and (2) above, via the presupposition argument — is a separate result from the coherence of the derivation target. Conditions (1) and (2) establish that the latent register *exists*: every foundational commitment at Level I or above has structural presuppositions, and those presuppositions are not themselves features of the five-level domain; they are what every primitive commitment stands on, whether or not they are derived. This existence claim does not depend on joint satisfiability. Condition (3)'s work is different and additional: it establishes that the domain whose existence has already been secured by (1) and (2) is a *coherent derivational target* — that the eight requirements characterising the structural shape of what a derivation must supply are mutually consistent and jointly satisfiable. The joint satisfiability proof therefore does not establish existence; it establishes that pursuing a derivation of the latent register is not an inquiry into an internal contradiction. Together, conditions (1), (2), and (3) establish that the latent register is a real and coherent structural target: it exists as the presupposed underside of every relational ontology's foundational commitment, and a derivation reaching it would not be self-defeating. What the present investigation does not establish is the positive character of that domain — what it actually is rather than what any derivation reaching it must supply. This is acknowledged as the task of the companion investigation, not of the critique that establishes its necessity.

Definition 10.4 (Foundational Constitution (Formal))**.** A framework $\mathcal{F}$ *foundationally constitutes* a structural feature F at level n if and only if:

(i) F occupies level n within $\mathcal{F}$'s own account;

(ii) $\mathcal{F}$ takes F as a primitive: F is not derived within $\mathcal{F}$ from conditions that are themselves derivable without presupposing F;

(iii) $\mathcal{F}$'s derivational resources are defined over features at level n or above — that is, the vocabulary, formal structure, and inferential apparatus of $\mathcal{F}$ are constituted with respect to the domain of features at and above level n.

Remark 10.2 (On condition (ii): what "presupposing F" means). Condition (ii) carries the diagnostic burden of the definition. A potential objection must be closed before the Horizon Lemma is proved: some frameworks appear to *derive* their foundational feature from prior principles rather than simply positing it — Spinoza's modes are derived from Substance by logical entailment; Ontic Structural Realism's real patterns are derived from constraints on structural representations of physical theory. Does such derivation show that condition (ii) is not met, so that the framework does not foundationally constitute F?

The answer is no, for a formally determinate reason. A derivation within $\mathcal{F}$ that purports to produce F from prior principles Π either (a) relies on Π that are themselves derivable without presupposing F — in which case F is genuinely derived and condition (ii) is not met — or (b) relies on Π whose own structural content, validity, or application to the domain already requires F as a background condition. The test for case (b) is precise: substitute the negation of F — remove F from the framework's structural commitments — and ask whether Π retains its derivational force. If Π without F either collapses (loses coherence), yields a vacuous result (fails to produce any structural differentiation), or fails to be applicable to $\mathcal{F}$'s domain (its vocabulary becomes indeterminate), then Π depends on F and the derivation of F from Π is circular within $\mathcal{F}$.

Applied to the two cases: Spinoza's logical derivation of modes from Substance relies on the entailment structure of the *ordo geometricus* — but that entailment structure presupposes the determinate distinction between the infinite attributes of Substance and the finite modes they express, which is precisely the structural content of the modes as a type. Remove modes as a structural category and the infinite attributes have nothing to express differentially; the entailment produces no structural differentiation. The derivation of modes from Substance does not derive the *mode-type distinction* from something more primitive — it applies it in constituted form. Condition (ii) is met. For Ontic Structural Realism, the derivation of real patterns from structural constraints on physical theory uses the concept of a "real pattern" as the criterion for what counts as a structural output worth retaining — the derivation selects for real patterns because that is what the framework is calibrated to identify. The selection criterion itself presupposes the concept of a real pattern as a positively specifiable structural type. Condition (ii) is met.

The general principle is therefore: a derivation of F from Π within $\mathcal{F}$ is not a genuine derivation of F in the sense of condition (ii) when Π's structural content, validity conditions, or domain applicability presuppose F. Condition (ii) is satisfied whenever the validation criteria of $\mathcal{F}$'s derivational apparatus are calibrated to the domain of features that includes F as a constituted element, whether or not $\mathcal{F}$ presents the introduction of F as a derived result.

Definition 10.5 (Structural Foreclosure)**.** A framework $\mathcal{F}$ *structurally forecloses level* m ($m < n$) if and only if $\mathcal{F}$ foundationally constitutes a feature at level n, and there is no derivational pathway within $\mathcal{F}$ — using only $\mathcal{F}$'s own vocabulary, formal structure, and inferential apparatus — that terminates in the satisfaction of the defining conditions of level m.

Lemma 10.1 (The Horizon Lemma)**.** Any framework $\mathcal{F}$ that foundationally constitutes a feature F at level n necessarily possesses derivational resources defined over features at level n or above — that is, condition (iii) of Definition 10.4 is not an additional stipulation but a structural consequence of conditions (i) and (ii).

Proof. The argument proceeds in three steps.

Step 1: Calibration defined. A framework's derivational resources are *calibrated to a domain* D — that is: the framework's conceptual apparatus was built to operate on features in D and was not developed for structural domains outside D — if and only if every inferential step the framework can execute — every derivational move available to it — either (a) produces conclusions about features within D, (b) takes features within D as premises, or (c) takes features within D as the standard against which derivational moves are validated. More precisely: a framework

$\mathcal{F}$ is calibrated to domain D if and only if there is no derivational move available to $\mathcal{F}$ that produces a conclusion about a feature outside D without that feature being derivable from features within D. This is a formal condition: it rules out that $\mathcal{F}$ can reach features outside D by any available inference, unless those features are shown to follow from D-features. Domain-specificity of derivational resources in this sense is not an optional constraint but is constitutive of what it means to be a framework with determinate derivational content. A system whose inferential moves are equally available for any domain whatsoever has no determinate domain; it is not a framework but a purely formal calculus without an interpretation. Every framework in relational ontology has determinate philosophical content, which means it has a determinate domain. The question is what fixes that domain.

Step 2: Constitution generates domain-adequate vocabulary. A framework's derivational resources consist of three components: (a) a *vocabulary* — a set of terms and their associated formal conditions — that applies to structural features within the framework's domain; (b) *inferential rules* governing transitions between claims about features within that domain; and (c) *validation criteria* against which any proposed derivational move is assessed as legitimate or illegitimate within the framework.

A framework's vocabulary is *adequate to* a structural domain D if the vocabulary's terms have determinate content with respect to features in D — that is, if the formal conditions associated with each term yield determinate verdicts (satisfied or not satisfied) when applied to features in D. Adequacy is not a global property; it is a domain-relative one. The same vocabulary may be adequate to features at level n and above while being indeterminate — yielding no verdicts rather than false ones — with respect to features at levels below n.

The question Step 2 must answer is: when $\mathcal{F}$ foundationally constitutes F at level n — conditions (i) and (ii) — what is the adequate domain of $\mathcal{F}$'s vocabulary? The answer follows from what foundational constitution consists in.

Conditions (i) and (ii) jointly specify that F occupies level n in $\mathcal{F}$'s account and that $\mathcal{F}$ takes F as its structural terminus: F is not derived within $\mathcal{F}$ from conditions that are themselves derivable without presupposing F. The intellectual work of constructing $\mathcal{F}$ has therefore consisted in characterising F and its structural consequences — in developing vocabulary adequate to F, to the features at level n whose specification requires F, and to the features above level n that F generates or makes possible. This is not merely an empirical claim about how frameworks happen to have been built: it is a logical consequence of condition (ii). Condition (ii) states that F is not derived from conditions derivable independently of F — which means that $\mathcal{F}$'s derivational direction runs from F outward to its structural consequences, not from conditions below F upward to F. A framework whose derivational direction is outward from F cannot have developed, within that framework, vocabulary adequate to what lies below F: the below-F direction is precisely the direction condition (ii) closes off. The calibration of $\mathcal{F}$'s vocabulary to features at level n or above is therefore a structural consequence of what condition (ii) specifies, not a post-hoc description of construction history. This is what the framework has done: it has developed the apparatus adequate to the domain of features at level n or above, because condition (ii) ensures that the derivational direction constitutive of $\mathcal{F}$ runs in that direction and no other.[1]

A potential objection must be addressed before proceeding. One might grant that $\mathcal{F}$'s vocabulary was developed for features at level n or above while maintaining that the

[1]The domain-adequacy account advanced in Step 2 is not an a priori claim about the impossibility of frameworks extending their vocabulary in general: it is a structural claim about what the *act of foundational constitution* specifically produces, confirmed by the diagnostic survey of Part II. In each of the thirteen traditions examined, the apparatus developed around the foundational commitment does not possess instruments for descending below the entry level. The symptom is uniform across traditions of radically different methodology and content: at the point where the diagnostic framework predicts the horizon, every tradition either falls silent, generates a regress whose resolution it cannot supply, or imports an external principle whose derivation from the framework's own resources it cannot execute. This convergent symptom is the empirical signature of what Step 2 characterises structurally: a framework whose derivational resources are calibrated to a domain and thereby indeterminate below it. The Lemma describes a structural pattern whose reality the survey confirms case by case.

same vocabulary could, in principle, be applied downward — that a framework built for one domain might extend its apparatus to a domain it was not built for, without abandoning that construction. This objection receives a precise answer by considering what the apparent extension would consist in.

Suppose $\mathcal{F}$ produces, from its existing level-n vocabulary, an inference whose conclusion appears to concern a structural feature X at level $m < n$. There are exactly two cases. Either (a) the inference derives X from features at level n or above — that is, it reaches the below-n feature as a consequence of above-n features — or (b) the inference reaches X by a move that cannot be validated by $\mathcal{F}$'s existing vocabulary at all.

Case (b) is not a $\mathcal{F}$-internal move: a move whose conclusion is not reachable by any chain of inferences available within $\mathcal{F}$'s validating apparatus is not a move *within* $\mathcal{F}$. It is an assertion from outside the framework, which is consistent with the Lemma's result.

Case (a) is more significant, because it represents what a sophisticated objector would claim: that the framework has derived a below-n feature from above-n resources and has therefore crossed the horizon. But this is precisely the misidentification the five-level hierarchy is designed to diagnose. Features at level $m < n$ are, by the structure of the hierarchy, *structurally prior* to features at level n: level-n

features depend on them, not conversely. An inference that derives a below-n feature from above-n features is therefore not crossing the horizon downward into the latent register — it is doing something formally different: *applying* level-n vocabulary to a domain it was built to characterise only from above. The below-n feature that emerges from such an inference has been characterised in level-n terms; it has not been derived as what it is in its own right, prior to and independently of any level-n characterisation. The distinction is between *redescribing* a below-n feature in borrowed above-n vocabulary and *deriving* that feature's own structural conditions. A framework that characterises its ground by applying event-level vocabulary downward would be performing the former operation: assigning a level-n description to something structurally prior to level n, not deriving the below-n domain's internal structure. The result is a description of the latent register in borrowed vocabulary, not an entry into it. Case (a) therefore confirms rather than threatens the Lemma: what appears to be derivation of a below-n feature is, on inspection, redescription from above. The latent register is not reached; it is named at a distance.

The consequence for features below level n is now derivable rather than stipulated. A derivational move within $\mathcal{F}$ is a move within $\mathcal{F}$ if and only if its conclusion is evaluable by $\mathcal{F}$'s validation criteria — if the criterion can yield a determinate verdict on whether the conclusion follows from the premises. For a conclusion about a structural feature X below level n to be evaluable within $\mathcal{F}$, $\mathcal{F}$'s vocabulary must be adequate to X: the formal conditions associated with $\mathcal{F}$'s terms must yield a determinate verdict when applied to X. But $\mathcal{F}$'s vocabulary was developed to characterise F and its structural domain — the domain of features at level n or above. The vocabulary is adequate to that domain by construction. It is indeterminate with respect to features below level n because $\mathcal{F}$ has not characterised those features: they are not features the framework's apparatus has been developed to handle.

Could $\mathcal{F}$ extend its vocabulary to features below level n from within its own resources? Only if $\mathcal{F}$'s existing vocabulary — adequate to features at level n or above — could generate characterisations of features below level n by application of its inferential rules to its existing domain. But features below level n are, by the structure of the five-level hierarchy, those upon which features at level n *depend*: they are structurally prior to level-n features, not derivable from them. An inferential move that derived the features upon which level-n features depend from those level-n features would be an inference from consequent to antecedent in the structural order — not a derivation but a reversal of derivation. $\mathcal{F}$'s inferential rules were developed to operate on the domain of level-n features and above; they do not include resources for such a reversal, because the reversal would require characterising exactly the domain that $\mathcal{F}$'s foundational constitution placed outside its characterised territory.

The difference is between lacking a tool and being structurally unable to manufacture one. $\mathcal{F}$ lacks vocabulary adequate to features below level n. It cannot manufacture that vocabulary from within its existing resources because manufacturing it would require characterising the domain below level n — and characterising that domain requires taking it as an object of framework-construction, which is precisely to step outside $\mathcal{F}$ and begin constructing a different framework. Remaining within $\mathcal{F}$ and manufacturing below-n vocabulary are not jointly possible: $\mathcal{F}$-internal moves are moves within $\mathcal{F}$'s characterised domain, and the characterised domain is the domain of features at level n or above. The foreclosure is therefore of this precise form: *no derivational pathway within* $\mathcal{F}$ reaches below level n — not because $\mathcal{F}$ prohibits the question, but because a move whose conclusion is unevaluable by $\mathcal{F}$'s apparatus is not a move within $\mathcal{F}$; it is a move in a different inquiry, one that has already stepped outside $\mathcal{F}$'s characterised domain.

Step 3: The horizon. The domain fixed by the foundational constitution in Step 2 is precisely the domain of features at level n or above. By the formal definition of calibration in Step 1, $\mathcal{F}$'s derivational resources are calibrated to this domain: every available inferential move either produces conclusions about features at level n or above, takes such features as premises, or is validated against them. No available inferential move reaches features outside this domain — specifically, no available move reaches features at levels $m < n$ — unless those features are derivable from features at level n or above. But the features at levels $m < n$ are, by the structure of the five-level hierarchy, those upon which features at level n depend. They are not derivable from level-n features; level-n features presuppose them. The framework therefore cannot reach levels $m < n$ by any of its available derivational moves. Condition (iii) of Definition 10.4 — that $\mathcal{F}$'s derivational resources are defined over features at level n or above — is the formal expression of this structural horizon: it names, in the definition, what the constitution produces as a structural consequence.

The three steps together establish that condition (iii) follows from conditions (i) and (ii) via the formal definition of calibration: foundational constitution of F at level n entails that $\mathcal{F}$'s derivational resources are defined over features at level n or above. Condition (iii) is therefore not a stipulation added to the definition but the structural consequence of constitutive calibration made explicit.

Remark 10.3. The Horizon Lemma addresses a potential objection from traditions that claim access to a pre-thematic or pre-ontological understanding that is not itself a feature at the level the tradition takes as its explicit foundational commitment. Heidegger's claim that Dasein has a pre-ontological understanding of being is the most prominent instance: one might argue that the pre-ontological understanding gives Heidegger's framework a resource below Level V's possibility dimension that the Lemma fails to credit. The reply is precise. The pre-ontological understanding in Heidegger is not a resource that operates independently of Dasein's foundational constitution: it is described as Dasein's own mode of being, the always-already of its thrown existence. It is, in other words, a feature constituted at Level V's possibility dimension — not a derivational resource that reaches below that level. The Horizon Lemma's point is not that no feature of the framework concerns the pre-thematic, but that any such feature, if constituted by the framework as part of its foundational domain, is thereby at or above the framework's entry level — and therefore confirms rather than escapes the horizon the Lemma establishes.

A more refined version of the objection presses one step further: the pre-ontological understanding, it might be said, is not a *feature* of Dasein's constitution but a *condition* of it — something structurally prior to Level V, not posterior to it. Heidegger himself describes the pre-ontological understanding as the condition that makes Dasein's explicit ontological inquiry possible, not as a product of that inquiry. If so, the understanding is not constituted at Level V; it is what makes Level V possible. Does the Horizon Lemma fail to reach this prior condition?

The response requires distinguishing two kinds of structural priority. A condition can be prior to a feature in the sense of *genealogical priority* — it is what generated or makes possible the feature — or in the sense of *derivational priority* — it is what a derivation of the feature must proceed from. These are not the same. The pre-ontological understanding is genealogically prior to Dasein's explicit ontological projection, but it is not derivationally prior to Dasein's being-in-the-world — it is constituted as part of Dasein's thrown existence, as Heidegger's own analysis of thrownness, the "there," and the disclosed horizon makes explicit. What a Heideggerian framework has derivational access to is what it can operate on with Dasein's constituted apparatus: the structures of care, temporality, understanding, and projection — all of which presuppose Dasein as their site. A derivational step that descends below Dasein — to a pre-Daseinal structural domain from which Dasein itself emerges — is not a step available to the Heideggerian framework's own instruments. The pre-ontological understanding, however it is characterised, discloses being through Dasein's mode of being-in-the-world; it does not supply a derivational pathway to what is structurally prior to any world at all. The Horizon Lemma therefore applies to Heidegger's framework with full force: the entry level is Level V's possibility dimension, and the framework's derivational resources, including the pre-ontological understanding, are calibrated to that level and above.

A further objection concerns not *Being and Time* but the later Heidegger. Blattner has argued that ecstatic-horizonal temporality cannot ground the time-space (*Zeit-Raum*) of the Ereignis analyses in the *Contributions to Philosophy* and the later work.[2] On Blattner's reading, the structural depth of Ereignis exceeds what the Dasein-centred analysis of *Being and Time* can supply. Heidegger himself acknowledged the shift: the approach through Dasein was preparatory, and Ereignis names what that

[2] See Blattner, *Heidegger's Temporal Idealism*, ch. 9.

preparation was oriented towards — the event of mutual appropriation by which Being and Dasein are brought into their belonging-together.[3]

This might seem to pose a challenge: if Ereignis moves below Dasein-centrism, the later Heidegger's framework might operate below Level V, constituting a derivational resource the earlier analysis cannot recognise. The response is precise. Ereignis is not derived from conditions more primitive than itself within the *Contributions*' own framework; it is the structural event the inquiry names and characterises as its fundamental starting point. The *Contributions* proceed *from* Ereignis — the task is to think from within Ereignis, not to derive it from something more primitive.[4] The later Heidegger's shift is therefore a shift *within* Level V — from the possibility-dimension sub-variant (being-in-the-world) to the relational event structure sub-variant (Ereignis as the happening of mutual appropriation) — not a descent below Level V. The Horizon Lemma applies to both phases of Heidegger's thought: in both cases the foundational constitution is at Level V and the framework's derivational resources are calibrated to the domain that constitution characterises.

A methodological objection requires separate treatment. Heidegger's *formale Anzeige* — formal indication — is his methodological term for the mode of disclosure that operates prior to conceptual determination: a pointing-towards that leaves the structure of the directed-at formally undetermined rather than

[3]See Heidegger, *Contributions to Philosophy (From Enowning)*, §131–§135.
[4]Id., §1–§5.

conceptually fixing it in advance. One might argue that formal indication is not a feature constituted at Level V but a method that operates prior to any level, and that therefore the Horizon Lemma — which tracks foundationally constituted features — simply does not apply to Heidegger's method as such. The reply is precise and distinguishes two dimensions of formal indication. As an *epistemic* procedure, formal indication is prior to conceptual determination: it discloses structures without yet determining them. But as a *derivational* resource — as a procedure that could supply steps in a derivation reaching below Level V — formal indication is calibrated to the domain its pointing-towards discloses, and that domain is always disclosed through and within Dasein's constituted apparatus of care, temporality, understanding, and projection. Formal indication points towards the structures of Dasein's existence; it does not supply a derivational pathway to what is structurally prior to any Daseinal existence at all. The method's pre-conceptual character is an epistemic feature (it does not pre-determine), not a derivational feature (it does not descend below the domain Dasein's constitution makes accessible). The Horizon Lemma is a claim about derivational resources, not about epistemic modes; formal indication, precisely as Heidegger employs it, confirms the Lemma's point by being consistently deployed to disclose structures within the Level V domain it is designed to approach without pre-determining.

One textual resource in the *Contributions* requires acknowledgment. Heidegger's references to the *Abgrund* — the abyss — and to the last god have been read by some commentators as gesturing towards something more primordial than Ereignis itself.[5] On this reading, Ereignis would not be the *Contributions*' structural terminus but a result derived from the Abgrund. The dominant reading, however, treats the Abgrund as a feature of Ereignis's own structure — the groundlessness internal to the event of mutual appropriation — rather than a structural level below it.[6] On this reading the Abgrund does not supply a derivational pathway below Level V; it characterises the kind of ground Ereignis is, from within Ereignis's structural domain. The Level V assignment holds on either reading: if the Abgrund is internal to Ereignis, the entry level is Level V; if the Abgrund is a more primitive level from which Ereignis is derived, the entry level shifts downward within the hierarchy — but the structural foreclosure result applies at whatever level the framework enters. Engaging the minority reading on its own terms: even if the Abgrund is taken as more primordial than Ereignis rather than internal to it, the Abgrund is itself *constituted* as such within the framework of the *Contributions* — it is encountered through and within Heidegger's own thinking of Ereignis, not derived from conditions more primitive than the *Contributions*' own conceptual-existential apparatus. The Abgrund, on the minority reading, is a structural feature that the *Contributions* take as what grounds Ereignis; it is not derived from conditions derivable without presupposing the framework's governing attunements (Angst, Erschrecken, Verhaltenheit). It is therefore

[5]See Vallega-Neu, *Heidegger's Contributions to Philosophy: An Introduction*, 107–130.

[6]See Polt, *The Emergency of Being: On Heidegger's Contributions to Philosophy*, ch. 6.

itself a foundational constitution — one that enters at a lower level than Ereignis but that is no less subject to the Horizon Lemma's result at that lower level of entry.

10.3 The Structural Insufficiency Theorem

Remark 10.4 (On the Logical Status and Method of the SIT). A referee may observe that the SIT is analytic: once foundational constitution is defined to include condition (iii) (resources defined over features at level n or above), the foreclosure follows by definition. The Horizon Lemma derives (iii) from (i) and (ii), but a deeper objection presses further: does the triviality reappear at the level of the Lemma? If "foundational constitution" simply means "taking as a starting point from which one builds upward," then it follows by definition that one does not build downward. Tautologies cannot establish philosophical conclusions.

The response is precise. The SIT is not offered as a tautology. Its non-trivial content lies in what the Horizon Lemma establishes about the philosophical practice of foundational constitution. The Lemma does not merely say: frameworks with restricted resources cannot reach outside their restriction. It says: the *act* of philosophical foundational constitution — taking a structural feature as the terminus of explanation — *produces* a framework with domain-restricted resources, in the sense that the framework's conceptual apparatus, developed to operate on that feature and what it enters into, does not possess instruments for descending below it. The non-trivial claim is that this domain restriction is not a contingent limitation that could be remedied by additional effort within the framework, but a structural consequence of what foundational constitution *is*: an act of establishing a terminus. Termini do not have what comes before them; they are constituted as the point before which one does not inquire, within the framework that constitutes them.

The SIT is therefore not the tautology "frameworks cannot reach outside their domain." It is the substantive claim that the philosophical act of foundational constitution of a structural feature at level n is identical with the act of establishing level n as the terminus of inquiry within $\mathcal{F}$, and that this identity is not contingent but structural: one cannot foundationally constitute F at level n within $\mathcal{F}$ and simultaneously conduct an inquiry within $\mathcal{F}$ that treats F as non-terminal. The SIT names this structural identity as "foreclosure" and proves that it holds for every framework satisfying Definition 10.4. The proof's content is not the tautology; it is the demonstration that the definition's conditions jointly constitute the foreclosure. The non-trivial content is therefore this: the formal definition captures a real feature of how philosophical frameworks are actually built. The Horizon Lemma shows that any framework constructed by the method of foundational constitution — as all thirteen traditions surveyed actually were — inherits the domain restriction as a consequence of how their conceptual apparatus was developed, not merely as a consequence of how we choose to define 'foundational constitution.' The definition is not stipulative; it is descriptive of a practice, and the Lemma establishes that the practice produces the restriction. That the definition is descriptive rather than stipulative is established by the diagnostic survey of Part II: each of the thirteen traditions examined satisfies the three conditions of Definition 10.4, and each exhibits precisely the domain restriction that the Horizon Lemma predicts for a framework satisfying those conditions. The convergence of the survey's results with the Lemma's prediction is what converts the definition from a merely possible characterisation into a demonstrated one.

A second objection, distinct from the tautology worry, holds that the eliminative transcendental method is conceptual analysis under a different name — that identifying what the relational thesis requires for coherence is precisely what conceptual analysis does: it unpacks the conceptual commitments of a concept and identifies the necessary conditions for its application. Three structural differences distinguish the methods.

First, conceptual analysis proceeds from a concept of X to the conditions for X's correct application; eliminative derivation proceeds from the structural commitments of X — what X requires in order to remain structurally coherent — to a residue that is not derivable from anything else. The direction is from inside the structure outward, not from a concept inward.

Second, conceptual analysis operates on a fixed conceptual stock and characteristically produces analytic truths — truths that hold in virtue of the meanings of terms involved. Eliminative derivation operates on a candidate set of structural features and produces a residue whose status is not analytic but structural: it is what remains after everything eliminable has been eliminated. The residue is not a conceptual necessity but a structural one, established by the eliminative procedure rather than by the analysis of any concept.[7]

Third, the SIT's conclusion is not the analytic truth 'foundationally constituted frameworks cannot reach below their foundational level' — a claim about what the concept of 'foundational constitution' analytically contains. It is the structural claim that the philosophical act of foundational constitution produces a domain restriction as a structural consequence of how conceptual apparatus is constructed. This is a claim about a structural fact about frameworks, established by the Horizon Lemma, not a claim about the content of any concept.

Theorem 10.1 (Structural Insufficiency)**.** Let $\mathcal{F}$ be any framework in relational ontology that foundationally constitutes a feature F at level n ($1 \leq n \leq 5$). Then $\mathcal{F}$ structurally forecloses the latent register: there is no derivational pathway within $\mathcal{F}$ that derives Level I as a structural consequence of conditions belonging to the latent register.

Proof. By Definition 10.4, $\mathcal{F}$ foundationally constitutes F at level n if and only if: (i) F is at level n within $\mathcal{F}$; (ii) F is not derived within $\mathcal{F}$ from conditions derivable independently of F; (iii) $\mathcal{F}$'s derivational resources are defined over features at level n or above.

By the Horizon Lemma (Lemma 10.1), condition (iii) is not a stipulation but a structural consequence of conditions (i) and (ii): any framework that foundationally constitutes a

[7]Compare Olenius, "Causality Without Time."

feature at level n is thereby calibrated to the domain of features at level n or above, and inherits a formal horizon below which its derivational instruments cannot resolve.

Suppose for contradiction that there exists a derivational pathway P within $\mathcal{F}$ that derives Level I from conditions in the latent register — that is, from conditions below Level I and not themselves features of the five-level structural order.

Since $n \geq 1$, the latent register is below level n (by Definition 10.2, the latent register is below Level I, and Level I is at or below level n). Therefore P must pass through conditions below level n — specifically, through conditions in the latent register, which are not features at any level of the structural order.

But by the Horizon Lemma, $\mathcal{F}$'s derivational resources — whose domain is established by conditions (i) and (ii), not imposed externally — do not extend below level n. A derivational pathway that passes through conditions below level n requires the execution of inferential steps for which $\mathcal{F}$'s calibrated apparatus has no instruments. The pathway cannot be executed within $\mathcal{F}$.

Therefore no such pathway P exists within $\mathcal{F}$. By Definition 10.5, $\mathcal{F}$ structurally forecloses the latent register.

Corollary 10.1 (Foreclosure of Levels Below Entry)**.** Under the conditions of Theorem 10.1, if $\mathcal{F}$ foundationally constitutes a feature at level n, then $\mathcal{F}$ structurally forecloses every level

$m < n$: for each $m \in \{1, \ldots, n-1\}$, there is no derivational pathway within $\mathcal{F}$ that satisfies the defining conditions of level m.

Proof. By the same argument as Theorem 10.1, replacing the latent register with level m ($m < n$). Since level m is below level n, it is outside the domain over which $\mathcal{F}$'s derivational resources are defined. No derivational pathway within $\mathcal{F}$ can satisfy the conditions of level m.

10.4 The Eliminative Corollary

Theorem 10.1 establishes a negative result: no framework that foundationally constitutes a non-latent feature can reach the latent register from within its own resources. The eliminative corollary is the positive counterpart: it specifies the only type of derivation not structurally foreclosed from the latent register, and establishes that any such derivation, conditional on termination, reaches the latent register.

Corollary 10.2 (The Eliminative Corollary)**.** The eliminative transcendental method is the only type of derivation that is not structurally foreclosed from the latent register by its own commitments. Specifically: a derivation D reaches the latent register only if D proceeds by the eliminative method: the systematic derivation of each structural feature, including every feature that any tradition takes as its foundational primitive, from conditions that are themselves formally derivable without constituting any such feature as a terminus of explanation. The eliminative transcendental method is not guaranteed to succeed; it is uniquely compatible with the latent register's accessibility, in the sense that it alone is not structurally foreclosed before it begins.

Proof. Necessity: Suppose D reaches the latent register. Then D does not foundationally constitute any feature at any level $n \geq 1$ as a primitive; if it did, Theorem 10.1 would immediately foreclose the latent register from within D's framework (via the Horizon Lemma). Therefore D derives every structural feature it employs from conditions that are themselves derivable without foundational constitution. This is precisely the eliminative method. The necessity direction is therefore established.

Structural compatibility (in place of sufficiency): Suppose D proceeds by the eliminative method. Then D imposes no foundational constitution of any feature at any level; by Theorem 10.1 and the Horizon Lemma, D inherits no formal horizon below which its instruments cannot resolve. D's derivational resources are not restricted in advance to any level of the structural order or above: no structural foreclosure has been produced by D's own commitments.

This does not establish that D will succeed in reaching the latent register — absence of structural foreclosure is a necessary condition for success, not a sufficient one. A derivation can lack the internal contradiction that would foreclose it from the start while still lacking the structural resources to produce a successful result. What the elim-

> inative method guarantees is that the derivation is not self-defeating before it begins: it is uniquely structurally compatible with the latent register's accessibility, in the sense that any other method — any method that constitutes a non-latent feature as its terminus of explanation — forecloses the latent register as a consequence of that constitution. The eliminative method is the only open pathway; it is not guaranteed passage.

Remark 10.5. The eliminative corollary establishes two results. First, the eliminative transcendental method is the necessary form of any derivation that could reach the latent register: any derivation that reaches the latent register must be eliminative (the necessity direction). Second, the eliminative transcendental method is uniquely compatible with the latent register's accessibility: it is the only type of derivation that is not structurally foreclosed by its own commitments before it begins. These two results together identify the eliminative transcendental method as the only open pathway to the latent register — without guaranteeing passage. Whether the eliminative method, applied to the presuppositions of relational ontology, in fact produces a derivation that satisfies the Level I conditions from latent register conditions — and thereby satisfies Levels II through V as derived structural consequences — is a further question. The corollary establishes the necessary form of any derivation that could succeed and the unique structural compatibility of that form with the latent register's accessibility. It does not guarantee that a derivation of this form exists or, if it exists, that it has been executed. The following chapters establish what such a derivation would need to supply.

Remark 10.6 (On the Self-Application of the SIT to the Eliminative Method). A methodologically alert reader may press the following objection: the eliminative method governs all derivational moves by the criterion of coherence — the systematic identification of what, among a candidate set of structural features, cannot be shown to be derivable from others. If coherence is the governing criterion, is it not a foundational constitution in the sense that the Horizon Lemma specifies? And if so, is the eliminative method itself subject to the SIT — calibrated to a coherence-governed domain and thereby foreclosed from whatever is structurally prior to coherence conditions?

The objection trades on an ambiguity between two kinds of structural involvement. The Horizon Lemma establishes that a framework is calibrated to domain D when it foundationally constitutes a *structural feature F at level n* within the five-level order — that is, when F is a feature within the object domain that the framework takes as its structural terminus of explanation (Definition 10.4, conditions (i) and (ii)). The calibration effect arises because the intellectual work of constructing the framework consists in characterising F and its structural consequences, which generates vocabulary adequate to F's domain and indeterminate below it. This is the mechanism Step 2 of the Horizon Lemma proof identifies.

Coherence — understood as the procedural constraint that every structural claim must be evaluable for consistency against the others — is not a structural feature at any level of the five-level domain. It is not Level I, II, III, IV, or V; it does not appear in Definition 1.1, Definition 1.2, or the conditions of any structural level. It is a meta-level norm governing the form of inquiry, not an object-level commitment about the content of the structural domain being investigated. The distinction is categorical: a framework foundationally constitutes a structural feature when it takes that feature as the terminus of explanation of *what exists* at the relevant level; a framework operates under a procedural norm when that norm governs *how* claims about what exists are evaluated for validity. The Horizon Lemma applies to the former. It does not apply to the latter, because a procedural norm does not occupy any level within the five-level structural order and therefore does not trigger the constitution of domain-adequate vocabulary and the corresponding calibration effect that Step 2 of the Horizon Lemma's proof establishes.

The point can be stated directly: the SIT's scope condition is "any framework $\mathcal{F}$ that foundationally constitutes a feature F at level n ($1 \leq n \leq 5$)" (Theorem 10.1). The governing criterion of the eliminative method is not a feature at any level n within that range. It is therefore not a foundational constitution in the SIT's sense, and the SIT does not apply to it.

This does not mean the eliminative method escapes all constraint. R8 (the self-exemplification requirement) explicitly addresses the condition under which the ETM itself must be derived as the unique open pathway rather than taken as given — and acknowledges that W10 satisfies only the local form of R8, leaving the global form for the companion investigation. The distinction between the SIT's non-application to meta-level procedural norms and R8's requirement that the ETM itself be derivatively justified is therefore not a tension but a structural articulation: the SIT rules out object-level foundational constitutions; R8 addresses the methodological self-justification of the eliminative procedure itself.

The non-trivial content of the SIT is therefore precisely this: Definition 10.4 is not a stipulative definition of a concept but a descriptive characterisation of a philosophical practice, and the Horizon Lemma establishes that the practice produces the restriction as a structural consequence of how conceptual apparatus is constructed — not merely as a consequence of how we choose to define foundational constitution. The distinction matters because a purely stipulative reading would allow a framework to escape the foreclosure by simply declining to call its starting point a "foundational constitution" in the defined sense. The Horizon Lemma closes this escape route: any framework that (i) treats a structural feature at level n as the terminus of its account and (ii) does not derive that feature from conditions derivable independently of it will, in fact, have built its apparatus to characterise features at level n and above — because there is no other apparatus to build when the intellectual work of constructing a framework consists in characterising the consequences of a terminus. This is not definitional; it is a claim about the structure of explanation, verified by independently examined instances in Part II. A framework that wanted to dispute the Horizon Lemma's result would need to show that foundational constitution can be combined with apparatus adequate to the domain below the constituted level — which is precisely to show that the constituted feature is not in fact the terminus of explanation within the framework. The Lemma's result stands wherever the foundational commitment is genuine: wherever explanation stops, apparatus adequate to what is below the stopping-point is not built.

Remark 10.7 (On Non-Classical Logics and the SIT's Scope). The SIT proof proceeds by reductio ad absurdum. A reader familiar with paraconsistent logic — particularly the dialetheist programme of Garfield and Priest,[8] already engaged in the Madhyamaka discussion of Section 10.5 — may observe that the proof's final inference (from the contradiction generated by supposing a derivational pathway through the latent register to the conclusion that no such pathway exists within $\mathcal{F}$) is valid in classical but not in paraconsistent logic. This observation is correct, and the scope footnote acknowledges it. Two responses are available.

First, the SIT's proof structure can be reformulated without reductio. The direct formulation proceeds: a derivational move within $\mathcal{F}$ is by definition a move whose conclusion is evaluable by $\mathcal{F}$'s validation criteria (Step 1 of the Horizon Lemma proof). The Horizon Lemma establishes that $\mathcal{F}$'s criteria do not extend below level n (Step 3). Therefore no move within $\mathcal{F}$ produces a conclusion evaluable by $\mathcal{F}$'s apparatus about a domain below level n; a move that produced such a conclusion would not be evaluable and would therefore not be a move *within* $\mathcal{F}$. This is not a proof by contradiction but a direct proof from the Lemma's result and does not invoke excluded middle or ex falso quodli-

[8]Garfield and Priest, "Nāgārjuna and the Limits of Thought."

bet. The reductio formulation was chosen for perspicuity; the direct formulation is the logically more basic version and survives in paraconsistent contexts. The SIT's content is therefore not hostage to classical logic.

Second, Garfield and Priest's paraconsistent analysis of Madhyamaka establishes that the true contradictions arising at the limits of thought in Nāgārjuna's system are contradictions about the *content* of śūnyatā — about whether emptiness has svabhāva, and what this entails at the boundary of expressibility — not contradictions about the formal structure of derivability within a given framework. The SIT concerns not what is true or false about the latent register's content but what derivational pathways are available within a given framework given its calibrated domain. These are questions about the structure of inquiry, not about the truth-values of structural claims. A paraconsistent logic that permits true contradictions about whether X has property P does not thereby license a framework's apparatus to operate on a domain the apparatus was not built to address. The paraconsistent move addresses content; the SIT addresses the boundaries of derivational capacity. The two operate at different levels, and the dialetheist position does not constitute a counter-instance to the theorem.

10.5 The Scope of the Theorem

Theorem 10.1 is a general formal result. Its proof relies on the definitions of foundational constitution, structural foreclosure, and the latent register — not on the specific content of any tradition surveyed in Part II. The theorem therefore applies to any framework in relational ontology that satisfies the conditions of Definition 10.4, whether or not that framework was surveyed in the preceding chapters.

This is the precise sense in which the pattern identified in Chapter 9 is not an inductive generalisation but a formal result. The thirteen traditions surveyed in Part II are instances of a class defined by the conditions of Definition 10.4. Theorem 10.1 applies to every instance of that class, regardless of whether it has been surveyed. Any tradition in relational ontology that foundationally constitutes a non-latent feature — any tradition at all, in any domain of inquiry, whether philosophical, scientific, or otherwise — inherits the structural foreclosure of the latent register as a necessary consequence of its foundational commitment.[9]

The scope claim has a further consequence. There is no tradition in relational ontology that has not been surveyed but might, if surveyed, be found to have already reached the latent register from within its own foundational commitments. The theorem forecloses this possibility for every framework satis-

[9]The scope claim should be stated with precision. The theorem applies to frameworks that satisfy the formal conditions of Definition 10.4. It is conceivable that a framework might resist those conditions in some way — by adopting a non-classical logic, by rejecting the derivational structure the definition assumes, or by some other formal move. The theorem does not claim invulnerability to every conceivable philosophical manoeuvre. It claims that any framework satisfying the three conditions of the definition — which are the conditions that every framework surveyed in Part II satisfies, and which are the conditions that any framework must satisfy if it is to constitute something as a foundational primitive in the sense relevant to relational ontology — inherits the foreclosure. This is a wide but not universal scope.

fying its conditions. The latent register is not a territory that some tradition has reached without our having noticed. It is a territory that no tradition *can* reach from the direction from which every tradition has approached it.

The theorem's scope, however, should be stated with a qualification that intellectual honesty requires. The survey of Part II instantiates the theorem through thirteen traditions from the Western philosophical record. Whether the theorem's conditions are satisfied by frameworks in other traditions is a question this volume does not investigate with the depth those traditions deserve. What can be said is this: the theorem applies to any framework satisfying the three conditions of Definition 10.4, and the application of those conditions to any specific tradition is a substantive philosophical question that requires engagement with that tradition's primary texts and internal logic.

The strongest candidate for scrutiny from outside the Western survey is Nāgārjuna's Madhyamaka. Nāgārjuna's *Śūnyatā* (emptiness) is deployed to establish that no entity has intrinsic existence and that all entities arise through dependent origination (*pratītyasamutpāda*) — a structural move that, from the outside, resembles the eliminative method this volume identifies as the only path to the latent register. Whether Madhyamaka foundationally constitutes *śūnyatā* as a positive structural domain, thereby satisfying condition (ii) of Definition 10.4 and inheriting the structural foreclosure the theorem establishes — or whether *śūnyatā* functions precisely as the refusal to constitute anything as foundational — is a question that requires direct engagement with Madhyamaka's own internal logic. Garfield and Priest[10] establish that Nāgārjuna discovers "true contradictions arising at the limits of thought" and that the emptiness of emptiness — *śūnyatā* being itself empty — represents a genuine limit contradiction rather than a positive structural primitive. If this reading is correct, Madhyamaka does not constitute *śūnyatā* as a foundational domain in the sense Definition 10.4 requires — it refuses that constitution as its central move. This would mean Madhyamaka satisfies neither condition (ii) nor condition (iii) of the definition and therefore falls outside the theorem's scope: it would be not a counterexample but an independent instance of the structural refusal that the eliminative transcendental method itself embodies. The *Vigrahavyāvartanī* (VV), Nāgārjuna's direct engagement with the self-referential challenge, supplies the decisive textual evidence for resolving the Candrakīrti–Bhāviveka question. In VV 22–29, the interlocutor presses the following objection: if the Mādhyamika's own statement "all dharmas are empty" possesses svabhāva, it contradicts the thesis; if it lacks svabhāva, it cannot establish anything. Nāgārjuna's response is structurally decisive: he accepts without qualification that the statement lacks svabhāva and argues that this acceptance is not self-defeating but precisely what the thesis entails. The statement functions at the conventional level — it is a valid linguistic act within a field where all things (including the act of stating) lack inherent existence — without asserting *śūnyatā* as a positive structural domain that the act itself exemplifies or inhabits.

[10] Id.

Westerhoff's analysis of the VV establishes that Nāgārjuna's response does not constitute *śūnyatā* as a meta-level positive feature: the reflexive application of emptiness to the emptiness thesis does not assert that *śūnyatā* has its own svabhāva-free mode of existence as a structural entity. Rather, *śūnyatā* functions as what Westerhoff calls a "tool notion" — a conventional designation whose use is part of the eliminative procedure rather than a metaphysical commitment to the existence of emptiness as a domain.[11] The VV response thereby confirms the Garfield–Priest (2003) reading: *śūnyatā* generates limit results without constituting a positive structural primitive.

The Candrakīrti–Bhāviveka debate supplies the systematic form of this insight. Bhāviveka holds that Mādhyamikas must formulate independent autonomous inferences (*svatantrānumāna*): prasaṇga arguments are insufficient; positive theses must be asserted. Candrakīrti responds that Mādhyamikas operating through prasaṇga alone assert no positive thesis of their own — they demonstrate incoherence in the opponent's position and let the logical consequence stand. The VV supports the Candrakīrti reading: Nāgārjuna never

[11]Jan Westerhoff, *Nāgārjuna's Madhyamaka: A Philosophical Introduction*, The most rigorous recent analytic treatment of Nāgārjuna's texts. Chapter 7 analyses the *Vigrahavyāvartanī* (VV) in detail, establishing that śūnyatā functions as a "tool notion" in Nāgārjuna — a conventional designation whose use is part of the eliminative procedure — rather than a meta-level positive structural domain. This reading confirms that the reflexive application of emptiness to the emptiness thesis in VV 22–29 does not constitute śūnyatā as a foundational structural primitive in the sense of Definition 10.4; it is an application of the eliminative procedure to the procedure's own operator, not an assertion of the operator's positive metaphysical status (Oxford: Oxford University Press, 2009), ch. 7.

asserts a positive content for *śūnyatā*; he uses it to dissolve the opponent's svabhāva claims without constituting it as the foundational content from which further structural conclusions are drawn.

On the Candrakīrti reading, Madhyamaka does not satisfy condition (ii) of Definition 10.4: no structural feature is taken as a primitive terminus of explanation, because the framework is constituted specifically as the systematic refusal of such constitution. Madhyamaka therefore falls outside the SIT's scope on this reading — not as a counterexample but as an independent instantiation of the eliminative orientation the Eliminative Corollary identifies as the only open pathway to the latent register.

If the Bhāviveka reading were correct, Madhyamaka would fall within the SIT's scope: the autonomous inference form would constitute dependent origination as a positive structural feature at Level II–III, and the structural foreclosure would follow at whichever entry level the independent thesis occupies. The diagnostic result is therefore conditionally determinate: the Candrakīrti reading places Madhyamaka outside the SIT's scope (confirming the Eliminative Corollary); the Bhāviveka reading places it within the SIT's scope (confirming the theorem). Neither reading constitutes a counter-instance. A conditionally determinate result is not an absence of a verdict: it is the correct form of verdict when the relevant determining question is genuinely contested within the tradition's own scholarship. A grid that could produce only one verdict regardless of which reading of the primary texts is correct would be insensitive to the textual evidence — which is a failure of precision, not its demonstration. That the grid produces different verdicts for different readings, with both verdicts logically compelled by the grid's formal conditions and the respective readings' structural commitments, is the signature of formal precision, not of indecision. The grid yields a determinate conditional result for Madhyamaka, and the selection criterion is thereby fully vindicated: the strongest available external candidate either confirms the theorem directly or instantiates the eliminative orientation the theorem identifies as its only alternative.

Non-Madhyamaka Buddhist traditions — including Theravāda dependent origination and Yogācāra consciousness-only accounts — are not individually surveyed here; their structural analysis requires engagement with primary texts at the level the Madhyamaka treatment demands and is reserved for a subsequent treatment. Intentionality — the directedness of consciousness towards its objects — is similarly not included among the eight requirements, because the investigation concerns the structural ground of relation as such, which is prior to and generative of the specific relational form intentional acts take: intentionality is a form of relation, not its structural ground.

10.6 The Survey as Instantiation

With the theorem established, the relationship between the theorem and the diagnostic survey can be stated precisely.

The survey was not conducted to prove the theorem. Formal proofs do not proceed by example. The survey was conducted to make the structural pattern legible — to show, with the precision that full engagement with each tradition requires, that the pattern holds with the specificity and consistency that the theorem must generalise. The survey functions as a set of instantiations: concrete illustrations of what the theorem asserts in the abstract, richly enough developed that the theorem's formal conditions can be checked against each tradition and found to be satisfied.[12]

The result is a double verification. The formal proof establishes the theorem's necessity. The survey establishes its applicability: the conditions of Definition 10.4 are not vacuous — they are satisfied by real traditions, developed with genuine rigour, in the history of philosophy. The theorem has force beyond the survey; the survey demonstrates that the theorem applies to actual philosophical work, not merely to possible but unrealised frameworks.

[12]The relationship between formal theorem and empirical survey here is similar to the relationship between a mathematical theorem about the convergence of a class of sequences and a sequence of worked examples illustrating convergence in specific cases. The examples do not prove the theorem; the theorem does not make the examples redundant. Each serves a function the other cannot.

10.7 What the Theorem Does Not Claim

The Structural Insufficiency Theorem must not be read as claiming more than it establishes. Three misreadings are possible and should be excluded.

First misreading: the theorem claims that relational ontology is impossible or incoherent. This is false. The theorem claims only that relational ontology, as it has been and can be conducted from within any foundational commitment, structurally forecloses the latent register. It says nothing about whether relational ontology is possible from a different starting point — namely, from a derivation that does not foundationally constitute any non-latent feature. The theorem forecloses a way of doing relational ontology; it does not foreclose relational ontology as such. Indeed, the Eliminative Corollary identifies the form of a derivation that is not foreclosed.

Second misreading: the theorem claims that the traditions surveyed have nothing of philosophical value to offer. This is false and is the inverse of what the investigation has established. The survey has documented genuine formal achievements at every level that each tradition occupies. The theorem explains why those achievements are bounded by the traditions' entry levels, not why they are worthless. Bounded achievement is still achievement.

Third misreading: the theorem claims that the latent register is real and that it has a specific character that the positive derivation has already discovered. The theorem makes no such claim. It establishes that the latent register exists as the presupposition of every foundational commitment, and that it is formally reachable by the eliminative method. It does not characterise the latent register positively. That characterisation is the work of the Stage II proposition and falls outside the scope of the present volume.

Fourth misreading: the joint satisfiability proof in Chapter 11 guarantees that the present investigation satisfies all eight requirements in full. This is false. Theorem 11.1 establishes that a derivation satisfying all eight requirements is logically possible and internally coherent; it does not establish that the present investigation is that derivation. Requirement R8 in its global form — requiring that the inquiry-as-a-whole proceed through the structural sequence as its own method, instantiating in its own formal movement the dynamics it establishes — is satisfied by the present investigation only in the local sense: the logical-witness model $\mathcal{M}$ exhibits the structural sequence R8 specifies. The global satisfaction of R8 requires that a completed derivation of the latent register enact that sequence in the order of its own inquiry. That derivation falls outside the scope of the present volume, which is explicitly a Stage I investigation. This is an architectural decision, not a defect. The function of Stage I is to establish with formal rigour what Stage II must supply, so that Stage II can be assessed by a standard derived independently of it. A Stage I investigation that also claimed to satisfy R8 globally would be indistinguishable from a system that designed its own validation criteria and then applied them to itself — precisely the self-validation circularity the General Validation Framework of Volume IV is designed to prevent. The joint satisfiability proof establishes that the eight requirements are coherent and mutually consistent; it does not constitute their collective satisfaction.

Summary of Chapter 10. The Structural Insufficiency Theorem establishes formally what Chapter 9 established inductively: every framework in relational ontology that foundationally constitutes a non-latent structural feature structurally forecloses the latent register as a consequence of foundational constitution itself. The proof proceeds from four definitions — structural level, latent register, foundational constitution, and structural foreclosure — to the theorem and its two corollaries by formal entailment, without appeal to any specific tradition surveyed in Part II. The Eliminative Corollary identifies the necessary form of any derivation that can reach the latent register: the eliminative method, the systematic derivation of every structural feature from conditions more primitive than the feature itself, without foundational constitution of any non-latent feature as a terminus of explanation. What such a derivation would need to supply, stated as eight jointly necessary conditions, is the subject of Part IV.

PART IV

The Formal Requirements

CHAPTER 11

What the Ground Requires

When you have eliminated the impossible, whatever remains, however improbable, must be the truth.

Arthur Conan Doyle,
The Sign of the Four

The detective's maxim is a statement about derivation under constraint. When the space of possibilities has been formally bounded — when every possibility except one has been shown to be foreclosed by the structural conditions in play — what remains is not a guess but a derivational result. The investigation that Part II conducted and Part III formalised has done precisely this: it has established, with the precision of the Structural Insufficiency Theorem, that no existing framework can derive the structural ground of relational ontology from within its own foundational commitments. What remains is a question that has been formally bounded: not "what might the ground be?" but "what must a derivation that reaches the ground be able to supply?"

Scope of this chapter. The present chapter specifies the conditions any satisfying derivation must meet. It does not execute the derivation those conditions govern, and it does not describe the positive character of what the latent register contains. What the ground is — its content, its structural texture, what it generates and how — belongs to the companion investigation. What the ground *must be* in order to satisfy the eliminative constraints the diagnostic survey has established: that is the question this chapter answers. A reader who arrives at R1–R8 expecting to learn what lies below every tradition's entry level will find instead the formal specification of the shape any answer must take. That specification is a genuine result; it is not a deferral. Knowing exactly what a derivation must supply — with the precision these requirements provide — is knowing something substantive about the structural domain the derivation must reach.

The present chapter answers this question as an answer to the first ought to be answered: not by describing what the ground is (that is the work of the derivation, not of the investigation that establishes its necessity) but by deriving, from the Structural Insufficiency Theorem and its corollaries, the eight conditions that any derivation of the ground must satisfy. These eight conditions are not imposed from outside. They are the exact inverse image of what the thirteen traditions surveyed in Part II have failed to supply. They are what the structural pattern demands.

Why exactly eight? The answer is that each requirement is the structural complement of a specific foreclosure documented in the diagnostic survey, and the total follows from the structure of the five-level grid plus two cross-cutting patterns. Six of the eight requirements are level-specific: R1 (derived ground) corresponds to the foreclosure of Level I; R3 (relational individuation without regress) corresponds to Level II; R4 (encounter source) corresponds to Level III; R5 (relational yield) corresponds to Level IV; and Level V generates two requirements because its direction dimension has two formally distinct sub-conditions — R6 (co-instantaneous relational event structure) addresses the full event architecture at Level V, while R7 (temporal asymmetry derived from the event's structure) addresses the specific foreclosure of the temporal derivation that every tradition at Level V fails. Two further requirements are cross-cutting — not complements of any single level's foreclosure but of patterns that hold across all levels: R2 (pre-temporal latent register) is the complement of the universal condition, shared by every tradition surveyed, of constituting the temporal or relational register as the starting point rather than deriving it; and R8 (self-exemplification) is the complement of the diagnostic finding that no tradition's derivational method enacts the structural dynamics it formally establishes — a failure that applies regardless of entry level. Six level-specific requirements plus two cross-cutting requirements equals eight. The count is not arbitrary and not retroactively constructed; it is the formal consequence of the diagnostic results.

R1–R8 as conditions, not descriptions. Before the eight requirements are stated, one structural point must be placed on record. R1–R8 are formal conditions on any derivation that would satisfy them, not descriptions of any specific derivation that does. A reader who arrives at the definitions below and finds them reading as a description of a system the author already possesses should treat that impression as an artefact of the requirements' specificity, not as evidence that they were designed for a particular system. The test of this is formal: each requirement is stated as a condition that admits of multiple possible satisfiers. R1 requires a derived structural ground; it does not specify the ground's content. R2 requires that the derivation begin from a domain formally prior to the temporal order; it does not specify what that domain is. R3–R6 specify what the derivation must produce and how; they do not specify the route. R7 and R8 specify conditions on the derivation's operating context and its self-exemplification; they do not specify what a derivation meeting those conditions would reach. A derivation could satisfy R1–R8 in multiple ways. If only one derivation is possible, that would be a further result requiring demonstration — it is not entailed by the requirements themselves. The requirements are neutral with respect to any particular satisfying derivation, and any reader wishing to assess a proposed derivation against them has all the resources to do so independently of anything the present volume asserts about any candidate.

11.1 The Method of Derivation

Each requirement is derived by the same procedure. The Structural Insufficiency Theorem establishes that a specific kind of structural result — deriving a specific level from conditions below it — is foreclosed by every existing framework. Each requirement names what a derivation that is not foreclosed must be able to demonstrate. The requirement is therefore not an arbitrary desideratum but a formally derivable consequence of the theorem: if a derivation were to fail this condition, it would either have foundationally constituted a non-latent feature (inheriting the theorem's foreclosure) or would have left unsatisfied a necessary condition of Level I's four defining criteria (failing to reach the ground in the first place).

The derivation of each requirement from the diagnostic failure analysis proceeds by a specific formal step: for each tradition, the diagnostic survey establishes what that tradition fails to supply at its structural entry level. The requirement is the structural complement of that failure — what any account that is not foreclosed in the way the tradition is foreclosed must supply.

The determinacy of the complement. The complement operation is not a choice among several possible formulations. Each structural level has exactly one gap: what must be supplied for a derivation to satisfy that level's defining conditions without foundationally constituting a non-latent feature. Level I's gap is the derived ground (R1); removing the foundational constitution of any Level I feature leaves exactly one structural void, and R1 names that void. Level II's gap is relational individuation without the regress (R3); the two available alternatives — pre-given units and purely relational constitution — are both shown to fail, leaving exactly one structural condition as the complement: derivation of relata from the ground's own dynamics. This pattern of two-sided elimination — showing both that the foundational constitution forecloses the level and that the available alternatives each generate the regress or the regress's complement — is what makes each requirement the *unique* structural consequence of its corresponding failure. The procedure is eliminative identification: each requirement is the residue left when every alternative to it is shown to generate the failure it was introduced to resolve. The requirements are therefore not the products of reflection on the failures but of elimination over the structural alternatives at each level. R1 through R8 are what remain when everything else is removed.

R1 (derived ground) is the complement of every tradition's failure to derive Level I from conditions more primitive: every tradition that foundationally constitutes its ground rather than deriving it fails what R1 requires. The parallel across the classical and contemporary analytic traditions is exact: Spinoza foundationally constitutes Substance as *causa sui* — self-caused, requiring no external principle, admitting no exterior — and thereby forecloses the question of what generates Substance's self-sustaining structure; Schaffer foundationally constitutes the cosmos as the unexplained intrinsic whole from which finite things derive their properties by dependence, and thereby forecloses the same question at the same structural level. Both positions reach Level I; neither derives it. R1 names what any account that does not repeat this failure must demonstrate. R2 (pre-temporal latent register) is the complement of every tradition's beginning within the temporal or relational order: every tradition enters the temporal register from within, so what a non-foreclosed account must supply is a pre-temporal domain. R3 (relational individuation) is the complement of the relational regress identified at Level II in every tradition. R4 (encounter source) is the complement of the external perturbation principle identified at Level III. R5 (relational yield) is the complement of the failure to derive irreducibility and encounter-specificity identified at Level IV. R6 (relational event structure) is the complement of the black-box character of the relational event in every tradition. R7 (temporal asymmetry) is the complement of every tradition's presupposition of temporal directionality rather than derivation of it. R8 (self-exemplification) is the complement of the observation that no tradition's derivational method enacts the structural dynamics it formally establishes.

One of the eight requirements — R8, the eighth — does not correspond to a single structural level but to the diagnostic finding that every tradition's derivational method is mismatched to the structural dynamics it establishes: it describes what the ground must supply without itself instantiating those dynamics. This is a structural asymmetry, not a stylistic observation, and its complement — a derivation that exemplifies what it formally establishes — is as structurally load-bearing as the level-specific requirements R1 through R7.

Each requirement is presented in four components that follow the consistency architecture established at the outset. The first component states the requirement as a question — the form in which the diagnostic survey has made it urgent. The second states it as a formal condition. The third provides a concrete illustration from a non-philosophical domain. The fourth states explicitly why no existing tradition meets the condition, cross-referencing the specific diagnostic results of Part II.

11.2 Requirement One: A Formally Derived Absolute Ground

The question it answers. What would it mean to begin before every tradition's starting point? Not to stipulate a ground more persuasively than prior stipulations, or to posit a ground more fundamental than prior posits, but to *derive* the ground — to show that it is the unique structural result of what every relational ontology must presuppose, derived by the eliminative method without foundational constitution of any non-latent feature.

Definition 11.1 (Requirement R1: Derived Ground)**.** A derivation satisfies Requirement R1 if and only if it demonstrates that the structural ground — the feature that meets all four conditions of Level I: totality, generativity, self-sustenance, and formal derivability — is the unique structural result of the eliminative method applied to the presuppositions of relational ontology, without foundational constitution of any feature at Level I or above as a primitive terminus of explanation.

Concrete illustration. The derivation of the natural numbers from pure logic in Frege's *Grundlagen* supplies the structural model. Frege did not posit numbers as primitive objects. He showed that the concept of a natural number, including all its formal properties, can be derived from logical concepts more primitive than arithmetic — from the concept of equinumerosity, the concept of a concept falling under another, and the formal apparatus of second-order logic. The derivation did not describe what numbers are from the outside; it showed that, given the logical apparatus, numbers must be what they are.[1] R1 requires the same operation for the structural ground of relational ontology: not a description of what the ground is but a derivation that shows it must be what it is, given the formal presuppositions that every relational account must make. That any adequate account of the ground must proceed by elimination rather than assertion — and that structural constraints on what the ground must be are derivable rather than merely asserted — has been established independently by Lucero,[2] who uses grounding theory to derive that the ultimate ground must have aseity (it cannot depend on anything external), simplicity (it can have no parts, since wholes are grounded in their parts), immutability and atemporality (since dependence structures that change would introduce are incompatible with ultimacy). Lucero's grounding-theoretic derivation converges on the same structural conclusion as R1 from the direction of the philosophy

[1] Frege's derivation, its successes, and its failure at the level of Basic Law V are documented in Frege (*Die Grundlagen der Arithmetik*). The contemporary reconstitution that rescues the derivation from Russell's paradox — Frege's Theorem, as proved by Wright (*Frege's Conception of Numbers as Objects*) — is relevant here: it shows that the core of the derivation survives even the failure of the specific axiom that generated the contradiction. What matters for the present illustration is the structural model, not the technical details of Frege's own execution.

[2] Lucero, "What the Ground of Beings Must Be: Aseity, Simplicity, Immutability, Atemporality."

of religion: what grounds the structural order cannot itself occupy a position within that order, and the ground's structural character admits derivation — cannot be merely stipulated.

Why no existing tradition meets R1. Spinoza comes closest. His Substance meets three of the four Level I conditions. But the fourth — formal derivability, the requirement that the ground be shown to be the unique structural result of the eliminative method — is explicitly refused: Substance is the terminus at which derivation stops, not a result that derivation produces. Schaffer's priority monism, the strongest contemporary analytic candidate for a Level I account, meets the same three conditions — totality (G_1), partial generativity (G_2), and self-sustenance (G_3) — and fails the same fourth: the cosmos is the terminus of explanation, not a structural result derived by the eliminative method. The convergence with Spinoza is structural, not merely rhetorical: both achieve what can be achieved by a ground that is posited as total and self-sustaining without being shown to be the unique consequence of eliminating every non-primitive feature that relational ontology presupposes. Every other tradition foundationally constitutes a non-latent feature at a level above Level I, thereby inheriting, by the Structural Insufficiency Theorem, the foreclosure of Level I derivation entirely.

11.3 Requirement Two: A Pre-Temporal Latent Register

The question it answers. Every tradition surveyed — Whitehead, Bergson, Heidegger, relational quantum mechanics, and the rest — begins within the temporal or relational order. Their accounts of relation, encounter, and yield are accounts of what happens within a structural domain already in operation. None asks what is structurally prior to the relational order's being in operation at all: what must be the case about reality before any relation occurs, before any relatum exists, before any encounter is possible.

Definition 11.2 (Requirement R2: Pre-Temporal Latent Register)**.** A derivation satisfies Requirement R2 if and only if it establishes a structural domain formally prior to the temporal and relational orders — a domain in which no structural feature presupposes temporal succession, relational encounter, or the existence of individuated relata — and derives the temporal and relational orders as structural consequences of the dynamics of that prior domain.

Pre-temporal and atemporal distinguished. A referee may press the following objection: an *atemporal* domain — one that exists outside time in the manner of abstract objects, mathematical structures, or the Lewisian plurality of possible worlds — also satisfies the minimal negation of the temporal presupposition. It is not temporal. Why does R2 require "pre-temporal" specifically, rather than the weaker and more familiar "atemporal"?

The distinction carries structural weight. An atemporal domain is one whose content is fixed and self-identical across all possible temporal contexts — it has no internal dynamic. Abstract objects and mathematical structures are atemporal precisely because they undergo no process: their "content" is what it is necessarily, without activity internal to them. A ground that is merely atemporal therefore satisfies G1 (totality, construed as logical necessity spanning all possibilities), G3 (self-sustaining, since no external principle maintains it), and arguably G4 (formally derivable as the realm of mathematical necessity), but decisively fails G2 (generativity). An atemporal domain cannot produce structural differentiation from within its own dynamic because it has no dynamic: it is the same at every moment, or rather has no moments at all. The temporal order would not emerge from an atemporal ground as a structural consequence; it would simply be unexplained from within such a ground, since a domain with no internal activity cannot produce a domain with an internal dynamic as its structural output.

The Lewis objection presses this most sharply: the Platonic realm of mathematical structures is atemporal, total, and self-sustaining. Does it satisfy R2? The answer is no, for the reason just given: G2 is not met. Mathematical structures do not generate the specific configuration of relata and their relational web from within a differentiating dynamic; they simply *are* the structures they are, necessarily. R2 specifies a pre-temporal domain rather than an atemporal one because "pre-temporal" names what is required: a domain with an internal dynamic from which the temporal structure of encounter events emerges as a structural consequence — not a domain that is simply outside time while time exists alongside it as a brute complement. "Pre-temporal" is the minimum that satisfies both G2 and the non-presupposition condition: it names a structurally prior dynamic, not a structurally inert domain that co-exists with what it supposedly grounds.

Concrete illustration. The distinction between a quantum field and a quantum particle is the nearest scientific analogue. A quantum field is not itself a particle; it is the structural domain from which particles emerge as local excitations under specific conditions. The field is prior to the particles not in time but in structural order: the field must be in place — characterised by its vacuum state, its symmetry group, its propagation dynamics — before any particle can exist or any particle interaction can occur. R2 requires an analogous structural priority for the latent register over the relational domain.[3]

Why no existing tradition meets R2. Whitehead, Bergson, and Heidegger each begin within the temporal order: actual occasions are temporal events; duration is irreversible temporal flow; being-in-time is the condition of all understanding. Spinoza's Substance is atemporal but is not formally prior to the relational order in the required sense — it does not generate the temporal and relational orders as structural consequences of its own internal dynamic. Ontic structural realism and relational quantum mechanics take the physical relational domain as their starting point and do not consider what is structurally prior to

[3]The analogy between the latent register and the quantum vacuum field is structural, not physical. The claim is not that the latent register is a quantum field or that it behaves like one. The claim is that the structural relationship between the latent register and the relational order is analogous to the structural relationship between a quantum field and its particle excitations: the former is prior to and generative of the latter, not in temporal sequence but in structural order of derivation.

that domain's being in operation. No tradition surveyed has established a pre-temporal latent register.

11.4 Requirement Three: A Formal Derivation of Relational Individuation

The question it answers. The relational regress identified in Chapter 1 cannot be resolved by any tradition that takes relata as pre-given units or constitutes them entirely through the relations they enter into. The regress demands a third option: a formal derivation that shows how structurally distinct relata emerge from the ground without being pre-given and without dissolving entirely into the relational web.

Definition 11.3 (Requirement R3: Relational Individuation)**.** A derivation satisfies Requirement R3 if and only if it derives, from the latent register and the Level I ground, a formal account of how structurally distinct relata emerge as locally determinate expressions of the ground's own dynamic — each uniquely individuated by its specific position within that dynamic — without presupposing them as pre-given units and without reducing their structural identity to the relations they contingently enter into.

Concrete illustration. The emergence of standing waves in a resonant cavity provides the structural model. A resonant cavity — a drum, a laser, a microwave oven — is a structural domain with specific boundary conditions. The cavity does not contain standing waves as pre-given units. The standing waves emerge from the cavity's own dynamics — from the reflection and superposition of travelling waves under the specific geometric and material constraints of the cavity — and each standing wave is uniquely determined by the cavity's mode structure. Remove the cavity and there are no standing waves; change the cavity's dimensions and the entire mode structure changes. R3 requires an analogous emergence of relata from the structural dynamics of the latent register and Level I ground.

Why no existing tradition meets R3. Whitehead's actual occasions are primitives: their numerical distinctness is given, not derived. Spinoza's modes follow from Substance by logical entailment, but logical entailment does not generate the kind of structural differentiation R3 requires — modes are differentiated by their attributes, not by an internal dynamic of weighted structural individuation from the ground. Ontic structural realism dissolves the object into the structure but cannot re-derive structurally distinct nodes from within a purely structural account without generating a new version of the regress it sought to resolve. No tradition surveyed provides a derivation of the kind R3 specifies.

11.5 Requirement Four: A Formally Derived Account of Encounter Source

The question it answers. What brings any two specific relata into relational contact? Not all relata encounter each other; encounter is selective, structured, and — in a non-trivial sense — necessary in its specific form. Yet every tradition that has an explicit account of encounter source either stipulates an external principle (Whitehead's God) or defers the question (relational quantum mechanics, which specifies the form of encounter but not what brings specific systems into contact at specific moments). The source of encounter must be derived from the same dynamic that produces the relata, without importing an external agent.

Definition 11.4 (Requirement R4: Encounter Source)**.** A derivation satisfies Requirement R4 if and only if it derives, from the dynamics of the latent register and the structural properties of the relata it has produced, a formal account of what brings specific relata into relational contact — an account that identifies encounter source as a structural consequence of the same dynamic that produces the relata, without external perturbation principle, without randomness stipulated as primitive, and without necessitation so complete that no encounter is genuinely novel.

Concrete illustration. Enzyme-substrate encounters in biochemistry provide the structural model. An enzyme does not encounter every molecule in its environment; it encounters specific substrates determined by its active site geometry — the specific three-dimensional configuration of binding residues that creates a structural affinity for molecules of complementary shape. The encounter source is not external to the enzyme; it is a structural consequence of the enzyme's own molecular architecture. And the encounter is not predetermined: the active site creates a *structural affinity*, not an *inescapable bond*. R4 requires an analogous structural account of encounter source:

a formal derivation of why specific relata have specific structural affinities for specific encounters, derived from the same dynamic that produces the relata themselves.

Why no existing tradition meets R4. The most explicit existing account of encounter source is Whitehead's: God's provision of an initial aim to each new occasion selects the prehensive pattern from the infinite field of possibilities. But the initial aim is an external perturbation principle — it imports, at the heart of the account of encounter, a source that is itself not derived from the dynamic that produces the occasions. Relational quantum mechanics specifies that encounter is defined by the interaction of physical systems but does not derive what brings specific systems into specific interactions. No tradition surveyed provides the derivation R4 specifies.

11.6 Requirement Five: A Formally Derived Account of Relational Yield

The question it answers. What does a genuine relational encounter produce that neither relatum contained before the encounter? This is the question of emergence without mystery — the question of how genuine novelty can be derivable rather than merely asserted. Every tradition acknowledges that relational encounter produces something new; few can say formally why. The novelty is stipulated as a feature of the encounter rather than derived as a structural consequence of the encounter's internal organisation.

Definition 11.5 (Requirement R5: Relational Yield)**.** A derivation satisfies Requirement R5 if and only if it derives, from the formal structure of the encounter and the structural properties of the relata, a result that is:

(i) *Formally new*: not a rearrangement of the structural content of the relata but a structural result that neither relatum contained and that no formal operation on either relatum alone produces;

(ii) *Formally bounded*: conserved within the structural resources that the Level I ground provides, without addition from outside and without the disappearance of structural content into nowhere;

(iii) *Formally generative*: itself capable of becoming a structurally distinct relatum in further encounters, carrying the formal mark of the encounter that produced it as part of its structural constitution;

(iv) *Asymmetrically novel*: the yield of the encounter between R_1 and R_2 is formally distinct from the yield of any encounter between R_1 and R_3 — the novelty is encounter-specific, not merely a generic property of all relational yield, so that the structural mark carried by the yield uniquely indexes the particular encounter that produced it.

Concrete illustration. Mycorrhizal networks in forest ecology provide the structural model. When the root system of a tree forms a mycorrhizal relationship with a fungus, the yield of the encounter is a new structural entity — the mycorrhizal network — that possesses properties neither the tree root nor the fungus separately possessed: the capacity to transfer nutrients across distances neither could reach independently, the ability to distribute carbon and water according to network-level demand signals, the emergence of a systemic resilience that belongs to the network as a whole rather than to any of its nodes. The yield is formally new (the network is not a sum of root and fungus), formally bounded (no energy is created; the network exchanges what is structurally available), and formally generative (the network is itself capable of further mycorrhizal connections).[4] R5 requires that the yield of relational encounter be derivable with this precision — not just asserted as emergent but derived as the formal structural consequence of a specific encounter between structurally specific relata.

Why no existing tradition meets R5. Whitehead achieves the most formally developed account of yield in the tradition: the satisfaction of a new occasion is genuinely novel, bounded by the material of its prehensions, and generative of new occasions through its own perishing. But the novelty of satisfaction is not formally derived — it is a property of the occasion's creative synthesis, which is itself not derived but given as the character of actual occasions as such. The formal derivation of why yield has these four properties — why it is necessarily new, bounded,

[4]The mycorrhizal network's structural properties are described in Simard (*Finding the Mother Tree: Discovering the Wisdom of the Forest*). The present use is illustrative; no claim is made about the metaphysics of fungal networks.

generative, and asymmetrically novel rather than merely being described as such — is not achieved. In particular, no tradition has derived why yield is encounter-specific in the indexed sense required by condition (iv): that the yield of the encounter between R_1 and R_2 be formally distinct from the yield of any encounter between R_1 and R_3 — so that novelty is relational-pair-specific, uniquely indexing the particular encounter that produced it, rather than a generic property of all relational yield.

11.7 Requirement Six: A Formally Derived Account of the Relational Event

The question it answers. Every tradition has an implicit account of the relational event — the structured moment in which relata come into contact, produce a yield, and are changed by the encounter. But the internal structure of that event — why it has the specific formal organisation it has, why it has exactly the dimensions it has rather than more or fewer, why those dimensions are simultaneous rather than sequential — is treated in every tradition as a given feature of encounter rather than a derived structural result.

Definition 11.6 (Requirement R6: Relational Event Structure)**.** A derivation satisfies Requirement R6 if and only if it derives, from the formal structure of the relata and the dynamics of the encounter source, a formal account of the relational event that:

(i) Identifies the internal dimensions of the event and demonstrates that exactly those dimensions, in that number, are necessary and jointly sufficient for the event to produce formally new, bounded, and generative yield;

(ii) Demonstrates why those dimensions are co-instantaneous rather than sequential — why the event cannot be decomposed into stages without ceasing to be the kind of event it is;

(iii) Derives the event's internal organisation from the same dynamic that produces the relata and the encounter source, without importing any structural principle specific to the event that is not already present in the prior derivation;

(iv) *Self-exemplifies*: the formal account of the relational event's structure is itself an instance of the structural dynamic it formally establishes, so that the account's own organisation instantiates the co-instantaneous multi-dimensional structure it specifies for the event. This is a local condition on the event-level account and is distinct from the global self-exemplification required by Requirement R8.

Concrete illustration. A fugue subject's first exposition in a Bach fugue illustrates the structural co-instantaneity that R6 requires. The moment the subject is presented is a co-instantaneous event: the melodic contour, the harmonic implication, and the formal function as statement-before-answer are not three sequential features that the listener receives one after another — they are three dimensions of a single structured event, each constituted in and through the others. Separate them and there is no fugue subject; there are only melodic fragments, harmonic intervals, and formal gestures that have lost the relational structure that constitutes them as what they are.[5] R6 requires that the relational event have a demonstrably co-instantaneous internal structure, derived from the prior derivation rather than asserted as a phenomenological given.

Why no existing tradition meets R6. Whitehead comes closest: the phases of concrescence — conformal, conceptual, and comparative feeling — are described as logically distinct but temporally co-instantaneous. But this co-instantaneity is asserted as a feature of actual occasions rather than derived from the structural dynamics that produce occasions. The formal derivation of why the relational event must have co-instantaneous internal structure — rather than merely being described as having it — is not achieved by any tradition surveyed. Nor does any tradition achieve self-exemplification of the event-level account (iv): the account of the relational event is invariably delivered from an analytical standpoint external to the structural dynamic it describes — itself a consequence of treating the relational event as a given rather than deriving it from prior relational dynamics, so that the account's own organisation cannot instantiate the co-instantaneous structure it specifies for the event.

[5]The analysis of fugal structure as a co-instantaneous formal event is developed in the contrapuntal theory of Westergaard (*An Introduction to Tonal Theory*) and, in a more analytically accessible form, in Mann (*The Study of Fugue*).

11.8 Requirement Seven: A Formally Derived Account of Temporal Asymmetry

The question it answers. Time flows in one direction. The past is determinate; the future is open. Before and after are not interchangeable. This asymmetry is experienced as the most fundamental structural feature of lived reality. Yet every tradition that addresses it — Bergson most deeply, Heidegger most formally — treats temporal asymmetry as a given feature of the structural domain within which their accounts operate, not as a derived structural consequence of something more primitive. The question of *why* time has a direction at all — not merely the description of what that direction consists in — remains formally unaddressed in every tradition surveyed.

Definition 11.7 (Requirement R7: Temporal Asymmetry)**.** A derivation satisfies Requirement R7 if and only if it derives temporal asymmetry — the irreversibility of the structural ordering before and after — as a structural consequence of the relational event's internal organisation, without presupposing temporal directionality as a primitive feature of any level of the structural order or of the latent register. In particular, the derivation must show how the co-instantaneous event structure established by R6 generates temporal asymmetry as a consequence of what the event produces: why the yield of every relational event becomes part of the structural conditions that determine subsequent events, while the reverse direction of influence is structurally foreclosed.

Concrete illustration. The irreversibility of sedimentation in geology provides the structural model. When sediment deposits on a riverbed, the deposit becomes part of the bed's physical constitution — it shapes the flow dynamics that will govern subsequent deposition, but subsequent deposition cannot undo what has already been deposited. The deposit is not reversible: there is a structural asymmetry between the layer that deposited first and the layer that deposits next, not because of anything in the sediment particles themselves but because of the structural role each layer plays in constituting the conditions of subsequent deposition. R7 requires an analogous account of temporal asymmetry: the yield of each relational event deposits into the structural conditions of subsequent events — and that deposition is irreversible, not as a contingent physical fact but as a structural consequence of what relational yield is.[6]

Why no existing tradition meets R7. Bergson describes temporal irreversibility with more phenomenological precision than anyone else in the tradition: duration is asymmetric because the past accumulates in the present in a way that cannot be undone. But Bergson does not derive this irreversibility — he begins within it. Heidegger formally analyses the structure of temporality as ecstatic, with each temporal ekstasis related asymmetrically to the others. But this asymmetry is presupposed as a feature of the being-structure of Dasein, not derived

[6]The irreversibility of sedimentary deposition is used here as a structural analogy, not as a physical explanation of temporal asymmetry. For the physics of thermodynamic irreversibility, see Price (*Time's Arrow and Archimedes' Point*). For the philosophical analysis of temporal asymmetry, see Callender (*The Oxford Handbook of Philosophy of Time*).

from a more primitive structural dynamic. No tradition surveyed provides the derivation R7 specifies.

11.9 Requirement Eight: A Self-Exemplifying Derivation

The question it answers. A derivation of the structural ground of relational ontology makes the strongest possible structural claim: that it has reached the level from which the structural order itself emerges. Such a claim invites, and must survive, the following test: does the derivation itself exhibit the structural properties it claims to derive? Is the derivation an instance of the dynamic it formally establishes? If the derivation claims to have reached a ground that generates all structural order from within its own dynamic, then the derivation must itself be a structural event of the kind it describes — not a description of that kind of event from outside it.

Definition 11.8 (Requirement R8: Self-Exemplification)**.** A derivation D satisfies Requirement R8 if and only if the following three conditions jointly hold:

(i) *Method continuity*: The derivational method by which D proceeds is an instance of the eliminative method that D formally establishes as the only method capable of reaching the latent register (Eliminative Corollary, Corollary 10.2): every step of D derives what it establishes from conditions more primitive than the established result, without foundational constitution of any non-latent feature as a terminus.

(ii) *Sequence self-instantiation*: The order in which D proceeds — the sequence in which it derives ground, relata, encounter source, yield, event structure, and temporal asymmetry — is itself an instance of the structural sequence that D establishes as the necessary ontological order. The derivation does not proceed by a method that contradicts or is neutral with respect to the order it establishes; it enacts that order.

(iii) *No residual methodological posit*: D does not foundationally constitute its own derivational method as a primitive from which conclusions are drawn. The method itself must be derived as the unique structurally open pathway (by the Eliminative Corollary) rather than taken as a starting point whose justification is external to the derivation.

Conditions (i) and (ii) specify what self-exemplification positively requires. Condition (iii) specifies the residual failure mode that (i) and (ii) do not individually exclude: a derivation could exhibit method continuity and sequence self-instantiation while still constituting the eliminative method itself as a foundational primitive — taking it as given rather than deriving it as the unique open pathway. Condition (iii) closes this. A derivation satisfying all three is one whose structure is throughout an instance of what it formally establishes.

Concrete illustration. Gödel's incompleteness theorems provide the structural model — not for the content of R8 but for its logical form. Gödel's proofs are themselves instances of the formal systems whose properties they establish: each proof is a sentence within the system's formal language, constructed by the arithmetic operations whose properties the proof characterises. The self-referential architecture is not a paradox — it is the mechanism by which the proof achieves its result. R8 requires the analogous self-referential architecture for the derivation of the structural ground: the derivation must exhibit, in its own structure, the properties it formally establishes for the relational order as such.[7]

Why no existing tradition meets R8. No tradition surveyed has attempted to demonstrate that its derivational method is itself an instance of the structural dynamics it analyses. This is not a failure that any tradition within the existing landscape could have noticed from within its own foundational commitments — R8 becomes a requirement only once the full structural order has been derived, including the co-instantaneous event structure (R6) and its temporal asymmetry (R7), from which the self-exemplification requirement emerges as the condition of the derivation's own internal consistency. R8 is the requirement that no tradition could have satisfied before the others were in place.

Remark 11.1 (The Distinct Derivation Locus of R8). Requirements R1–R7 share a common derivation structure: each is derived as the structural inverse of a specific failure mode iden-

[7]Gödel's use of arithmetisation — the encoding of proof-theoretic properties as arithmetic properties — is the technical mechanism. The philosophical significance is discussed in Nagel and Newman (*Gödel's Proof*) and, in a more extensive treatment, in Hofstadter (*Gödel, Escher, Bach: An Eternal Golden Braid*).

tified in the diagnostic survey. The failure at Level I (no internally differentiated generative ground) inverts to yield R1 (the structural ground must be internally differentiated and self-generative). The failure at Level II (no invariant relational identity prior to the foundational constitution) inverts to yield R2. And so on through R7, each of which directly inverts the failure mode at or cross-cutting the level that generates it.

R8 is derived differently. It does not invert a specific Level-I-through-V failure mode identified in the survey; it is derived from a *meta-structural* source: the requirement that the derivation of the structural ground be internally consistent with what it derives. Specifically: once R1–R7 are in place, a derivation of the structural ground satisfying all seven is constrained by the result it derives. If the structural ground is what the derivation claims — a dynamic whose relational structure generates all structural order from within its own constitutive tension — then the derivation of it must itself be a structural event of the kind it formally establishes. This is not a requirement that can be generated by looking at the traditions' failures; no tradition surveyed attempted a derivation that reached the latent register, so no tradition was in a position to exhibit or violate self-exemplification as a constraint on that derivation.

This means R8's derivation locus is genuinely distinct from R1–R7: it is derived not from the failure analysis but from the internal consistency requirement that any derivation satisfying R1–R7 must itself meet. The source is a formal argument of the following form: *if* a derivation succeeds in reaching a ground that satisfies R1–R7, *then* the derivation, considered as a structural event, must be an instance of what it has derived (by Condition (ii) of Definition R8) and must not foundationally constitute its own method as a primitive (by Condition (iii)). These two consequences are forced on any derivation that has genuinely reached the latent register; they are not added as further desiderata but follow from what it means to have reached it. R8 is therefore derived from the structural conditions imposed on any derivation satisfying R1–R7, not from an additional requirement imported independently.

The formal proof that R1–R8 are jointly satisfiable — that there exists a possible derivation meeting all eight conditions simultaneously — is given in Appendix C. That proof is load-bearing: without it, R1–R8 could be an inconsistent set of conditions whose joint satisfaction is impossible. The appendix demonstrates joint satisfiability by exhibiting a sufficiency structure — a schematic derivation that meets all eight conditions without producing the full positive account, thereby confirming that the requirements are coherent and achievable without presupposing the companion investigation that will execute the full derivation.

The universality of self-exemplification failure across the thirteen traditions is not a pattern claimed inductively from the survey and extrapolated. It is documented tradition-by-tradition in Table 11.1 below.

Table 11.1: Self-Exemplification Failure: Thirteen Primary Traditions plus Merleau-Ponty

Tradition	Method employed	Self-exemplification failure
Whitehead	Speculative philosophy; hypothetico-deductive system	The method of speculative hypothesis is not itself an instance of prehensive constitution: speculative hypothesis is how abstract systems are proposed, not how occasions become.
Spinoza (relational reading)	Geometrical demonstration from definitions	The *more geometrico* is not derived as a consequence of Substance's self-causation; it is imported from mathematical practice.
Bergson	Intuition as philosophical method	Intuition is proposed as adequate to duration; it is not shown to be itself an unspatialisable durational event. The method claims to overcome intellectual spatialisation without demonstrating that it escapes it.

(continued)

Tradition	Method employed	Self-exemplification failure
Heidegger	Phenomenological ontology; formal indication	Formal indication is not derived as an instance of the possibility-dimension structure it specifies; the method proceeds by Dasein's disclosed possibilities without showing that the procedure itself is a relational event in the Level V sense.
OSR	Structural realism; no-miracles inference	The structural realist methodology (abduction to best structural explanation) is not derived as an instance of the relational structures it takes as its ontological result.
RQM	Operationalism; relational interpretation of QM formalism	The operationalist methodology is not itself a relational fact in RQM's sense: it is a procedure for an observer-system pair, not a derivation of the observer-relative structure from within the relational dynamic.
Stengers	Cosmopolitical constructivism	The cosmopolitical method is proposed; it is not shown to enact the ecology of practices it describes.

(continued)

Tradition	**Method employed**	**Self-exemplification failure**
Emirbayer	Relational transaction as analytical unit	The analytical method of taking transactions as the unit is not itself a transaction in Emirbayer's sense; it is adopted as a theoretical commitment, not derived transactionally.
Kant (relational reading)	Transcendental critique	The critical method is not derived as itself a case of the conditions of possible experience it identifies; it stands apart from what it conditions rather than instantiating it.
Simondon	Ontogenetic analysis; transductive reasoning	Transductive reasoning is not shown to be itself an individuating event in the sense Simondon's account specifies.
Deleuze (*DR*, *LS*)	Transcendental empiricism; concept creation	Concept creation is not derived as an instance of the differential-relational structure it specifies; difference is said to be productive but the method does not show itself to be produced by difference.

(continued)

Tradition	**Method employed**	**Self-exemplification failure**
Schaffer (priority monism)	Grounding theory; inference to best structural explanation	Priority monism proposes the cosmos as fundamental; the method of theoretical abduction is not itself a structural instantiation of whole-to-part priority.
Fine (essence-grounding)	Essence-analysis; constitutive dependence	The essence-analytic method identifies essences and their grounding consequences; it is not itself derived as a case of essence-grounding.
Merleau-Ponty (flesh ontology)	Phenomenological ontology; chiasmic description	The chiasmic method of describing the reversibility of flesh proceeds by first-personal disclosure; it is not shown to be itself a chiasmic event in the Level V sense.

Note. *DR* = *Difference and Repetition* (1968); *LS* = *The Logic of Sense* (1969). The parenthetical scope qualifiers indicate which of Deleuze's texts the self-exemplification analysis primarily addresses.

Remark 11.2 (The Exhaustiveness of Exactly Two Cross-Cutting Patterns). The count of two cross-cutting patterns — temporal presupposition (basis of R7) and self-exemplification failure (basis of R8) — requires an exhaustiveness argument. A cross-cutting pattern satisfies three conditions: (i) it is independent of entry level, appearing in traditions at any level; (ii) it is distinct from the level-specific failures; (iii) its complement specifies something not already required by R1–R6.

Both identified patterns satisfy all three conditions. Temporal presupposition is level-independent (every tradition inherits an underived temporal or quasi-temporal order regardless of entry level), distinct from level-specific failures (it concerns the framework's operating context, not its structural content), and specifies a distinct requirement (the derivation of the temporal order, which no level-specific requirement addresses). Self-exemplification failure is similarly level-independent (Table 11.1), distinct from level-specific failures (it concerns the framework's derivational method, not its structural content), and specifies a distinct requirement (the method-structure isomorphism).

Why exactly two? A framework has two dimensions that are not level-specific: its *operating conditions* (the structural context within which the framework functions, including the temporal order it presupposes) and its *derivational method* (the structural procedure by which it generates its results). Both dimensions generate cross-cutting patterns because both are features of every framework regardless of where in the hierarchy its foundational commitment falls. These are the only two such dimensions. The exhaustiveness of the pair is established by showing that every candidate third pattern either reduces to one of the two identified patterns or fails one of the three conditions that qualify a pattern as cross-cutting.

Axiological presupposition: every framework takes some normative standards as primitive — validity, coherence, relevance.

This does not satisfy condition (iii): the complement of axiological presupposition would require a derivation of normative standards from the framework's structural results, but this is not a distinct requirement on any derivation of relational conditions. Whether the normative standards a framework employs are derived from its structural results is a question about the framework's epistemological self-understanding, not a structural condition of the derivation's target. It therefore reduces to self-exemplification failure (R8): the framework's normative method is not itself an instance of what the framework structurally establishes. It is a sub-case of R8, not an independent cross-cutting pattern.

Representational presupposition: every framework takes some correspondence between its claims and the domain it analyses as primitive rather than deriving it. This fails condition (i): it is not fully independent of entry level. A framework that enters at Level I and derives the relational dynamic has stronger resources for grounding the claim-domain correspondence than a framework that enters at Level IV; the presupposition's weight varies with entry level. It is therefore partly level-specific — a gradient rather than a uniform cross-cutting feature. To the extent it is level-independent, it again reduces to R8: the framework's representational method is not itself derived as an instance of the relational dynamic it establishes.

Modal presupposition: every framework takes the possible/actual distinction as available rather than deriving it. This is the most serious candidate for an independent third pattern. Its reduction is as follows. The modal presupposition is the presupposition of the temporal order in the dimension of possibility: to treat the actual/possible distinction as primitive is to take the possibility space as structurally given — which is precisely what temporal presupposition (R7's basis) requires to be derived. Modal presupposition is not a separate cross-cutting pattern but the aspect of temporal presupposition that concerns the possibility dimension of the framework's operating context. The Level V structure of the relational event includes both an actuality and a possibility dimension; R7 requires the derivation of the temporal order that structures both dimensions. The modal presupposition is absorbed into R7.

No candidate third pattern survives this elimination. The count of two cross-cutting patterns is therefore exhaustive by elimination, not by stipulation, and the count of eight requirements is derived.

The asymmetry of the present volume. Before Relation does not satisfy R8. This is not a defect of the present volume but an architectural consequence of its scope. The present volume derives R8 as the requirement that any adequate response to the structural gap must meet. Satisfying R8 — producing a derivation that exhibits in its own formal structure the dynamics it formally establishes — is the task of the companion derivation, not of the critique that specifies the conditions for that derivation. The critique cannot satisfy R8 without ceasing to be a critique and becoming the positive derivation itself. Before Relation establishes the conditions of the very standard it cannot, within its own scope, meet. That is what it means to have identified the problem precisely before having supplied the solution.

The apparent tension — Before Relation applies the SIT to every tradition including itself, and is therefore subject to structural foreclosure — resolves as follows. The SIT forecloses the *derivation* of the latent register from within a framework that foundationally constitutes a non-latent feature. It does not foreclose the *specification of what that derivation must supply*. Before Relation foundationally constitutes the five-level diagnostic grid and therefore cannot, from within its own framework, derive the latent register. It can, and does, derive what any derivation of the latent register must accomplish. The specification and the derivation are structurally distinct acts: one establishes the formal requirements for a target; the other reaches the target. The SIT's foreclosure is a foreclosure of the second act. Specifying the requirements for the second act is itself a first act that the diagnostic framework's own structural apparatus is equipped to perform — precisely because specifying requirements does not require executing the derivation they govern. A theorist who identifies the conditions that any adequate ground theory must satisfy has not satisfied those conditions; they have bounded the inquiry that will. This distinction between the specification of requirements and their satisfaction is itself an instance of the pattern the SIT formalises: the present investigation constitutes the requirements foundationally, and thereby forecloses reaching their satisfaction from within its own framework. The companion investigation is addressed to the satisfaction, not the specification.

Remark 11.3 (On the Joint Satisfiability of R1–R8). Requirements R1 through R8 are necessary conditions: each specifies what any adequate derivation of the structural ground must supply. They are not trivially satisfiable together. A requirement at a higher level of structural organisation may conflict with conditions at a lower level; a requirement about the derivation's method may conflict with conditions on what the derivation derives. The joint satisfiability of R1–R8 is therefore not self-evident and requires demonstration: without it, the requirements might collectively specify an impossible object, and the latent register would be not a derivable structural domain but an internally contradictory demand. Appendix C provides the full formal demonstration: a schematic model $\mathcal{M}$ is constructed from what R1 through R8 independently specify, and its coherence — the absence of any formal contradiction among the eight conditions jointly — is established by a non-contradiction proof. $\mathcal{M}$ is a logical witness, not a completed derivation: it demonstrates that the eight requirements admit a common realisation without supplying the positive structural characterisation of that realisation. The positive derivation is the object of the Stage II investigation.

11.10 The Joint Satisfiability of the Eight Requirements

The eight requirements have been derived, each from the Structural Insufficiency Theorem and the specific diagnostic results of Part II. They specify, collectively, the conditions that any derivation claiming to have reached the latent register and thereby to have satisfied the Level I conditions must demonstrably meet.

The most important result the present section can establish is that these eight requirements are mutually consistent and collectively characterise a unified structural target — a complete eliminative derivation of the relational ground — that is not self-defeating. The result is a joint satisfiability demonstration in the following precise sense: the eight requirements specify conditions on a single structural object, no subset of these conditions generates a contradiction, and a schematic structural outline ($\mathcal{M}$) in which all eight are simultaneously represented is constructible. The section does not provide the derivation whose conditions R1–R8 specify; it establishes that pursuing such a derivation is not an inquiry into an impossibility.

Remark 11.4 (On the Two Forms of R8 Satisfaction). R8 has a global and a local form. The global form requires that the derivation-as-a-whole be self-exemplifying: the inquiry must itself proceed through the structural sequence — ground, relata, encounter source, yield, event structure, temporal asymmetry — as its own method, not merely specify a model whose elements stand in that order. The local form requires that the logical-witness model $\mathcal{M}$ exhibit the structural sequence R8 specifies, without requiring the inquiry itself to be an instance of that dynamics.

The joint satisfiability theorem below demonstrates the local form: the conditions R1–R8 collectively permit a realisation of R8 in $\mathcal{M}$, and no internal contradiction prevents it. Whether the global form is satisfiable — whether a completed derivation can be self-exemplifying in the full sense — is established by the theorem's scope: the conditions jointly permit it. The global form is not satisfied by the present investigation, which is explicitly a Stage I inquiry; this is not a defect but an architectural consequence of the book's scope, as the Preface notes.

The distinction between *specifying the conditions of a derivation* and *executing that derivation* is the structural hinge on which the global form's non-satisfaction rests, and it can be stated precisely in terms of the Eliminative Corollary (Corollary 10.2). The Corollary establishes the necessary *form* of any derivation capable of reaching the latent register: it must proceed exclusively by the eliminative method, without foundational constitution of any non-latent feature. This establishing of the form is precisely what the present investigation does — it specifies what a derivation satisfying R1–R8 would have to be. The R8 global form requires something structurally distinct: that the inquiry *proceed through* the relational structural sequence as its own method, not merely characterise the sequence and confirm it is coherent. A theorist who establishes the formal shape of the target has not thereby hit the target; a theorist who specifies that any adequate derivation must satisfy R1–R8 has not thereby executed a derivation satisfying R1–R8. These are structurally distinct acts — the first operates on the space of possible derivations from outside; the second is an instance of the kind of derivation the first identifies — and no amount of specification-precision converts the first into the second. The present investigation is an instance of the first. That it does not satisfy the global form of R8 is therefore the correct architectural consequence of what it is, not a failure of what it attempts.

Theorem 11.1 (Joint Satisfiability)**.** Requirements R1 through R8 are jointly satisfiable: there exists no formal contradiction among their defining conditions, and the existence of a derivation satisfying all eight simultaneously is logically consistent.

Proof. We proceed by direct examination of the defining conditions of each requirement pair. The strategy is pure non-contradiction: for each pair of requirements, we verify that the defining conditions of one do not logically entail the negation of the defining conditions of the other.

R1 and R2: R1 requires a formally derived absolute ground — total, generative, self-sustaining, formally derivable. R2 requires that the latent register be pre-temporal. These conditions are not contradictory: a total field that is self-sustaining and generative is not thereby temporal; temporal asymmetry can be a structural consequence of operations within such a field rather than a feature of the field itself. There is no entailment from R1's conditions to the negation of R2's, nor from R2's conditions to the negation of R1's.

R2 and R3: R2 requires pre-temporality of the latent register. R3 requires a formal derivation of structurally distinct relata. These conditions are not contradictory: structurally distinct relata can emerge from a pre-temporal ground as locally determinate expressions of that ground's internal dynamic, with temporal asymmetry generated as a consequence of their encounters rather than presupposed by the ground that produces them.

R3 and R4: R3 requires formally derived structurally distinct relata. R4 requires a formally derived encounter source — a mechanism that brings relata into contact that is not itself one of the relata. These conditions are not contradictory: the same dynamic that produces relata as invariant configurations of a total field can produce an encounter source as the variant expression of that same field pressing towards invariant configurations from the complementary structural dimension. Distinctness of relata (R3) and the derivability of an encounter source (R4) are compatible because both can be derived from a common structural ground whose own internal tension generates both simultaneously.

R4 and R5: R4 requires a formally derived encounter source. R5 requires that encounter produce a genuinely novel structural surplus. These conditions are not contradictory: the novelty required by R5 is not a violation of conservation — a derivation satisfying R5 can specify both that the yield is genuinely new (not a rearrangement of what the relata contained) and that it is conserved within the total structural field. There is no entailment from R4's derivability requirement to the impossibility of genuine novelty, nor from R5's novelty requirement to the impossibility of a structurally derived encounter source.

R5 and R6: R5 requires genuinely novel relational yield. R6 requires that the relational event have a co-instantaneous internal structure — that its dimensions obtain simultaneously rather than sequentially. These conditions are not contradictory: the co-instantaneity of the relational event's dimensions does not preclude the production of a novel yield; it specifies the structure of the event through which yield is produced. An event in which ground, relata, encounter, and yield dimensions obtain simultaneously is not thereby an event in which no yield occurs — it is an event in which all the conditions of yield are in place at once rather than in sequence.

R6 and R7: R6 requires co-instantaneous event structure. R7 requires formally derived temporal asymmetry. These conditions are not contradictory: temporal asymmetry can arise from the irreversibility of the yield's deposition in the total structural field, which is a consequence of events that are internally co-instantaneous. The co-instantaneity of the event's structure (R6) does not imply that the event leaves no asymmetric trace in the structural distribution of the field (R7). The asymmetry is between the field's state before and after the event, not between stages within the event.

R7 and R8: R7 requires formally derived temporal asymmetry. R8 requires that the derivation be self-exemplifying — that it exhibit in its own formal structure the dynamics it establishes. These conditions are not contradictory: a derivation that proceeds eliminatively (removing each assumption derivable from more primitive conditions) and that thereby exhibits the same structural sequence — ground, relata, encounter source, yield, event structure, temporal asymmetry — as the ontological order it derives is formally continuous with what it establishes, and the temporal asymmetry it derives is a consequence of the sequence the derivation itself exhibits. R8 does not require that the derivation *be* a temporal event in the sense R7 characterises; it requires that the derivation's formal structure be *continuous with* the structural dynamics it establishes.

Since no pair of requirements among R1 through R8 produces a formal contradiction — since for every pair, the defining conditions of one can simultaneously obtain with the defining conditions of the other — no pairwise incompatibility exists. The pairwise argument establishes that no two requirements conflict. To confirm this extends to the joint case — that all eight can be simultaneously satisfied — we construct a schematic model $\mathcal{M}$ in which each element satisfies its corresponding requirement given the prior elements.

Why pairwise non-contradiction is sufficient here. In general, pairwise non-contradiction of all pairs within a set of conditions does not guarantee joint satisfiability: it is conceivable that three or more conditions jointly produce a contradiction even when no two do. The joint satisfiability theorem is therefore not established by the pairwise argument alone; the schematic model $\mathcal{M}$ constructed below provides the independent positive confirmation. The pairwise argument is included because it makes each pair's compatibility explicit and traceable; $\mathcal{M}$ then confirms that these pairwise compatibilities obtain simultaneously.

That said, a structural observation reduces the logical gap between the two components. R1 through R8 are conditions on *different aspects of a single structural object* — the complete eliminative derivation of the relational ground. R1 specifies the character of the ground, R2 its temporal status, R3 the emergence of relata, R4 the encounter mechanism, R5 the yield, R6 the event structure, R7 the temporal asymmetry derived from that structure, and R8 the self-exemplification of the derivation as a whole. Because each requirement governs a structurally distinct aspect of the derivational object, no requirement's satisfaction conditions overlap with another's in a way that could generate a higher-order interaction contradiction. The satisfaction of R1 does not occupy the same structural position as the satisfaction of R5; their conditions are domain-disjoint. In a case where requirement-satisfaction conditions are domain-disjoint in this structural sense, pairwise non-contradiction is a reliable indicator of joint satisfiability — the $\mathcal{M}$ construction below confirms rather than supplies this.

The following outline is a standard logical-witness construction: it does not claim to execute the derivation whose conditions R1–R8 specify, only to show that such a derivation is not self-contradictory. It is constructed solely from what R1 through R8 independently specify and posits no architectural element not entailed by those requirements. Any derivation exhibiting the structure $\mathcal{M}$ schematically specifies would satisfy R1–R8; $\mathcal{M}$ is not itself that derivation.

Schematic model $\mathcal{M}$*:* Let $\mathcal{M}$ consist of:

(a) A structural ground G posited as total (no exterior), generative (differentiation from within its own dynamic), self-sustaining (no external regulator), and formally derivable as the unique result of the eliminative method applied to the presuppositions of relational ontology. G satisfies R1 in the schematic model because: G has no exterior by stipulation, satisfying G1 (totality); G differentiates structurally from within its own dynamic — this is what distinguishes a schematic G from a Spinozistic Substance, which is total and self-sustaining but not generatively differentiating — satisfying G2 (generativity); G requires no external principle, since its generativity is not dependent on anything outside it, satisfying G3 (self-sustenance); and G is posited as the unique result of applying the eliminative method to relational presuppositions — its existence is schematically derivable rather than arbitrary — satisfying G4 (formal derivability). The four conditions are simultaneously satisfied by G as schematically specified.

(b) A dynamic P internal to G that operates without reference to temporal succession, oriented becoming, or already-constituted relata. P satisfies R2 because: (i) G is stipulated as pre-temporal, so P, being internal to G, operates within a domain in which temporal succession is not yet constituted; (ii) the non-temporality of P is not in conflict with any element of G, since G's conditions (totality, generativity, self-sustenance, formal derivability) do not presuppose temporal succession. The compatibility of R1 and R2 is thereby demonstrated constructively: a ground meeting R1 does not preclude having an internal dynamic meeting R2.

(c) A differentiation function ∂ on P that produces, at structurally distinct positions within G, structural identities $r_1, r_2, \ldots$ each uniquely individuated by its position. ∂ satisfies R3 because: (i) the structural identities are derived from G's dynamic P rather than pre-given; (ii) the numerical distinctness of positions within G is a consequence of G's generativity: a ground that generates structural differentiation from within its own dynamic necessarily generates distinct outputs at distinct points of its internal activity — if it did not, it would not have generated differentiation at all, since indistinguishable outputs constitute no differentiation. Distinctness of positions is therefore not an independent presupposition imported alongside ∂ but a structural consequence of G's satisfying the generativity condition (G_2) of R1; (iii) individuation by position within a generative field does not reintroduce the relational regress, because the positions are not individuated by standing in relations to one another but by being the distinct loci that the ground's own differentiating dynamic produces. The compatibility of R1, R2, and R3 is thereby constructively demonstrated without circularity.

(d) An encounter mechanism ε that brings structurally compatible pairs (r_i, r_j) into contact, where compatibility is determined by a complementarity relation derivable from the structural properties of the relata as produced by ∂. ε satisfies R4 because: (i) it is not one of the relata; (ii) its selectivity is derived from the structural properties that ∂ assigns to relata, making it internal to the same dynamic P that produces the relata; (iii) it is not depleted by enabling encounter, since its operation is a function of the structural field rather than a consumable resource. The compatibility of R4 with R1–R3 is constructively demonstrated: a differentiation function that assigns structural properties to relata by locus provides the basis for a complementarity-based encounter mechanism without circularity.

(e) A yield function υ mapping compatible pairs (r_i, r_j) under ε to a structural result y_{ij}. υ satisfies R5 because: (i) y_{ij} is formally irreducible to any operation on r_i or r_j individually. The irreducibility is structural, not merely asserted: each relatum r_i is individuated by its position in G, which is a single-locus structural fact. The encounter between r_i and r_j produces a structural relationship that is a two-locus fact — it involves the joint structural configuration of both loci, which is not present in either locus individually. No function f defined solely over r_i's structural properties (i.e., over the single-locus facts at r_i's position) can produce a result that includes information about r_j's position, and vice versa. y_{ij} contains that joint structural information; it is therefore not expressible as $f(r_i)$ or $f(r_j)$ for any such f. Whether it is expressible as $f(r_i, r_j)$ jointly depends on whether joint function application introduces no new structural content — but ε produces the encounter by bringing the structural configurations of both loci into contact, and this contact produces a configuration at the encounter locus that was not present at either individual locus. y_{ij} is this encounter-locus configuration: formally irreducible to either relatum because it is a structural fact about their joint configuration under encounter, not about either separately; (ii) the total structural resources of G, conserved through P, bound what v can produce (boundedness); (iii) y_{ij} is assigned a position within G by the same differentiation function ∂, making it capable of further encounter (generativity); (iv) since different pairs (r_i, r_j) and (r_i, r_k) have structurally distinct r_j and r_k occupying distinct positions in G, the joint structural configuration produced by ε for each pair is distinct, so υ produces formally distinct yields for distinct pairs (asymmetric novelty). No conflict with R1–R4 is introduced.

(f) An event structure σ governing the encounter as three co-instantaneous dimensions derivable from the prior elements. σ satisfies R6 because: (i) the three dimensions (initiation, sustained tension, resolution) are determined by the structure of the encounter as specified by ε and υ together; (ii) their co-instantaneity follows from the fact that the encounter is a single structural event within G's dynamic, not a temporal sequence (since P is pre-temporal); (iii) the account of σ instantiates a three-element co-present structure in its own organisation, satisfying the local self-exemplification condition.

(g) An irreversibility function τ mapping each application of υ to a permanent structural deposit in G's distribution. τ satisfies R7 because: (i) each yield y_{ij} is deposited as a structural fact within G, modifying the conditions of subsequent encounter; (ii) G's total structure, conserved by P, does not provide a mechanism for undoing deposits — reversal would require a structural resource not present in G's economy, so irreversibility follows from G's self-sustaining character; (iii) temporal asymmetry is thereby derived as a structural consequence of deposition rather than presupposed.

(h) A derivational sequence Δ proceeding $G \to P \to \partial \to \varepsilon \to \upsilon \to \sigma \to \tau$, where each stage is derived from all prior stages. Δ satisfies R8 in the logical-witness sense specified in the remark preceding this theorem: (i) at each stage, the element introduced follows from the structural conditions established by all prior stages, not constituted as a foundational primitive; (ii) no stage imports a non-latent feature as its terminus; and (iii) the sequence of Δ itself exhibits ground, relata, encounter source, yield, event structure, and temporal asymmetry in the same order that $\mathcal{M}$ establishes as the necessary ontological order of relational constitution.

The preceding element-by-element demonstration establishes that each element of $\mathcal{M}$ satisfies its corresponding requirement given the prior elements, without circularity. R1–R8 are not eight independent constraints that happen to be pairwise compatible; they are eight specifications of aspects of a single structural object — the complete eliminative derivation of the relational ground — and their joint satisfiability follows from the structural unity of that object. A derivation exhibiting the structure $\mathcal{M}$ schematically specifies would satisfy all eight requirements simultaneously, because all eight are conditions on the same derivational project.

$\mathcal{M}$ is a logical witness in the precise sense: its elements are specified by what the requirements jointly demand, not by a positive account of what the structural ground is. It demonstrates that the requirements admit a common realisation without characterising the latent register. The model's function is purely logical, not ontological.

Remark 11.5. Theorem 11.1 establishes that the eight requirements are jointly satisfiable, not that a derivation satisfying all eight has been executed. The proof proceeds in two stages: a pairwise non-contradiction argument confirming that no two requirements conflict, followed by the construction of a schematic model $\mathcal{M}$ in which all eight requirements are simultaneously satisfied. $\mathcal{M}$ is derived entirely from what R1 through R8 independently specify; it is not a positive characterisation of the latent register but a logical witness to the joint coherence of the requirements. The ground is reachable. The eight conditions for reaching it are jointly coherent. Whether a completed derivation satisfying all eight has been executed is a question the present investigation declines to answer.

What a Satisfying Derivation Would Look Like

The non-contradiction proof above establishes that a derivation satisfying all eight requirements is logically possible. The following characterisation does not execute that derivation; it maps the structural shape it must take, derived solely from what

R1 through R8 jointly demand. It is offered as a forward specification — a statement of the formal target — not as a positive result.

R1 requires that the derivation begin from a structural ground $\mathcal{G}$ that is total (admitting no exterior), generative (producing structural differentiation from within its own dynamic), self-sustaining (requiring no external principle), and formally derivable (shown to be the unique result of applying the eliminative method to the presuppositions of relational ontology). $\mathcal{G}$ therefore cannot be a feature within the structural order it grounds; it must be at the level of the latent register, prior to every tradition's foundational commitment.

R2 requires that $\mathcal{G}$ be pre-temporal: the temporal and relational orders must emerge as structural consequences of $\mathcal{G}$'s own internal dynamic, not be presupposed by it. This means the internal dynamic of $\mathcal{G}$ must be specifiable without reference to temporal succession, oriented becoming, or already-constituted relata.

R3 requires that $\mathcal{G}$'s internal dynamic produce structurally distinct relata — entities with determinate structural identities that are derived from within $\mathcal{G}$ rather than pre-given or merely relational nodes. These relata must be individuated without reintroducing the relational regress.

R4 requires that $\mathcal{G}$'s dynamic also produce an encounter source internal to itself — a structural mechanism that brings specifically these relata into specifically this encounter without external perturbation and without necessitation so complete that no encounter is genuinely novel.

R5 through R7 require that the derivation produce, from the formal structure of compatible encounter, a yield that is genuinely new, bounded, generative, and encounter-specific; that the encounter event have a co-instantaneous internal structure whose dimensions are derived rather than described; and that temporal asymmetry arise as a structural consequence of the yield's irreversible deposition rather than being presupposed.

R8 requires that the derivation's own formal sequence — ground, relata, encounter source, yield, event structure, temporal asymmetry — exhibit the same structural order it establishes as ontologically necessary, so that the method enacts rather than merely describes the dynamic it derives.

A derivation satisfying R1 through R8 simultaneously would need to begin below every tradition's entry level, proceed by the eliminative method exclusively, and arrive at a formal account of the structural order of relational possibility that is complete, self-consistent, and self-exemplifying. Whether such a derivation has been executed is a question the present volume does not answer. That it must be pursued — and that its formal target is here specified with sufficient precision to evaluate any candidate — is the positive contribution this chapter makes.

Note on the companion series. Whether a derivation satisfying R1 through R8 has been executed — and in what structural sequence — is addressed in the note on the companion series at the close of Chapter 11.10.

Summary of Chapter 11. *Eight requirements — derived from the Structural Insufficiency Theorem and the specific diagnostic results of Part II — specify the conditions that any derivation claiming to have reached the latent register must satisfy. Each requirement is the structural complement of a specific failure documented in the diagnostic survey: not an arbitrary desideratum but the formal demand made by a specific structural gap across the entire tradition of relational ontology. The derivation of R1–R8 is formally mapped to the five-level diagnostic results and the two cross-cutting patterns (the universal temporal presupposition and the universal failure of self-exemplification). The eight requirements are jointly satisfiable — as the schematic model $\mathcal{M}$ in the proof of Theorem 11.1 demonstrates — meaning that the ground the investigation has established as formally necessary is also formally reachable. The entailment chain that the present investigation establishes is the following. Before Relation proves that no tradition in relational ontology has derived the structural ground of relation, specifies the eight requirements any such derivation must satisfy, and proves those requirements jointly satisfiable. A companion investigation must now satisfy R1 through R8 simultaneously — deriving the ground (R1), establishing the pre-temporal latent register (R2), deriving relational individuation (R3), the encounter source (R4), relational yield (R5), the co-instantaneous event structure (R6), temporal asymmetry (R7), and the self-exemplifying derivational sequence (R8). The present investigation is structurally prior to that companion: it establishes the necessity of the derivation and the formal shape of its target. Whether such a derivation has been executed is a question the present investigation declines to answer. That it must be attempted is the conclusion the investigation cannot avoid.*

Conclusion: Before Relation

The question is not whether a ground exists. The question is whether thought can reach it.

Before Relation

The investigation has been completed. Part I established that general ontology necessarily arrives at relational ontology and that relational ontology carries five structural commitments it has never collectively discharged. Part II demonstrated, with the precision that full engagement with each tradition demands, that thirteen traditions spanning the seventeenth through twenty-first centuries — across the major traditions of Anglophone, continental, process, and analytic metaphysics — have each achieved formal results within their domain and each been structurally unable to reach the level below their foundational entry point. Fine's essence-grounding achieved the most formally precise analytic account of encounter specificity; Schaffer's priority monism achieved the highest Level I entry point in the analytic tradition. Part III formalised this pattern as the Structural Insufficiency Theorem and identified the only type of derivation that is not structurally foreclosed from the start. Part IV derived, from the theorem and the diagnostic results, eight requirements that any satisfying response must meet, and proved that those requirements are jointly satisfiable.

What remains is to say, as plainly as the argument permits, what has been established and what has not.

What has been established. The structural ground of relational ontology — the latent register from which the capacity for relation, the structural identity of relata, the dynamic of encounter, the irreducibility of yield, and the co-instantaneous organisation of the relational event all emerge as derived structural results — has not been reached by any tradition in the field. This is not a historical accident. It is the formal consequence of the way every tradition has entered the field: by constituting a non-latent structural feature as its foundational primitive, and thereby rendering the latent register formally unreachable from within its own framework.

The pattern of failure is exact. It maps precisely onto the five-level structural order established in Chapter 2. Every tradition occupies a specific level; every tradition forecloses what is below it. Spinoza came closest among the classical traditions, reaching Level I but failing its generativity and formal derivability conditions. Schaffer's priority monism, the strongest contemporary analytic candidate for a Level I account, meets three of the same four conditions and fails the same fourth one. Every other tradition entered higher, achieving real results within its domain but leaving the ground entirely uncharted. Taken together, the thirteen traditions — together with the supplementary engagement with Merleau-Ponty's flesh ontology as a post-Heideggerian coda (see §6.6) — constitute a remarkably comprehensive map of the non-latent register — the most thorough exploration of what relation does, what relata are through their relations, how encounter produces novelty, and how the relational event is temporally directed — that philosophy has yet produced. The comprehensiveness of this map is not incidental: it is the diagnostic consequence of the SIT. When every possible entry point has been occupied by a tradition that achieves formal results there, the fact that none of them reaches the ground is not a gap in the literature's ambition. It is structural. What they do not constitute, and cannot constitute from within their current commitments, is a map of what grounds the non-latent register itself.

The Structural Insufficiency Theorem is the formal expression of why this is so. The Eliminative Corollary is the formal identification of the only path that is not foreclosed. The eight requirements of Chapter 11 are the formal specification of where that path must go.

What has not been established. The present investigation has not derived the ground. It has established that the ground must be derived, specified the formal conditions any derivation of the ground must satisfy, and proved that those conditions are jointly satisfiable. The positive derivation — the execution of the eliminative method applied to the presuppositions of relational ontology, proceeding from the latent register to Level I to Levels II through V in strict ontological order, satisfying each of the eight requirements at each step — falls outside the scope of the present volume.

This is not evasion. The decision to separate the critique from the proposition reflects a commitment to the intellectual discipline that the argument itself demands. A derivation that claims to reach the latent register will be evaluated by the standards the investigation has established: Does it derive the ground without foundational constitution? Does it establish a pre-temporal latent register from which the temporal and relational orders emerge as structural consequences? Does it derive relational individuation without the relational regress? Does it account for encounter source without external perturbation principle? Does it derive yield as formally new, bounded, and generative? Does it establish the co-instantaneous structure of the relational event? Does it derive temporal asymmetry from within? And does it exemplify in its own structure the dynamics it formally establishes? These eight questions are the critical apparatus the present volume has built. They apply to any candidate derivation — including any that the reader of this volume might attempt.

What the inquiry has been. The eliminations performed in this volume — across traditions, across structural levels, across the arguments by which each tradition's horizon is identified — have not been arbitrary. They have followed a single formal procedure: remove every assumption that can be shown to be derivable, and record what the removal cannot touch. The eight requirements derived in Part IV are not conclusions appended to an independent argument. They are the formal residue of the entire eliminative movement: what remains when every structural assumption that can be set aside has been set aside.

The reader who has followed that movement from the beginning has been participating in an eliminative inquiry. The method whose results have been accumulated across fourteen chapters is the same method whose formal identity will be established in the companion derivation. The inquiry has enacted what it cannot yet name.

Note on the companion series. Such a derivation has been executed in the companion series *The Triadic Ontological System*, comprising five articles (*Kawana'on*; *What Relation Cannot Do Without*; *The Lakeborne*; *The Lakewater*; *The Waterlake*) and four formal volumes (*Volume I: The Operation of Relational Initiation*; *Volume II: The Operation of Relational Regulation*; *Volume III: The Operation of Relational Direction*; *Volume IV: The General Validation Framework*). The five articles are available as preprints on PhilArchive;[8] the four formal volumes are forthcoming.[9] Whether that series satisfies R1 through R8 — and in what structural sequence — is a question that the present investigation leaves to the reader's independent evaluation of the companion works against the criteria established here.

[8]Suano, "Kawana'on: The Collective Name for the Generative First Philosophy"; Hector Caliso Suano, "What Relation Cannot Do Without: Compatibility as Structural Necessity" (Unpublished preprint. Article in the companion series *The Triadic Ontological System*. Derives compatibility as the structural condition under which two paired sides open a field between them, by eliminative derivation of three jointly necessary conditions, 2026); Hector Caliso Suano, "The Lakeborne: On the Structural Necessity of Novelty" (Unpublished preprint. Article in the companion series *The Triadic Ontological System*. Derives surgens — the irreversible structural novelty that compatible encounter produces — as a necessary consequence of compatible interlocking, 2026); Hector Caliso Suano, "The Lakewater: Activity as the Intrinsic Dynamic of the Total Field" (Unpublished preprint. Article in the companion series *The Triadic Ontological System*. Derives activity as the intrinsic dynamic of the total field, showing that a uniformly latent total domain fails to be a ground for relational constitution, 2026); Hector Caliso Suano, "The Waterlake: Totality as Structural Necessity" (Unpublished preprint. Article in the companion series *The Triadic Ontological System*. Derives the total field — the Waterlake — as the structural fact that cannot be denied without presupposing itself, 2026).

[9]Hector Caliso Suano, "The Triadic Ontological System, Volume I: The Operation of Relational Initiation" (Forthcoming volume in the companion series to *Before Relation*, 2026); Hector Caliso Suano, "The Triadic Ontological System, Volume II: The Operation of Relational Regulation" (Forthcoming volume in the companion series to *Before Relation*, 2026); Hector Caliso Suano, "The Triadic Ontological System, Volume III: The Operation of Relational Direction" (Forthcoming volume in the companion series to *Before Relation*, 2026); Hector Caliso Suano, "The Triadic Ontological System, Volume IV: The General Validation Framework" (Forthcoming volume in the companion series to *Before Relation*, 2026).

What is at stake. The structural ground of relational ontology is not only a philosophical question. Every domain that has undergone a relational turn — quantum physics, molecular biology, social theory, cognitive science — has built its relational framework on a foundation whose structural properties have not been formally derived. The frameworks are powerful and productive; their results are real. But a framework built on an unexamined foundation cannot know what its foundation supports or how far its support extends. When relational quantum mechanics reaches the limits of its account of observer-relative quantum states, it cannot determine whether the limit is contingent — a feature of the specific relational structure it has adopted — or structural — a feature of what any relational account of quantum mechanics can achieve without a derivation of the ground. When social relational ontology faces the question of what individuates social actors as the specific actors they are, independently of the transactions that constitute them, it cannot resolve the regress from within its own resources without knowing what the structural ground from which actors emerge could provide. The thirteen traditions surveyed have laid open the full extent of what relation does. The formal derivation of the structural ground from which relation emerges would establish, for the first time, why it does what it does — and what lies beyond the horizon of every framework that has worked without that derivation.

The question as it now stands. The present investigation opened with a question that every reader, philosophical or general, will have recognised as genuinely important: what is a relation? Not what relations do. Not what they produce. Not how they structure reality. What they are.

That question has been transformed by the investigation. The reader who asks it now does not ask it as a naive inquiry whose answer is surely already known. They ask it as a formally bounded question: What is a relation, given that the structural ground from which the capacity for relation emerges has not been derived? What is a relation, given that the latent register from which relata, encounter, yield, and event structure all emerge as structural consequences remains, formally and rigorously, uncharted?

And the question opens, as it must, onto another: What is *before* relation? Not prior in time — time itself, the present investigation has established, is a structural consequence of the relational event and not its ground. Prior in structural order: the level from which the relational order itself emerges, presupposed by every foundational commitment any tradition has ever made, reached by no derivation any tradition has yet executed.

That is where the question still lives. That is where the answer waits.

APPENDIX A

Formal Statement of the Five Structural Requirements

The five structural requirements of relational ontology are formally defined in Chapter 2. The present appendix restates them with the maximum formal precision, as criteria against which any putative account of relational ontology can be assessed.

Definition A.1 (Level I: The Ground — Formal Specification)**.** An account satisfies the Level I requirement if and only if it provides a structural feature G such that:

(G1) *Totality*: G admits no exterior — there is no structural domain outside G from which additional structural resources could be imported into G's own dynamic.

(G2) *Generativity*: G produces structural differentiation from within its own dynamic — the diversity of relata and the character of their relational capacity are structural consequences of G's internal architecture, not additions to it from outside.

(G3) *Self-sustenance*: G requires no external principle to maintain its own structural existence or dynamic — G is, in the formal sense of the term, self-grounding.

(G4) *Formal derivability*: G is the unique structural result of the eliminative method applied to the presuppositions of relational ontology — G is derived, not stipulated; it is the terminus of elimination, not an arbitrary starting point.

Definition A.2 (Level II: Relational Identity — Formal Specification)**.** An account satisfies the Level II requirement if and only if it derives from Level I a formal account of the structural identity of relata such that:

(RI1) Each relatum is uniquely individuated by its specific position within the Level I ground's generative dynamic — no two relata are formally identical.

(RI2) The structural identity of each relatum is neither pre-given (independent of the ground) nor purely relational (exhausted by the relations the relatum contingently enters into) — it is a derived invariant that persists through encounter without being either unaffected by encounter or dissolved by it.

(RI3) The derivation of relational identity from Level I does not import the relational regress: the account does not require that the relatum have already been in relations in order to be individuated.

Definition A.3 (Level III: Encounter Source — Formal Specification)**.** An account satisfies the Level III requirement if and only if it derives from the Level I ground and the Level II account of relational identity a formal account of what brings specific relata into specific relational contact such that:

(ES1) The encounter source is a structural consequence of the same dynamic that produces the relata — it is not an external perturbation principle, a random event stipulated as primitive, or a necessitation so complete that it forecloses genuine novelty.

(ES2) The encounter source is structurally selective — not every relatum can enter into productive encounter with every other relatum, and the structure of this selectivity is formally derivable from the Level I and Level II results.

(ES3) The derivation of encounter source does not introduce any new structural principle not already present in the Level I and Level II derivations.

Definition A.4 (Level IV: Relational Yield — Formal Specification)**.** An account satisfies the Level IV requirement if and only if it derives from Levels I through III a formal account of the yield of relational encounter satisfying conditions (i)–(iv) of Definition 11.5 (Requirement R5): formal newness, formal boundedness, formal generativity, and asymmetric novelty.

Definition A.5 (Level V: Relational Event — Formal Specification (Revised))**.** A framework $\mathcal{F}$ has **entry level V** (relational event structure) if and only if $\mathcal{F}$ foundationally constitutes the formal organisation of the relational event — the internal structure of how encounter between relata is initiated, sustained, and oriented towards the production of relational yield — without deriving this organisation from conditions at level IV or below. Formally, $\mathcal{F}$ is at Level V if and only if:

(i) $\mathcal{F}$ specifies a formal account of the relational event's internal structure $\mathcal{E}$, characterising (a) the conditions of encounter initiation, (b) the conditions of encounter continuation, and (c) the structural relation between encounter continuation and relational yield;

(ii) $\mathcal{F}$ takes $\mathcal{E}$ as a foundational commitment: $\mathcal{E}$ is not derived within $\mathcal{F}$ from conditions derivable without presupposing $\mathcal{E}$; and

(iii) $\mathcal{F}$ does not derive conditions (i)(a)–(i)(c) from any structural account at Level IV (yield specification) or below.

Remark A.1. The self-exemplification sub-condition (iv) that appeared in earlier formulations of the Level V criterion — requiring that the account's own derivational method instantiate the event structure it specifies — has been removed from the assignment criterion and relocated. It is the content of Requirement R6(iv) (self-exemplification of derivational method), which is a condition on whether a framework *satisfies* R6, not a condition on whether a framework is *assigned* to Level V. The Level V assignment criterion asks: where does $\mathcal{F}$'s foundational commitment fall in the hierarchy? The R6(iv) satisfaction criterion asks: does $\mathcal{F}$'s derivational method instantiate the event structure it specifies? These are formally distinct questions. The revised definition makes Level V assignment formally determinate without importing an interpretive demand appropriate only to requirement-satisfaction.

Remark A.2. The five level requirements are jointly exhaustive of the structural commitments of relational ontology: any account that satisfies all five has formally discharged every structural obligation that taking relation as foundational imposes. They are also jointly minimal: removing any one of the five leaves the remaining four insufficient to constitute a complete relational ontology — either the ground remains uncharacterised, relata remain individuated by fiat, encounter source remains inexplicable, yield remains asserted rather than derived, or the event structure remains a black box. The five requirements are therefore both necessary and jointly sufficient.

APPENDIX B

The Structural Insufficiency Theorem: Full Statement and Proof

The Structural Insufficiency Theorem is stated and proven in Chapter 10. The present appendix restates the full formal apparatus — definitions, theorem, corollaries, and proof — in one place, for ease of reference and citation.

Definition A2.1 (*Structural Level*). A structural level is a formally specifiable domain of structural analysis characterised by a set of necessary and jointly sufficient conditions. A structural feature F *occupies level* n iff F satisfies the defining conditions of Level n and is not derivable from those of any level $m > n$. (See Definition 10.1.)

Definition A2.2 (*Latent Register*). The latent register is the structural domain below Level I: the totality of conditions presupposed by any foundational commitment to any feature at Level I or above, not themselves features of the structural order. (See Definition 10.2.)

Definition A2.3 (*Foundational Constitution*). A framework $\mathcal{F}$ foundationally constitutes F at level n iff: (i) F occupies level n in $\mathcal{F}$; (ii) F is not derived within $\mathcal{F}$ from conditions derivable independently of F; (iii) $\mathcal{F}$'s derivational resources are defined over features at level n or above. (See Definition 10.4.)

Definition A2.4 (*Structural Foreclosure*). $\mathcal{F}$ structurally forecloses level m iff $\mathcal{F}$ foundationally constitutes a feature at level $n > m$, and no derivational pathway within $\mathcal{F}$ satisfies the defining conditions of level m. (See Definition 10.5.)

Theorem A2.1 (*Structural Insufficiency*). Let $\mathcal{F}$ be any framework in relational ontology that foundationally constitutes a feature F at level n ($1 \leq n \leq 5$). Then $\mathcal{F}$ structurally forecloses the latent register.

Proof. By Definition A2.3, conditions (i) and (ii) specify that F occupies level n in $\mathcal{F}$ and that F is not derived within $\mathcal{F}$ from conditions derivable independently of F. We derive condition (iii) via the domain-adequacy of $\mathcal{F}$'s vocabulary (full argument: Chapter 10, Lemma 10.1, Step 2).

Define: $\mathcal{F}$'s vocabulary $\mathcal{V}_{\mathcal{F}}$ is *adequate to* a structural domain D iff the formal conditions associated with each term of $\mathcal{V}_{\mathcal{F}}$ yield determinate verdicts when applied to features in D. By conditions (i) and (ii), $\mathcal{F}$ has taken F at level n as its structural terminus and developed its apparatus to characterise F and its structural consequences. $\mathcal{V}_{\mathcal{F}}$ is therefore adequate to features at level n or above by construction. For features below level n: a derivational move within $\mathcal{F}$ requires that its conclusion be evaluable by $\mathcal{F}$'s validation criteria, which requires adequacy. $\mathcal{V}_{\mathcal{F}}$ is not adequate to features below level n (since $\mathcal{F}$ has not characterised those features). No derivational move within $\mathcal{F}$ produces an evaluable conclusion about features below level n.

Manufacturing below-n vocabulary would require characterising the below-n domain — stepping outside $\mathcal{F}$. Therefore $\mathcal{F}$'s derivational resources are defined over features at level n or above: condition (iii). The latent register is below Level I, hence below level n for all $n \geq 1$. Any derivational pathway within $\mathcal{F}$ reaching the latent register would require vocabulary adequate to a domain outside the domain over which $\mathcal{F}$'s resources are defined. No such pathway exists within $\mathcal{F}$. By Definition A2.4, $\mathcal{F}$ structurally forecloses the latent register. □

Remark B.1. The compressed derivation above records the structure of the full argument. The domain-adequacy account is developed in full at Chapter 10, Lemma 10.1, Step 2, which is the primary proof of record. Readers verifying Step 2 should consult the Chapter 10 version.

Corollary A2.1 (*Foreclosure of Lower Levels*). If $\mathcal{F}$ foundationally constitutes a feature at level n, then for each $m \in \{1, \ldots, n-1\}$, $\mathcal{F}$ structurally forecloses level m. *(Proof as Theorem A2.1.)*

Corollary A2.2 (*Eliminative Corollary*). A derivation D reaches the latent register iff D proceeds by the eliminative method: systematic derivation of each structural feature from conditions more primitive than the feature itself, without foundational constitution of any non-latent feature. *(Proof: as Corollary 10.2.)*

APPENDIX C

The Eight Requirements and Their Joint Satisfiability

The eight requirements are formally stated in Chapter 11, Definitions 11.1 through 11.8, where the reason for exactly eight is also derived (see Chapter 11, opening section). The present appendix provides a consolidated reference table and a schematic representation of their derivational relationships.

Req.	What it demands	Why no tradition meets it
R1	Ground derived by eliminative method; no foundational constitution	Spinoza meets G1, G3, G4 (qualified); all others foundationally constitute non-latent feature
R2	Pre-temporal latent register; temporal order derived as consequence	Every tradition begins within time or takes temporal features as given
R3	Relational individuation without regress; relata derived from ground	Pre-given units (Whitehead); structural dissolve without re-derivation (OSR)
R4	Encounter source derived from same dynamic producing relata	External principle (Whitehead's God); question deferred (RQM)
R5	Yield formally new, bounded, generative, and asymmetrically novel	Asserted as property of occasions (Whitehead); not addressed (Spinoza, Bergson, Heidegger)
R6	Co-instantaneous relational event; internal structure derived; event-level account self-exemplifying	Asserted, not derived (Whitehead); not addressed (all others)
R7	Temporal asymmetry derived from relational event	Presupposed as given (Bergson, Heidegger); not addressed (all others)
R8	Derivation self-exemplifying	Requires R1–R7 as prerequisite; not attempted in any tradition

The non-contradiction proof of Theorem 11.1, supplemented by the schematic model $\mathcal{M}$ constructed from what R1 through R8 independently specify, establishes joint satisfiability without constituting a positive characterisation of the latent register. $\mathcal{M}$ is a logical witness — it demonstrates that the eight requirements admit a common realisation, derived entirely from the requirements themselves — not a completed derivation. The positive derivation that would give $\mathcal{M}$'s elements their precise structural content — and thereby derive each of R1 through R8 not merely as consistent with each other but as jointly necessitated by the eliminative method — is the object of the Stage II investigation.

APPENDIX D

Diagnostic Failures and Their Formal Requirements

Each of the eight requirements derived in Chapter 11 is the structural complement of a specific diagnostic failure documented in Part II. The table below maps every requirement to its originating failure and to the traditions in which that failure is most fully documented. The mapping is not merely illustrative: each requirement is the unique structural consequence of its corresponding failure type, derived by the complement procedure described in Section 11.1. Traditions are listed by the name used throughout Part II; full chapter and section references appear in the text.

Table D.1: Structural Failures and the Requirements They Generate

Req.	**Structural failure type**	**What the failure forecloses**	**Traditions documenting it**
R1	Foundational constitution of the Level I ground: the ground is posited as a primitive terminus rather than derived as the unique result of the eliminative method	Any derivation of the ground as a necessary structural consequence; the ground is stipulated rather than reached	Spinoza, Whitehead, Bergson, Heidegger, OSR, RQM, Stengers, Emirbayer, Kant, Simondon, Deleuze, Schaffer, Fine
R2	Temporal presupposition at the entry level: the framework begins within the temporal or relational order and does not establish what is structurally prior to that order's being in operation	Derivation of the temporal and relational orders as structural consequences of a pre-temporal dynamic; every tradition inherits temporal structure as a given operating condition	Whitehead, Bergson, Heidegger, OSR, RQM, Stengers, Emirbayer, Kant, Simondon, Deleuze

(continued)

Req.	Structural failure type	What the failure forecloses	Traditions documenting it
R3	Level II failure: relata are either pre-given as primitive units or dissolved entirely into the relational web, generating the relational regress in either direction	Derivation of structurally distinct relata from the ground's own dynamic without presupposing them and without dissolving their identity into the relations they enter	Whitehead, OSR, RQM, Emirbayer
R4	Level III failure: the source of encounter is either imported as an external perturbation principle or left formally unaddressed	Derivation of encounter source as a structural consequence of the same dynamic that produces the relata, without external agent and without stipulated randomness	Whitehead (God as initial aim), RQM (interaction specified, source deferred), Spinoza, Bergson, Heidegger

(continued)

Req.	Structural failure type	What the failure forecloses	Traditions documenting it
R5	Level IV failure: the yield of relational encounter is asserted as an intrinsic property of the encounter type rather than derived from the specific structural character of the relata and their encounter	Derivation of yield as formally new, bounded, generative, and asymmetrically novel — a result uniquely indexing the particular relata and their particular encounter	Whitehead (novelty given as character of occasions), Spinoza (modes follow by logical entailment, not dynamic yield), Bergson, Heidegger, Simondon
R6	Level V failure: the internal structure of the relational event is treated as a given phenomenological or formal feature rather than derived; the dimensions and their co-instantaneity are described, not accounted for	Derivation of the event's co-instantaneous internal structure from the prior derivation; self-exemplification of the event-level account	Whitehead (phases of concrescence asserted, not derived), Bergson, Heidegger, all others

(continued)

Req.	**Structural failure type**	**What the failure forecloses**	**Traditions documenting it**
R7	Cross-cutting temporal presupposition: temporal asymmetry is presupposed as a feature of the structural domain within which the framework operates, not derived from a structural dynamic more primitive than the temporal order itself	Derivation of temporal asymmetry — the irreversibility of before and after — as a consequence of the relational event's internal organisation and the irreversible deposition of yield	Bergson (most deeply), Heidegger (most formally), all traditions operating within the temporal order
R8	Cross-cutting self-exemplification failure: the framework's derivational method does not instantiate the structural dynamics it formally establishes; the account is delivered from a standpoint external to the dynamic it describes	A derivation exhibiting method continuity, sequence self-instantiation, and no residual methodological posit: the derivation's own formal sequence enacts what it formally establishes	All thirteen traditions; documented systematically in Table 11.1

Reading the table. The "Structural failure type" column names the failure as identified in Part II; the "What the failure forecloses" column names the structural operation that the failure makes impossible from within the framework that exhibits it; the "Traditions" column lists the traditions in which the failure is most extensively documented, not every tradition in which it is present. R1 and R2 appear in all thirteen traditions because the SIT establishes them as universal: any framework satisfying the theorem's antecedent conditions fails both, regardless of entry level. R3 through R6 are level-specific and appear in the traditions whose entry level makes the corresponding structural operation most directly legible. R7 and R8 are cross-cutting; the table lists those where the diagnostic treatment is most extended.

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